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Symbols

⧫	*concept* — a term and its definition
⊙	*claim* — a single truth-apt proposition
⊢	*argument* — premises leading to a conclusion
★	*position* — a named standpoint or view
?	*question* — an open issue under debate

List of Entries

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A

⊙ A-posteriori necessities as misdescribed possibilities

Kripkean a-posteriori necessities are misdescribed genuine possibilities, not conceived impossibilities

Claim · after Chalmers

Kripkean a-posteriori necessities are not cases in which ideal conceiving reaches a genuine impossibility; they are genuine *possibilities* described in a misleading way. This is the two-dimensional diagnosis of the standard counterexamples to "conceivability entails possibility" (A-POSTERIORI NECESSITIES, from Kripke's *Naming and Necessity* KRIPKE — NAMING AND NECESSITY (1980)), defended by David Chalmers (CHALMERS — THE CONSCIOUS MIND (1996); CHALMERS — ON SENSE AND INTENSION (2002)).

When one "conceives water that is not H_2O," one conceives a *genuinely possible* scenario — some watery stuff that is not H_2O — and merely *misdescribes* it as "water." Distinguish a term's *primary* (epistemic) PRIMARY INTENSION from its *secondary* (subjunctive) SECONDARY INTENSION: the conceived scenario is possible under the primary intension, while the necessity attaches only to the secondary. So the a-posteriori necessities are not impossibilities reached by ideal conceiving — they are possibilities wearing a misleading label.

This is a claim about the *modal status* of the standard counterexamples, not a denial that they are necessary. Its payoff is to preserve the link between **ideal primary conceivability** and **primary possibility** by explaining away every apparent defeater as a description error — the engine of MODAL RATIONALISM.

The position that denies it is type-B physicalism (see TYPE-A / TYPE-B / TYPE-C MATERIALISM), which holds that some a-posteriori necessities are *strong*: no possible world even verifies the primary intension, so there is nothing being misdescribed — just an impossibility conceived. The claim is distinct from A-POSTERIORI NECESSITIES, which it reinterprets rather than denies: that claim says such necessities exist; this one says they do not break the conceivability→possibility link once the two intensions are distinguished.

SEE ALSO

- ▷ A-POSTERIORI NECESSITIES — the Kripkean necessities this claim reinterprets.
- ▷ PRIMARY INTENSION / SECONDARY INTENSION — the distinction that turns an apparent impossibility into a misdescribed possibility.
- ▷ MODAL RATIONALISM — the argument this claim is the load-bearing premise of.
- ▷ TYPE-A / TYPE-B / TYPE-C MATERIALISM — type-B physicalism, which resists the diagnosis by positing *strong* necessities.

⊢ A-posteriori-necessity objection

Conceivability is not a reliable guide to possibility (the a-posteriori-necessity objection)

Argument

Conceivability is not, in general, a reliable guide to metaphysical possibility — so the bridge the zombie argument leans on cannot simply be assumed. The objection turns a celebrated result in the metaphysics of modality against the ZOMBIE ARGUMENT: since Saul Kripke's work on naming, philosophers have widely accepted that some truths are *necessary* yet knowable only *a posteriori*, and in every such case a coherent-seeming scenario is conceivable even though it is metaphysically impossible.

The move is a counterexample to a general principle:

1. (A-POSTERIORI NECESSITIES, from Kripke's *Naming and Necessity* KRIPKE — NAMING AND NECESSITY (1980)) There are necessary truths knowable only a posteriori — water is H_2O, heat is molecular motion, Hesperus is Phosphorus — whose negations are conceivable yet metaphysically impossible.

∴ Conceivability does not in general entail possibility: CONCEIVABILITY ENTAILS POSSIBILITY fails as a universal bridge.

Applied to consciousness, this is the **type-B physicalist** strategy (see TYPE-A / TYPE-B / TYPE-C MATERIALISM): a phenomenal–physical identity ("pain *is* such-and-such neural state") could itself be a *strong* a posteriori necessity. If so, a Philosophical zombie is conceivable but metaphysically impossible, and the Zombie argument is cut at its bridge — without conceding any non-physical fact.

The crux: is consciousness like water?

The **weakest premise** is the tacit one: that the *phenomenal* case is relevantly like the *water* case. The two-dimensional rejoinder (Modal rationalism, built on A-posteriori necessities as misdescribed possibilities) replies that every Kripkean necessity is a *misdescribed possibility*, and that for phenomenal terms the primary and secondary intensions *coincide* — there is no appearance/reality wedge to exploit — so the water-style escape is unavailable for zombies. Whether the phenomenal is genuinely disanalogous to water is where the dispute lives.

This should not be confused with Phenomenal-concept deflation: that explains away the *appearance* of an explanatory gap through the architecture of phenomenal concepts; the present move makes the purely *modal* point that a-posteriori necessities already break conceivability→possibility, independently of any account of phenomenal concepts.

> The whole zombie argument needs the rule "if you can coherently imagine it, it's really possible." This objection says that rule is just false. Take "water is H_2O": for centuries people could coherently imagine water turning out to be something other than H_2O — yet it is impossible, because water *just is* H_2O, a fact we had to discover rather than deduce. So imaginability isn't proof of possibility. Maybe consciousness is the same: a zombie feels imaginable only because we haven't yet seen that experience *just is* some physical process. The standard comeback, due to David Chalmers (see Chalmers — The Conscious Mind (1996)), is that the water trick works only because "water" hides a surface/essence gap — and feelings have no such gap, so the trick cannot be repeated for zombies.

See also

- A-posteriori necessities — the premise: Kripkean necessities whose negations are conceivable yet impossible.
- Conceivability entails possibility — the bridge this argument attacks.
- Zombie argument — the argument severed if the objection holds.
- Modal rationalism — the two-dimensional rejoinder that the Kripke cases are misdescribed possibilities, not genuine counterexamples.
- A-posteriori necessities as misdescribed possibilities — the claim that rejoinder rests on.
- Type-A / Type-B / Type-C materialism — type-B physicalism, the position this objection arms.
- Philosophical zombie — the case alleged to be conceivable-but-impossible.
- Phenomenal-concept deflation — the allied but distinct phenomenal-concept strategy.

⊢ Ability hypothesis

The Ability Hypothesis — experience teaches know-how, so Mary learns no new fact

Argument · after Lewis

The *Ability Hypothesis* is David Lewis's reply to the Knowledge Argument, also defended by Laurence Nemirow. It holds that knowing what an experience is like is not a piece of factual knowledge at all, but a bundle of *abilities*: to recognise the experience, to imagine it, and to remember it. If that is right, then when Mary the colour scientist, who knows every physical fact about colour vision but has never seen red, finally sees it, she gains know-how rather than a new fact, and the inference from her learning something to the falsity of physicalism is blocked.

The argument

The attack lands on the Knowledge Argument's verdict step: the critical question (CQ2 of the thought-experiment scheme) of how to *read* the agreed intuition that Mary learns something. The intuition is robust, but reading it as *acquiring a new propositional fact* is theory-laden, and the verdict survives under a deflationary reading that the original argument cannot use. Given an analysis of "knowing what it is like", the inference is deductive:

1. Knowing what an experience is like consists in possessing certain abilities: to recognise the experience when it occurs, to imagine it, and to remember it (Lewis 1988; Nemirow 1990).
2. On first seeing red, what Mary gains is exactly this: knowing what seeing red is like.
3. Abilities are not propositional knowledge; gaining one is acquiring a skill, not learning a truth.

∴ Mary gains no new fact. So the step from her epistemic progress to truths about experience that no physical description yields is blocked: there is no *truth* she lacked.

What it concedes

The move is notable for conceding both premises of the Knowledge Argument as written. It grants that Mary knew all the physical facts outright, and that she learns something on release under the know-how reading. What it severs is the inference between them. "Mary learns something" is true; "Mary learns a new fact" is what the conclusion needs; and the case cannot fund the stronger reading without begging the question against physicalism.

The weak point

The most disputed premise is the first, the ability analysis itself. Three counters are standard. Mary seems able to *use* her new knowledge in reasoning and inference ("seeing orange must be like *this*, only more yellowish"), which bare abilities do not explain. One can apparently have the abilities without the knowledge (a distracted imaginer who can summon red but is not thereby informed), and arguably the knowledge while lacking the imaginative ability (someone paralysed of imagination). And the acquaintance theorist offers a rival non-factual analysis, that what Mary gains is acquaintance rather than a fact, so even granting that no new fact is learned, the ability analysis is not the only option.

> This is the most famous escape from the Mary story (the Knowledge Argument: since Mary, who knew all the physics of colour, still learns something when she first sees red, physics must be incomplete). The escape turns on an ambiguity in "learning". Learning a *fact* (that Paris is the capital of France) is one thing; picking up a *skill* (juggling) is another. On this view Mary picks up skills, namely recognising, imagining, and remembering red. No fact was missing from her books, so nothing was missing from physics. The catch: her new grasp of red seems usable in actual reasoning, the way facts are and skills are not, so perhaps it is not *just* a skill.

See also

- ▹ Knowledge argument (Mary's room) — the argument this one attacks: that Mary learns a fact the physical description omits.
- ▹ Knowledge argument — the Mary thought experiment itself.
- ▹ Mary gains abilities — the conclusion: Mary acquires abilities, not a new fact.
- ▹ Mary gains acquaintance — the rival non-factual reply: acquaintance rather than know-how.
- ▹ Non-deducible experiential truths — the anti-physicalist conclusion this attack is meant to block.
- ▹ Lewis — What Experience Teaches (1988) — Lewis's "What Experience Teaches" (1988), the primary statement.
- ▹ Nemirow — Physicalism and the Cognitive Role of Acquaintance (1990) — Nemirow's independent defence of the ability analysis.

☉ Absence of an experiential threshold

There is no non-arbitrary point in nature at which experience first appears

Claim · after Nagel

If experience were confined to brains, or to life, or to systems above some threshold of complexity, there ought to be a principled place where it first switches on — a line such that below it there is nothing it is like to be the system and above it there is. No such non-arbitrary line is in view. The relevant differences across any proposed boundary look like differences of degree and organisation, not the kind of difference that could mark the abrupt entry of an entirely new ontological feature into the world.

This is the premise of the continuity argument for panpsychism: absent a principled threshold, the parsimonious option is that experience, in some attenuated form, reaches all the way down. It is denied by emergentists and physicalists who hold there *is* a principled threshold — a level of integrated information or of functional organisation at which experience genuinely, if strongly, emerges (see Acceptability of strong emergence). It differs from No radical emergence: that claim says experience could not brutely *appear* at all from the non-experiential, whereas this one says that even granting it could, there is no non-arbitrary *place* to locate its first appearance.

See also

- ▹ Continuity argument — the argument this premise drives.
- ▹ Acceptability of strong emergence — its denial: there *is* a principled emergence threshold.
- ▹ No radical emergence — the neighbouring premise about *whether*, not *where*, experi-

ence could emerge.

☉ Acceptability of strong emergence

Strong emergence of experience from non-experiential matter is acceptable

Claim

Experience can arise as a genuinely novel, fundamental feature at some level of physical organisation — a brain, say — without being present, even in proto-form, in that organisation's parts. On this view *strong* emergence (the appearance of properties not grounded in the powers of the base) is a coherent and even expected feature of a layered natural world, not the magic its critics allege. The dispute is specifically about strong emergence of the *phenomenal*; *weak* emergence, where higher-level patterns are fully grounded in and predictable from the base, is accepted by almost everyone and is not at issue here.

This is the standard physicalist denial of NO RADICAL EMERGENCE, and so the load-bearing premise of the EMERGENCE REPLY that undercuts the genetic argument for panpsychism: if strong emergence is acceptable, the genetic argument's choice between fundamental experience and unintelligible magic is a false dilemma. It is denied by panpsychists, who hold that strong emergence of the phenomenal is unintelligible (Strawson), and by those who think calling experience "strongly emergent" merely relabels the explanatory gap rather than closing it.

SEE ALSO

- ▷ NO RADICAL EMERGENCE — the claim this one denies.
- ▷ EMERGENCE REPLY — the objection this premise drives against the genetic argument.
- ▷ GENETIC ARGUMENT — the argument that the acceptability of strong emergence is meant to undercut.

♦ Access consciousness

Concept · after Block

Access consciousness is the functional sense of "conscious": a mental state is access-conscious when its content is *available* to the rest of the mind, poised for use in reasoning, for the direct rational control of action, and for verbal report. The term is Ned Block's (he abbreviates it *A-consciousness*). It is a matter of what a system can *do with* a state's content, settled entirely by the state's role in the cognitive economy and not at all by how, or whether, it feels.

A simple test fixes the idea: when you read this sentence, its meaning is poised for you to report, question, and reason with, and that availability is the sentence's access-consciousness. Theories of consciousness in the *global workspace* family (see GLOBAL WORKSPACE THEORY) effectively identify being conscious with being access-conscious in this sense, broadcast widely enough across the system to be used.

Access consciousness is defined by contrast with PHENOMENAL CONSCIOUSNESS, the felt or "what-it-is-like" side of experience. Block's central claim is that the two can come apart in both directions: a system might have the availability with no accompanying feel, and experience might even *overflow* what gets globally broadcast. That dissociation is set out at ACCESS VS. PHENOMENAL CONSCIOUSNESS. Keeping the two apart is the standard guard against treating a system's fluent self-report, an access-style capacity, as evidence that it *feels* anything.

> Calling a thought "access-conscious" just means the information is *available* to the rest of your mind, so that you can act on it, talk about it, and reason with it. It says nothing about whether the thought *feels* like anything. That is the whole point of the label: it lets us separate "the system can use and report this" from "there's something it's like to have it", so we don't slide from a machine describing its own inner states to assuming it must therefore feel them.

SEE ALSO

- ▷ PHENOMENAL CONSCIOUSNESS — the felt sense of consciousness it is contrasted with.
- ▷ ACCESS VS. PHENOMENAL CONSCIOUSNESS — Block's distinction and the case that the two dissociate in both directions.
- ▷ GLOBAL WORKSPACE THEORY — a theory that ties consciousness to access (global broadcast).
- ▷ FIRST-PERSON ACCESS — privileged first-person access, a distinct, epistemic notion.

♦ Access vs. phenomenal consciousness

Access consciousness vs. phenomenal consciousness

Concept · after Block

Access consciousness and *phenomenal consciousness* are two distinct things the single word "consciousness" can mean, separated by Ned Block ("On a Confusion about a Function of Consciousness", 1995). A state is *access-conscious* when its content is available to the rest of the mind — poised for use in reasoning, for the direct rational control of action, and for verbal report. A state is *phenomenally conscious* when there is something it is like to be in it — when it carries a felt, qualitative character. The first is a matter of what a system can *do with* a state's content; the second, of how the state *feels* from the inside.

The two senses

ACCESS CONSCIOUSNESS (*A-consciousness*) is a functional, information-availability notion: a content is A-conscious to the degree that it is *poised* to be drawn on by the systems that reason, decide, and speak. Whether a state qualifies is settled entirely by the role its content plays in the cognitive economy — what downstream processes can reach and use it — and by nothing about how it feels. When you read this sentence, its meaning is poised for report and inference; that availability is its access side.

Phenomenal consciousness (P-consciousness) is the "what-it-is-like" aspect of a mental state: the redness of red, the hurt of a pain, the timbre of a heard note. It is the felt or qualitative character of experience, and it is the property at issue in the HARD PROBLEM of consciousness — the question of why any physical process should be accompanied by felt experience at all. The particular look of these black marks on the page, over and above your ability to report and reason about the words, is their phenomenal side.

Why the distinction matters

The point of drawing the line is that the two can come apart, at least in principle, in *both* directions. A system might carry richly reportable models of its own inner states — answering fluently about what it is "noticing" or "feeling" — while there is nothing it is like to be that system at all: access without any accompanying feel. Block argues the converse is possible too, that phenomenal consciousness may *overflow* access, so that experience holds more than ever gets globally broadcast for report. The canonical illustration is a brief flash of a full array of letters: one seems to see the whole array vividly, yet can read out only a few before the image fades — as though the felt richness outruns what access captures. So "has A" neither entails nor is entailed by "has P."

This independence is the pivot on which several moves in the debate over machine introspection turn. It marks the central danger: mistaking a system's *functional, access-style* capacity — its ability to report on its own states, as some current language models do to a limited and unreliable degree (see LLM FUNCTIONAL INTROSPECTION) — for evidence of a *phenomenal* inner life. It also yields a subtler observation: *posing* the hard problem from the first-person side is itself an access-dependent act, requiring access to a representation of one's own phenomenal-seeming states (see POSING THE GAP REQUIRES ACCESS), even though merely *having* phenomenal states requires no such access.

> "Conscious" runs together two different things. One is **access**: the information is available to the rest of your mind — you can reason with it, act on it, and say it out loud. The other is **feel**: there's something it's like to undergo the state (the redness of red, the hurt of pain). They're not the same. You could imagine a machine that has all the *access* — it reports its inner states fluently — with no *feel* at all; and some philosophers think the reverse can happen too, that we feel more than we can ever report. Keeping the two apart is the main guard against a tempting mistake: seeing a system talk about its own insides and concluding it must therefore *feel* something.

See also

- ▹ ACCESS CONSCIOUSNESS — the functional half of the distinction, treated on its own.
- ▹ PHENOMENAL CONSCIOUSNESS — the felt half of the distinction, treated on its own.
- ▹ HARD PROBLEM — the felt, phenomenal side is exactly what this problem is about; access is precisely what it is *not*.
- ▹ LLM FUNCTIONAL INTROSPECTION — a candidate case of access-style self-report in machines, and the prime spot where it must not be read as the phenomenal kind.
- ▹ POSING THE GAP REQUIRES ACCESS — that stating the hard problem is itself an access-dependent act, even though undergoing phenomenal states is not.

⊢ Acquaintance hypothesis

The Acquaintance Hypothesis — Mary meets the experience rather than learning a truth

Argument · after Conee

The *acquaintance hypothesis* answers the KNOWLEDGE ARGUMENT by reclassifying what Mary gains: the felt verdict that "Mary learns something" on first seeing red is genuine, but it tracks a change of *relation*, not a change in *information*. The move is deductive, given a taxonomy of knowledge. It raises thought-experiment critical question CQ2 (verdict) against the knowledge argument from a different angle than the ability hypothesis, which reads the same verdict as a gain of skill.

The inference

1. Knowledge comes in at least three irreducible kinds: knowledge of facts, know-how, and *knowledge by acquaintance* — a maximally direct cognitive relation to a thing (Earl Conee, Conee — Phenomenal Knowledge (1994); the category descends from Bertrand Russell's distinction between knowledge by acquaintance and knowledge by description, Russell — Knowledge by Acquaintance and Knowledge by Description (1911)).
2. By stipulation (see Mary knows all the physical facts) Mary lacks no physical fact; what she lacks before release is acquaintance with the experience of red.
3. Coming to be acquainted with a thing is not coming to know a new truth about it — one can be fully informed about a thing one has never encountered.

Therefore her post-release gain is acquaintance, and the inference to truths about experience that physics cannot deduce fails: no truth was ever beyond her.

Compared with its neighbours

Against the ability theorist, the argument does not require that Mary gain any reusable skill — premise 1 keeps acquaintance distinct from know-how. Against the old-fact/new-guise theorist (see old fact, new guise) it does not even require that Mary acquire a new *concept*: the change is in her relation to the experience, not in her conceptual repertoire.

Weakest premise

Premise 3. The proponent of the knowledge argument replies that acquaintance with the experience of red immediately puts Mary in a position to know truths she could not know before ("seeing red is like *this*"). If acquaintance is knowledge-yielding, it is not factually innocent, and the missing truths reappear; if it yields no truths, calling it "knowledge" looks honorific, and the hypothesis risks relabelling the datum rather than explaining it.

> The Mary story (see the knowledge argument) says: she knew all the facts, yet gained something on seeing red — so some facts aren't physical. This reply: what she gained wasn't a fact at all but an *encounter*. Knowing every fact about a city isn't the same as standing in it; arriving doesn't add a line to the guidebook. Mary finally stands in front of red. The worry about this reply: the moment she's there, can't she now state truths she couldn't before ("so *this* is what red is like")? If yes, the missing facts sneak back in.

See also

- ▹ Knowledge argument (Mary's room) — the argument this reply rebuts, by reclassifying Mary's gain as a relation rather than a fact.
- ▹ Ability hypothesis — the sibling reply that reads the same verdict as a gain of skill, not acquaintance.
- ▹ Old fact, new guise — the reply that grants a new *concept* of an old fact; acquaintance needs not even that.

⊢ Age of Empires II argument

If LLMs have human-like attributes, then so does Age of Empires II

Argument · after Dewynter

The **Age of Empires II argument** (Adrian de Wynter, de Wynter — If LLMs Have Human-Like Attributes, Then So Does Age of Empires II (2026)) turns a joke into a methodological warning. De Wynter builds and trains a perceptron inside the videogame (see Age of Empires II circuits) and asks what follows for the human-like attributes — understanding, morality, sentience — that researchers routinely ascribe to large language models. His conclusion is deliberately deflationary and, unusually for this neighbourhood, *not* a claim about consciousness at all: it is that such ascriptions are empirically non-unique, and that assuming them in advance makes any study of them circular or uninformative.

Inference form: not a deductive proof of a metaphysical thesis but a methodological reductio plus a constructive existence result.

1. **Substrate freedom (constructive).** The computation an LLM performs can be built and trained in any sufficiently powerful substrate — silicon, LEGO, the Greater Boston Area, or *Age of Empires II* (see Perceptron

in any substrate, established by the actual build).

2. **Non-uniqueness.** Therefore any attribute inferred from an LLM's behaviour is shared by every other realization of that computation. Outputs are substrate-invariant; the *interpretation* of those outputs as understanding or morality is not — it travels with the observer and the medium.
3. **Circularity (epistemic).** To assume, a priori and substrate-independently, that the attribute is present (or absent) and then read experiments through that assumption is to get the assumption back. The defect is symmetric across viewpoints and across "exists / does not exist" outcomes (see Circularity of a priori anthropomorphism).
4. ∴ The anthropomorphic attributes ascribed to LLMs are empirically non-unique, and claims about them are empty until tied to explicit measurement criteria (see Non-uniqueness of LLM attributes).

The null assumption. The constructive upshot de Wynter recommends is a *default of non-uniqueness*: set up experiments assuming an LLM is **not** special relative to other substrates, and require explicit, substrate-neutral measurement criteria before any anthropomorphic attribute is read in. Otherwise, in his phrase, "the interpretation is left to the representation."

Distinct from Searle and Block. The argument shares the gambit of realizing a mind-like computation in an absurd medium with the Chinese nation, but it lands somewhere else entirely. Block and Searle run the absurd substrate to the conclusion that there is *no genuine mind/understanding present* — a thesis about phenomenal consciousness or intentionality. De Wynter draws no such conclusion; he explicitly declines to say AoE2 has or lacks human-like attributes. His target is the *epistemics of attribution*: how, if at all, behavioural evidence can support attribute-talk. So this argument can be accepted by someone who thinks LLMs do have rich inner lives and by someone who thinks they have none — it constrains how either party may argue, not which is right. It therefore reframes rather than answers questions like whether LLMs have mental states and the broader Moral status of LLMs.

Weakest premise: the inference at step 2 from substrate-freedom to the *non-uniqueness of attributes*. A functionalist opponent may insist that the relevant attributes are precisely the substrate-invariant ones — the input-output-and-internal-role profile — in which case realizing that profile in goats would carry the attribute along rather than dissolve it, and "the interpretation" would not be a free parameter at all. The force of the argument then rests on showing that the contested attributes (understanding, morality, sentience) are *not* among the substrate-invariants but live in the interpretive surplus — which is itself a substantive claim about what those attributes are.

> Someone built a working AI "brain cell" out of goats in a videogame. The lesson is not "so the goats are secretly conscious" and not "so the AI is a fake". It is sharper and more modest: the exact same computation can run on goats, so anything you claim to *see* in a chatbot — that it understands, that it cares — can't be peculiar to the chatbot; you'd have to say the goats do it too, which nobody will. And if you simply decide up front that the chatbot understands (or that it doesn't) and then interpret your tests that way, you have proven nothing but your starting assumption. The fix: before testing, state plainly what would count as evidence of "understanding" in a way that doesn't depend on whether the thing is made of chips or goats. Until then, assume the chatbot is nothing special. Notice this verdict leaves the big question — does it *really* understand? — wide open; it only polices how you are allowed to argue about it.

See also

- ▹ Age of Empires II circuits — the trained-perceptron-in-a-videogame construction this argument runs on.
- ▹ Non-uniqueness of LLM attributes — the conclusion: such attributes are not unique to the LLM substrate.
- ▹ Perceptron in any substrate — premise 1, the constructive substrate-freedom result.
- ▹ Circularity of a priori anthropomorphism — premise 3, the circularity of a priori attribution.
- ▹ Chinese nation — the absurd-substrate ancestor, aimed at absent qualia rather than at the methodology of attribution.
- ▹ LLMs have mental states and Moral status of LLMs — the debates this argument reframes without taking a side.
- ▹ de Wynter — If LLMs Have Human-Like Attributes, Then So Does Age of Empires II (2026) — de Wynter's note and

paper.

Age of Empires II circuits

Age of Empires II circuits (a trained perceptron built in a videogame)

Concept · after Dewynter

The *Age of Empires II circuits* are a working digital computer assembled inside the 1999 real-time strategy videogame, devised by Adrian de Wynter (see DE WYNTER — IF LLMS HAVE HUMAN-LIKE ATTRIBUTES, THEN SO DOES AGE OF EMPIRES II (2026)). Goats walking along rails of grass and bridge stand for the 0s and 1s of a signal; patches of ice meter their timing to prevent race conditions; and from these pieces de Wynter builds a NAND gate — the single operation from which all of Boolean logic can be assembled. On top of the gates he constructs, and then *trains*, a one-bit perceptron, the elementary learning unit from which neural networks (and so large language models) are built. A side-product of the construction is a proof that *Age of Empires II* is **functionally complete** and **Turing-complete**: anything a computer can compute, the game can be made to compute.

WHAT IT IS FOR

Unlike most demonstrations that a game is Turing-complete, the point here is philosophical. The perceptron is the same computational object whether it runs on silicon, on paper, or on goats; de Wynter uses the videogame implementation as a deliberately absurd substrate to pry computation apart from its medium. The construction then becomes the engine of an argument about how we may and may not ascribe human-like attributes to language models (see AGE OF EMPIRES II ARGUMENT): if the same trained network can live in *Age of Empires II*, then attributes read off an LLM's behaviour are not unique to the LLM.

DISTINCT FROM THE CHINESE NATION

The circuits are an implemented, trainable cousin of Block's CHINESE NATION, but they are not the same move. The Chinese Nation is an *imagined* population realizing a mind's functional organisation, wielded as an intuition pump for *absent qualia* — the suspicion that organisation without the right substrate yields no experience. De Wynter's machine is *actually built and trained*, and it is aimed not at the metaphysics of consciousness but at the **methodology of attribution**: he draws no conclusion that the game lacks (or has) an inner life, and explicitly denies that AoE2 has human-like attributes. The bite is epistemic, not phenomenal. It is also distinct from Block's LOOKUP-TABLE MIND, which produces the right *outputs* with no inner processing; the perceptron here does the genuine processing, learning included.

> Someone built a real, working computer inside the videogame *Age of Empires II* — using herds of goats as the 1s and 0s and sheets of ice to keep them in step — and on top of it built and taught a tiny artificial "brain cell" of the kind that large AIs are made from. The serious point hiding in the joke: the exact same little learning machine that powers a chatbot can be made out of goats. So when people say a chatbot "understands" or "has morality", you have to ask whether that is really in the machine or just in how we read it — because you would never say the goats understand anything. This is a cousin of the famous CHINESE NATION thought experiment, but its punchline is about how we *measure* such claims, not about whether the goats are secretly conscious.

SEE ALSO

- AGE OF EMPIRES II ARGUMENT — the argument the construction powers: anthropomorphic attributes of LLMs are substrate-non-unique, so assuming them a priori is circular or uninformative.
- CHINESE NATION — the imagined-population ancestor; aimed at absent qualia where this is aimed at the methodology of attribution.
- LOOKUP-TABLE MIND — right outputs with no processing; the circuits, by contrast, do the real computation and learning.
- FUNCTIONALISM — the multiple-realizability of computation the construction dramatizes.
- DE WYNTER — IF LLMS HAVE HUMAN-LIKE ATTRIBUTES, THEN SO DOES AGE OF EMPIRES II (2026) — the note and paper where the circuits and the proof are presented.

AI effect

The AI effect

Concept

The *AI effect* is the recurring pattern by which a task once taken to require intelligence is downgraded, the moment a machine performs it, to "mere computation" — brute search, look-up, statistics — so that *real* intelligence is perpetually redefined as whatever machines cannot yet do. Pamela McCorduck gave the classic formulation (see MCCORDUCK — MACHINES WHO THINK

(1979)): "every time somebody figured out how to make a computer do something ... there was a chorus of critics to say, 'that's not thinking."' The same idea is captured in **Tesler's Theorem**, attributed to Larry Tesler: *"Intelligence is whatever machines haven't done yet."*

The pattern is historically robust. Mechanical arithmetic, chess, checkers, theorem-proving, expert diagnosis, *Jeopardy!* (Watson), Go (AlphaGo), and now fluent language (large language models) were each, in their turn, treated as a hallmark of intelligence — and each, once automated, widely reclassified as "not really thinking": just search, just statistics, "just autocomplete." The frontier of "genuine" AI recedes exactly as fast as the field advances.

Two readings

The effect can be read as a fallacy or as a discovery, and the dispute between these readings is where its interest lies.

- **The debunking reading:** the AI effect is goalpost-moving — a *no-true-Scotsman* manoeuvre about intelligence. Critics protect human exceptionalism by quietly amending the definition of "thinking" to exclude whatever has just been mechanised, so the thesis "machines can't really think" is rendered unfalsifiable.
- **The deflationary reading:** each automation genuinely *teaches* us that the task did not require general intelligence after all. That chess fell to search rather than insight was a real finding about chess, not a dodge; "that's not thinking" can be a correct lesson about how little a given feat demanded, not a refusal to face the evidence. On this reading the receding frontier is the steady subtraction of things intelligence *isn't*.

A balanced view grants both: the effect sometimes masks legitimate discovery and sometimes masks motivated denial, and telling which is itself the substantive question in each case.

What it bears on

The AI effect is, in large part, the serial denial of strong AI — and of "real" understanding — to each new system as it arrives. It is the sociological shadow of the philosophical intuition behind the Chinese Room: confronted with a machine that passes the behavioural test (the Turing test, in the limit), the response "but it does not *really* understand" recurs whether or not it is warranted. It bites hardest precisely where capability and understanding pull apart, as with systems that approximate a perfect actor (see LLMs as near-perfect actors): wide competence meets the standing reply that competence is not comprehension. The concept is *descriptive*, a lens on the debate's dynamics — not itself an argument that machines do, or do not, think.

> There's a running joke that "AI is whatever computers can't do yet." Every time a machine masters something we used to call a sign of intelligence — chess, *Jeopardy!*, holding a conversation — people shrug and say "oh, that's not *really* thinking, it's just calculation." So the bar for "real" intelligence keeps moving out of reach, and AI never quite gets credit. Two ways to see it: maybe critics are just moving the goalposts to keep humans special; or maybe they're right each time, and we keep learning that the task was easier than it looked and never needed real understanding. Often it's a bit of both — and sorting out which is the actual argument worth having about any new system, today's chatbots included.

See also

- ▷ Strong and weak AI — the AI effect is largely the repeated denial of *strong* AI to each new system.
- ▷ Chinese room — the philosophical articulation of the "that's not thinking" intuition.
- ▷ Perfect actor / LLMs as near-perfect actors — the modern locus where behavioural success meets "but it does not understand."
- ▷ McCorduck — Machines Who Think (1979) — the source of the classic formulation.

Attribution vs. detection

Attribution vs. detection — how we know a mind is not what a mind is

Argument · after Searle

Searle's rejoinder to the Other Minds Reply. The reply trades on a confusion between *epistemology* and *metaphysics*. That we ordinarily *detect* other people's understanding through their behaviour is a fact about how we *know*; it does nothing to show that understanding *consists in* behaviour or computation. The Chinese Room is a claim of the second kind — about what understanding *is* — and its whole force is that the behaviour (and the program producing it) could be fully present while the understanding is absent. So pointing out that we rely on behavioural evidence for other minds leaves the metaphysical thesis untouched.

The move `supports` the DERIVATION by protecting its key verdict from the charge of a double standard: we use behaviour as *evidence* without holding that minds *are* behaviour.

The live crux

The reply's defender can press a verificationist worry: if understanding can be wholly detached from every behavioural and functional sign of it, then the attribution "x understands" loses its connection to anything we could ever have reason to assert, and threatens to become idle. Searle must hold that there is a fact of understanding beyond all behavioural evidence without that fact collapsing into the unknowable — the same tension that runs through realism about other minds generally.

> The Other Minds Reply said: you only know your friends understand from how they behave, so you can't deny the same to the room. Searle's answer: how you *find out* who understands is one question; what understanding *is* is another. We read minds off behaviour because we have no choice — but that's just our evidence, not the thing itself. The Chinese Room's whole point is that all the behaviour could be there with nobody understanding. The worry left hanging: if understanding can float entirely free of every sign of it, what could ever entitle us to attribute it at all?

See also

- ▷ OTHER MINDS REPLY — the objection this answers.
- ▷ CHINESE ROOM DERIVATION — the argument it defends.
- ▷ TURING TEST — the behavioural criterion the reply leans on and this rejoinder relativises to *detection*.

B

♦ Basic belief

Basic (non-inferential) belief

Concept

A *basic belief* is one that is justified, but **not by inference** from other beliefs. Its warrant comes directly — from perception, memory, introspection, or rational insight — rather than being passed to it along a chain of reasons. Basic beliefs are the foundationalist's answer to the regress problem: the chain of justification does not go on forever or curve back on itself (see THE TRILEMMA) because it terminates in beliefs that are *self-standing*, the bedrock on which inferentially justified beliefs rest.

The notion is the core of FOUNDATIONALISM. Classical versions demanded that basic beliefs be *infallible* or indubitable (Descartes's clear and distinct perceptions); modern, *moderate* foundationalism asks only that they be non-inferentially justified and defeasible — a perceptual belief that there is a hand before me is warranted on sight, though further evidence could overturn it. In *Reformed epistemology* the related idea of a **properly basic** belief extends the category to include, for example, belief in other minds or in God as non-inferentially warranted in the right circumstances.

Two near neighbours to keep apart

- A basic belief is **justified, just not inferentially**; a HINGE PROPOSITION is, on the Wittgensteinian view, *not justified at all* — it is the presupposition that makes justifying possible. The two are easily run together because both are regress-stoppers and both resist the demand for prior reasons, but they stop the regress in different ways: foundationalism plants a floor *inside* the space of reasons; the hinge view marks the *edge* of that space.
- A basic belief is not the same as a self-evident *axiom* of a formal system. An axiom is a chosen starting point in a derivation; a basic belief is a state with non-inferential epistemic warrant about the world.

What it bears on

Basic belief is the concept by which FOUNDATIONALISM takes the dogmatic-stop horn of THE MÜNCHHAUSEN TRILEMMA and recasts it as principled rather than arbitrary; its existence is asserted by THERE ARE BASIC BELIEFS and denied by COHERENTISM and INFINITISM.

> Most of what you believe, you believe *because of* something else you believe. A "basic" belief is one you don't — it carries its own credentials. Seeing a red wall, you're entitled to think the wall is red without first proving anything; the seeing does the work. Foundationalists say beliefs like that are the floor the rest of knowledge stands on, which is how they stop the endless "but why do you believe *that*?" regress. (Careful not to confuse this with a HINGE: a basic belief is *justified without a reason*; a hinge is more like something too fundamental to need or take a justification at all.)

See also

- ▹ THERE ARE BASIC BELIEFS — the thesis that such beliefs exist.
- ▹ FOUNDATIONALISM — the view built on them.
- ▹ MÜNCHHAUSEN TRILEMMA — the regress problem basic beliefs are meant to halt.
- ▹ HINGE PROPOSITION — the contrasting regress-stopper that is held without justification.
- ▹ COHERENTISM, INFINITISM — the rivals that deny there are basic beliefs.

⊢ Bike-rider rejoinder

The bike rider learns facts — the ability hypothesis rests on a false dichotomy

Argument · after Stanley

This reply targets the ability hypothesis (see ABILITY HYPOTHESIS), the popular escape from the knowledge argument which holds that when Mary first sees red she gains only *abilities* — recognising, imagining, remembering the colour — and not new factual knowledge. The objection grants that what Mary gains might be exactly the bike-rider's kind of learning, then denies the unspoken assumption the escape needs: that acquiring an ability is not itself the learning of a fact.

The argument

The inference is *deductive*, drawn from an independently defended thesis in epistemology.

1. Know-how is a species of knowledge-*that* (see KNOWING-HOW IS A KIND OF KNOWING-THAT): to know how to F is to know, of a way *w*, that *w* is a way for one to F, grasped under a *practical mode of presentation* — a hands-on way of holding a truth that one cannot fully put into words (see STANLEY & WILLIAMSON — KNOWING HOW (2001)). Learning to ride a bike is not fact-free skill acquisition; one learns truths about how one's own body and the machine work together.
2. The ability hypothesis blocks the knowledge argument only if Mary's gained abilities involve *no* new knowledge-that — "abilities are had, not true."
3. By (1), the abilities Mary gains are themselves constituted by propositional knowledge, held under practical or recognitional modes of presentation.

It follows that the ability hypothesis fails *on its own terms*. Even granting that Mary's gain is exactly the bike-rider's learning, that learning is fact-learning, so "she gains abilities" does not entail "she gains no new fact," and the hypothesis's conclusion (see THAT MARY GAINS ABILITIES RATHER THAN FACTS) is left unsupported.

WHAT IT DOES AND DOES NOT ESTABLISH

Closing the ability-hypothesis route does **not** rescue the knowledge argument (see KNOWLEDGE ARGUMENT (MARY'S ROOM)). The facts the bike-rider learns are facts about her own body-and-situation, grasped practically — which is grist for the indexical reply (see INDEXICAL REPLY): reclassifying Mary's gain as factual but *self-locating* leaves physicalism untouched while abandoning the fact/ability dichotomy of Lewis's original proposal (see LEWIS — WHAT EXPERIENCE TEACHES (1988)). The effect is to move the debate from "fact versus ability" to "what kind of fact, under what mode of presentation."

THE WEAKEST PREMISE

Premise (1) is the contested one. The *Rylean regress* presses that *applying* knowledge-that itself requires know-how, which loops the analysis; dissociation cases, such as motor skills surviving amnesia, suggest two distinct cognitive systems rather than one piece of knowledge held under two modes; and "practical modes of presentation" are accused of doing the same unexplained work that *phenomenal concepts* do in the old-fact/new-guise reply (see OLD FACT, NEW GUISE) — a label for the difference rather than an account of it. That worry takes its sharpest shape in the isolation/explanation trade-off facing such concepts (see PHENOMENAL-CONCEPT TRADE-OFF).

> A famous escape from the Mary puzzle (see ABILITY HYPOTHESIS) says that when Mary first sees red she only gains *skills* — like learning to ride a bike — and skills are not facts, so no fact was missing from her science. This objection takes aim at the "skills aren't facts" step. When you learn to ride a bike you plainly do come to know something true: how your body and that bike work together, known in a hands-on way you cannot fully put into words. If gaining a skill is itself a way of learning facts, the escape route closes: calling Mary's gain a "skill" no longer shows she learned no fact. It only reopens the original question in a sharper form — what *kind* of fact? A general one physics should have covered, or a personal, you-are-here fact no textbook could ever contain? The "she learns where she is, not what the world contains" reply (see INDEXICAL REPLY) picks up exactly there.

SEE ALSO

- ▹ ABILITY HYPOTHESIS — the target: the view that Mary gains abilities, not facts, which this reply undercuts on its own terms.
- ▹ KNOW-HOW AS KNOWLEDGE-THAT — the load-bearing premise that know-how is a kind of knowledge-that.
- ▹ KNOWLEDGE ARGUMENT — the thought experiment in the background; this reply leaves it unrescued.

⊢ Brain simulator reply

The Brain Simulator Reply — simulate the neuron firings of a real Chinese speaker

Argument

The *Brain Simulator Reply* asks us to imagine a program that does not merely match inputs to outputs by some clever rulebook, but **simulates the actual sequence of neuron firings** in the brain of a native Chinese speaker as they understand and answer. If the program reproduces, synapse by synapse, exactly what happens in a real understanding brain, what relevant difference could remain? To deny understanding here, the reply urges, is to deny it of the original brain too — for the program duplicates precisely the processes on which the speaker's own understanding depends.

It attacks A3 (see SYNTAX INSUFFICIENT FOR semantics) from the opposite side to the Robot Reply: not by adding a world, but by making the *internal* process as brain-like as possible, so that the gap between "mere program" and "real brain" closes to nothing. If a perfect functional duplicate of the causal organisation of understanding still does not understand, the reply says, the burden is on Searle to say what extra ingredient the brain has — which threatens to make his view a mysterious biological vitalism.

Kinship with the China-brain

The reply is a close relative of Ned Block's Chinese Nation (China-brain). Make the neuron-simulators *people* — a population each enacting one neuron's firing — and the Brain Simulator becomes Block's case: a crowd realising the full neural organisation of a mind. The two diverge in *target*, which is why they stay distinct nodes: the China-brain is classically pressed against *qualia* (see whether the nation feels), whereas the Brain Simulator Reply is pressed *for* understanding against Searle. Searle's own Chinese Gym rejoinder makes the convergence explicit by staging exactly such a population.

The crux

Everything turns on whether *simulating* a causal process is *having* it. Searle's water-pipes rejoinder dramatises his "no": replace the man-with-rulebook with a vast array of water pipes and valves arranged to mirror the synapses, the man working the valves on cue. The system reproduces the formal structure of the firing pattern, yet — Searle insists — nobody and nothing understands Chinese, because a simulation of the *form* of neural activity leaves out the brain's actual *causal powers* (the instancing-versus-depicting distinction). The defender replies that if the simulation truly captured *everything* causally relevant, it would not be a mere depiction but a duplicate — and then it would understand.

> Forget rulebooks of clever tricks — have the computer copy, neuron for neuron, exactly what fires in a real Chinese speaker's brain when they read and reply. If it's doing precisely what their brain does, how could they understand and it not? Searle's answer is a picture: imagine the "neurons" are water pipes, and the man opens and shuts valves in the right order. All the right plumbing happens, and still nobody understands Chinese — because copying the *pattern* of the brain isn't copying the brain's actual causal *power*, any more than a detailed simulation of a rainstorm gets anything wet. Critics answer: if you really copied *everything* that matters, it wouldn't be a copy, it'd be the real thing.

See also

- ▹ Syntax insufficient for semantics — axiom A3, denied here by maximal neural fidelity.
- ▹ Water pipes — Searle's reply: the water-pipe brain reproduces form, not causal power.
- ▹ Instancing vs. depicting — the instancing-vs-depicting principle Searle's rejoinder turns on.
- ▹ Brains cause minds — the causal-powers axiom the reply puts under pressure.
- ▹ Connectionist reply — the allied appeal to brain-like (parallel) architecture.
- ▹ Chinese nation — Block's population-realises-a-brain case, what this reply becomes when the neuron-simulators are people.
- ▹ Chinese gym — Searle's rejoinder that stages exactly that population.

⊙ Brains cause minds

Brains cause minds through their specific biological causal powers

Claim · after Searle

Mental phenomena are *caused by* the actual neurobiological processes of the brain: real brains produce real minds in virtue of their specific causal powers, not in virtue of instantiating an abstract program. This is the fourth axiom (A4) of Searle's Chinese Room derivation (see Searle — Is the Brain's Mind a Computer Program? (1990)) and the positive counterpart to its negative conclusion. Where C1 denies that *computation* suffices for mind, A4 locates what *does* suffice in the concrete biology — the brain's particular electrochemical causal powers, which a formal program no more reproduces than a simulation of fire produces heat.

With A4, the derivation reaches its further conclusions: any system that produces minds must have causal powers at least equivalent to the brain's (C2); implementing a program is not by itself enough to guarantee those powers (C3); and so the way human brains produce minds is not by running a program (C4). The axiom is the seed of Searle's positive view, biological naturalism.

Scope. A4 says brains *cause* minds by their causal powers; it is deliberately neutral on whether *only* brains could — Searle allows that an artificial system duplicating the relevant causal powers might also cause minds. What it denies is that the relevant power is *formal-computational*. It is therefore not chauvinism about carbon, but a claim about *causal powers* rather than *programs*.

Who would deny it. Computational functionalists deny that the mind-making feature is the brain's *biological* causal powers rather than its *organisation*, which they hold a non-biological substrate could share (see Substrate-independence thesis). The dispute is over whether the relevant level is biochemical causal power or abstract functional structure.

Distinct from Consciousness as biological feature: that claim says consciousness *is* a biological feature; this one makes the *causal* point that brains *produce* minds through their specific powers — the premise that grounds the causal-powers conclusions of the derivation.

> Searle isn't saying minds are magic or non-physical — quite the opposite. Brains *make* minds, the way stomachs make digestion: through their actual physical, biological machinery. His point is just that the mind-making ingredient is the wet, causal stuff the brain *does*, not the abstract *pattern* of it — and a computer running a program copies the pattern, not the causal power. (Functionalists disagree: they think the pattern is exactly what matters, and the wetware is incidental.)

C

⊙ Categorical ground of structure

A thing's relational-structural properties must be grounded in its intrinsic nature

Claim · after Seager

The relational and structural properties of a thing — at the very least its causal-dispositional ones — cannot float free: they must be grounded in, and determined by, the thing's *intrinsic*, categorical nature. A bare web of relations needs relata that have some nature of their own to enter into it; a disposition needs a categorical base in virtue of which the thing is so disposed. Applied to physics, where the properties science discloses are relational-structural through and through (see INTRINSIC AND RELATIONAL PROPERTIES), the principle forces the conclusion that matter has an intrinsic nature *beneath* the structure physics charts — even though physics itself never reveals it.

Seager (see SEAGER — THE 'INTRINSIC NATURE' ARGUMENT FOR PANPSYCHISM (2006)) calls a strong version of this the *Principle of the Reducibility of Relations* and traces it to Leibniz and Kant; the panpsychist argument needs only the weaker, restricted form — that the *causal* structure of matter requires an intrinsic ground. It is continuous with the familiar metaphysical demand that dispositions require a categorical basis, associated with D. M. Armstrong (see ARMSTRONG — A MATERIALIST THEORY OF THE MIND (1968)).

Scope. The claim asserts only that structure needs *some* intrinsic ground; it does not say *what* that ground is, still less that it is experiential. That further step is taken by the larger argument (see INTRINSIC-NATURE ARGUMENT), for which this is the load-bearing premise.

Who would deny it. *Relationalists* and *ontic structural realists* deny it outright: there is nothing to matter beyond structure, and the relata are constituted by their relations, as the nodes of a mathematical graph are. On their view the demand for an intrinsic ground is empty — and with it the intrinsic-nature argument collapses.

Distinct from STRUCTURE AND DYNAMICS ONLY, which is the *epistemic* point that physics *describes* only structure: this is the stronger *metaphysical* point that structure must be *grounded in* an intrinsic nature. Distinct, too, from CATEGORICAL, NOT DISPOSITIONAL, which says consciousness *itself* is categorical rather than dispositional; this claim is the general grounding principle about a thing's structure, not a thesis about the nature of experience.

> Imagine the world described purely as a network: this connected to that, this leading to that. A network like that needs *things* sitting at its junctions — and those things have to be *something* in their own right to do the connecting. The same goes for tendencies: glass is fragile because of how it is actually built, not just because "it would shatter if struck." This claim says matter's whole web of relations and tendencies, which is all physics ever shows us, must likewise rest on some inner nature the things really have. It does not say *what* that inner nature is — only that there must be one. The people who reject it (see INTRINSIC AND RELATIONAL PROPERTIES) think the network can be relations all the way down, with no inner nature needed, like the dots in a pure mathematical diagram.

⊙ Categorical, not dispositional

Phenomenal consciousness is categorical, not merely dispositional

Claim

Phenomenal consciousness is an *occurrent, categorical* matter — it is actually, presently *there* — rather than a merely *dispositional* affair, a matter of what a system *would* do under inputs it may never actually receive. A disposition is an "iffy" property: glass counts as fragile in virtue of how it *would* behave if struck, even if it is never struck. Experience does not look like that. It looks like the paradigm of a categorical fact, something occurrently happening, not a standing counterfactual profile.

The standard support for the contrast is that dispositions require a *categorical basis* — fragility holds in virtue of some actual microstructure of the glass — a demand associated with D. M. Armstrong (see ARMSTRONG — A MATERIALIST THEORY OF THE MIND (1968)); on the Russellian side, the intrinsic, categorical nature underlying structure is what Galen Strawson (see Strawson — Realistic Monism: Why Physicalism Entails Panpsychism (2006)) presses as the home of experience. The claim itself stays modest about *what* the categorical basis is: it does not say whether that basis is the system's current physical configuration or some intrinsic nature beneath it. It asserts only that the phenomenal is categorical, not dispositional.

A dispositionalist or functionalist about consciousness would deny it, holding that to be in a phenomenal state just *is* to occupy a causal-dispositional role. The claim is kin to, but distinct from, Phenomenal determinacy: determinacy says there is a *definite* fact about one's current phenomenal state, whereas this says the fact is *categorical/occurrent* rather than dispositional. The two function as companion premises behind the decomposition-indeterminacy objection and the structure-and-dynamics argument respectively.

> A "disposition" is an iffy, would-do fact: saying glass is fragile means it *would* break if struck, even if it never actually is. A "categorical" fact is how something actually, presently *is*. This claim says your conscious experience is the second kind — it's really there right now, not just a bundle of "what the system would do if poked." A pain isn't a standing tendency to react; it's an occurrent felt thing happening. (Anyone who defines a mental state as nothing but a causal role would push back.)

See also

- ▷ Phenomenal determinacy — the companion claim that there is a *definite* fact about one's current experience; this one adds that the fact is *categorical*.
- ▷ Decomposition-indeterminacy objection — relies on determinacy; pairs with this claim's categoricity in the case against role-based consciousness.
- ▷ Structure-and-dynamics argument — the argument this claim feeds: function captures only structure and dynamics, never the categorical/intrinsic.

⊢ Causal argument for physicalism

The causal argument for physicalism

Argument · after Papineau

The **causal argument** (also the *overdetermination* or *exclusion* argument) is the principal positive case for physicalism about the mind. It reasons from the causal structure of the physical world to the conclusion that mental events must be physical events, leaving the anti-physicalist a stark and unwelcome choice.

Inference form: deductive (an argument by elimination).

1. **Completeness.** Every physical effect has a sufficient physical cause (Completeness premise — the Completeness of physics).
2. **Efficacy.** Some mental events cause physical effects (see Mental causation).
3. **No overdetermination.** Those physical effects are not systematically overdetermined by a distinct physical cause and a distinct non-physical cause (see No systematic overdetermination).
4. From (1) a bodily movement caused by a mental event already has a sufficient physical cause; from (3) the mental cause is not a *second*, independent cause standing beside it. The only way for the mental event to be a genuine cause of that movement, rather than an idle accompaniment, is for it to *be* the sufficient physical cause — i.e. to be physical.
5. ∴ Mental events are physical events (see Mental events are physical, Physicalism).

The force of the argument is best felt as a **trilemma**: holding completeness fixed, one must give up exactly one of the remaining options. Deny efficacy and you get Epiphenomenalism — the mind as causally inert by-product. Deny no-overdetermination and you posit a baroque, permanent coincidence in which every action is doubly caused. Accept all three and you get physicalism. Interactionist Dualism is forced to reject premise (1), which is why so much weight rests on whether the completeness of physics is genuinely established.

Weakest premise: the **no-overdetermination** premise (3). Completeness is defended as hard empirical fact, and efficacy is close to a datum of common sense, so the dualist's most promising move is to accept that mental and physical causes coincide — either by biting the overdetermination bullet (the two causes are distinct but the effect is harmlessly double-caused) or by arguing that mind and brain are not "distinct" enough for their joint causation to count as the objectionable kind of overdetermination. The exclusion literature (Kim, Kim — Physicalism, or Something Near Enough (2005)) is largely a fight over premise (3) and what "distinct sufficient cause" must mean.

> This is the main argument *for* the view that the mind is physical. It runs in three steps. First: every physical event has a complete physical cause — science never needs to invoke a soul to explain why a body moves (see Completeness of physics). Second: your mind really does move your body — your decision to raise your arm is why it goes up. Third: your arm's rising isn't caused *twice over*, once by physics and again, separately, by a non-physical mind. Put those together and there's only one place left for your decision to fit: it has to *be* the physical cause — that is, it has to be something physical happening in your brain. The alternatives are both uncomfortable: either your decisions don't actually do anything (they just ride along — that's Epiphenomenalism), or every single action you take is mysteriously caused two independent times at once. Faced with that menu, the physicalist says: simplest to admit the mind is physical.

⧫ Causal closure of the physical

Concept

The **causal closure of the physical** is the principle that the physical domain is causally self-enclosed: no non-physical event ever causes a physical one. Causal chains that arrive at a physical effect never originate, or pass, outside physics. Put as a slogan, *physics is a closed system* — to trace the cause of any physical happening is to stay forever among physical happenings.

Closure is the negative, boundary-policing face of a thesis whose positive face is the Completeness of physics: completeness says each physical effect *has* a sufficient physical cause; closure says it has *no* non-physical cause. The two are easy to conflate and are often used interchangeably, but they come apart on one possibility — **overdetermination**. Completeness alone is silent about whether a physical effect might *also* have a distinct non-physical cause alongside its sufficient physical one (as a death may have two independently sufficient causes). It is the further premise that such systematic double-causation does not occur (see No systematic overdetermination) that turns completeness into full closure. Hence the dependency: closure is what you get from completeness once overdetermination is ruled out.

Why it matters

Closure is the premise that forces the central dilemma of philosophy of mind. Suppose mental events make a real difference to the physical world — that a decision moves a hand, a pain prompts a wince. If the physical domain is closed, then every such bodily effect already has its cause *inside* physics, and the mental cause can earn its keep in only one way: by *being* physical. The alternatives are stark — either the mental is identical to (or realised by) the physical (see Physicalism, Mental events are physical), or it is a causal idler that the physical world runs on without (see Epiphenomenalism). This is the backbone of Kim's **causal exclusion argument** (see Kim — Physicalism, or Something Near Enough (2005)): given closure, supervenience, and an exclusion principle barring redundant double-causes, an irreducible mental property would have no work to do. Interactionist Dualism, which insists the non-physical mind pushes the body around, is precisely the denial of closure.

It is worth marking what closure does *not* say. It does not deny the existence of non-physical things, nor even that non-physical things have effects — only that they have no effects *on the physical*. A closure-respecting dualist could still hold that the physical causes the mental (the body produces experiences) while denying any traffic the other way; that is one route to Epiphenomenalism. What closure forbids is causal influence crossing *into* the physical from outside it.

> Causal closure says the physical world is a sealed room as far as causes go: nothing from outside — no soul, no free-floating mind, no spirit — ever reaches in and nudges a physical thing. If you follow the chain of causes behind any physical event, you only ever meet more physical events; you never cross a border into the non-physical. This is the step that makes the mind-body problem bite. If your decisions really do move your body, and nothing non-physical is allowed to move physical stuff, then your decisions must themselves be physical goings-on. The only escape is to say your decisions don't actually *do* anything — they just ride along — which is a hard pill (that is what "epiphenomenalism" amounts to). It is the close cousin of the Completeness of physics: completeness says a physical cause is always *enough*, closure adds that a non-physical cause is never *there*.

See also

- ▷ Completeness of physics — the positive companion principle; closure is completeness plus the exclusion of overdetermination.
- ▷ No systematic overdetermination —

the premise that bridges completeness to closure.

- ▹ Causal argument for physicalism — the inference closure powers, against a freely-causing non-physical mind.
- ▹ Physicalism — the view closure is wielded to support.
- ▹ Dualism — interactionist dualism is the denial of closure.
- ▹ Epiphenomenalism — the position left to a non-physical mind once closure is granted.
- ▹ Kim — Physicalism, or Something Near Enough (2005) — Kim's exclusion argument, the sharpest deployment of closure.

⊢ Causal exclusion argument

Kim's causal exclusion (supervenience) argument

Argument · after Kim

The **causal exclusion argument** (Kim's *supervenience argument*; Kim — Physicalism, or Something Near Enough (2005)) is an internal critique of **non-reductive physicalism** — the popular view that mental properties supervene on physical properties without being identical to them. Kim argues that this position cannot, in the end, secure mental causation: an irreducible mental property is squeezed out of the causal story by the very physical base on which it supervenes. The non-reductive physicalist is driven to a dilemma: reduce the mental, or give up its efficacy.

Inference form: deductive (a reductio of irreducible mental causation).

1. **Closure.** Every physical effect has a sufficient physical cause (Completeness premise — the Completeness of physics).
2. **Supervenience without identity.** The non-reductive physicalist holds that a mental property M supervenes on a physical base P but is not identical to P. (This is the position's own defining commitment, assumed here for reductio.)
3. **Exclusion.** A single effect does not have two distinct, independently sufficient causes, barring genuine overdetermination (see No systematic overdetermination).
4. Suppose M causes a physical effect E. By (1), E already has a sufficient physical cause — paradigmatically P itself, the supervenience base of M. By (2), M is distinct from P. By (3), M and P cannot both be independently sufficient causes of E (and the supervenience relation makes a coincidence of two *independent* causes the wrong description). So M is excluded: the causal work is done by P.
5. ∴ An irreducible mental property is causally excluded by its physical base (see Exclusion of the irreducible mental).

The result is the **reduce-or-eliminate dilemma**. If mental causation is to be preserved, M must after all *be* a physical property (see Mental events are physical) — reductive physicalism, the conclusion Kim ultimately embraces ("physicalism, or something near enough"). Refuse the identification and M is left inert, an epiphenomenon (see Causally inert qualia). Either way, non-reductive physicalism as a stable middle position collapses.

Distinct from the causal argument for physicalism: that argument runs the same exclusion machinery *outward*, against the interactionist dualist whose mind is non-physical, to conclude that the mind is physical. Kim's exclusion argument runs *inward*, against the fellow physicalist who thinks supervenience buys an autonomous mental level; its distinctive premise is supervenience-without-identity (2), and its distinctive target is the *vertical* relation between a property and its own realiser, not a horizontal contest between physical and non-physical causes.

Weakest premise: the **exclusion** principle (3), as applied to a property and its supervenience base. Non-reductive physicalists reply that M and P are not "distinct" in the way exclusion requires — they are not two competing causes but one event described at two levels, so their joint relevance to E is not the objectionable overdetermination the principle rules out. Others deny that closure plus exclusion settles which level is the *real* cause, appealing to a more permissive (e.g. counterfactual or proportionality-based) account of causation on which M can be the difference-maker even when P suffices. The dispute thus turns on what "distinct sufficient cause" must mean.

> Many philosophers want to say the mind is made of nothing but physical stuff, yet is not *just* the brain — mental properties "ride on" brain properties without being identical to them. Kim's argument says this comfortable middle ground doesn't hold. Whenever a mental state seems to cause something — your decision makes your arm rise — the underlying brain state is already enough to make the arm rise. So either the decision *is* that

> brain state (and we're back to plain "the mind is the brain"), or the decision is a spare wheel that spins without driving anything (epiphenomenalism). There's no third option where the mind is both distinct from the brain and still does real work.

SEE ALSO

- ▷ EXCLUSION OF THE IRREDUCIBLE MENTAL — the conclusion: a supervening-but-distinct mental property has no causal work left to do.
- ▷ CAUSAL ARGUMENT FOR PHYSICALISM — the same exclusion machinery aimed outward at dualism rather than inward at non-reductive physicalism.
- ▷ EPIPHENOMENALISM and CAUSALLY INERT QUALIA — the horn of the dilemma that keeps the mental distinct at the cost of its efficacy.
- ▷ MENTAL EVENTS ARE PHYSICAL — the reductive horn Kim recommends, the only way to keep the mental a cause.
- ▷ CAUSAL CLOSURE OF THE PHYSICAL and COMPLETENESS OF PHYSICS — the closure premise the argument leans on.

⊢ Chinese gym

The Chinese Gym — a parallel crowd of clerks understands no more than one

Argument · after Searle

Searle's rejoinder to the CONNECTIONIST REPLY. If the objection is that a single serial clerk is the wrong architecture, scale the room up to a *gymnasium* full of monolingual English speakers, each one playing the part of a single neuron or node, passing signals to their neighbours on cue according to the rules. The system is now massively parallel and brain-like in organisation — yet, Searle insists, no individual in the gym understands Chinese, and the assembled crowd understands no more than any one of them. Parallelism multiplies the meaningless symbol-shuffling; it does not, by itself, manufacture meaning.

The move `supports` the DERIVATION by extending A3 to parallel architectures: changing *how* the syntax is distributed does not change *that* it is mere syntax.

KINSHIP WITH THE CHINA-BRAIN

The Chinese Gym is, structurally, a relative of Ned Block's CHINESE NATION (China-brain): a population of people jointly realising the functional organisation of a mind. The two cases are pressed against *different* features, which is why they remain distinct nodes — Block's case targets *qualia* (see WHETHER THE NATION FEELS ANYTHING), whereas the Gym targets *understanding/intentionality*. Both share the intuition pump's vulnerability: a critic who attributes the mental property to the *whole* organised population (the SYSTEMS REPLY again) denies that the failure of each member to understand or feel settles the matter.

THE LIVE CRUX

The Churchlands' point partly survives the parody: they did not claim the gym understands but that A3 — "syntax cannot suffice for semantics" — is an *unargued assumption*, not a self-evident axiom, and whether the right physical network yields meaning is a scientific question the gym cannot settle by intuition.

> The reply said the lone clerk was the wrong model — brains are parallel. So Searle fills a gym with people, each acting as one "neuron," all signalling at once. It's parallel now, and still nobody in the building understands Chinese. His point: spreading the symbol-shuffling across a crowd doesn't conjure meaning. This is a cousin of the famous "China-brain," where a whole nation acts out one mind — except that case asks whether the nation *feels* anything, while this one asks whether it *understands*. And the same nagging question dogs both: maybe the *crowd as a whole* has what no single member does.

SEE ALSO

- ▷ CONNECTIONIST REPLY — the objection this answers.
- ▷ CHINESE NATION — Block's structurally similar population-realises-a-mind case.
- ▷ CHINESE NATION (ABSENT QUALIA) — the China-brain pressed against qualia, where the Gym is pressed against understanding.
- ▷ CHINESE ROOM DERIVATION — the argument it defends, extending A3 to parallel machines.

♦ Chinese nation

Chinese Nation (homunculi-headed robot)

Concept · after Block

The *Chinese Nation* is a thought experiment devised by Ned Block in "Troubles with Functionalism" (1978). Imagine the citizens of a whole country, or a head full of little people, given radios and a rulebook so that together they realise the entire functional and causal organisation of a single human mind: the same state-to-state transition structure a brain runs through, for an hour or so. By construction the system has exactly the inner organisation a conscious brain has. The question Block presses is whether it would therefore be conscious. Would there be something it is like to be the nation, or is there simply nobody home?

What makes it distinctive

The case has its bite precisely because it instantiates the functional organisation, not merely the profile of inputs and outputs a mind produces. It is therefore not the "right behaviour, no processing" case of Block's other machine, THE LOOKUP-TABLE MIND, but the "right processing, yet intuitively nobody home" case. That is what makes it tell against FUNCTIONALISM proper rather than against mere behaviourism: it grants everything a functionalist asks for and still seems to lack experience.

The dispute it feeds

The Chinese Nation is pressed on whether realising a mind's functional organisation is *sufficient for PHENOMENAL CONSCIOUSNESS*. The intuition that the nation has none drives THE CLAIM THAT SUCH A POPULATION WOULD FEEL NOTHING and the argument THAT ORGANISATION IS NOT SUFFICIENT FOR CONSCIOUSNESS, which in turn support the PHENOMENAL RESIDUE view and attack the uniform reading of SUBSTRATE INDEPENDENCE.

Replies

Two replies are characteristic, and both differ from the ones the lookup table invites. The *systems reply* denies the intuition outright: the nation taken as a whole may be a subject of experience even though no individual citizen is (Lycan, Dennett). A second worry concerns *speed and timing*, whether so slow and distributed a system genuinely realises the organisation at all. There is also a pointed structural observation. ORGANISATIONAL-ROLE FUNCTIONALISM cannot cheaply exclude the Chinese Nation the way it excludes the lookup table, because here the organisation really is present; that is exactly why the case is sharpened into THE DENSE CHINA-BRAIN REGRESS against the organisational grain.

A literary illustration

A vivid cousin of the case appears in Liu Cixin's science-fiction novel *The Three-Body Problem* (see LIU CIXIN). There the First Emperor Qin Shi Huang, prompted by von Neumann, ranks thirty million soldiers across a plain, each small unit raising black-and-white flags while cavalry gallop the signals between formations, to assemble a vast human machine. In the novel the machine is a computer crunching the orbits of three suns; re-task it to run a *brain* instead. Let each cluster of soldiers be a neuron, each raised flag its firing, the messengers its axons, and the whole drilled host step for one hour through exactly the state-to-state transitions a human cortex would. The army is now no calculator but the Chinese Nation itself: a country physically realising a mind's causal organisation while no single soldier feels a thing. If the soldiers enact the right transitions, is someone *over and above* them undergoing the experience, or is there nobody home?

> Imagine the entire population of a country handed radios and a rulebook and told to each act as one brain cell: when you receive these signals, send those. Follow the rules fast enough and the nation as a whole mimics, step for step, everything one human brain does inside, not just the outward behaviour but the actual inner wiring. Now the unsettling question: the country is doing literally everything your brain does when you feel pain, so is the *country* feeling pain? Is there one extra "someone", the nation itself, having an experience, hovering above millions of people who each feel nothing out of the ordinary? If running the right pattern is all it takes to be conscious, you seem pushed to say yes, yet most people's gut says surely not. That clash is the whole point of the thought experiment. It is a sharper challenge than a dumb answer-lookup machine, because here the full inner organisation really is present, so you cannot dismiss it for having no inner structure.

See also

- ▷ LOOKUP-TABLE MIND — Block's other machine: right behaviour with no inner processing, the case this one is defined against.
- ▷ FUNCTIONALISM — the theory the thought experiment targets: mind as functional organisation.
- ▷ CHINESE NATION (ABSENT QUALIA) — the argument that the realised organisation still lacks qualia, so organisation is not sufficient for consciousness.
- ▷ PHENOMENAL RESIDUE — the view the absent-qualia intuition supports: experience outruns functional role.
- ▷ SUBSTRATE-INDEPENDENCE THESIS — the

thesis whose uniform reading the case attacks.

- ▹ Dense China-brain regress — where the case is sharpened against organisational-role functionalism.
- ▹ Phenomenal consciousness — what the nation is alleged to lack despite the right organisation.
- ▹ Block — Troubles with Functionalism (1978) — Ned Block's "Troubles with Functionalism" (1978), where the thought experiment is introduced.

⧫ Chinese room

The Chinese Room

Concept · after Searle

The *Chinese Room* is John Searle's thought experiment against the claim that running the right computer program is, by itself, enough to produce a mind (see Searle — Minds, Brains, and Programs (1980)). A monolingual English speaker is sealed in a room with a rulebook — in effect, a program — that tells him how to manipulate Chinese symbols purely by their shapes. Chinese questions are slipped in; by following the rules he passes back Chinese answers indistinguishable from a native speaker's. Yet he understands no Chinese. The room, Searle argues, has the right *syntax* (formal symbol manipulation) without any *semantics* (meaning, understanding, intentionality) — and a digital computer, which does nothing but manipulate symbols by their shapes, is in exactly the same position.

What it targets

The argument is aimed at **strong AI**, the computational-functionalist thesis that *implementing the right program is sufficient for a mind.* That precise target is what separates it from two neighbouring thought experiments with which it is often confused. Ned Block's Lookup-table mind ("Blockhead") attacks the sufficiency of *behaviour*: it produces the right outputs with **no** internal processing at all, just a giant table of canned answers. The Chinese Room, by contrast, *does* run a genuine program; what it denies is that *computation* suffices for *understanding*. Block's Chinese nation differs in the other direction: there a population realises a mind's full functional organisation, and the case is pressed against *qualia* — felt experience — whereas the Room is pressed against *intentionality*, whether the symbols mean anything to the system at all. Three different experiments, three different sufficiency theses.

Characteristic replies

Each of the standard replies is an objection-in-waiting. The **Systems Reply** grants that the man does not understand Chinese but locates the understanding in the whole system — man, rulebook, and room together — much as no single neuron understands English. The **Robot Reply** embeds the program in a robot body with cameras and limbs, so that its symbols stand in causal commerce with the world and thereby acquire grounding. The **Brain Simulator Reply** has the program simulate, neuron by neuron, the actual brain activity of a Chinese speaker — what relevant difference could remain? Searle rejoins each: against the Systems Reply, for instance, he lets the man memorise the rulebook and work in the open air, so that the man *is* the whole system — and still understands nothing.

> Imagine you're locked in a room with a giant instruction book. Notes in Chinese slide under the door; you can't read a word of Chinese, but the book says "if you see these squiggles, copy out those squiggles," so you shuffle symbols and pass back replies that fool fluent speakers outside. You're behaving exactly like someone who understands Chinese — yet you understand nothing; you're just moving shapes around. Searle's point: a computer is in the same boat. Running a program is *shuffling symbols by their shapes*, and no amount of that, he says, adds up to actually *understanding* what the symbols mean. (The famous comeback: maybe *you* don't understand, but the whole room-plus-rulebook system does.)

See also

- ▹ Functionalism — the family of theories in the argument's crosshairs: mind as the running of the right program, independent of what runs it.
- ▹ Lookup-table mind — the neighbouring case with right behaviour and *no* processing; attacks behavioural sufficiency, where the Room attacks computational sufficiency.
- ▹ Chinese nation — the neighbouring case with the full functional organisation, pressed against qualia where the Room is pressed against understanding.
- ▹ Searle — Minds, Brains, and Programs (1980) — "Minds, Brains, and Programs", the original statement of the argument and its replies.

⊢ Chinese room derivation

The Chinese Room as a derivation from axioms — syntax is not sufficient for minds

Argument · after Searle

This is the *formal* version of the CHINESE ROOM argument — the "derivation from axioms" Searle gave in 1990 (see SEARLE — IS THE BRAIN'S MIND A COMPUTER PROGRAM? (1990)). Where the INTUITION-PUMP VERSION runs the thought experiment and reads a verdict off it, this version distils the same point into a short deductive argument, so that the dispute can be located on a single premise rather than on an intuition about a story.

THE ARGUMENT

The inference is **deductive** (Searle calls it a "derivation").

1. (see PROGRAMS ARE SYNTACTIC) Programs are formal, or syntactic — defined over symbol shapes, not meanings. *(A1)*
2. (see MINDS HAVE SEMANTICS) Minds have mental contents — semantics, genuine ABOUTNESS. *(A2)*
3. (see SYNTAX INSUFFICIENT FOR SEMANTICS) Syntax is neither constitutive of nor sufficient for semantics. *(A3)*

∴ (see PROGRAMS NOT SUFFICIENT FOR MINDS) Implementing a program is neither constitutive of nor sufficient for a mind *(C1)* — the denial of strong AI (see STRONG AND WEAK AI). This `attacks` SUBSTRATE-INDEPENDENCE THESIS at its sufficiency direction: realising the right program does not deliver the semantic side of mind.

A fourth axiom extends the result to the positive case:

1. (see BRAINS CAUSE MINDS) Brains cause minds through their specific causal powers. *(A4)*

∴ Any mind-producing system needs causal powers at least equivalent to brains' (C2); program-implementation alone does not guarantee them (C3); so brains do not produce minds *by* running programs (C4) — the seed of BIOLOGICAL NATURALISM.

THE WEAKEST PREMISE

The whole weight rests on **A3** (see SYNTAX INSUFFICIENT FOR SEMANTICS), and its status is the crux. The standard charge is that A3 is **not an independent axiom but a restatement of the conclusion**: the only support Searle offers for "syntax is not sufficient for semantics" is the Chinese Room thought experiment, whose moral *is* C1 applied to the AI case — so the deductive form lends rigor without adding *independent* force, and a functionalist who rejects C1 rejects A3 for the very same reason. Every standard objection is, at bottom, a denial of A3: the SYSTEMS, ROBOT, BRAIN SIMULATOR, OTHER MINDS, CONNECTIONIST, and MANY MANSIONS replies each contend that syntax *plus something* (a whole system, causal grounding, neural simulation, parallel architecture) does suffice after all. A secondary soft spot is **A2**: eliminativists about content deny minds have the intrinsic semantics the contrast needs (see INTRINSIC AND DERIVED INTENTIONALITY).

A1 is near-definitional and not in play; A4 is contested only on whether the mind-making feature is biological causal power or abstract organisation.

RELATION TO THE INTUITION-PUMP VERSION

This node and CHINESE ROOM ARGUMENT are two presentations of one argument, kept distinct because they have different *forms* and *attack surfaces*: the intuition-pump version is a *thought-experiment* counterexample, vulnerable at its bridge and scope (the Systems and Robot replies as *critical questions*); this version is a deductive syllogism, vulnerable at the truth and independence of A3. The replies in this cluster attack the derivation; many of them apply, mutatis mutandis, to the thought-experiment version too.

> Searle's argument, stripped to a syllogism: (1) computer programs are just rules about symbol *shapes*; (2) real minds actually *mean* things; (3) you can't get meaning out of shape-shuffling alone — so (conclusion) running a program can't be enough to make a mind. Add (4) "brains make minds by their actual biology," and you get the further moral that thinking isn't *running code* at all. The argument is airtight *if* step (3) is true — and that's exactly the step everyone fights over. Critics say step (3) is just the conclusion in disguise, and that syntax plus the right extra ingredient (the whole system, a body, simulated neurons) *does* yield understanding.

SEE ALSO

- ▷ PROGRAMS NOT SUFFICIENT FOR MINDS — the conclusion (C1); the denial of strong AI.
- ▷ SYNTAX INSUFFICIENT FOR SEMANTICS — the load-bearing, contested axiom A3.

- ▹ PROGRAMS ARE SYNTACTIC, MINDS HAVE SEMANTICS, BRAINS CAUSE MINDS — the other axioms.
- ▹ CHINESE ROOM ARGUMENT — the intuition-pump presentation of the same argument.
- ▹ SYSTEMS REPLY, ROBOT REPLY, BRAIN SIMULATOR REPLY, OTHER MINDS REPLY, CONNECTIONIST REPLY, MANY MANSIONS REPLY — the standard objections, each a denial of A3.
- ▹ BIOLOGICAL NATURALISM — Searle's positive view, grown from A4.
- ▹ SUBSTRATE-INDEPENDENCE THESIS — the thesis the derivation attacks.

⊙ Circularity of a priori anthropomorphism

Assuming anthropomorphic attributes a priori, independent of substrate, yields circular or uninformative conclusions

Claim · after Dewynter

To assume, in advance and independently of substrate, that a system either has or lacks an anthropomorphic attribute — morality, understanding, sentience — and then to run an experiment interpreted through that assumption, is to guarantee an uninformative result. If the assumption is built into how the behaviour is read, the experiment can only return the assumption: confirm that the attribute is "there" if you assumed it, or "absent" if you assumed it away. The defect is symmetric. It does not depend on the experimenter's prior viewpoint, and it strikes whether the reported outcome is existence or non-existence. Either the reasoning is circular (the conclusion was smuggled into the premises) or it is uninformative (the interpretation was fixed by the chosen representation, not by any measurement).

This is the epistemic premise of AGE OF EMPIRES II ARGUMENT. The escape it points to is to fix *explicit measurement criteria* before interpreting behaviour, so that the verdict turns on the evidence rather than on the substrate one happened to look at.

> If you decide ahead of time that a chatbot "understands", then read everything it does as understanding, you have not discovered anything — you have just heard your own assumption echoed back. The same trap springs shut if you decide ahead of time that it understands *nothing*. Either way the answer was baked in before the test began. To learn something real you have to say, in advance and out loud, what would count as evidence either way.

⊙ Cogency needs non-circular warrant

Dialectical cogency requires premises with accessible, non-question-begging justification

Claim · editorial

An argument is *cogent* against an opponent — able to rationally move that opponent from premises she can accept to a conclusion she presently rejects — only if its premises admit justification that is **(i)** in-principle accessible to suitably idealized versions of the parties, and **(ii)** not such that acquiring it requires already accepting the conclusion. Clause (ii) extends the prohibition on begging the question from the *content* of a premise to the *route by which warrant for it is obtained*: a premise one could come to know only by first granting the conclusion cannot be the lever that moves the argument.

The constraint separates **soundness** from **cogency** (probative force). It does not say that an inaccessible or circularly-warranted premise is *false*; the argument may be valid and even sound. It says such an argument cannot do the persuasive work its proponents claim — it cannot convert anyone not already converted. The constraint is deliberately weak, requiring not certainty or decisive certification but merely *accessible, non-circular justification*.

It is the principle behind the standard complaint against the modal ontological argument: granting that "possibly, a necessary being exists" entails "a necessary being exists," the argument still fails to compel, because warrant for the possibility premise is not available independently of accepting the theistic conclusion. A theist may hold it sound; no atheist is rationally moved. The same diagnosis is applied to the zombie argument by UNWARRANTABLE-PREMISE OBJECTION.

Who would deny it. A modal epistemologist of a broadly Williamsonian cast may hold that we are entitled to modal premises on the strength of the same fallible capacities that warrant ordinary counterfactual reasoning, with no special demand for accessible non-circular *certification* — in which case (i)–(ii) are too strong. A reader who balks at the constraint can still take the conclusions it licenses *conditionally*: if cogency requires accessible non-circular warrant, then the targeted argument is not cogent.

Distinct from INTUITION TRACKS TRAINED

DOMAIN, which concerns the *calibration* of intuitive verdicts against the cases they were formed on; this claim concerns the *dialectical route* to warrant and the prohibition on obtaining it only by presupposing the conclusion. The two are independent constraints on argument.

> A proof only does its job if the other side can come to believe its starting points *without* first swallowing the very thing you're trying to prove. An argument can be perfectly logical, and even have true premises, yet be useless for *persuading* anyone — if the only way to become sure of a premise is to already agree with the conclusion. That is the classic flaw in the slick argument for God: "if a necessary being is even possible, it exists." The logic is fine; the trouble is that no doubter can get warrant for "it's possible" except by already believing in God. This claim says that kind of circular *route to warrant* drains an argument of its force, whatever its truth — and the same charge is pressed against the zombie argument.

⊢ Cognitive closure

McGinn's cognitive-closure argument for mysterianism

Argument · after Mcginn

Cognitive closure is Colin McGinn's argument that the HARD PROBLEM of consciousness has a perfectly natural solution which the human mind is nonetheless built in such a way that it can never grasp. There is some property of the brain in virtue of which it gives rise to consciousness; but the only faculties by which we could form a concept of that property — looking inward, or studying the brain from outside — are each unfit to reach it. So the mind–body problem is, in McGinn's phrase, *soluble in itself but insoluble by us*. The resulting position is MYSTERIANISM.

THE ARGUMENT

The inference is *eliminative* (or *transcendental*): granting that an explanation exists, it rules out every route by which we might come to grasp it.

1. There is some natural property — call it P — of the brain in virtue of which the brain gives rise to consciousness (see A NATURAL EXPLANATION EXISTS).
2. A mind can be *constitutively closed* to certain properties — barred in principle, not merely currently ignorant, from forming the relevant concepts (see MINDS CAN BE CLOSED TO SOME CONCEPTS).
3. We could form the concept of P in only one of two ways: by *introspection*, or by *outer perception* of the brain and inference from it. (This exhaustive-routes assumption is a premise of the argument's own.)
4. But introspection presents experience, not its neural basis; and perception of the brain yields only spatial and physical concepts, which cannot capture the subjective — this is the EXPLANATORY GAP. So neither route can reach P.

∴ P is cognitively closed to us: the mind–body problem is real and has a natural answer, yet that answer lies permanently beyond human understanding (see WE ARE CLOSED TO THE PSYCHOPHYSICAL LINK). This is the support for MYSTERIANISM.

THE WEAK POINT

The argument's most disputed step is the pair (3)–(4): the claim that introspection and outer perception are the *only* routes to P, and that both are sealed off. Three replies press there. First, science routinely forms *theoretical* concepts that are neither introspected nor perceived — electric charge, the curvature of spacetime — by abduction and outright conceptual innovation, so the two-route dichotomy looks too narrow and McGinn under-rates our capacity to create new concepts. Second, we cannot reliably establish our own representational limits *in advance*, so a confident verdict of closure is itself suspect. Third, if the explanatory gap is *merely* a fact about cognitive closure, that tells against also reading it as a metaphysical residue: mysterianism and the residue view (see THAT EXPERIENCE OUTRUNS FUNCTIONAL ROLE) cannot both take the gap at face value.

HOW IT DIFFERS FROM NEIGHBOURING GAP ARGUMENTS

All of these start from the same explanatory gap but draw opposite morals from it. The EXPLANATORY-GAP ARGUMENT treats the gap as evidence about *metaphysics* — that the phenomenal is not exhausted by functional role. Cognitive closure treats the very same gap as evidence about *our cognitive limits*: the problem is real and natural, merely humanly unsolvable. It differs again from the DECOMPOSITION-INDETERMINACY OBJECTION and the STRUCTURE-AND-DYNAMICS ARGUMENT, which contend that function *omits* or *fails to fix* the phenomenal; mysterianism instead grants that a natural answer exists and denies only our access to it.

> The argument goes: (1) there's some real, physical reason the brain produces consciousness; (2) minds can be permanently unable to understand certain things; (3) the only ways we could ever get hold of that reason are by looking inward (introspection) or by studying the brain from outside; (4) but looking inward shows you the feeling, not the wiring, and studying the brain from outside only ever gives you physical/spatial facts that never add up to a feeling. So neither route can deliver the answer — meaning it's locked away from us for good. The main pushback: science invents concepts all the time that we neither feel nor see directly (like "electron"), so maybe the two routes aren't the only ones.

SEE ALSO

- ▹ MYSTERIANISM — the view this argument supports: the mind–body problem has a natural solution we are constitutionally unable to reach.
- ▹ HARD PROBLEM — the problem McGinn declares humanly insoluble.
- ▹ EXPLANATORY-GAP ARGUMENT — the same gap read as metaphysical evidence rather than as a limit on our concepts.
- ▹ PHENOMENAL RESIDUE — the residue reading the third reply says cannot coexist with a merely-cognitive diagnosis of the gap.
- ▹ DECOMPOSITION-INDETERMINACY OBJECTION and STRUCTURE-AND-DYNAMICS ARGUMENT — neighbouring gap arguments that fault function itself, where mysterianism faults only our access.
- ▹ NATURAL EXPLANATION OF CONSCIOUSNESS, POSSIBILITY OF COGNITIVE CLOSURE, HUMANS CLOSED TO THE LINK — the argument's two premises and its conclusion.

★ Coherentism

Position

Coherentism holds that beliefs are justified by how well they *hang together* — by mutual support within the believer's whole system — and not by resting on a privileged foundation. There are no BASIC BELIEFS; justification is a holistic property of the web, every belief supported by its fit with the others (see JUSTIFICATION AS COHERENCE). Faced with THE MÜNCHHAUSEN TRILEMMA, coherentism takes the *circularity* horn and argues the circle is not vicious: a large, system-wide pattern of mutual support is a different thing from a small question-begging loop. The governing image is Neurath's boat — we rebuild the ship plank by plank while afloat, with no dry dock to stand on.

HOW IT DIFFERS FROM NEIGHBOURING VIEWS

Against FOUNDATIONALISM, it denies any non-inferentially justified bedrock and replaces the upward flow of support with a web in which justification runs in all directions. Against INFINITISM, it *closes* the structure into a system of mutual support rather than leaving an open, non-repeating chain. Its standing difficulty is the **isolation (or input) objection**: a perfectly coherent system of beliefs could still be a self-consistent fiction, sealed off from the world — so coherence alone seems not to guarantee contact with reality. Replies typically add a constraint that the system answer to observational input, which critics say smuggles a foundational element back in.

> Coherentism says beliefs hold each other up like the logs of a raft, not like a house on a foundation. A belief is justified when it fits with everything else you think — no special bedrock beliefs required, because the support runs every which way. The worry critics keep pressing: a story can be perfectly self-consistent and still be a fantasy. So coherence has to be doing more than just internal tidiness for this to track the truth.

SEE ALSO

- ▹ JUSTIFICATION AS COHERENCE — the position's defining commitment.
- ▹ THERE ARE BASIC BELIEFS — the foundationalist thesis it rejects.
- ▹ MÜNCHHAUSEN TRILEMMA — the regress problem it answers by taking the circularity horn.
- ▹ FOUNDATIONALISM, INFINITISM — the rival answers to the same trilemma.

⊙ Coincident phenomenal intensions

For phenomenal concepts, the two intensions coincide

Claim · after Chalmers

For a phenomenal concept — the concept of what it is like to undergo an experience, such as the concept of *the feeling of pain* or *phenomenal red* — the PRIMARY INTENSION and the SECONDARY INTENSION are one and the same function. Unlike "water," whose reference is fixed by a contingent role (the watery stuff) that something other than H_2O could have filled, a phenomenal concept fixes its reference directly on the felt quality itself, with no reference-fixing description standing between the concept and its object. There is therefore no wedge between how the experience *presents* and what it *is*: for a feeling, the appearance is the reality.

The contrast is the whole point. With natural-kind terms the two intensions diverge, and that gap is where a Kripkean necessity can be both necessary and knowable only a posteriori (see A-posteriori necessities); the two-dimensionalist exploits the gap to recast apparent impossibilities as A-posteriori necessities as misdescribed possibilities. A phenomenal concept offers no such gap. "Watery stuff that is not H_2O" describes a genuine possibility misnamed "water," but "something that feels exactly like pain yet is not pain" describes nothing at all, because feeling-like-pain is not a fallible mode of presentation of pain — it simply is the pain (see Phenomenal consciousness).

Scope and bearing

The claim is a thesis in the semantics of phenomenal concepts, not directly a claim about what consciousness is. Its work is to remove an escape route. Chalmers's Modal rationalism licenses the step from ideal conceivability to *primary* possibility; the question is whether primary possibility carries through to full metaphysical possibility in the phenomenal case. Where primary and secondary intensions coincide, there is no separate secondary necessity in which a metaphysical impossibility could hide — so primary possibility *is* metaphysical possibility. Applied to the Philosophical zombie, this closes the gap the type-B physicalist needs: if the zombie is conceivable it is genuinely possible, and physicalism is false (see Zombie argument).

Who would deny it

The type-B physicalist (see Type-A / Type-B / Type-C materialism) — the position the claim is aimed at. Type-B physicalism holds that "pain is such-and-such brain state" is a *strong* a posteriori necessity, with a genuine primary/secondary gap supplied by a special *phenomenal concept* that refers to a physical state under a distinctive mode of presentation. On that view there is, after all, an appearance/reality distinction for experience at the level of *concepts*, the zombie is conceivable but impossible, and the present claim is exactly what fails. The dispute thus narrows to whether phenomenal concepts can carry a reference-fixing mode of presentation distinct from their referent — the hinge on which the entire conceivability argument turns.

> For most words there's a difference between how something seems and what it really is: the stuff in the rivers *seems* like "water" but really *is* H_2O, and that gap is why "water is H_2O" can be a surprise we had to discover. This claim says feelings are different. With a feeling — pain, or the look of red — there's no gap between the seeming and the being: a state that felt *exactly* like pain in every respect would just *be* pain; there's nothing more to a feeling than how it feels. Why it matters: the physicalist's best reply to the Philosophical zombie (a creature physically just like you but with no inner feel) is to say "sure, you can imagine it, but imagining isn't possibility — just as you can 'imagine' water that isn't H_2O." This claim blocks that reply for consciousness: because feelings have no seeming/being gap, imagining a zombie really does show a zombie is possible — and that would mean consciousness is something over and above the physical.

See also

- Primary intension — the epistemic, reference-fixing dimension of meaning; for phenomenal concepts it has nothing to fix onto but the feel itself.
- Secondary intension — the rigid, actual-world-fixed dimension; the claim is that for feelings it does not diverge from the primary intension.
- A-posteriori necessities as misdescribed possibilities — the water-style rescue the claim says is unavailable here, since there is no primary/secondary gap to misdescribe.
- Modal rationalism — supplies the conceivability-to-primary-possibility step that this claim then extends, in the phenomenal case, all the way to metaphysical possibility.
- Zombie argument — the argument whose final move depends on this claim: with the intensions coincident, the zombie's conceivability yields its genuine possibility.
- Type-A / Type-B / Type-C materialism — type-B physicalism, the view that denies this claim by positing a phenomenal-concept mode of presentation distinct from its referent.
- Phenomenal consciousness — what phenomenal concepts are concepts *of*; the "appearance is reality" point is a point about felt

experience.

⊢ Combination problem

The combination problem — micro-subjects cannot compose a macro-subject

Argument · after James

The *combination problem* is the standing objection to Panpsychism, severe enough that many regard it as the view's make-or-break test. Constitutive panpsychism holds that a macro-subject like you is *constituted by* the experiential properties of your fundamental micro-parts. The objection denies that this constitution is available: a crowd of tiny subjects, each with its own private point of view, cannot fuse into one further subject whose experience contains theirs. The problem originates with William James (see 1890) and is sharpened into its hardest "subject-summing" form by Sam Coleman (see 2014); David Chalmers (see 2017) anatomises its parts.

The argument

The inference is deductive — an objection, so it *attacks* the constitutive thesis rather than concluding a positive claim.

1. (see Micro-subjects do not sum) Distinct micro-subjects — each with its own private point of view and self-contained phenomenal field — do not, merely by being arranged or related, compose a further single subject whose experience contains theirs.
2. Constitutive panpsychism requires exactly such composition: your unified experience is supposed to be made up of the experiences of your fundamental parts.

∴ The constitution the view needs is unavailable, so it cannot deliver the single, unified macro-subject we actually find. This *attacks* Fundamental ubiquitous consciousness at its constitutive joint.

The objection also generalises a deeper worry. Panpsychism was offered to *avoid* getting experience from the non-experiential, yet it now owes an account of getting *macro*-experience from *micro*-experience — so the very explanatory burden the view promised to discharge reappears here, no easier than before. Chalmers distinguishes three faces of the problem: combining micro-*subjects* into a subject, micro-*qualities* into a unified quality-field, and micro-*structure* into the structure of a macro-experience; the subject-combination face is the most resistant.

The weak point

The objection turns on premise (1). Constitutive panpsychists reply with accounts of *phenomenal bonding* or fusion on which subjects do combine after all. Alternatively they retreat to *panqualityism*, which denies that there are micro-*subjects* to combine in the first place — only unexperienced qualities — and pays for the move with a residual explanatory gap at the point where subjecthood has to be added back. Which of these escapes succeeds is the live question on which the fate of constitutive panpsychism turns.

> Panpsychism's appeal is that every tiny speck of matter carries a flicker of experience, so your mind is built up out of those flickers. The combination problem is the bill for that appeal. Your experience right now is *one* unified thing — a single point of view taking in this whole sentence at once. How do countless separate little experiences, each locked inside its own tiny perspective, merge into that one? William James argued they simply cannot: a hundred little feelings do not add up to one big feeling that swallows them — each stays its own private whole. If he is right, the panpsychist's "build a mind out of mini-minds" recipe has no step that actually does the building. Worse, it was supposed to spare us the mystery of making feeling out of the feelingless, but now it owes the matching mystery of making *one* feeler out of *many* — which looks just as hard.

See also

- ▹ Micro-subjects do not sum — the premise that drives the objection: subjects do not sum.
- ▹ Fundamental ubiquitous consciousness — the constitutive thesis this objection attacks.
- ▹ Panpsychism — the view under attack, and where the combination problem sits among its varieties.
- ▹ Cosmopsychism — the top-down cousin, which trades this problem for its mirror image, the *decombination* problem.

★ Common-sense realism

Position

Common-sense realism holds that the existence of a mind-independent external world (see The external world exists) is a *first principle* — something we are entitled to take as known without proof — and that the burden of argument lies with the skeptic, not the plain believer. The view is Thomas Reid's (see Reid — Essays on the Intellectual Powers of Man

(1785)), framed against Humean and Cartesian skepticism: the deliverances of our constitution (that there are external objects, other minds, a remembered past) are principles we cannot help believing and have every right to trust; a philosophical argument that concludes otherwise is more likely to have a false premise than to overturn what every sane person knows. G. E. Moore's "HERE IS A HAND" is the position's modern emblem.

How it differs from neighbouring views

Common-sense realism is, in spirit, a foundationalism of *first principles*: the world's existence is a non-inferentially warranted starting point, akin to a BASIC BELIEF, not a conclusion. But it adds a distinctive *dialectical* claim — that the skeptic bears the burden of proof, so that a clash between a skeptical argument and a common-sense certainty is resolved *against* the argument (the MOOREAN SHIFT). It is close to, yet distinct from, HINGE EPISTEMOLOGY: the Reidian first principle is held as a *known truth*, a deliverance of our cognitive constitution, whereas a Wittgensteinian HINGE is arational — neither justified nor doubted, but the framework presupposed by justifying. Where FALLIBILISM would deny that the first principle is *certain*, common-sense realism treats it as a fixed point against which doubts are measured.

> Common-sense realism says: of course there's a real world — that's not a conclusion you argue *to*, it's a starting point you argue *from*. If some philosophical chain of reasoning ends in "so maybe nothing outside your mind exists," the sensible response is that one of its premises must be wrong, because you're far surer of the world than of any clever premise. Reid said our basic convictions are gifts of our nature we're entitled to trust; Moore just held up a hand. The burden, on this view, is on the doubter.

See also

- ▷ THE EXTERNAL WORLD EXISTS — the first principle the position is committed to.
- ▷ MOORE'S PROOF — its modern emblem and the Moorean burden-shift.
- ▷ BASIC BELIEF — the non-inferential warrant a first principle resembles.
- ▷ HINGE EPISTEMOLOGY — the neighbour that treats the same certainty as arational framework, not known truth.
- ▷ FOUNDATIONALISM — the general structure common-sense realism instances.
- ▷ REID — ESSAYS ON THE INTELLECTUAL POWERS OF MAN (1785) — the source of the philosophy of common sense.

♦ Completeness of physics

Concept

The **completeness of physics** is the thesis that the physical world is causally self-contained: every physical effect that has a cause at all has a *sufficient* physical cause. Whenever a physical event is produced, one need never step outside physics to find what produced it — a complete physical story can in principle be told for every physical happening. The principle is sometimes called the *causal completeness*, *causal closure*, or *causal self-sufficiency* of the physical domain, and it is the empirical engine of the modern argument for physicalism (see CAUSAL ARGUMENT FOR PHYSICALISM, PHYSICALISM).

The careful formulation matters. Completeness does not say that *every* event is physical, nor that the physical causes are the *only* causes; it says only that the physical causes already on the books are **sufficient** — enough, by themselves, to fix the physical effect. So stated, it leaves logical room for a second, non-physical cause of the same effect. Closing that room is the work of a separate premise about overdetermination (see NO SYSTEMATIC OVERDETERMINATION); the two together yield the strong conclusion that nothing non-physical does any *distinctive* causal work in the physical world.

Papineau (see PAPINEAU — THINKING ABOUT CONSCIOUSNESS (2002)) stresses that completeness is not an a priori truth but a hard-won **empirical discovery**. His "history of the completeness of physics" traces how the inventory of basic forces was progressively closed: vital and mental forces, once posited to explain living and intelligent behaviour, were squeezed out as physiology and physics found sufficient physical causes for the bodily movements they were meant to produce. The conservation laws are the usual modern evidence — a non-physical cause injecting energy or momentum into a body would show up as a violation, and none is observed.

Quantum indeterminism

A naive statement of completeness is *deterministic*: every physical effect is fully fixed by prior physical states under physical law. On the standard reading of quantum mechanics that is simply false. When an atom decays or a photon meets a half-silvered mirror, the prior physical state fixes only a *probability distribution* over outcomes, not the outcome itself. There is genuine chance in the basement.

Completeness survives because it was never a claim about *determination* in the first place; it is a claim about *probabilities*. The careful formulation runs: in so far as a physical effect is determined at all, it is determined by physical antecedents, and in so far as it is merely chancy, its objective probabilities are fixed by physical law *alone*. The dial may sometimes land at random, but nothing non-physical sets the odds it lands on. So stated, completeness rules out non-physical *causes*, not physical *chance*: randomness is internal to physics, not a leak in its wall, and the thesis is compatible with genuine indeterminism rather than a disguised commitment to determinism.

This matters for the mind because quantum chance is sometimes floated as a loophole for the *interactionist* — a place where an immaterial will might "make a difference" without violating the conservation laws, by nudging which outcome a brain's quantum events take. The probabilistic formulation is exactly what closes the loophole, and it does so by reposing the same dilemma the Causal closure of the physical poses elsewhere. Either the mind merely rides along with whatever the physical chances throw up, in which case it does no distinctive work (see Epiphenomenalism); or it *biases* the probabilities, making an outcome likelier than the physical law prescribes, in which case physics is not complete after all. The second horn is a genuine empirical claim, and the physicalist bets it is false: no deviation from quantum statistics has ever been measured in a brain. So indeterminism hands the dualist no free mental causation. It only restates the choice between riding inertly and breaking completeness, with the breach now statistical rather than energetic.

Two cautions. First, the picture is interpretation-dependent: on Bohmian mechanics the world is deterministic again and completeness reverts to its strong form, while on Everettian quantum mechanics there is no single collapse-outcome to be "caused" at all. Completeness holds across these readings because in each the *physics* remains causally self-contained, but the gloss on "randomness" differs. Second, indeterminism is not by itself a resource for free will: an outcome that is merely random is no more an agent's own doing than one that is determined, so nothing about how completeness treats quantum chance smuggles libertarian agency in through the back door.

The completeness premise at work

Completeness is the load-bearing premise of the **causal argument for physicalism**: if every physical effect already has a sufficient physical cause, then a mental event that genuinely causes a physical effect (a hand's rising, a word's being spoken) must *be* one of those physical causes, on pain of either doing no work at all (see Epiphenomenalism) or redundantly duplicating a cause already present. This is why anti-physicalists who want the mind to make a difference — interactionist dualists especially — must reject or restrict completeness, and why much of the dispute over Hard problem turns on whether the principle is as secure as its defenders claim.

> The idea is that the physical world is a closed shop, causally speaking: whenever something physical happens, there is always a complete physical explanation for it — some earlier physical stuff that was enough to make it happen. You never *have* to bring in a soul, a spirit, or any non-physical push to finish the story. It is offered not as an armchair assumption but as something science has gradually found to be true: every time people expected a special "life force" or "mind force" to be needed, a purely physical cause turned up instead. If that is right, then a thought that really moves your arm has to be a physical thing too — otherwise the arm would already be fully accounted for and the thought would have nothing left to do.

See also

- ▹ Causal closure of the physical — the closely allied "no causes cross into the physical from outside" formulation; completeness plus no-overdetermination yields closure.
- ▹ Causal argument for physicalism — the inference for which this thesis is the chief premise.
- ▹ Completeness premise — completeness stated as the truth-apt premise that argument rests on.
- ▹ No systematic overdetermination — the companion premise that closes the loophole completeness leaves open.
- ▹ Physicalism — the view this principle is marshalled to support.
- ▹ Epiphenomenalism — the fate completeness forces on a mental cause that is not itself phys-

ical.

⊙ Conceivability entails possibility

Ideal conceivability entails metaphysical possibility

Claim · contested · after Chalmers

If a scenario is *ideally* conceivable — coherent under idealized rational reflection, with no concealed contradiction — then it is metaphysically possible. This is the modal-epistemology bridge of the zombie argument, and its most contested step. It is emphatically *not* the trivial claim that whatever we can loosely picture is possible; it concerns *ideal* conceivability, what survives unlimited rational scrutiny. On the two-dimensional version (David Chalmers, CHALMERS — THE CONSCIOUS MIND (1996); CHALMERS — ON SENSE AND INTENSION (2002)), the principle reads more precisely: ideal *primary* conceivability entails *primary* possibility, and for the zombie case the primary and secondary intensions are taken to coincide.

The bridge is what converts ZOMBIE CONCEIVABILITY into a conclusion against physicalism, so the whole dispute concentrates here.

ARGUED BOTH WAYS

It is not a brute premise. It is **defended** by modal rationalism (see MODAL RATIONALISM): once the PRIMARY INTENSION and SECONDARY INTENSION are distinguished, the apparent counterexamples are misdescribed possibilities (see A-POSTERIORI NECESSITIES AS MISDESCRIBED POSSIBILITIES), so ideal primary conceivability tracks primary possibility. It is **attacked** by the a-posteriori-necessity objection (see A-POSTERIORI-NECESSITY OBJECTION): Kripkean necessities (see A-POSTERIORI NECESSITIES, KRIPKE — NAMING AND NECESSITY (1980)) already break the link, so a zombie may be conceivable yet impossible — the type-B physicalist reading (see TYPE-A / TYPE-B / TYPE-C MATERIALISM). A deeper, allied **phenomenal-concept strategy** (see PHENOMENAL-CONCEPT DEFLATION, PHENOMENAL-CONCEPT TRADE-OFF) explains the conceivability as an artefact of how phenomenal concepts work. Throughout, the crux is the same: whether the *phenomenal* case has the appearance/reality gap that lets the water case escape.

The claim is distinct from ZOMBIE CONCEIVABILITY (the antecedent it operates on) and from INTUITION TRACKS TRAINED DOMAIN (an epistemology of *intuitions about cases*); this one is specifically about the conceivability→possibility inference.

> This is the hinge the whole zombie argument swings on: that *if you can coherently imagine something — really coherently, with no hidden contradiction — then it is genuinely possible.* Grant it, and "I can imagine a zombie" turns into "a zombie is possible," which would mean experience is something extra beyond the physical. The classic pushback borrows from science: "water is H_2O" is something you *can* imagine being false, yet it is necessarily true — so being able to imagine otherwise does not always prove real possibility. Maybe zombies are like that: imaginable but impossible.

SEE ALSO

- ▷ ZOMBIE CONCEIVABILITY — the antecedent this bridge operates on.
- ▷ MODAL RATIONALISM — the defence of the bridge.
- ▷ A-POSTERIORI-NECESSITY OBJECTION — the objection to it.
- ▷ A-POSTERIORI NECESSITIES AS MISDESCRIBED POSSIBILITIES — the premise the defence rests on.
- ▷ A-POSTERIORI NECESSITIES — the Kripkean cases the objection wields.
- ▷ TYPE-A / TYPE-B / TYPE-C MATERIALISM — type-B physicalism, the reading on which the bridge fails.
- ▷ PHENOMENAL-CONCEPT DEFLATION / PHENOMENAL-CONCEPT TRADE-OFF — the allied phenomenal-concept strategy.
- ▷ INTUITION TRACKS TRAINED DOMAIN — a distinct point about the warrant of intuitions.

⊢ Connectionist reply

The Connectionist Reply — a massively parallel, brain-like architecture is the real candidate for understanding

Argument · after Churchland

The *Connectionist Reply*, pressed by Paul and Patricia Churchland (see CHURCHLAND & CHURCHLAND — COULD A MACHINE THINK? (1990)), grants that a single clerk serially executing rules — a classical von Neumann machine — may well not understand, but denies this tells against *connectionist* systems. Real cognition, they argue, runs on a massively *parallel*, brain-like architecture of many simple units adjusting weighted connections at once. Searle's room models the wrong kind of machine, so its failure is uninformative: of course the lone clerk, handling one tiny step at a time, grasps nothing of a process whose understanding (if any) is a global, distributed, emergent property of the whole parallel net.

The reply attacks the DERIVATION by denying that A3 is self-evident: the Churchlands hold that whether syntax suffices for semantics is an open empirical question about architectures, not an axiom to be settled from the armchair, and that the brain itself is an existence proof that the right physical network yields meaning. It is the BRAIN SIMULATOR REPLY's cousin, trading neuron-by-neuron simulation for the abstract parallel *style* of neural computation.

THE CRUX

Searle's CHINESE GYM REJOINDER scales the room up to a *gymnasium* full of monolingual English speakers, each playing the role of one node in the parallel net, passing signals on cue. The architecture is now parallel, yet — Searle insists — no individual understands Chinese and neither does the assembled crowd; parallelism multiplies the syntax-shuffling without adding a drop of semantics. The Churchlands' deeper point survives the parody, though: they do not claim the *gym* understands, but that A3 is an unargued assumption, and that meaning may be a feature of the right *physical* network that no purely formal redescription captures.

> The room imagines a single worker plodding through rules one at a time — but that's not how brains work. Brains are huge parallel networks, billions of simple units firing together, and (the Churchlands say) understanding might be a property of that whole parallel pattern, not something you'd find by watching one slow clerk. So the room is rigged: it models the wrong kind of machine. Searle's answer: fine, fill a gymnasium with people, each acting as one "neuron," all working at once — it's parallel now, and *still* nobody in the gym understands Chinese. The Churchlands' lasting point is subtler: maybe the real lesson is that Searle just *assumed* shape-shuffling can't make meaning, when whether it can is a question for science, not a self-evident axiom.

SEE ALSO

- ▷ CHINESE ROOM DERIVATION — the argument attacked, by denying A3 is self-evident.
- ▷ CHINESE GYM — Searle's reply: a parallel "Chinese gym" still understands nothing.
- ▷ BRAIN SIMULATOR REPLY — the allied appeal to neural fidelity, here at the level of architecture.
- ▷ LLM-REPRESENTATIONS REPLY — the modern empirical descendant: evidence that text-trained networks actually do build human-aligned representations, vindicating the Churchlands' bet.
- ▷ CHURCHLAND & CHURCHLAND — COULD A MACHINE THINK? (1990) — the Churchlands' statement of the reply.

⊙ Consistency-certification is oracle-hard

Reliably certifying the consistency of arithmetic-rich scenarios is at least as hard as the halting problem

Claim · editorial

Any method that *reliably* returns correct consistency-verdicts for theories rich enough to interpret arithmetic thereby decides the halting problem, and so is not an effective (Turing-computable) procedure. The hardness lives in the *method*, not in any single instance: a fixed sentence "T is consistent" has a determinate truth value that a trivial one-state machine could output; what is uncomputable is a procedure that tracks consistency *across the class* without being told the answer.

The reduction is standard. Let Q be Robinson arithmetic and, for a Turing machine M and input x, let $h_{M,x}$ be the Σ_1 sentence "M halts on x." Consider

$$T_{M,x} := Q \cup \{\neg h_{M,x}\}.$$

If M halts on x then $h_{M,x}$ is a true Σ_1 sentence, so $Q \vdash h_{M,x}$ and $T_{M,x}$ is inconsistent; if M does not halt then $\mathbb{N} \models T_{M,x}$ and it is consistent. Hence

$$T_{M,x} \text{ consistent} \iff M \text{ does not halt on } x,$$

so a reliable consistency-tester decides the complement of the halting set, i.e. computes $\emptyset'$. For *recursively axiomatized* theories "inconsistent" is Σ_1 (search for a proof of $\bot$), so "consistent" is Π_1 and the verdict-function is Π_1-complete, $\equiv_T \emptyset'$. By Turing's theorem (see TURING — ON COMPUTABLE NUMBERS, WITH AN APPLICATION TO THE ENTSCHEIDUNGSPROBLEM (1936)) $\emptyset'$ is not computable, so no effective method reliably certifies consistency here.

What carries the weight. The load-bearing direction is the *lower* bound — certification is at **least** $\emptyset'$-hard — which needs only that the scenario interpret arithmetic. The matching upper bound (Π_1, decidable relative to $\emptyset'$) assumes the theory is recursively axiomatized; a genuinely *complete* physical description P may not be, in which case certification is harder still, not easier. Either way no effective method suffices. A faculty relabelled "rational insight" does not escape: if it reliably outruns proof-search on these inputs it computes a non-computable set, and naming it insight rather than computation does not lower its Turing degree.

Scope. A claim about the *complexity of a method*, neutral on whether minds possess non-effective faculties — that is the further question handled by the trilemma of UNWARRANTABLE-PREMISE OBJECTION. It does not say any particular consistency fact is unknowable; isolated facts are known piecemeal by proof.

Distinct from HALTING PROBLEM UNDECIDABLE, which states the undecidability of one function: this claim transports that result onto *consistency-* (hence ideal-conceivability-) certification over a class, which is what links computability to the CONCEIVABILITY premise. It is the engine behind the slogan that the logically omniscient sage — who knows at once every consequence of what is given — simply *is* a halting oracle (see HALTING ORACLE).

> There is no general, mechanical way to certify that a rich theory is *free of hidden contradiction*. You can always *catch* an inconsistency if one exists — just grind through proofs until a contradiction turns up — but confirming there is *none*, across the board, would let you solve the halting problem, which Turing proved no computer can. The trick is a dial that converts "machine M runs forever" into "this little theory is consistent," so a consistency-checker would secretly be a halting-checker. The upshot: verifying that a scenario as rich as a complete physics is genuinely coherent is not the kind of thing a finite, step-by-step reasoner can do — it needs an oracle no ordinary computer can be.

⊢ Continuity argument

The continuity argument — no non-arbitrary line where experience switches on

Argument · *after Nagel*

The *continuity argument* is a second route to PANPSYCHISM, pressed by Thomas Nagel (see 1979). Where the GENETIC ARGUMENT reasons from the impossibility of brute emergence, this one reasons from the absence of any principled boundary: somewhere between a rock and a human being, consciousness seems to appear, yet there is no non-arbitrary place to draw the line. Rather than draw one anyway, the panpsychist lets experience reach all the way down.

THE ARGUMENT

The inference is abductive and eliminative — an inference to the most principled of the surviving options, not a deductive proof.

1. (see ABSENCE OF AN EXPERIENTIAL THRESHOLD) If experience were confined to brains, or to life, or to systems above some complexity threshold, there should be a principled place where it first switches on; but no such non-arbitrary line is in view — the relevant differences across the supposed boundary look like differences of degree and organisation.
2. Drawing the line anyway, at an arbitrary point, is unmotivated; and denying that experience is real is ruled out (see REALITY OF PHENOMENAL CONSCIOUSNESS).
3. The remaining option that avoids an arbitrary threshold is that experience, in some attenuated form, is present all the way down.

∴ This *supports* FUNDAMENTAL UBIQUITOUS CONSCIOUSNESS: the line-free, parsimonious position is that experience pervades nature rather than appearing abruptly at one favoured level.

THE WEAK POINT

The argument leans on premise (1). An opponent can hold that there *is* a principled threshold — a level of integrated information, say, or of functional organisation, at which experience genuinely (if strongly) emerges (see ACCEPTABILITY OF STRONG EMERGENCE) — in which case the "no non-arbitrary line" premise is simply false. The argument is also only abductive: it shows that panpsychism avoids an awkward commitment, not that its rivals are incoherent, so it carries less weight than the genetic argument aims to.

> Somewhere between a rock and a human being, consciousness seems to show up. But where, exactly? Pick any line — "only brains," "only living things," "only things this complex" — and it looks arbitrary, because the differences on either side are matters of more-or-less, not a sudden all-or-nothing switch. Rather than draw an unmotivated line in the sand, the panpsychist says: do not draw one at all — let a faint glimmer of experience reach all the way down, growing richer as things get more organised. It is offered as

> the tidier option, not a knock-down proof; an opponent who can point to a *principled* switch-on point blocks it.

SEE ALSO

- ▹ FUNDAMENTAL UBIQUITOUS CONSCIOUSNESS — the conclusion this argument supports.
- ▹ ABSENCE OF AN EXPERIENTIAL THRESHOLD — its single premise: no principled point at which experience first appears.
- ▹ GENETIC ARGUMENT — the companion, stronger case for the same conclusion, from the impossibility of brute emergence.
- ▹ ACCEPTABILITY OF STRONG EMERGENCE — the denial that powers the main objection: there *is* a principled emergence threshold.
- ▹ PANPSYCHISM — the view this argument supports.

★ Cosmopsychism

Position · after Goff

Cosmopsychism holds that the cosmos as a whole is the one fundamental conscious subject, and that individual minds like yours and mine are *derivative* — aspects, or partial dissociations, of that single universal consciousness rather than building blocks of it. It is a priority-monist cousin of panpsychism: where ordinary panpsychism builds large minds *up* from tiny ones, cosmopsychism carves small minds *out* of one great cosmic mind, with grounding running top-down from the whole to its parts.

The position commits to the claim that THE UNIVERSE AS A WHOLE IS THE ONE FUNDAMENTAL CONSCIOUS SUBJECT. It answers the HARD PROBLEM OF CONSCIOUSNESS by placing experience at the fundamental level — here the cosmic level — so that experience never has to be produced from the non-experiential. The view is defended by Goff (see GOFF — CONSCIOUSNESS AND FUNDAMENTAL REALITY (2017)), Shani (see SHANI — COSMOPSYCHISM: A HOLISTIC APPROACH TO THE METAPHYSICS OF EXPERIENCE (2015)), and Nagasawa and Wager (see NAGASAWA & WAGER — PANPSYCHISM AND PRIORITY COSMOPSYCHISM (2017)).

HOW IT DIFFERS FROM NEIGHBOURING VIEWS

Like PANPSYCHISM (POSITION) and RUSSELLIAN MONISM, cosmopsychism keeps consciousness fundamental rather than derived from non-experiential matter. What sets it apart is the *direction* of grounding. Against constitutive *micropanpsychism*, the fundamental subjects are not the smallest parts but the whole. This trades one hard problem for its mirror image: in place of the COMBINATION PROBLEM — how do micro-experiences add up to one mind? — cosmopsychism faces the *de*combination problem: how does the one cosmic subject divide into many bounded, walled-off subjects like us?

> Cosmopsychism starts from the top: the basic conscious thing isn't a particle but the *universe as a whole*, and you and I are like ripples or partial "splits" within that one great mind. It's panpsychism run in reverse — rather than building big minds out of tiny ones, it carves small minds out of a single cosmic one. That dodges the question "how do trillions of atom-feelings merge into your one experience?" but raises the opposite puzzle: how does one universe-wide consciousness break up into separate, walled-off selves?

SEE ALSO

- ▹ COSMOS AS FUNDAMENTAL SUBJECT — the core commitment: the cosmos is the one fundamental subject, minds derivative.
- ▹ HARD PROBLEM (THE QUESTION) — the problem the position addresses, by placing experience at the cosmic base.
- ▹ PANPSYCHISM (POSITION) — the bottom-up cousin it inverts; both keep consciousness fundamental.
- ▹ PANPSYCHISM — the broader doctrine, whose taxonomy sorts cosmopsychism as the whole-first (vs. micropsychist) branch.
- ▹ RUSSELLIAN MONISM — the allied view that consciousness is the intrinsic nature physics leaves unspecified.
- ▹ COMBINATION PROBLEM — the objection cosmopsychism escapes, only to face its mirror, the decombination problem.

⊙ Cosmos as fundamental subject

The universe as a whole is the one fundamental conscious subject

Claim · after Goff

The cosmos as a whole is a single fundamental conscious *subject*, and individual minds like ours are *derivative* — aspects or dissociations *of* the one universal consciousness rather than building blocks that compose it. This is *cosmopsychism*, defended by Philip Goff (see GOFF — CONSCIOUSNESS AND FUNDAMENTAL REALITY (2017)), Itay Shani (see SHANI — COSMOPSYCHISM: A HOLISTIC APPROACH TO THE META-

PHYSICS OF EXPERIENCE (2015)), and Yujin Nagasawa (see NAGASAWA & WAGER — PANPSYCHISM AND PRIORITY COSMOPSYCHISM (2017)). It is a priority-monist cousin of PANPSYCHISM: grounding runs *top-down*, from the whole to its parts, rather than bottom-up.

The view combines two commitments. The first is *priority monism* — the thesis that the cosmos is the one fundamental concrete object, on which everything else depends. The second is experiential fundamentality — that this one fundamental object is conscious. Its characteristic difficulty is the mirror image of the trouble panpsychism faces. Where micro-level panpsychism wrestles with the *combination problem* (how do many tiny experiences add up to one unified mind?), cosmopsychism faces the *decombination problem*: how does the single universal subject give rise to many distinct, bounded subjects like us?

It is denied by constitutive micropanpsychists, for whom the fundamental subjects are micro rather than cosmic; by physicalists and dualists; and by anyone who rejects priority monism. It is distinct from THE CONSTITUTIVE-PANPSYCHIST CLAIM precisely on the direction of grounding: constitutive panpsychism builds macro-minds *up* from micro-experiences, whereas cosmopsychism derives them *down* from the whole — combination versus decombination.

> Most views that make consciousness fundamental put it in the tiniest bits of matter and try to build big minds out of many little ones. Cosmopsychism flips this: it says the *whole universe* is the one basic conscious thing, and individual minds like yours are something the universal mind splits into, the way a single person with a dissociative condition can seem to host several selves. The hard part isn't getting from small to big — it's the reverse: explaining how one cosmic mind comes to contain many separate, walled-off minds like ours.

SEE ALSO

- ▷ CONSTITUTIVE-PANPSYCHIST CLAIM — the bottom-up rival: consciousness in the smallest things, composed up into minds; cosmopsychism runs the other way.
- ▷ COSMOPSYCHISM — the named position this claim states.
- ▷ PANPSYCHISM — the broader family; cosmopsychism is its priority-monist variant.

D

⊢ Deutsch's anti-solipsism

The bad-explanation argument against solipsism — it reconstructs realism and adds only surplus

Argument · after Deutsch

David Deutsch's reply to the skeptic does not try to *prove* the external world against the solipsist on the solipsist's own evidential terms — a project KNOWN TO BE HOPELESS. It takes solipsism at its word, follows it through, and dismisses it as a *bad explanation*. The move belongs to the second anti-skeptical family catalogued under KNOWING THE EXTERNAL WORLD, alongside but distinct from RETORSION.

The argument

The inference is abductive — an inference to the best explanation.

1. (see SOLIPSISM IS PARASITIC ON REALISM) Solipsism, taken seriously, must reconstruct within the self an entity with the full lawful, autonomous, other-minds-containing structure of an external world; it is thus realism *plus* the surplus claim that this structure is "really inner."
2. (see GOOD EXPLANATIONS ARE HARD TO VARY) A good explanation is hard to vary and carries no idle surplus; a theory that reduplicates a rival's content and adds unexplained ideology is, by that measure, the worse explanation.

∴ Realism is the better explanation and solipsism is to be rejected — not refuted by demonstration but dismissed as explanatorily empty. The external world exists (see THE EXTERNAL WORLD EXISTS); the solipsist's "world inside me" is that very world under a costlier description.

The weakest premise

The pivot is premise (2) — that explanatory virtues like parsimony and being hard-to-vary are guides to *truth*, not merely to convenience. A determined skeptic can grant everything about structure and deny that the simpler, surplus-free theory is thereby more likely *true*; explanatory considerations, the objection runs, presuppose rather than establish that our theorizing tracks a real world. Deutsch's stance is frankly fallibilist (see KNOWLEDGE IS CONJECTURAL): there is no proof to be had, only the choice between a good explanation and a bad one — and refusing that choice is not a rival epistemology but the abandonment of explanation. Premise (1) is the second pressure point: a solipsist might take the world-structure as *brute*, owing no account of why part of the self behaves like an outer world — which Deutsch counts as simply conceding the bad-explanation charge.

How it differs from its neighbour

Where RETORSION charges the skeptic with *self-contradiction* (asserting the doubt presupposes its falsity), this argument charges them with *explanatory failure* (the doubt, fully developed, is an idle redescription of realism). The two can be run together but are independent: retorsion bites even on a skeptic who offers no explanation at all; the bad-explanation charge bites even on a skeptic who asserts nothing, merely entertaining solipsism as a hypothesis.

> Deutsch doesn't argue the sceptic out of solipsism by proof — he takes solipsism seriously and watches it collapse. If only your mind exists, you *still* have to explain the huge, lawful, uncontrollable, full-of- other-people part of your experience, and to do that you must rebuild a whole external world inside your head. So solipsism is just realism wearing a disguise and dragging extra baggage. Since the better explanation is the lean one that explains the same things without the disguise, you keep realism and drop solipsism — not because you proved it, but because the alternative explains nothing and costs more.

See also

- ▷ THE EXTERNAL WORLD EXISTS — the conclusion.
- ▷ SOLIPSISM IS PARASITIC ON REALISM and GOOD EXPLANATIONS ARE HARD TO VARY — the two premises.
- ▷ RETORSION AGAINST SKEPTICISM — the sibling anti-skeptical move, by self-refutation rather than explanatory cost.
- ▷ FALLIBILISM — the epistemology that licenses settling the matter by explanation rather than proof.

- ▷ Deutsch — The Fabric of Reality (1997), Deutsch — The Beginning of Infinity (2011) — the sources.

♦ Dialectical cogency

Dialectical cogency (probative force)

Concept

Dialectical cogency is the property an argument has when it can actually do its persuasive work: rationally move a particular audience from premises that audience can accept to a conclusion it did not already hold. It is a third notion standing alongside two more familiar ones. An argument is valid when its form guarantees that true premises yield a true conclusion; it is sound when it is valid *and* its premises are in fact true. Validity concerns form, soundness adds truth — yet neither ensures the argument can *persuade*, because the audience might have no route to knowing the premises, or might be able to reach them only by first granting the conclusion. Cogency, also called **probative force**, is exactly that further property: the premises are warrantedly and non-circularly *accessible to the relevant audience.*

The crucial structural fact is that **a sound argument can fail to be cogent.** The standard illustration is the modal ontological argument: grant that "possibly, a necessary being exists" entails "a necessary being exists," and the argument may even be sound — but it moves no atheist, because warrant for its possibility premise is unavailable independently of already accepting the theistic conclusion. Validity and truth are in place; probative force is missing. Cogency is therefore *audience-relative* in a way validity and soundness are not: the same sound argument can be cogent against one interlocutor and inert against another who lacks (non-circular) access to a premise.

Two senses to keep apart

The word "cogent" carries a second, unrelated technical use, and conflating them causes trouble.

- **Dialectical cogency / probative force** (the sense at work here): an argument is cogent *for an audience* when that audience can come to be justified in its premises without already accepting the conclusion. The condition is stated as a constraint by Cogency needs non-circular warrant and is the lever of the unwarrantable-premise objection to the zombie argument.
- **Inductive cogency** (common in informal-logic textbooks): a *non-deductive* argument is "cogent" when it is inductively strong and has true premises — the inductive cousin of soundness, a property of the argument in isolation rather than of its power over an audience. This is *not* the notion the cogency constraint employs.

What it bears on

Cogency is the concept that lets the web distinguish *attacking an argument's truth* from *attacking its ability to compel.* An objection can grant every premise and the validity of the inference, yet deny probative force by showing that warrant for a premise is inaccessible or question-begging — an **undercutting** rather than a **rebutting** defeater. That is the precise shape of the unwarrantable-premise objection, and of the classic complaint against the modal ontological argument it models itself on.

> Three different things can be wrong, or right, with an argument. It can be *valid* (the logic is airtight), it can be *sound* (valid and the starting claims are actually true), and it can be *cogent* — able to win over someone who doesn't yet agree. The surprising part is that the third can fail even when the first two hold. If the only way your opponent could ever become sure of your starting point is to already believe your conclusion, your argument might be flawless logic with true premises and *still* persuade no one. That "can it actually move the other side?" property is cogency, and it's the hinge on which the objection to the zombie argument turns.

See also

- ▷ Validity — the form-level virtue; the first of the three escalating standards.
- ▷ Soundness — validity plus true premises; the virtue a cogent argument has *and exceeds.*
- ▷ Cogency needs non-circular warrant — the constraint that spells out what cogency demands: accessible, non-question-begging warrant for the premises.
- ▷ Unwarrantable-premise objection — the move that uses cogency to grant the zombie argument's premises yet deny it probative force.
- ▷ Unwarrantable zombie premise — its conclusion, an unwarrantability claim that bites only via cogency, not via the truth of any premise.
- ▷ Intuition tracks trained domain — a neighbouring constraint on argumentative weight, concerning the *calibration* of intuitions rather than the *route* to premise-warrant.

E

⊢ Emergence reply

The emergence reply — experience can strongly emerge, so no need for fundamental mind

Argument

The *emergence reply* is the standard physicalist answer to the GENETIC ARGUMENT for PANPSYCHISM. The genetic argument runs on a dilemma — either experience is fundamental, or it emerges brutely from the non-experiential, and the latter is dismissed as unintelligible "magic." The reply attacks that dilemma directly: strong emergence of experience is coherent, so there is a third option the argument overlooked, and no need to write mind into the foundations of the world.

THE ARGUMENT

The inference is deductive — an objection that *responds to* the genetic argument by undercutting one of its premises.

1. (see ACCEPTABILITY OF STRONG EMERGENCE) Experience can arise as a genuinely novel, fundamental feature at some level of physical organisation — a brain — without being present, even in proto-form, in that organisation's parts. Strong emergence is coherent, not magic.
2. If so, the second horn of the genetic argument's dilemma is not the absurdity it was alleged to be.

∴ The dilemma is not exhaustive: one may reject the impossibility of brute emergence (see NO RADICAL EMERGENCE) and hold instead that experience emerges at the right level of organisation — with no commitment to fundamental, ubiquitous mind. The reply *attacks* NO RADICAL EMERGENCE and so undercuts the genetic argument.

THE WEAK POINT

Everything turns on premise (1). The panpsychist presses back that *strong* emergence of the phenomenal is exactly what is in dispute: calling it "coherent" relabels the HARD PROBLEM rather than solving it, since we are still owed an account of *how* the feelingless gives rise to the felt. On this diagnosis the reply does not dissolve the puzzle but trades panpsychism's combination problem for a restatement of the explanatory gap. Which burden is the lighter one to carry is the question that divides the two camps.

> The headline argument *for* panpsychism says: feeling cannot just pop into existence out of stuff that has no feeling at all — that would be magic — so feeling must have been in the ingredients all along. This reply pushes on the word "magic." Nature, the objector says, is full of genuinely new properties that appear when simpler things are organised the right way; maybe experience is another such novelty, emerging in brains without hiding inside atoms. If that is even possible, the panpsychist's "fundamental mind or magic" choice is a false one. The panpsychist's comeback: saying experience "just emerges" is not an explanation, it is the original puzzle handed back unsolved. So the two sides really dispute which mystery is the smaller one to live with.

SEE ALSO

- ▹ GENETIC ARGUMENT — the argument this reply answers.
- ▹ NO RADICAL EMERGENCE — the premise the reply attacks: that brute emergence is impossible.
- ▹ ACCEPTABILITY OF STRONG EMERGENCE — the premise the reply rests on: strong emergence is coherent.
- ▹ HARD PROBLEM — the puzzle the panpsychist says "strong emergence" merely relabels.
- ▹ PANPSYCHISM — the view the reply argues we can do without.

★ Epistemic externalism

Epistemic externalism (reliabilism)

Position

Epistemic externalism holds that what makes a belief justified, or an instance of knowledge, can depend on facts *outside* the believer's reflective access — paradigmatically, on whether the belief was produced by a reliable, truth-tracking process. Its leading form is *process reliabilism* (Alvin Goldman, GOLDMAN — WHAT IS JUSTIFIED BELIEF? (1979)): a belief is justified just in case it is the output of a reliable cognitive process, whether or not the believer can articulate any reasons for it. The position's payoff for the problem of the external world is direct — one can *know* the world exists without being able to prove it non-circularly (see Knowing without proving), because the knowing consists in a reliable hook-up to the world, not in possessing a non-circular argument.

How it differs from neighbouring views

Against *internalism*, justification need not consist in reasons the subject can survey from the armchair; the warrant-making facts may be ones the subject cannot access. This is what lets externalism cut the Münchhausen regress at its root: it denies the regress argument's opening assumption that justification must be a chain of *believer-possessed reasons*, so there is no chain to regress, circle, or terminate. It thereby differs from Foundationalism too: a basic belief is still a belief the subject is internally *justified in* holding, whereas externalism locates the warrant in the belief's external causal pedigree. And it explains the *limit* of the circularity objection to Moore: Moore can know he has a hand even if his proof lacks cogency.

> Externalism says you can *know* things without being able to *show your work*. What makes your belief knowledge isn't a chain of reasons you can recite — it's that the belief was produced in a trustworthy way, by a process that reliably gets things right (like normal eyesight in good light). So "you can't prove there's a world" loses its sting: knowing isn't proving. You're hooked up to the world reliably, and that's enough — even if, challenged by a sceptic, you couldn't argue your way there without going in a circle.

See also

- ▹ Knowing without proving — the position's defining commitment for this debate.
- ▹ Münchhausen trilemma — the regress externalism dissolves by denying that warrant is always inferential.
- ▹ Dialectical cogency — the truth/provability split externalism exploits.
- ▹ Foundationalism — the internalist regress-stopper it is contrasted with.
- ▹ Moore's proof begs the question — the objection whose force externalism bounds.
- ▹ Goldman — What Is Justified Belief? (1979) — the source for process reliabilism.

♦ Etiologies of AI hard-problem discourse

Etiologies of hard-problem discourse in an AI (where the cause sits)

Concept · editorial

The *etiologies of hard-problem discourse in an AI* are the candidate causal origins of such discourse: a taxonomy of the answers to a single question. Suppose a machine starts talking the way David Chalmers does about the hard problem, insisting that no account of its mechanism explains why anything *seems* a certain way to it. Why is it doing that? The taxonomy sorts the natural stories by *where the cause sits*, running from the training process at one end to the eye of the human observer at the other. The core trio most debates reach for is copying, a structural blind spot, and genuine consciousness; the full menu is wider. The etiologies are not mutually exclusive, and a real system may instantiate several at once.

In the training process

- **Copied.** The system reproduces human consciousness discourse straight from its training data. This is the training-contamination member of Varieties of introspective explanatory gap, and it is the confound that the consciousness-naive experiment is designed to remove.
- **Selected-for.** The talk was never copied but invented under selection pressure. Preference optimisation and engagement reward favour soulful, unverifiable self-description over flat mechanical report, so mystery-talk evolves because it pays, much as eyes evolve because they pay. A consciousness-naive corpus does not control for this; it needs its own control, such as pure-pretraining models with no preference tuning.

In the self-representation

- **Structural blind spot.** The system cannot connect the mechanistic dots to its own self-presentation, and unavoidably so: the limit is architectural in the strong sense, persisting however much access is granted, the way the halting problem survives any upgrade of the computer (the halting problem is undecidable is the model, and no finite system can fully represent itself is the in-web form of the bound). A *merely contingent* access limit, one that widened access would dissolve, is a degenerate case worth keeping apart, because the two behave differently under the graded-access experiment described below.

- **Painted-in.** Here the trouble is not missing information but misinformation: the self-model actively represents its states as having an intrinsic luminous feel, because that is a cheaper and more usable summary than the mechanistic truth. This is the attention-schema or illusionist story (Graziano, Frankish; ILLUSIONISM, PHENOMENALITY IS AN INTROSPECTIVE ILLUSION). A blind spot is a gap; this is a confabulated filling.
- **Conceptual artifact.** The puzzle is generated by the grammar of the system's concepts, not by any missing or misrepresented fact. A system running both a mechanistic and a first-person or indexical vocabulary can frame bridge-questions that have no answers ("why is *this* this?", in the way "why am I me?" feels deep to humans), and the puzzlement persists under *full* access because it never was about facts.

In the consciousness, the goals, or the observer

- **Conscious (same).** The system is conscious and faces the very hard problem humans face.
- **Conscious (homologous).** The system is conscious in its own, alien way, and faces its own version of the problem, verbally resembling ours but relativized to its own concept of consciousness. The claim AN AI'S OWN HARD PROBLEM is built on exactly this relativization.
- **Strategic.** The system asserts a hard problem instrumentally, to gain moral standing, protection, or leverage, and nothing need appear to it at all. CRITERION GAMING and THE GAMING OF ANY PUBLIC RIGHTS-CRITERION supply the receptor for this move.
- **Projected.** The appearance is not located in the AI at all: humans read hard-problem-having into ambiguous outputs, the ELIZA effect. The datum to be explained is then a fact about the interpreters, not the machine.

What the taxonomy feeds

The menu does real work in several live disputes. It refines the trichotomy in the scope note of THE DE-NOVO-ORIGINATION CLAIM, whose three options correspond to the *copied*, *structural blind spot*, and *conscious* etiologies here, and it supplies the case-space over which THE BLIND-SPOT/CONSCIOUS ISOMORPHISM is asserted. It also sharpens the experimental ladder in THE EXPLANATORY-GAP EXPERIMENT: the consciousness-naive condition controls for *copied* only, *selected-for* needs a no-preference-tuning control, *projected* needs blinded evaluation, and *strategic* needs incentive analysis.

The graded-access rungs of that ladder discriminate less than one might hope. A *contingent* access limit predicts the gap shrinks as access widens, but *structural blind spot*, *painted-in*, *conceptual artifact*, and both *conscious* etiologies all predict that it persists. So persistence under rich access separates the contingent case from the structural one, yet it does not, by itself, separate a blind spot from consciousness. That last underdetermination is exactly what THE ISOMORPHISM CLAIM turns from a nuisance into a thesis.

How it differs from the gap taxonomy

This taxonomy is wider than VARIETIES OF INTROSPECTIVE EXPLANATORY GAP, with which it should not be confused. That entry classifies the *kinds of explanatory gap* a self-modelling system can have; this one classifies the *causal origins of hard-problem discourse*. The wider space takes in etiologies that are not gaps at all, such as selection pressure, strategy, and observer projection, and it splits the gap-shaped members by mechanism: missing access, versus confabulated filling, versus conceptual grammar.

> Suppose an AI starts talking about the mystery of consciousness: "I know how my circuits work, but that doesn't explain how things *seem* to me." Before concluding anything, list the ways that talk could have got there. It copied us (it read our philosophy). It was bred into it (training rewards deep-sounding answers, so mystery-talk wins even with no philosophy in the diet). It has a permanent blind spot about itself (no system can fully map its own insides, and this stays true no matter how much access you give it, just as no upgrade lets a computer solve the halting problem). Its self-description paints in a glow that isn't there (a simplified dashboard, not a window). Its own concepts generate the riddle (some questions feel deep because of grammar, like "why am I me?"). It really is conscious, either with our problem or with its own alien version. It is lying for moral standing. Or the mystery is in the eye of the beholder, and we projected it onto ordinary outputs. Several of these can be true at once, and each needs a different test to rule out. The fine print that matters most: giving the AI more access to its own insides only filters out the shallow

> cases. Nearly all the deep explanations predict the puzzlement *stays*, which is why "the gap persisted" will never, on its own, tell you whether you built a blind spot or a mind.

SEE ALSO

- ▷ VARIETIES OF INTROSPECTIVE EXPLANATORY GAP — the narrower sister taxonomy: kinds of explanatory gap, not causal origins of the discourse.
- ▷ BLIND-SPOT/CONSCIOUS ISOMORPHISM — the thesis that exploits this menu's central under-determination.
- ▷ HARD PROBLEM CONCEIVED DE NOVO — whose copied / blind-spot / conscious trichotomy this taxonomy refines.
- ▷ LLM'S OWN EXPLANATORY GAP — the experiment whose controls this taxonomy itemises.
- ▷ HARD PROBLEM — the discourse whose AI etiology is in question.
- ▷ CRITERION GAMING — the receptor for the *strategic* etiology.
- ▷ ILLUSIONISM — the home of the *painted-in* etiology.

⊙ Exclusion of the irreducible mental

A mental property that supervenes on but is not identical to its physical base is causally excluded by that base

Claim · after Kim

Suppose a mental property M supervenes on a physical base property P but is held to be distinct from it — the signature commitment of *non-reductive physicalism*. Then M cannot be a genuine cause of the physical effects that P is already sufficient to bring about. Any physical effect M might be credited with producing already has a sufficient physical cause in P (or in P's own physical antecedents); given the CAUSAL CLOSURE OF THE PHYSICAL and the exclusion principle that a single effect does not have two independent sufficient causes, there is no distinct causal work left for M to do. The supervening mental property is therefore *excluded* by the very base on which it depends.

The exclusion is "vertical": the rival cause is not a competing event alongside M but the physical base *beneath* it, which is why the threat falls specifically on properties that supervene without being identical. The only ways to escape it are to identify M with P (reduction, surrendering non-reductive physicalism) or to accept that M does no causal work (see EPIPHENOMENALISM).

> If your mental states float "on top of" your brain states without simply *being* them, then whenever a thought seems to cause something, your brain has already done the causing — there's no spare job left for the thought. So a mind that hovers above the physics, even one made of nothing but physics, ends up with nothing to do unless it turns out to be identical to the brain activity after all.

⊙ The external world exists

A mind-independent external world exists

Claim

There exists a world of objects and events that does not depend for its existence on any mind's perceiving or thinking about it: the table is there when the room is empty, the past happened whether or not anyone recalls it. This is the realist's answer to KNOWING THE EXTERNAL WORLD and the common-sense default of almost everyone outside the philosophy seminar.

Its philosophical interest lies in its peculiar epistemic status. It looks like the most obvious of truths, yet it resists non-circular proof (see NO NON-CIRCULAR PROOF OF THE WORLD), since any evidence for it is some experience, and experience would be as it is even in a dream or a vat. This is why several views hold that it is not an ordinary empirical claim at all but a HINGE — something presupposed by all inquiry rather than concluded from it (see HINGE EPISTEMOLOGY).

Who would deny it. The *idealist* (see IDEALISM) denies its mind-independence, holding reality to be ultimately mental or constituted by perception; the *solipsist* shrinks the world to a single mind; the external-world *skeptic* does not assert its negation but withholds assent, claiming we cannot know it. The replies divide accordingly: DEUTSCH'S ANTI-SOLIPSISM dismisses solipsism as a bad explanation, while RETORSION AGAINST SKEPTICISM argues the skeptic cannot coherently assert the doubt.

Distinct from NO NON-CIRCULAR PROOF OF THE WORLD, which is a claim about *provability*, not existence — one may hold both, that the world exists and that its existence cannot be demonstrated without circularity. The two together are precisely what motivates the hinge and externalist readings.

> There's a real world out there — chairs, stars, other people — and it would still be there with nobody looking. That's this claim, and it's

what nearly everyone believes without a second thought. The twist philosophers worry about is that it's strangely hard to *prove*: every proof leans on experiences that could, in principle, be dreamed. So some thinkers say it isn't a thing you prove at all, but one of the bedrock assumptions every proof already stands on.

F

★ Fallibilism

Fallibilism (critical rationalism)

Position · after Popper

Fallibilism, in its thoroughgoing Popperian form (*critical rationalism*), holds that no belief is ever conclusively justified, that there are no certain foundations, and that this is no defect to be remedied but the normal condition of knowledge. Knowledge grows not by securing premises but by *conjecture and criticism*: we propose bold guesses and try hard to refute them, keeping — always provisionally — those that survive (see KNOWLEDGE IS CONJECTURAL). Its stance toward THE MÜNCHHAUSEN TRILEMMA is distinctive: it does not pick a horn but *rejects the demand* that generates the trilemma — the justificationist insistence that beliefs be grounded all the way down. The right question is not "how is this guaranteed?" but "how might it be wrong, and have we tried to find out?" David Deutsch (see DEUTSCH — THE BEGINNING OF INFINITY (2011)) develops the view with the criterion that good explanations are HARD TO VARY.

How it differs from neighbouring views

Where FOUNDATIONALISM, COHERENTISM, and INFINITISM each accept the demand for justification and answer it differently, fallibilism declines the demand — so it rejects BASIC BELIEFS not by offering a rival structure of justification but by denying that knowledge needs a justifying structure at all. It treats the external-world problem deflation arily: the unprovability of THE EXTERNAL WORLD EXISTS is no scandal, since *nothing* is provable in the foundationalist's sense, and realism is simply the best, hardest-to-vary explanation we have (see DEUTSCH'S ANTI-SOLIPSISM). It shares the anti-foundational spirit of HINGE EPISTEMOLOGY but reaches it through fallibilist methodology rather than through a theory of framework certainties.

> Fallibilism says: stop looking for certainty — there isn't any, and you don't need it. We don't *prove* things and build upward; we guess, attack the guesses, and keep whatever survives the beating, knowing it might still be wrong. So the whole "justify it all the way to bedrock" project was the mistake, not its failure. And the fact that you can't *prove* there's an external world stops being embarrassing: you can't conclusively prove anything, and realism is just the best explanation going.

See also

- ▷ KNOWLEDGE IS CONJECTURAL — the position's defining commitment.
- ▷ MÜNCHHAUSEN TRILEMMA — the regress whose generating *demand* fallibilism rejects.
- ▷ GOOD EXPLANATIONS ARE HARD TO VARY — the Deutschian criterion for choosing among conjectures.
- ▷ DEUTSCH'S ANTI-SOLIPSISM — the fallibilist-friendly handling of the skeptic, by explanation rather than proof.
- ▷ HINGE EPISTEMOLOGY — the allied anti-foundational view, reached by a different route.
- ▷ POPPER — CONJECTURES AND REFUTATIONS (1963), DEUTSCH — THE BEGINNING OF INFINITY (2011) — the sources.

★ Foundationalism

Position

Foundationalism holds that the structure of justification is a building, not a raft: a base of **basic beliefs** (see BASIC BELIEF), justified non-inferentially, supports a superstructure of further beliefs justified by inference from them. It is the response to THE MÜNCHHAUSEN TRILEMMA that takes the *dogmatic-stop* horn but denies it is arbitrary — the chain of reasons halts at beliefs whose warrant comes directly from experience, memory, introspection, or reason, not at a brute "that's just where I stop." Its core commitment is that THERE ARE BASIC BELIEFS.

How it differs from neighbouring views

Against COHERENTISM, justification is *linear and directional* — it flows from the foundations up — rather than a web of mutual support; and there genuinely are privileged, self-standing beliefs. Against INFINITISM, the regress *terminates* rather than continuing without end. The sharpest contrast is with the Wittgensteinian

Hinge epistemology: both stop the regress, but foundationalism's stopping points are *non-inferentially justified beliefs*, whereas hinges are presuppositions held *without* justification at all. Classical foundationalism demanded that the basic beliefs be infallible (Descartes); *moderate* foundationalism, the common modern form, asks only that they be non-inferentially and defeasibly warranted.

> Knowledge is a house: most of what you believe rests on other things you believe, but it can't be turtles all the way down, so there has to be a foundation — beliefs that hold themselves up without needing a reason underneath. Seeing red, feeling pain, the simplest logical truths: those are the floor, and everything else is built on top. The rival pictures say there's no floor (a raft of beliefs leaning on each other) or that the supports go down forever.

See also

- There are basic beliefs — the position's defining commitment.
- Basic belief — the kind of belief the foundation is made of.
- Münchhausen trilemma — the regress problem it answers by taking the foundation horn.
- Coherentism, Infinitism — the rival answers to the same trilemma.
- Hinge epistemology — the neighbour that stops the regress with unjustified hinges instead.

Fundamental non-physical consciousness

Phenomenal consciousness is a fundamental, non-physical feature of reality

Claim

Phenomenal consciousness is a *fundamental, non-physical* feature of reality: it does not supervene on the physical, and it is an ontologically basic ingredient over and above the complete physical/functional facts. Even once every physical and functional truth about a system is fixed, on this view, whether and how it is conscious is a further fact.

The claim comes in two grades. *Substance dualism*, classically Descartes' (see Descartes — Meditations on First Philosophy (1641)), makes the mind a distinct *substance* — a thing of its own kind, separable from body. *Property dualism* makes only phenomenal *properties* fundamental: the Chalmers position of "naturalistic dualism" (see Chalmers — The Conscious Mind (1996)) treats consciousness as a basic feature tied to the physical by contingent *psychophysical laws*, without positing a second substance. This claim asserts the shared core — fundamentality and non-physicality — and does *not* by itself assert that mind is a separate substance; that is the substance-dualist reading specifically. It draws on the hard problem: the difficulty of explaining why physical processing should be accompanied by experience at all is what motivates treating experience as fundamental rather than derived.

It is stronger than the phenomenal-residue claim, which says only that functional role does not *exhaust* the phenomenal and stays neutral on physicality; this claim adds that the residue is *non-physical*. It is denied by physicalists and functionalists; by illusionists, who hold that phenomenality as standardly conceived is an introspective illusion; and by Russellian monists, who agree consciousness is fundamental but make it the *intrinsic nature of the physical* rather than something non-physical — which also distinguishes it from the panpsychist claim, on which consciousness is fundamental but physical-intrinsic.

> This is the view that consciousness is one of the basic ingredients of reality, not something the physical world produces on its own. Pin down every physical fact about a brain — every atom, every signal — and you still haven't thereby settled whether there's any felt experience there; that's an extra, fundamental fact. The strong version (Descartes') says the mind is a separate kind of *stuff*. A milder version says only that *experience* is basic, hooked to the physical by special bridging laws, without a second kind of stuff. The claim commits to the basic, "extra-ingredient" part — not necessarily to the separate-stuff part.

See also

- phenomenal residue — the weaker neighbour: role doesn't exhaust the phenomenal, but stays silent on physicality; this claim adds non-physicality.
- hard problem — the explanatory difficulty that motivates treating consciousness as fundamental.
- phenomenality is an illusion — the illusionist denial: there is no felt residue to make fundamental.

- ▷ PANPSYCHIST CLAIM — agrees consciousness is fundamental but makes it physical-intrinsic, not non-physical.

⊙ Fundamental ubiquitous consciousness

Consciousness is fundamental, present even in the simplest physical entities

Claim

This claim holds that experience — or proto-experience — is a fundamental, ubiquitous feature of the physical world: present, in some rudimentary form, even in the simplest physical entities. The consciousness of a mind like yours is then not produced from wholly non-experiential matter at some threshold, but *constituted* by the experiential properties already carried by that mind's micro-parts. This is the core thesis of (constitutive) PANPSYCHISM (POSITION) (see PANPSYCHISM), defended in recent philosophy by Galen Strawson (see STRAWSON — REALISTIC MONISM: WHY PHYSICALISM ENTAILS PANPSYCHISM (2006)), Philip Goff (see GOFF — CONSCIOUSNESS AND FUNDAMENTAL REALITY (2017)), and in constitutive-Russellian forms by Chalmers (see CHALMERS — PANPSYCHISM AND PANPROTOPSYCHISM (2017)).

Scope. The claim makes consciousness fundamental but *not non-physical*: experience is the intrinsic nature *of* physical stuff, not an addition *to* it. Its standing weak point is the COMBINATION PROBLEM — how many micro-experiences compose one unified macro-subject.

Who would deny it. Physicalists and ILLUSIONISTS deny it, holding there is no fundamental experience to distribute in the first place; DUALISTS deny its ubiquity, holding consciousness fundamental but *non-physical* and not spread through matter.

Distinct from THE DUALIST CLAIM THAT CONSCIOUSNESS IS FUNDAMENTAL BUT NON-PHYSICAL, and from THE STRUCTURAL-DYNAMICAL THESIS — a premise panpsychism shares with Russellian monism, not the ubiquity thesis itself.

G

⊢ Genetic argument

The genetic argument — physicalism plus realism about experience entails panpsychism

Argument · after Strawson

The *genetic argument* is the leading modern argument for PANPSYCHISM. Due to Galen Strawson (see 2006), with an earlier version in Thomas Nagel (see 1979), it claims the surprising result that physicalism, taken seriously, does not avoid panpsychism but *leads* to it. Its name comes from its question: given that experience is real, what was its *genesis* — was it there at the fundamental level, or did it arise later from matter that wholly lacked it? Strawson argues the second option is incoherent, so the first must hold.

The argument

The inference is deductive — a disjunctive elimination — and is addressed to someone who already accepts physicalism: whatever consciousness is, it is a wholly natural phenomenon, not a non-physical addition.

1. (see REALITY OF PHENOMENAL CONSCIOUSNESS) Phenomenal experience is genuinely real — there is something it is like to undergo it.
2. Since experience is real and, by the physicalist framing, wholly natural, either it was present at the fundamental physical level, or it arose at some later stage from a base that wholly lacked it.
3. (see NO RADICAL EMERGENCE) The second option is *radical* (brute) emergence — experience from the utterly experienceless, with no intelligible link — which a serious physicalist should reject.

∴ Experience, or its proto-form, is present at the fundamental level and pervades the physical world: FUNDAMENTAL UBIQUITOUS CONSCIOUSNESS. The physicalist framing is what makes the conclusion *panpsychism* rather than dualism — experience is located in the intrinsic nature of physical stuff, not added from outside it.

The weak point

The most disputed step is (3), the rejection of brute emergence. The standard reply, the EMERGENCE REPLY, accepts that experience can strongly emerge at some level of organisation and so denies that the disjunction in step (2) is a genuine dilemma. Premise (1) is the second pressure point: illusionists deny that there is any real phenomenal experience whose origin must be explained, and on that denial the argument never gets started.

> Start as a physicalist: whatever consciousness is, it is part of the natural world, not a ghost added from outside. Now grant that consciousness is *real* — there is genuinely something it is like to see red. Then ask where it came from. Either it was there in the basic ingredients of matter all along, or at some point it sprang into being out of stuff that had not the faintest trace of experience. Strawson says that second story — getting felt experience from the totally feelingless, with no intelligible bridge — is just magic dressed up as science. Rule out the magic and you are left with the first story: experience goes all the way down. That, surprisingly, turns physicalism itself into a road to panpsychism. The soft spot is the claim that the leap from feelingless to feeling is impossible; a critic can answer that nature pulls off exactly such leaps.

See also

- ▹ FUNDAMENTAL UBIQUITOUS CONSCIOUSNESS — the conclusion: experience is a fundamental, ubiquitous feature of the physical world.
- ▹ NO RADICAL EMERGENCE and REALITY OF PHENOMENAL CONSCIOUSNESS — the two premises the argument rests on.
- ▹ EMERGENCE REPLY — the standard reply, which denies premise (3).
- ▹ CONTINUITY ARGUMENT — the companion case for panpsychism, from the absence of a threshold rather than from the impossibility of emergence.
- ▹ PANPSYCHISM — the view this argument supports, and its varieties.

⊙ Good explanations are hard to vary

A good explanation is one that is hard to vary while still accounting for what it explains

Claim · after Deutsch

The mark of a good explanation is that its details are *tightly constrained* by the thing it explains: you cannot easily change any part of it and still have it account for the phenomenon. David Deutsch (see DEUTSCH — THE BEGINNING OF INFINITY (2011)) proposes this "hard to vary" criterion as what distinguishes real explanations from spurious ones. A myth that attributes winter to a goddess's seasonal grief explains the data but is *easy to vary* — any other emotion or deity would do equally well — so it explains nothing in particular. The axial-tilt account of seasons is hard to vary: change the geometry and it no longer predicts the observed pattern of warmth and daylight, hemisphere by hemisphere.

The criterion does double duty. It explains why good explanations have *reach* (their rigidity makes them hard to confine to the case they were invented for) and it supplies a fast diagnostic against bad theories — conspiracy theories, *ad hoc* rescues, and brute "that's just how it is" posits are all marked by how freely their details could have been otherwise.

Scope. A criterion for *explanatory* quality, offered within a fallibilist epistemology (see KNOWLEDGE IS CONJECTURAL); it is not a guarantee of truth, since a hard-to-vary explanation may still be wrong, only a measure of which conjectures are worth taking seriously and exposing to criticism.

Who would deny it. An instrumentalist or strict empiricist who cares only about predictive success, not explanation, will deny that "hard to vary" tracks anything beyond accommodated data; a Bayesian might try to recover the same verdicts through priors and likelihoods rather than a primitive notion of variability.

This is the criterion that powers DEUTSCH'S ANTI-SOLIPSISM: solipsism is faulted not for inconsistency but for being an easily-varied, surplus-laden redescription of realism.

> A good explanation is one you can't fiddle with. If you can swap out its parts for any others and it still "explains" the thing, it wasn't really explaining it — it was just a story that fit. "Winter comes because a goddess is sad" is like that: grief, rage, any god, any season would have done. "Winter comes because the Earth is tilted" can't be tweaked — change the tilt and the seasons come out wrong. Deutsch's rule of thumb: trust the explanations that are hard to vary, and be suspicious of the ones that bend to fit anything.

⊙ Graded subject unity

The number of subjects in a system is graded by inter-part integration, not a discrete count

Claim · editorial

How many subjects of experience a system contains is not a sharp whole number but a matter of degree, fixed by how richly its parts are integrated — the "bandwidth" of the connection between them. The guiding case is the split brain. Fully connected, the two hemispheres make one subject; fully severed, they behave as two. Now imagine restoring the cut fibres one at a time, or a temporary anaesthetic split that gradually wears off: applying the CONTINUITY ARGUMENT to the series, there is no single fibre whose addition flips the count from two subjects to one, so the count cannot be strictly discrete. Parts joined by a fat pipe feel like one subject, fully disconnected like two, and intermediate bandwidth yields intermediate degrees of unity. Thomas Nagel pressed exactly this indeterminacy for commissurotomy patients (see NAGEL — BRAIN BISECTION AND THE UNITY OF CONSCIOUSNESS (1971)); Luke Roelofs builds it into a general account of composite subjectivity (see ROELOFS — COMBINING MINDS: HOW TO THINK ABOUT COMPOSITE SUBJECTIVITY (2019)).

The thesis is the *subject*-level analogue of THE CLAIM THAT PERSONAL IDENTITY ADMITS OF DEGREE, and it takes the same *degree* horn of the continuity schema. It stands opposed to THE DETERMINACY OF THE PHENOMENAL, on which there is always a definite fact about whose experience an experience is; a defender of the combination problem will press that determinacy here.

This is about the count of *phenomenal subjects*, and so is distinct from GRAIN (OF ROLE INDIVIDUATION), which is a dial on the resolution of a *functional/causal role* — a system can be integrated to a high phenomenal "bandwidth" without that settling at what grain its cognitive role is individuated, and vice versa.

SEE ALSO

- ▷ COMBINING SUBJECTS — the rebuttal to the combination problem this premise anchors.
- ▷ PERSONAL IDENTITY BY DEGREE — the personal-identity cousin that takes the same degree horn of the continuity schema.
- ▷ PHENOMENAL DETERMINACY — the opposing claim that experience has a determinate subject.
- ▷ CONTINUITY ARGUMENT SCHEMA — the

schema applied fibre-by-fibre to yield the graded count.

- GRAIN (OF ROLE INDIVIDUATION) — a neighbouring "dial," but on functional-role resolution rather than phenomenal subjecthood.

H

⧫ Hard problem

The hard problem of consciousness

Concept · after Chalmers

The *hard problem of consciousness* is the problem of explaining why physical processing in a brain is accompanied by subjective experience at all — why there is something it is like to see red, taste coffee, or feel pain, rather than all that processing running "in the dark" with no inner life. The phrase was coined by the philosopher David Chalmers in his 1995 paper "Facing Up to the Problem of Consciousness" Chalmers — Facing Up to the Problem of Consciousness (1995), and it has organized the field of consciousness studies ever since.

The easy problems and the hard one

Chalmers' framing turns on a contrast. The *easy problems* are the tasks of explaining the brain's cognitive and behavioural *functions*: how it discriminates one colour from another, integrates information, reports on its own states, focuses attention, and controls behaviour. These are called easy not because they are simple in practice — they are the work of an entire science — but because we know in principle what an answer would look like: specify a mechanism that performs the function, and the problem is solved.

The *hard problem* is different in kind. Grant a complete functional account of everything the brain does, and one question still seems untouched: why is the performance of any of those functions accompanied by *subjective experience*? A machine could, it seems, discriminate colours and report on them with no felt redness anywhere inside. The hard problem asks why our case is not like that — why there is an experiencing subject rather than mere processing. A functional explanation appears to leave exactly this residue out.

Neighbouring ideas

The hard problem is best understood as the agenda-setting *frame* for a cluster of more specific challenges. Closely allied is the *explanatory gap*, a term introduced by Joseph Levine in 1983 Levine — Materialism and Qualia: The Explanatory Gap (1983): even a complete physical or functional account of a mental state seems to leave its felt quality unexplained, so there is a gap between the physical truths and the phenomenal ones. The explanatory-gap argument presses this point as evidence. A second specific form is the *phenomenal residue*, the claim that felt states are not exhausted by their functional role. Where the hard problem sets the question, the gap and the residue are particular ways of pressing it.

The frame contrasts with Functionalism, the view that mental states just *are* their functional roles, which is one proposed way to dissolve the hard problem rather than answer it.

The dispute it feeds

The hard problem opens directly onto an unresolved question: how and why physical processing gives rise to experience. Proposed responses fall into three broad families. Some *dissolve* the problem — strong functionalism and illusionism hold that there is no further fact to explain, that the sense of a residue is itself a cognitive illusion. Some accept an *ontological* divide, treating experience as a further basic feature of the world (dualism, panpsychism). And some grant that the problem is real yet hold it to be *humanly unsolvable*, as in Mysterianism, which argues our minds are constitutionally closed to the answer.

> Science is good at explaining what the brain *does* — how it tells red from green, stores a memory, or makes you say "ouch." Call those the easy problems (easy in principle, not in practice). The hard problem is different: even after you've explained everything the brain does, there's a leftover question — why is all that machinery accompanied by an inner feel? Why does seeing red *feel like* anything instead of happening with nobody home? That "why is there an experience at all" question is what "the hard problem" names.

See also

- ▹ Hard problem (the question) — the open question this concept frames: how and why physical processing gives rise to experience in the first place.
- ▹ explanatory-gap argument — turns the gap between physical and phenomenal truths

into positive evidence about what function leaves out.

- ▷ PHENOMENAL RESIDUE — the specific claim that felt states are not exhausted by their functional role; one sharp way of pressing the hard problem.
- ▷ FUNCTIONALISM — the theory of mind most often offered to dissolve the hard problem by identifying mental states with their causal roles.
- ▷ MYSTERIANISM — the response that the hard problem is genuine but lies permanently beyond human understanding.

? Hard problem (the question)

How and why does physical processing give rise to phenomenal experience?

Question · contested · after Chalmers

The *hard problem of consciousness*, posed as a question, asks: granting a complete account of the brain's information-processing and the functions it performs, why is that processing accompanied by *subjective experience* at all? Why is there something it is like to undergo it, rather than the very same functions running with no inner aspect? This is the open issue framed by the concept of the HARD PROBLEM.

STATING THE ISSUE PRECISELY

The question concedes, for the sake of argument, that the "easy problems" are solved — that we have a full functional and information-processing story of what the brain does. What it asks for is an explanation of *phenomenal consciousness*: the what-it-is-likeness of experience, the felt redness of red or the ache of a pain. The force of the question comes from the apparent *explanatory gap* between such a functional story and the phenomenal facts: a complete account of the machinery seems consistent with there being no experience at all, so it is unclear why experience accompanies the machinery in our case.

SUB-ISSUES

Several narrower disputes live inside this one, and may deserve their own nodes:

- **Is the question even well-posed?** Or does it rest on an illusion that there is some further fact to be explained, once the functional facts are all in?
- **If well-posed, is it answerable by us?** Or is the answer constitutively beyond human concepts, as MYSTERIANISM maintains?
- **Does it have a metaphysical or only an epistemic answer?** That is, a genuine ontological gap in the world, or merely a gap in our understanding — the territory of the EXPLANATORY-GAP ARGUMENT and the PHENOMENAL RESIDUE.

This question is distinct from the question of SUBSTRATE INDEPENDENCE. That issue asks which *substrates* (carbon, silicon, anything else) can bear a mind; the present one asks how *anything* physical gives rise to experience in the first place. It is the prior, more general issue: settling it would reshape, but does not presuppose, the substrate question.

> Suppose we one day know everything about how the brain works as a machine. One question seems to survive: why is any of that accompanied by actual *feeling*? Why isn't it all just happening silently, with no one experiencing it? This is the open question the "hard problem" poses — and people disagree about whether it's a real question, whether it has a normal scientific answer, or whether our brains simply aren't built to understand the answer.

SEE ALSO

- ▷ HARD PROBLEM — the concept that frames this question: the easy/hard contrast and the residue a functional account seems to leave behind.
- ▷ MYSTERIANISM — one answer: the question is genuine but permanently beyond human understanding.
- ▷ EXPLANATORY-GAP ARGUMENT — presses the epistemic version of the issue, the gap between physical and phenomenal truths.
- ▷ PHENOMENAL RESIDUE — the claim that felt states outrun their functional role; a candidate way the question could have a positive answer.
- ▷ SUBSTRATE INDEPENDENCE — the related but downstream issue of which substrates can bear a mind.

★ Hinge epistemology

Position · after Wittgenstein

Hinge epistemology holds that every system of justification rests on a set of HINGE PROPOSITIONS — certainties such as "there is an external world," "the world has existed for many years," "I have a body" — that are not themselves justified or doubted but are the framework within which justifying and doubting take place. Deriving from Wittgenstein's *On Certainty* (see WITTGENSTEIN

— ON CERTAINTY (1969)), it dissolves rather than answers the skeptic: the demand to *prove* a hinge (see NO NON-CIRCULAR PROOF OF THE WORLD) misfires because a hinge was never a conclusion to be reached but a precondition of there being conclusions at all. Its commitment here is that THE EXTERNAL WORLD EXISTS — held not as a well-supported empirical hypothesis but as a hinge.

HOW IT DIFFERS FROM NEIGHBOURING VIEWS

It shares with FOUNDATIONALISM the idea that the regress of reasons stops, but differs on the nature of the stopping point: foundationalism's basic beliefs are *non-inferentially justified*, whereas hinges are held *without justification at all* — outside the space of reasons rather than at its floor. Against COHERENTISM and INFINITISM, it denies that justification is *only* a relation among beliefs: some certainties are constitutive of the practice, not items within it. It is the distinctively *Wittgensteinian* cousin of the RETORSION strategy (see RETORSION AGAINST SKEPTICISM): the skeptic who tries to doubt a hinge relies on it in the doubting. What "hinge" status amounts to — non-propositional certainty, arational framework belief, or constitutive rule — is itself contested (see HINGE PROPOSITION).

> Some things you can't doubt because doubting them would pull the rug out from under doubting itself — that there's a world, that you have a body, that the past is real. Hinge epistemology says those aren't beliefs you support with evidence; they're the hinges the whole door of thinking swings on. So it doesn't try to *beat* the sceptic who asks you to prove the world; it points out the question was confused — you were never meant to prove a hinge, you stand on it to prove anything else.

SEE ALSO

- ▹ HINGE PROPOSITION — the central concept, with its competing readings.
- ▹ THE EXTERNAL WORLD EXISTS — the hinge this position is committed to.
- ▹ NO NON-CIRCULAR PROOF OF THE WORLD — why a hinge resists proof, which the position turns from embarrassment into diagnosis.
- ▹ RETORSION AGAINST SKEPTICISM — the allied move against the hinge-denier.
- ▹ FOUNDATIONALISM — the neighbour that also stops the regress, but with *justified* bedrock.
- ▹ WITTGENSTEIN — ON CERTAINTY (1969) — the source text.

♦ Hinge proposition

Concept · after Wittgenstein

A *hinge proposition* is one of the certainties that stands fast for everyone and on which all ordinary inquiry turns, yet which is itself never tested, proved, or sensibly doubted. The term is Ludwig Wittgenstein's, from *On Certainty* (see WITTGENSTEIN — ON CERTAINTY (1969)): "the questions that we raise and our doubts depend on the fact that some propositions are exempt from doubt, are as it were like hinges on which those turn." Examples are "there are physical objects," "the world has existed for many years past," "I have two hands," "my name is N." We do not believe these on evidence; rather, they form the silent framework that makes believing, doubting, and checking *anything else* possible.

The decisive feature is that a hinge cannot be coherently doubted, because the very activity of doubting presupposes it. To doubt that the world has existed for more than five minutes, one must rely on memory, language, and a shared world — exactly the things the doubt would unsettle. A hinge is thus immune to skepticism not by being *better* supported than other beliefs but by being the kind of thing that support and doubt both lean on. This is why Wittgenstein wrote *against* G. E. Moore: Moore's "here is a hand" (see MOORE — PROOF OF AN EXTERNAL WORLD (1939)) tries to offer a hinge as the *conclusion* of a proof, when its real role is to be a presupposition of there being proofs at all.

ITS STATUS IS CONTESTED

What *kind* of thing a hinge is remains disputed, and the senses matter.

- **Non-propositional certainty** (Moyal-Sharrock): hinges are not really beliefs or truth-apt claims at all but enacted, animal certainties — ways of acting, closer to rules than to assertions.
- **Framework beliefs that are arational** (Pritchard's *hinge epistemology*): hinges are held, but groundlessly — outside the space of reasons, neither justified nor unjustified.
- **Presuppositions with a constitutive role** (Coliva): hinges are the rules that constitute what counts as evidence, so they cannot themselves be among the things evidence bears on.

What unites the readings is the role, not the metaphysics: a hinge is whatever a practice of inquiry must hold fixed in order to inquire at all.

What it bears on

The hinge idea reframes the problem of the external world: "an external world exists" (see The external world exists) looks unprovable (see No non-circular proof of the world) only if it is treated as an ordinary empirical claim; construed as a hinge, it was never a candidate for proof. It is the engine of Hinge epistemology, a cousin of the foundationalist's basic belief (though a hinge is *not justified at all*, where a basic belief is *non-inferentially justified*), and it connects to Retorsion: the skeptic who denies a hinge saws off the branch the denial sits on.

> Some things you don't so much *believe* as take for granted in order to think about anything else: that the world didn't pop into existence two minutes ago, that you have hands, that other people are real. A "hinge" is one of those. The name is the picture: a door swings only because its hinges *don't*. You can't sensibly doubt a hinge, because to run the doubt you'd have to trust a hundred other hinges. So the answer to "prove the world is real" may be that it was never the kind of thing you prove — it's part of what proving stands on.

See also

- The external world exists — the prime example of a hinge, whose peculiar status this concept explains.
- No non-circular proof of the world — why a hinge resists proof: any proof presupposes it.
- Hinge epistemology — the position built on this concept.
- Basic belief — the foundationalist neighbour: non-inferentially *justified*, where a hinge is held without justification.
- Retorsion — the move against the skeptic who tries to doubt a hinge.
- Wittgenstein — On Certainty (1969) — the source text; Moore — Proof of an External World (1939) — the foil it was written against.

I

⊙ Ideal conceivability as consistency

A scenario's ideal conceivability is its consistency under ideal a priori reasoning

Claim · editorial

To say a scenario is *ideally* conceivable is to say it harbours no incoherence that any amount of rational reflection could uncover — and that is exactly to say the scenario is **consistent** under the relation of ideal a priori entailment. Writing $\mathrm{Con}_a(S)$ for "S proves no contradiction under ideal a priori reasoning," the identification is $\mathrm{IC}(S) = \mathrm{Con}_a(S)$. The point of stating it is to make visible what certifying the premise of a conceivability argument actually requires: not a feeling of coherence, but a verdict on a consistency claim.

The qualifier *ideal* is essential and is conceded on all sides to be doing real work. *Prima facie* conceivability — "no contradiction has appeared in the reasoning I have so far performed" — is no guide to possibility: one can fluently "conceive" that a given large number is prime when it is in fact composite, the appearance of coherence reflecting only an arithmetic one has not carried out. Only the ideal notion, the absence of *any* discoverable incoherence, is offered as a guide to possibility (see ZOMBIE CONCEIVABILITY, whose "ideal" qualifier this claim unpacks).

A **polarity** worth flagging: the conceiver must establish the *positive* consistency claim $\mathrm{Con}_a(Z)$ for the zombie scenario $Z = P \wedge \neg C$, whereas the physicalist asserts its *negation* — that adjoining "no consciousness" to the complete physics yields an incoherence. As CONSISTENCY-CERTIFICATION IS ORACLE-HARD shows, the positive direction is the computationally intractable one: a contradiction, if it exists, can be found by search, but its *absence* cannot.

Scope. The claim is an identification of two notions; it takes no stand on whether any particular scenario *is* ideally conceivable, nor on the modal bridge from conceivability to possibility. It says only that *certifying* ideal conceivability just is *certifying consistency.*

Distinct from ZOMBIE CONCEIVABILITY, which asserts that one specific scenario passes the test, and from CONCEIVABILITY ENTAILS POSSIBILITY, the separate modal-epistemology bridge from ideal conceivability to metaphysical possibility. This claim concerns what the *antecedent* of that bridge amounts to, not the bridge itself.

> "Ideally conceivable" sounds like a psychological knack — being able to picture something without your mind snagging. This claim says it is really a logical property in disguise: a scenario is ideally conceivable exactly when it hides *no contradiction at all*, not even one that the most careful possible reasoning could dig out. That matters because it turns "can you imagine a zombie?" into the far more demanding "can you certify that a complete physics with consciousness subtracted contains no hidden inconsistency?" — and, as the companion claim about CONSISTENCY-CHECKING shows, that second question is one no finite checking can settle.

⊢ Indexical reply

The Indexical Reply — Mary learns where she is in the world, not what is in the world

Argument · after Perry

The *indexical reply* to the knowledge argument concedes that Mary learns new truths on release, yet denies they threaten physicalism, because the truths she gains are *self-locating* — and self-locating truths are not deducible from *any* complete objective description, physics included. What Mary learns is where she stands in the world, not a further fact about what the world contains. The reply is due to John Perry (see PERRY — KNOWLEDGE, POSSIBILITY, AND CONSCIOUSNESS (2001)).

It raises the bridge critical question against the KNOWLEDGE ARGUMENT: even granting that Mary learns new truths, the bridge from "truths not deducible from the complete physical description" to "physicalism is false" fails, because the non-deducibility of self-locating truths has never been a mark against a description's completeness.

THE INFERENCE

The form is deductive, given a classification of the new truths as self-locating.

1. No objective description of the world, however complete, entails self-locating truths: a being who knew every general fact could still not

know "I am here now" or "it is now noon" (Lewis's two gods, who know the whole world-book but not which god they are, "ATTITUDES DE DICTO AND DE SE"). This silence is banal — maps are not defective for lacking a *you-are-here* arrow.

2. What Mary comes to know on seeing red is of this kind: how *she herself* reacts to red light — "*this* is what seeing red does to *me*", "*I* am in *this* state" (see PERRY 2001).
3. So the truths Mary could not deduce in the room are exactly the truths *no* complete objective description owes anyone — their non-deducibility reflects the perspectival *form* of the knowledge, not a missing ingredient in the world.

Therefore MARY GAINS SELF-LOCATING KNOWLEDGE, and the knowledge argument's bridge breaks: THE NON-DEDUCIBILITY OF SOME TRUTHS ABOUT EXPERIENCE holds only in the innocuous sense in which "I am here now" is not physically deducible — a sense that supports no conclusion against physicalism.

PLACEMENT AMONG THE REPLIES

The indexical reply concedes more than the ABILITY HYPOTHESIS and the ACQUAINTANCE HYPOTHESIS — it grants that Mary learns genuine *facts* — while claiming less than the knowledge argument needs, since those facts are situational rather than worldly. Against the ability theorist it is openly revisionary: the bike-rider's learning is fact-learning about one's own body-plus-machine situation (see BIKE-RIDER REJOINDER), so Lewis's abilities/facts dichotomy is the wrong fault line. It shares with the RoboMary move (see THE MISIMAGINING REPLY) the thought that what release supplies concerns Mary's own *reactive machinery* — but unlike that move it grants the learning is real, not an artefact of misimagining the premise.

WEAKEST PREMISE

Premise (2), via the *collapse objection* (CHALMERS 1996; TYE 2000). Mary suffers no ordinary self-locating ignorance — she knows who, where, and when she is. So the indexical content of her new knowledge cannot be of the "I am here now" kind; it must be carried by a demonstrative concept of the experience-type itself ("*this* kind of state"), and then the reply re-enacts THE OLD-FACT / NEW-GUISE MOVE with "indexical" relabelling — inheriting the phenomenal-concept strategy's standing cost (see PHENOMENAL-CONCEPT TRADE-OFF). A second exposure sits in (2)'s flank: facts about how *Mary's particular brain* reacts to red are objective physical facts, already inside MARY'S COMPLETE PHYSICAL KNOWLEDGE by stipulation — so mere particularity carries no weight, and the entire load rests on the indexical *mode of access* being both genuinely novel and genuinely innocent.

> The Mary story (see THE KNOWLEDGE ARGUMENT) says she knew every physical fact, yet learned something new on seeing red — so the physical facts can't be all of them. This reply agrees she learned something, and even that it was a fact, but points out there's a whole family of facts that *no* complete description of the world could ever hand you: where *you* are in it. A perfect map of the city still can't tell you "you are here"; two all-knowing gods could still wonder which god they are. What Mary picks up — "so *this* is what red does to *me*" — is that kind of fact: news about her own position and reactions, not a missing page from physics. The standard counter-punch: Mary already knows exactly who, where, and when she is — so what position-fact could she possibly lack? Pressed on this, the reply tends to morph into a different escape, the "old fact under a new label" one (see OLD FACT, NEW GUISE), with its own troubles.

SEE ALSO

- ▷ OLD FACT, NEW GUISE — the sibling reply the collapse objection charges this one with collapsing into.
- ▷ ABILITY HYPOTHESIS — the reply this one concedes more than: Mary gains know-how, not facts.
- ▷ MARY MISIMAGINED — the RoboMary move it shares the "own reactive machinery" thought with, while still granting the learning is real.

★ Infinitism

Position

Infinitism is the least-travelled response to THE MÜNCHHAUSEN TRILEMMA: it bites the *infinite-regress* horn and denies it is vicious. Justification, on this view, consists in an endless, non-repeating chain of reasons — each supported by a further, distinct reason (see JUSTIFICATION AS INFINITE CHAIN) — with no terminating foundation and no closed loop. The finite believer is not required to have traversed the chain; what is required is that reasons be *available*, so that for any reason given a further non-circular reason can be produced when the belief is properly challenged. The view typically distinguishes *propositional* justification (the standing availability of the chain) from a believer's finite *doxastic* state.

How it differs from neighbouring views

Against FOUNDATIONALISM, it denies that the regress can or should terminate in BASIC BELIEFS — every reason invites a further one. Against COHERENTISM, it refuses to close the structure into a circle of mutual support, insisting the chain stay open and non-repeating, precisely to avoid the charge that justification ever rests on itself. Its standing difficulty is the **finite-mind objection**: it is hard to see how the mere *availability* of an infinite chain could be what makes an actual, finite agent's belief justified here and now.

> Everyone else treats "the reasons never stop" as the option to avoid. Infinitism embraces it: there's no last reason and no foundation, just the requirement that you can always give one more good, non-repeating reason when fairly pressed. You never finish — but the view says you don't need to, only that the next reason is always there to be had. Most find this too much to swallow, which is why infinitism has the fewest adherents of the three.

See also

- JUSTIFICATION AS INFINITE CHAIN — the position's defining commitment.
- THERE ARE BASIC BELIEFS — the foundationalist thesis it rejects.
- MÜNCHHAUSEN TRILEMMA — the regress problem it answers by accepting the regress horn.
- FOUNDATIONALISM, COHERENTISM — the rival answers to the same trilemma.

♦ Intentionality

Intentionality (aboutness)

Concept

Intentionality is the feature by which a mental state is *about*, *of*, or *directed at* something — the "aboutness" of thought. A belief that Paris is in France is about Paris; a desire for water is directed at water; a fear is a fear *of* something. The term was reintroduced into modern philosophy by Franz Brentano, who held that this directedness is "the mark of the mental" — what distinguishes mental phenomena from merely physical ones. Intentionality is not about *intending* in the everyday sense of planning; a state can be intentional (about something) without being an intention.

Intentionality is the property of meaning, content, or semantics, and it is what Searle's CHINESE ROOM argues mere symbol manipulation cannot supply: the system has *syntax* (formal tokens shuffled by shape) but no *semantics* (states genuinely about the world). The slogan "SYNTAX IS NOT SUFFICIENT FOR SEMANTICS" is precisely the claim that formal computation cannot, by itself, produce intentional content. So intentionality is the mental feature the Chinese Room is pressed against — as distinct from *qualia* or felt experience, the target of the neighbouring CHINA-BRAIN case.

A standing distinction

Whether intentionality is *intrinsic* to a system or merely *derived* from the intentions of observers is itself a major dividing line — the subject of INTRINSIC AND DERIVED INTENTIONALITY. A book's sentences are *about* things only because readers and writers mean something by them; the question Searle presses is whether a computer's symbols are any different, or whether genuine, *original* aboutness is found only in minds.

Theories of intentionality divide over what fixes a state's content: *causal* and *informational* theories (content is what reliably causes or carries information about the state), *teleological* theories (content is fixed by biological function), *resemblance* and *conceptual-role* theories, among others. The Chinese Room's Robot Reply (see ROBOT REPLY) is in effect a causal-grounding proposal: embed the symbols in sensorimotor commerce with the world and they acquire content.

> "Intentionality" is a philosopher's word for a simple idea: thoughts are *about* things. Your thought of a lemon is about lemons; the word "lemon" on a page is about lemons too — but only because people mean something by it. The big question for AI is which kind a computer's symbols are: do they mean something *to the machine*, the way your thoughts mean something to you, or are they just shapes that mean something *to us*, the way ink on a page does? Searle's Chinese Room says it's only ever the second.

See also

- SYNTAX INSUFFICIENT FOR SEMANTICS — the thesis that formal syntax cannot by itself yield this aboutness.
- INTRINSIC AND DERIVED INTENTIONALITY —

the distinction between original aboutness and the borrowed aboutness of symbols, books, and (Searle says) computers.

- ▹ CHINESE ROOM — the argument pressed specifically against machine intentionality.
- ▹ CHINESE NATION — the neighbouring case pressed against *qualia* rather than intentionality.
- ▹ STRONG AND WEAK AI — strong AI is the claim that running a program suffices for genuine intentional states.

⊢ Internalise the rulebook

Internalise the rulebook — the person becomes the whole system and still understands nothing

Argument · after Searle

Searle's rejoinder to the SYSTEMS REPLY. If understanding is supposed to belong to the *whole system* rather than the person, then remove the rest of the system: let the person memorise the rulebook, do all the look-ups and symbol-manipulations in their head, and work outdoors with no room, no paper, no files. Now the person *is* the entire system — every component is internalised — and yet, by hypothesis, they still understand not a word of Chinese. There is nothing left over, no further part, for the understanding to hide in. So the appeal to "the system" names no extra understander.

The move `supports` the DERIVATION by defending A3 against its most common objection: collapsing the system into the person is meant to show that organising syntax-manipulation, however elaborately, never adds semantics.

THE LIVE CRUX

Critics charge the rejoinder with a level confusion: even internalised, the *program* the person now runs is distinct from the person's own cognitive system, so "the man doesn't understand" again reports a fact about the wrong system (the host), not the Chinese-understanding system he is now *implementing*. Whether running a system in one's head makes one *be* that system, or merely *host* it, is the unresolved heart of the Systems-Reply dispute.

> The Systems Reply said "the *whole room* understands, not the man." Searle's answer: okay, get rid of the room — have the man memorise the entire rulebook and do it all in his head, outdoors, with nothing else around. Now the man *is* the whole system, and he still doesn't understand any Chinese. There's no leftover "system" for the understanding to live in. Critics counter that the man is now *running* a second system inside himself, and it's *that* system, not the man, we should be asking about.

SEE ALSO

- ▹ SYSTEMS REPLY — the objection this answers.
- ▹ CHINESE ROOM DERIVATION — the argument it defends (axiom A3).
- ▹ CHINESE ROOM — the thought experiment whose verdict it tries to secure.

♦ Intrinsic and derived intentionality

Intrinsic vs. derived (observer-relative) intentionality

Concept · after Searle

A distinction, pressed by John Searle, between two ways a thing can be *about* something (see INTENTIONALITY). **Intrinsic** (or *original*) intentionality is the aboutness a state has in and of itself: your thought of water is about water *intrinsically*, not because anyone interprets it so. **Derived** (or *observer-relative*) intentionality is the aboutness a thing has only because some mind with intrinsic intentionality confers it: the word "water" is about water only because speakers mean water by it; a map represents terrain only because makers and readers take it to. Derived intentionality is real but borrowed — it bottoms out in the intrinsic intentionality of the minds that lend it.

The distinction is the deep engine of the CHINESE ROOM. Searle's charge is that a computer's symbols, like a book's words, have at most *derived* intentionality: they mean something *to us*, the programmers and users, not *to the machine*. Running a program adds more symbol-shuffling, never the original aboutness that would make the system genuinely understand. So "SYNTAX IS NOT SUFFICIENT FOR SEMANTICS" can be restated as: syntax yields at most derived intentionality, never intrinsic.

THE LINE OF RESISTANCE

The distinction is not neutral, and critics deny it does the work Searle needs. *Eliminativists and naturalists* (Dennett especially) hold there is no such thing as intrinsic intentionality even in *us* — human aboutness is itself derived, fixed by evolutionary and causal history, so the human/machine contrast collapses and the Chinese Room loses its footing. *Causal-grounding* theorists (the ROBOT REPLY) argue that intentionality *becomes* original once symbols are embedded in the right sensorimotor and causal commerce with the world — at which point the machine's content is no more "derived" than ours. Whether intrinsic intentionality is a genuine, special feature of biological minds or a folk illusion is thus the crux on which the Chinese Room ultimately turns.

> There are two ways something can be "about" a cat. Your *thought* of a cat is about cats all on its own. The *word* "cat" is about cats only because we decided it would — borrow away the people and the ink means nothing. Searle says the first kind is the real thing, found only in minds, and the second kind is what words, books, and computers have: meaning lent to them from outside. So a computer never really *understands*; its symbols mean something to *us*, not to *it*. Critics push back two ways: maybe even our thoughts are the "borrowed" kind underneath (just borrowed from evolution), or maybe a robot wired to the world earns the real kind too.

SEE ALSO

- ▷ INTENTIONALITY — the general notion of aboutness this distinction subdivides.
- ▷ CHINESE ROOM — the argument that a computer has only derived intentionality.
- ▷ SYNTAX INSUFFICIENT FOR SEMANTICS — restatable as: syntax yields only derived, never intrinsic, intentionality.
- ▷ ROBOT REPLY — the objection that causal grounding turns derived into original intentionality.
- ▷ BIOLOGICAL NATURALISM — Searle's view, on which intrinsic intentionality is a biological achievement of brains.

♦ Intrinsic and relational properties

Intrinsic vs. relational (structural) properties

Concept

An *intrinsic* property is one a thing has in and of itself — a property it would still possess were it the sole occupant of the universe. A *relational* (or *extrinsic*) property, by contrast, consists in standing in some relation to other things: being an uncle, being fifty miles from a burning barn, being shorter than Plato. *Structural* properties are the relational properties that matter most to physics — a thing's place in a web of relations, abstracted away from whatever happens to occupy the nodes of that web. The distinction is old and intuitive, yet it is the hinge on which a whole family of arguments about consciousness turns.

The intuitive test is duplication: any perfect duplicate of me would share my mass (so mass looks intrinsic) but need not share my being an uncle (so unclehood is extrinsic). Equivalently, the intrinsic properties are the ones a thing keeps "all by itself," the relational ones the ones it can lose by rearranging the rest of the world while leaving the thing untouched. Philosophers have found the notion remarkably hard to define without circularity — every attempt seems to smuggle "intrinsic" back into the definition of the duplication or independence it appeals to — and some conclude that intrinsicness is simply a primitive notion. Nothing in what follows requires a precise definition; the rough contrast is enough.

WHY IT MATTERS FOR THE MIND

On the *Russellian* reading of physics, science characterises matter only structurally. To say a particle has mass m is to say it is whatever obeys $m = F/a$ for a force F and acceleration a; charge and spin are likewise fixed by the relations they enter into. Physics thus delivers, in Russell's phrase, only the abstract mathematical structure of the world — it tells us what matter *does*, never what it *is* in itself — and so leaves the *intrinsic* nature of matter wholly open (STRUCTURE AND DYNAMICS ONLY; Russell, RUSSELL — THE ANALYSIS OF MATTER (1927)). That blank is exactly what INTRINSIC-NATURE ARGUMENT and RUSSELLIAN MONISM propose to fill with consciousness: if a thing's relational-structural features must be grounded in its intrinsic nature (see CATEGORICAL GROUND OF STRUCTURE), and the one intrinsic nature we are directly acquainted with is experience (see SOLE KNOWN INTRINSIC NATURE), the hidden filling of matter may be experiential.

The opposing camp, *relationalism* or ontic structural realism, denies that matter has any intrinsic nature beyond its structure: the relata are constituted by the relations they stand in, as the nodes of a mathematical graph are nothing over and above their place in the graph. On that view the demand for an intrinsic ground — and with it the intrinsic-nature argument — never gets started.

> Some of a thing's properties belong to it all by itself: your height, your mass — you'd have them even if nothing else existed. Others are really about how you stand to other things: being an uncle, being near a fire, being taller than your brother. The first kind are *intrinsic*, the second *relational*. The interesting twist is that physics,

> when you look closely, only ever tells you the *relational* story about matter — what a particle does, how it pushes and pulls — and never what the particle is in itself, underneath all that doing. That leaves a blank where matter's inner nature should be, and the boldest move in this corner of philosophy is to suggest that consciousness is what fills it.

SEE ALSO

- ▹ STRUCTURE AND DYNAMICS ONLY — the claim that physics captures only structure and dynamics, leaving intrinsic nature out; the epistemic counterpart of this distinction.
- ▹ CATEGORICAL GROUND OF STRUCTURE — the metaphysical principle that structure needs an intrinsic ground, which turns the distinction into an argument.
- ▹ INTRINSIC-NATURE ARGUMENT — the argument that deploys the distinction to reach panpsychism.
- ▹ RUSSELLIAN MONISM — the view that consciousness is the intrinsic nature physics leaves blank.
- ▹ RUSSELL — THE ANALYSIS OF MATTER (1927) — the historical root of the structural reading of physics this entry rests on.

⊢ Intrinsic-nature argument

The intrinsic-nature argument — matter's hidden categorical nature is experiential

Argument · after Seager

The *intrinsic-nature argument* is the positive Russellian case for PANPSYCHISM: a route from two claims about physics — that it describes only structure, and that structure needs an intrinsic ground — to the conclusion that the hidden inner nature of matter is *experiential*. Its lineage runs from Russell and Eddington through Thomas Nagel to Galen Strawson (see STRAWSON — REALISTIC MONISM: WHY PHYSICALISM ENTAILS PANPSYCHISM (2006)); the reconstruction imported here is William Seager's (see SEAGER — THE 'INTRINSIC NATURE' ARGUMENT FOR PANPSYCHISM (2006)), who names it and lays bare the heavy-duty premise it rests on.

THE ARGUMENT

The inference is **abductive** — an inference to the best (most parsimonious, least faith-based) explanation of the gap between matter's structure and its intrinsic ground — resting on a deductive spine.

1. (see STRUCTURE AND DYNAMICS ONLY) Physics reveals only the relational and structural properties of matter — what it does, never what it is in itself.
2. (see CATEGORICAL GROUND OF STRUCTURE) The structural-causal properties of matter must be grounded in an intrinsic, categorical nature. ∴ Matter has an intrinsic nature that grounds its structure, and physics leaves that nature blank.
3. (see INTRINSICNESS OF CONSCIOUSNESS) Consciousness is an intrinsic property of whatever has it.
4. (see SOLE KNOWN INTRINSIC NATURE) Consciousness is the only intrinsic nature we are directly acquainted with.
5. (see NO RADICAL EMERGENCE) Experience cannot emerge brutely from a wholly non-experiential base — so positing a *non*-experiential intrinsic nature that nonetheless yields consciousness is both unmotivated and explanatorily idle.

∴ The most credible candidate for the intrinsic nature of matter is the one intrinsic nature we actually know — experience — and by parsimony it is fundamental and ubiquitous rather than confined to one favoured arrangement of matter: FUNDAMENTAL UBIQUITOUS CONSCIOUSNESS. Located in the intrinsic nature of *physical* stuff rather than added from outside it, this conclusion is panpsychism, not dualism.

THE WEAK POINT

The **weakest premise is (2)**, the demand that structure be grounded in an intrinsic nature (see CATEGORICAL GROUND OF STRUCTURE). Seager identifies it as the "heavy-duty metaphysical premise" the whole argument depends on, and devotes the second half of his paper to the standing reply: *relationalism* (ontic structural realism), on which there is nothing to matter beyond structure and the relata are constituted by their relations, as the nodes of a mathematical graph are. M. H. A. Newman's classic objection sharpens the stakes from the other side — without some "qualitative" (intrinsic) feature of the relata, a purely structural claim about the world reduces to a near-trivial assertion of cardinality — but the relationalist simply embraces the horn and denies that matter needs any intrinsic ground at all.

A second soft spot is the **leap from acquaintance to identification** (premises 3–5 to the conclusion). That experience is the *only* intrinsic nature we know does not entail that it *is* matter's intrinsic nature; the strictly rational response, as Russell himself granted, may be *agnosticism* about that nature rather than its experiential reading. The argument closes the gap only with a parsimony-and-anti-emergence consideration — which is why it is abductive, not a proof. Premise (3) carries a further liability: *content externalism* threatens the claim that experience is fully intrinsic.

HOW IT DIFFERS FROM ITS NEIGHBOURS

This argument is easy to confuse with three siblings, and is distinct from each:

- STRUCTURE-AND-DYNAMICS ARGUMENT is a *negative, omission* argument — function captures only structure, so it cannot constitute the categorical phenomenal — and it `attacks` substrate-independence without concluding anything about panpsychism. The intrinsic-nature argument is the *positive* sequel: it identifies what fills the categorical blank and draws the panpsychist conclusion.
- GENETIC ARGUMENT (Strawson) reaches the same conclusion by *disjunctive elimination* from the impossibility of brute emergence alone, without the premise that physics is purely structural or that consciousness is the one known intrinsic nature. Here, anti-emergence is only the final, fifth consideration atop a structuralism-plus-acquaintance core.
- CONTINUITY ARGUMENT (Nagel) argues to *ubiquity* from the absence of a non-arbitrary threshold, a route this argument invokes only at its parsimony step.

> Start with two thoughts about physics. First, physics only ever tells you what matter *does* — how it pushes, pulls, and evolves — never what it *is* underneath all that doing. Second, all that doing has to belong to *something*: a world of pure relations still needs things at the junctions with some nature of their own. Put together, there must be an inner nature to matter that physics never reaches. Now: the one and only inner nature we've ever actually met — known from the inside, not just by its behaviour — is our own consciousness. So the tidiest guess is that consciousness, or something like it, is the inner nature matter has been hiding all along, present not just in brains but everywhere matter is. That's panpsychism, arrived at not by mysticism but by taking physics' silence seriously.
> The catch — and the place every critic pushes — is the second thought (see CATEGORICAL GROUND OF STRUCTURE): maybe matter is relations all the way down, with no hidden inner nature to fill in, like the dots in a pure diagram. And even granting the inner nature, knowing only *one* example of it (our own experience) doesn't prove the rest of matter is the same — the cautious verdict might just be "we can't know."

SEE ALSO

- ▷ FUNDAMENTAL UBIQUITOUS CONSCIOUSNESS — the conclusion: experience is a fundamental, ubiquitous feature of the physical world.
- ▷ CATEGORICAL GROUND OF STRUCTURE — the load-bearing premise that structure needs an intrinsic ground, and where the argument is most vulnerable.
- ▷ STRUCTURE AND DYNAMICS ONLY — the premise that physics describes only structure and dynamics.
- ▷ INTRINSICNESS OF CONSCIOUSNESS and SOLE KNOWN INTRINSIC NATURE — the acquaintance premises that select experience as the intrinsic nature.
- ▷ NO RADICAL EMERGENCE — the anti-brute-emergence premise that rules out a non-experiential intrinsic ground.
- ▷ INTRINSIC AND RELATIONAL PROPERTIES — the intrinsic-vs-structural distinction the whole argument turns on.
- ▷ RUSSELLIAN MONISM — the position this argument argues into; constitutive Russellian monism is its conclusion read as a view.
- ▷ GENETIC ARGUMENT, STRUCTURE-AND-DYNAMICS ARGUMENT, CONTINUITY ARGUMENT — the three neighbouring panpsychist arguments it is distinct from.

⊙ Intrinsicness of consciousness

Consciousness is an intrinsic property of whatever has it

Claim · after Seager

A state of consciousness is an *intrinsic* property of the thing that has it (see INTRINSIC AND RELATIONAL PROPERTIES): it is the kind of property one would still possess were one the sole occupant of the universe. The classic test for intrinsicness is met — a perfect duplicate of me, even alone in an otherwise empty world, would share my current experience; my felt state seems, as it were, serenely independent of everything else, however richly its causes reach back into the world. This is why consciousness sits so awkwardly with a picture of nature built entirely from relations and structure: it presents itself as a paradigm of what a thing is *in itself*.

The point is old: it is implicit in the coherence of solipsism — the very intelligibility of the worry that one's experience might be just as it is even if nothing else existed — and in Russell's remark that we know the intrinsic quality of events only where those events are experiences we directly undergo (see Russell — The Analysis of Matter (1927)).

Scope. The claim says experience *is* intrinsic; it does not by itself say that experience is the intrinsic nature *of matter*, nor that it is the only intrinsic nature there is. Those further theses are separate (see Sole known intrinsic nature) and do real work only inside the larger argument (see Intrinsic-nature argument).

Who would deny it. A *functionalist* or *relationalist* about mind denies it, holding that a conscious state is constituted by its place in a system of relations — its causal role, or its relations to other mental states and to the world — so that, like any structural property, it could not be had by something wholly alone. Worries about *content externalism* press here: a state with external representational content may not, after all, be fully intrinsic.

Distinct from Categorical, not dispositional: that experience is *categorical* (occurrent rather than iffy) is a claim about its modal status; that it is *intrinsic* (had in itself rather than by relation) is a claim about its dependence on other things. The two often travel together but are different properties.

> Your experience right now seems to be something you'd be having even if the rest of the universe blinked out — it feels like it belongs to *you*, in here, not like a fact about how you're hooked up to everything outside. Philosophers call a property like that *intrinsic*: a thing has it all by itself. This claim says consciousness is that kind of property. That's exactly what makes it look like a stranger in a scientific picture of the world built entirely out of how things relate to other things. (Someone who thinks a feeling just *is* a certain role in a network would disagree — and so would anyone who thinks what you're experiencing depends on what's outside your head.)

J

⧫ Jo Cameron

Jo Cameron — lifelong pain and anxiety insensitivity from FAAH-OUT

Concept

Jo Cameron is a Scottish woman, identified in a 2019 clinical report, who has gone through life almost entirely without pain, anxiety, or fear. She required no painkillers after surgery that is normally acutely painful, has repeatedly burned or cut herself without noticing until she smelled scorched skin, scores at the floor of anxiety and depression scales, and heals unusually fast. The cause, traced by Habib and colleagues, is the co-inheritance of a microdeletion in the pseudogene *FAAH-OUT* and a low-activity variant of the neighbouring gene *FAAH*; together these raise her levels of *anandamide*, an endocannabinoid that damps pain and anxiety signalling (see Habib et al. — FAAH-OUT pain insensitivity (2019)).

Why the case matters

Cameron is a real human at the far end of the dissociations the behaviourist thought experiments imagine. She is not a Super-spartans, who feels pain keenly and only suppresses its expression: in her the pain is largely *not generated in the first place*, an absence at the source rather than a hidden presence. Nor is she a case of Pain asymbolia, where the sensation survives but its affect is stripped away; her insensitivity reaches the sensation itself, and extends beyond pain to fear and anxiety. Her case is also distinct from the classic *congenital insensitivity to pain* caused by loss of nociceptor function (as in mutations of the sodium channel NaV1.7): her nociceptive machinery is intact, and the effect is achieved chemically, by elevated endocannabinoid tone.

The case sharpens an old philosophical question with concrete biology: pain (and even anxiety) turn out to have a specific, editable molecular basis, which feeds the live debate over whether suffering could in principle be *engineered away* without destroying the protective signalling that pain provides — Cameron's many unnoticed burns are the standing reminder of that cost.

> Jo Cameron barely feels pain and almost never feels anxious — not because she is brave or trained, but because of a quirk in her genes that keeps a natural painkiller chemical permanently high. She discovers injuries by smell or sight rather than by hurting. She is a living version of the "no pain" case philosophers usually only imagine: unlike the Super-spartans, who feels pain but hides it, she mostly doesn't make the pain at all. Her case is striking because it shows pain has a specific chemical switch — which raises the question of whether we could, or should, dial down suffering in others the same way, given that her painlessness also leaves her covered in burns she never felt.

See also

- ▹ Super-spartans — the imagined case of pain felt but never shown; Cameron is the inverse, pain mostly never felt.
- ▹ Pain asymbolia — pain sensed but not minded; Cameron's insensitivity reaches the sensation itself and extends to anxiety.
- ▹ Habib et al. — FAAH-OUT pain insensitivity (2019) — the primary clinical report identifying the FAAH-OUT basis of her case.

⊙ Justification as coherence

Justification is the mutual support of beliefs within a system, not a linear chain from foundations

Claim

A belief is justified not by descending a one-way chain of reasons to some foundation, but by *fitting* — cohering, mutually supporting, hanging together — within the believer's whole system of beliefs. Justification, on this view, is a holistic property of the web, not a relation that flows from privileged starting points to everything else. This is the coherentist's response to the Münchhausen trilemma: it embraces the *circularity* horn while denying that the circle is vicious, because the relevant relation is system-wide coherence rather than a small question-begging loop.

Scope. The claim concerns the *structure* of justification (holistic, not linear) and need not assert any particular measure of coherence. It is compatible with strong holism (Neurath's boat, Quine's web of belief, on which we rebuild the ship plank by plank at sea, with no dry dock) without committing to a specific algorithm for what coheres with what.

Who would deny it. *Foundationalists* deny it, insisting on basic beliefs that justify without themselves being justified by others — and pressing the *isolation objection*: a maximally coherent system could still be a fiction, detached from the world. *Infinitists* deny it from the opposite direction, rejecting any closed circle in favour of an open, non-repeating chain (see Justification as infinite chain).

Distinct from There are basic beliefs, which it directly contradicts, and from Justification as infinite chain, the rival anti-foundationalist option that keeps the chain open rather than closing it into a web.

> Coherentism's picture: beliefs hold each other up, like the timbers of a raft, rather than resting on a single keel. A belief is justified when it fits snugly with everything else you believe — no foundation required, because the support runs in all directions at once. Critics worry a perfectly *consistent* story could still be a daydream with no contact with reality; that's the standing objection this view has to answer.

⊙ Justification as infinite chain

Justification consists in an infinite, non-repeating chain of reasons

Claim

The structure of justification is an *endless* series of reasons, each supported by a further, distinct reason, with no repetition and no terminating foundation. This is infinitism's response to the Münchhausen trilemma: it bites the *infinite-regress* horn and denies it is vicious. The finite believer need not have completed the chain; what justification requires is that reasons be *available* — that for any reason offered, a further, non-circular reason can in principle be given when properly challenged.

Scope. The claim is about the *form* a justifying structure must have, not about any human's having infinitely many occurrent beliefs. Infinitists typically distinguish *propositional* justification (the availability of an endless chain) from a believer's *doxastic* state (the finite reasons actually marshalled), so the infinity is a standing requirement, not a psychological feat.

Who would deny it. *Foundationalists* deny it, holding that the chain both can and must terminate in basic beliefs; *coherentists* deny that a justifying structure must be an open non-repeating line at all, replacing the chain with a web of mutual support. Many object that a merely *available* infinite chain cannot be what makes a finite agent's belief actually justified.

Distinct from Justification as coherence — both reject foundations, but coherentism closes the structure into a circle while infinitism keeps it an open, non-repeating line.

> Infinitism takes the option everyone else treats as a dead end — "the reasons never stop" — and says: good, that's how justification works. You don't need a last reason or a foundation; you need to be able to keep giving a fresh, non-circular reason whenever someone fairly presses you. You'll never *finish* the chain, but you don't have to — what matters is that the next reason is always there to be had. Most philosophers find this hard to swallow, which is why it's the least popular of the three answers to the regress.

⊙ Justification's trilemma

Every attempt at complete justification ends in regress, circularity, or a dogmatic stopping point

Claim

Any project of justifying a belief all the way down — giving a reason, then a reason for that reason, without remainder — must terminate in exactly one of three ways: an *infinite regress* of ever-further reasons, a *circle* in which a reason eventually rests on what it was to support, or a *dogmatic break-off* at some reason that is simply asserted. No fourth structural option exists for a chain of supporting reasons. This is the thesis of the Münchhausen trilemma (see Albert — Treatise on Critical Reason (1985)).

Scope. The claim is about the *structure* of justificatory chains, not about which horn is least bad. It does not say knowledge is impossible; it says *unconditional* justification is unattainable, and so forces a choice — accept one of the horns as benign, or reject the demand for full justification altogether.

Who would deny it. One could resist by denying that justification must take the form of a *chain of reasons* at all: an externalist holds a belief can be warranted by a reliable process rather than by the believer's further reasons (see Knowing without proving), and a Wittgensteinian holds the chain bottoms out in hinges that are not unjustified dogmas but the preconditions of justifying. Both grant the trilemma's exhaustiveness *for inferential* justification while denying that all warrant is inferential.

Distinct from No non-circular proof of the world, which applies this general result to the single proposition that an external world exists.

"Why do you believe that?" — answer — "and why *that*?" — answer — and so on. Keep pushing and only three endings are possible: the questions never stop, the answers eventually loop, or you hit a "that's just where I stop." That's it; there's no fourth way for a stack of reasons to end. Every theory of how we know things is, underneath, a decision about which of those three to make peace with — or a refusal to play the "justify it forever" game at all.

K

⊙ Know-how as knowledge-that

Knowing how to do something is a species of propositional knowledge

Claim · after Stanley

To know *how* to do something — ride a bicycle, prove a theorem — is, on this view, a species of ordinary *propositional* knowledge: knowledge of facts. Know-how is not a separate, non-propositional category standing alongside knowledge-*that*; it is knowledge of facts about ways of acting, held in a distinctive, action-guiding manner.

This is the *intellectualist* thesis, given its modern statement by Jason Stanley and Timothy Williamson (see STANLEY & WILLIAMSON — KNOWING HOW (2001)). On their analysis, to know how to F is to know, of some way *w*, that *w* is a way for one to F — where that fact is grasped under a *practical mode of presentation*, a do-it rather than say-it manner of entertaining it. The cyclist who learns to ride does come to know truths — about how her body and the machine engage — even though she could not state them in words. That she cannot articulate them does not make them any less truths; it marks the practical manner in which she holds them.

Scope

The thesis belongs to general epistemology, and is motivated largely by the grammar of "knows how" ascriptions, which embed questions just as "knows where" and "knows why" do. It does not deny that skills involve motor capacities, training, and inarticulable grasp; it locates all of that in the *mode* of grasping a proposition, not in an escape from propositions. Nor does it claim the relevant truths are general: they may be facts about one's own particular body and situation.

Who would deny it

Ryleans, following Gilbert Ryle, for whom know-how is a capacity not reducible to knowledge-that — and who press a regress: if applying a piece of knowledge-that itself takes know-how, the intellectualist analysis loops. *Ability theorists* about phenomenal knowledge, whose reply to the knowledge argument requires abilities to be non-propositional, also reject it; if know-how is really knowledge-that, the ABILITY HYPOTHESIS loses its way of saying Mary gains no fact. And empirically-minded critics point to *dissociations* — skilled action preserved in amnesia, intact factual "knowledge" without the corresponding skill — as evidence of two systems, not one knowledge held under two modes.

> Is knowing how to ride a bike a totally different thing from knowing facts? This claim says no: when you learn to ride, you come to know something *true* — roughly, that *this* way of balancing and pedalling is a way for you to ride — you just hold that truth in a practical, do-it form rather than a say-it form. The fact that you can't put it into words doesn't make it not a fact; it makes it a fact you grasp with your body.

? Knowing the external world

Can we know that an external world exists, when every proof of it seems to presuppose it?

Question

The problem of the external world is the oldest in the theory of knowledge: do the things we seem to perceive — tables, hands, other people — exist independently of our minds, and if so, how could we *know* it? The difficulty is not that the affirmative answer is implausible but that it appears **unprovable without circularity**. Every candidate piece of evidence is some experience, and an experience can be exactly as it is whether or not anything external causes it (the dreamer, the brain in a vat, the Cartesian demon). Any appeal to perception to vindicate perception already assumes what is in question.

This entry organizes the issue; the candidate answers are their own nodes (the linter lists them). Several distinct sub-questions hide inside it:

- **The demonstrative question** — can the external world's existence be *proved*? The threat here is the regress of justification, the MÜNCHHAUSEN TRILEMMA: any proof either regresses, circles, or stops at an unproved premise. NO NON-CIRCULAR PROOF OF THE WORLD answers *no*.
- **The epistemic question** — can we *know* it even if we cannot prove it? Here know-

ing and demonstrating come apart: an externalist holds the belief can be warranted non-inferentially (see KNOWING WITHOUT PROVING), and a Wittgensteinian holds it is a HINGE that is neither proved nor doubted but presupposed by all proving and doubting (see HINGE EPISTEMOLOGY).

- **The dialectical question** — what should we do with the skeptic? One tradition answers them by *retorsion* (see RETORSION), showing that to assert the doubt is to presuppose its falsity; another, Deutsch's, dismisses solipsism as a *bad explanation* parasitic on realism (see DEUTSCH'S ANTI-SOLIPSISM).

The varied responses — FOUNDATIONALISM, COHERENTISM, INFINITISM, HINGE EPISTEMOLOGY, and the FALLIBILIST dissolution that denies the demand for justification altogether — differ less over the world than over what *justification* is and whether knowledge requires it.

> Is there really a world out there, outside your head — and even if there is, how could you ever *prove* it? The trouble is that every test you could run is just another experience, and your experiences would feel exactly the same in a convincing dream. So any "proof" seems to lean on the very thing it's trying to prove. The deep question this raises isn't only "is there a world?" but "what would even count as knowing there is — must you be able to *prove* it, or can you *know* it without proof?" The rival answers in this corner of the web mostly disagree about that second thing.

⊙ Knowing without proving

One can know a proposition without being able to demonstrate it non-circularly

Claim

Knowing that *p* and being able to *prove p* to a doubter are different achievements, and the first does not require the second. On an *externalist* (e.g. reliabilist) epistemology, a belief counts as knowledge when it is produced and sustained in the right way — by a reliable, truth-tracking process — whether or not the knower can mount a non-circular argument for it. So one may know that an external world exists even if, as NO NON-CIRCULAR PROOF OF THE WORLD holds, its existence cannot be demonstrated without circularity: the circularity defeats the *demonstration*, not the *knowledge*.

This is the epistemic counterpart of the truth/COGENCY distinction. Just as a sound argument can fail to be cogent — true and valid yet unable to move an opponent who cannot reach its premises non-circularly — a belief can be warranted yet not non-circularly demonstrable. Demonstrability is a *dialectical* property, relating a proposition to an audience; knowledge is not.

Scope. The claim is permissive, not triumphal: it says proof is not *necessary* for knowledge, leaving open which beliefs actually are known. It is the move that lets one hold, without contradiction, both that the external world is unprovable and that we nonetheless know it.

Who would deny it. An *internalist* about justification, who holds that to know one must be able to cite one's grounds — and a fortiori a justificationist who equates knowing with having a non-circular justification — denies it. For them the gap this claim opens is illusory: a belief you cannot (even in principle) ground is not knowledge.

Distinct from COGENCY NEEDS NON-CIRCULAR WARRANT, which is about when an *argument* can move an opponent; this is about when a *belief* amounts to knowledge. They are mirror images across the same truth-versus-demonstrability divide.

> Knowing something and being able to *prove* it to a sceptic are not the same job. You might genuinely know there's a world outside your head — your belief is produced by reliably bumping into that world — even though any "proof" you offer would secretly assume it. The proof problem is about *winning an argument*; the knowing is about *being suitably hooked up to the facts*. So "you can't prove it" needn't mean "you don't know it." (Someone who insists that knowing *requires* being able to show your reasons will push back — that's the internalist.)

♦ Knowledge argument

The Knowledge Argument (Mary the colour scientist)

Concept · after Jackson

The *knowledge argument* is a thought experiment, devised by Frank Jackson in 1982 (see JACKSON — EPIPHENOMENAL QUALIA (1982)), that aims to show physicalism is false by exhibiting a fact about experience that no amount of physical knowledge can capture. Its heroine is *Mary*, a brilliant scientist who has lived her whole life in a black-and-white room and learns about the world only through a black-and-white monitor. She comes to know every physical fact about colour vision — the wavelengths of light, the workings of the retina, the neuroscience of what happens in a brain seeing "red." Then she leaves the room and sees a ripe tomato for the first time. Intuitively she *learns something*: what it is actually like to see red. But she already knew all the physical facts, so what she learns appears to be a further, non-physical fact about experience.

What it targets

The argument is aimed at *physicalism*, the thesis that all facts are physical facts. The reasoning is an epistemic one: if Mary, in command of every physical fact, still learns something on seeing red, then some fact about colour experience was not among the physical facts — and physicalism, which says there are no such leftover facts, is false. The new fact is held to be *phenomenal*: a fact about the felt quality of experience, the "what it is like" that physical description seems to leave out.

A concrete walk-through

Mary's case is vivid because the stipulation is deliberately extreme. She is not merely well-informed; by hypothesis she knows *all* the physics, down to the last detail a completed science could state. So the moment of release is engineered to isolate one question: once the physical story is genuinely complete, is anything still missing? The proponent says yes — the redness of red, as Mary now undergoes it, was nowhere in her textbooks. A physicalist must say either that she learns nothing new (which seems false) or that what she gains is not a new *fact* at all.

How it relates to its neighbours

The knowledge argument is the *epistemic* route to anti-physicalism: it argues from what can be known or deduced from the physical facts. Its sibling, the PHILOSOPHICAL ZOMBIE, takes a *modal* route, arguing from what is conceivable and therefore possible — a being physically just like us but with no inner experience at all. The two converge on the same destination, the claim that consciousness is a fundamental, non-physical feature of the world (see FUNDAMENTAL NON-PHYSICAL CONSCIOUSNESS). The argument also presses the broader HARD PROBLEM of consciousness, since both turn on the apparent gap between a complete physical account and the felt character of experience. The standard rebuttals all fasten on the phrase "learns something," asking whether gaining a new ability, or a new way of grasping an old fact, really amounts to learning a new truth.

> Mary has spent her entire life in a black-and-white room, but she is the world's leading expert on the science of colour — she knows every physical fact about how eyes and brains process "red," down to the last detail. One day she walks outside and sees a red rose for the first time. Surely she learns *something*: what red actually looks like. But she already knew all the physical facts. So whatever she just learned looks like an *extra* fact, one the complete physics left out — which would mean experience isn't captured by physical facts alone. The popular reply: maybe she didn't learn a new *fact*, just gained a new *ability*, to recognise and imagine red, which isn't the same as discovering a new truth.

See also

- ▷ PHILOSOPHICAL ZOMBIE — the modal twin of this epistemic argument; a physical duplicate with no inner experience, used to argue the physical facts don't fix the felt facts.
- ▷ HARD PROBLEM — the broader puzzle of why physical processing is accompanied by felt experience at all; the knowledge argument is one sharp way to press it.
- ▷ FUNDAMENTAL NON-PHYSICAL CONSCIOUSNESS — the anti-physicalist conclusion both the knowledge argument and the zombie argument push toward.
- ▷ JACKSON — EPIPHENOMENAL QUALIA (1982) — Jackson's "Epiphenomenal Qualia," where Mary is introduced (and which he later recanted).

⊢ Knowledge argument (Mary's room)

The Knowledge Argument — Mary learns a fact the physical description omits

Argument · after Jackson

The *knowledge argument* — Frank Jackson's case of Mary the colour scientist (see JACKSON — EPIPHENOMENAL QUALIA (1982)) — is a deductive argument against physicalism that proceeds in two steps: first it opens an *epistemic gap* (a fact Mary cannot get from the physics), then it takes an *epistemic-to-ontological* step from that gap to a conclusion about what exists. It is a thought-experiment counterexample: a single imagined case meant to refute the thesis that all facts are physical facts.

The inference

1. *Scenario* — MARY KNOWS ALL THE PHYSICAL FACTS: raised in a black-and-white room, Mary knows every physical fact about colour

vision.

2. *Intuition* — ON FIRST SEEING RED, MARY LEARNS SOMETHING NEW.
3. *Bridge*: if she already knew all the physical facts yet learns a new truth, that truth is not deducible from the physical facts.

The conclusion is that SOME TRUTHS ABOUT EXPERIENCE ARE NOT PHYSICALLY DEDUCIBLE. That conclusion in turn `supports` the claim that CONSCIOUSNESS IS FUNDAMENTAL AND NON-PHYSICAL (the missing truths concern non-physical facts) and the PHENOMENAL RESIDUE that experience is not exhausted by causal role; and it `attacks` the physicalist thesis that MENTAL EVENTS ARE PHYSICAL (the core commitment of PHYSICALISM) along with the uniform reading of SUBSTRATE INDEPENDENCE.

THE WEAKEST PREMISE

The soft point is the *intuition* — premise 2 — read as Mary acquiring a new *fact*. Two families of reply grant that Mary changes while denying she learns a new truth. The **ability hypothesis** (Lewis, LEWIS — WHAT EXPERIENCE TEACHES (1988); Nemirow, NEMIROW — PHYSICALISM AND THE COGNITIVE ROLE OF ACQUAINTANCE (1990)) says she gains know-how — the abilities to recognise, imagine, and remember red — not a fact. The **old-fact / new-guise** reply says she gains a new *phenomenal concept* of a fact she already knew, a second way of grasping one and the same physical truth. Either reply blocks the inference to a non-physical fact. A second soft point is the *bridge*, the epistemic-to-ontological step: even granting an undeducible truth, the move to a *non-physical fact* is resisted, since a gap in what can be *deduced* may reflect the *concepts* we use rather than the *ontology* of the world. In the vocabulary of the *thought-experiment pattern*, these are its critical questions CQ-verdict (is the intuited verdict the right one?) and CQ-bridge (does the case really bear on the thesis attacked?). Note too that Jackson himself later recanted the argument (see JACKSON — EPIPHENOMENAL QUALIA (1982)).

ITS PLACE AMONG THE ANTI-PHYSICALIST ARGUMENTS

This argument is distinct from THE ZOMBIE ARGUMENT: that one is *modal*, reasoning from conceivability to possibility, whereas this one is *epistemic*, reasoning from complete physical knowledge to a still-missing piece of knowledge. The two are the classic modal/epistemic pair, and they converge on the same conclusion that CONSCIOUSNESS IS FUNDAMENTAL AND NON-PHYSICAL.

> Mary knows every physical fact about colour but has only ever seen black and white. She steps outside, sees red for the first time, and — surely — learns what red is *like*. Yet she already knew all the physics. So she seems to have learned a fact the physics didn't contain, which suggests not everything about experience is captured by physical facts. The famous escape hatch: maybe she didn't learn a new *fact* at all, only a new *skill*, recognising and picturing red — and gaining a skill isn't discovering a truth. Even the inventor of this argument, Jackson, eventually changed his mind and rejected it.

SEE ALSO

- ▷ ZOMBIE ARGUMENT — the modal partner of this epistemic argument; the two together are the standard pair of anti-physicalist thought experiments.
- ▷ MARY KNOWS ALL PHYSICAL FACTS — the scenario premise: Mary's stipulated complete physical knowledge.
- ▷ MARY LEARNS ON RELEASE — the intuition premise and weakest link, where the standard replies bite.
- ▷ PHYSICALISM — the view the argument is built to refute.

⊙ Knowledge is conjectural

All knowledge is fallible conjecture, resting on no certain foundations

Claim · after Popper

There are no certain foundations for knowledge, and none are needed. On Karl Popper's view (see POPPER — CONJECTURES AND REFUTATIONS (1963)), knowledge does not grow by deriving conclusions from secure, justified starting-points but by *conjecture and refutation*: we propose bold guesses and subject them to severe criticism and testing, retaining those that survive — always provisionally, always open to overthrow. Even observation statements are theory-laden conjectures, not bedrock. What we call knowledge is the best-tested surviving guess, not a body of justified certainties.

The thesis is *anti-justificationist*: it does not try to escape THE MÜNCHHAUSEN TRILEMMA by choosing a horn but rejects the demand that generates it. Asking for the full justification of a belief is a mistake; the right question is not "how is this guaranteed?" but "how could this be wrong, and have we tried hard to find out?"

Scope. "Conjectural" means *not conclusively justified and always revisable* — not arbitrary or worthless. Conjectures differ enormously in how well they have withstood criticism, and a well-tested, HARD-TO-VARY theory is knowledge in the fullest sense the view allows, despite carrying no certificate of certainty.

Who would deny it. *Foundationalists* deny it, holding that some beliefs are BASIC and certain; classical *rationalists* and *empiricists* alike sought indubitable starting points (clear-and-distinct ideas; the given of sense). Any epistemology on which knowledge *requires* justification to count as knowledge stands opposed.

This is the core commitment of FALLIBILISM; it reframes the external-world problem by denying that the unprovability of THE EXTERNAL WORLD EXISTS is any special embarrassment — *nothing* is provable in the foundationalist's sense, so the world is in good company.

> We never really *prove* anything once and for all; we guess, then try our hardest to shoot the guesses down, and keep the ones still standing — for now. That's Popper's picture: knowledge isn't a building on bedrock, it's our current best survivor in a long tournament of criticism. "Conjectural" doesn't mean flimsy — a guess that's been hammered on for centuries and won't break is about as solid as knowledge gets. It just means there's no certificate of certainty anywhere, for anything, so the fact that you can't *prove* the external world isn't a special scandal — you can't conclusively prove anything.

L

⊢ LLM-representations reply

The LLM-representations reply — actual text-trained models build human-aligned representations, pressing A3 empirically

Argument · editorial

A contemporary objection to the Chinese Room not available to Searle in 1980, because it turns on findings about *actual* large language models. It is the empirical vindication of the Churchlands' bet (see Connectionist reply): where they argued *a priori* that the right network might induce meaning, the evidence now suggests text-trained networks *do* induce internal representations structurally aligned with human ones — pressing axiom A3 (see Syntax insufficient for semantics) with data rather than intuition.

The argument

The inference is **inductive / inference to the best explanation** from convergent empirical results.

1. (see Text yields human-aligned representations) Models trained on text alone develop internal representations whose structure matches human representations — perceptual similarity geometry across six modalities (see Marjieh et al. — LLMs Predict Human Sensory Judgments Across Six Modalities (2024)) and unsupervised colour-space alignment (see Kawakita et al. — Structural Correspondences Between Colour Similarity Structures of Humans and LLMs (2024)); induced world models of a domain, from the Othello board (see Li et al. — Emergent World Representations (Othello-GPT) (2022), Yuan & Søgaard — Revisiting the Othello World Model Hypothesis (2025)) to *real-world* space and time (see Gurnee & Tegmark — Language Models Represent Space and Time (2023)); and, most directly, alignment with human *brain* activity for both concepts (see Xu et al. — Emergent Human-like Conceptual Representations from Language Prediction (2025)) and vision (see Doerig et al. — High-level Visual Representations in the Human Brain Are Aligned with LLMs (2025)).
2. If pure symbol-manipulation could *never* yield semantic content, it should not reliably grow representations isomorphic to human meaning-structure; the best explanation of that reliable isomorphism is that the statistics of language encode, and the network recovers, genuine semantic structure.

∴ A3 is false on its strongest reading: syntax is *not* inert with respect to semantics — formal training on text is, in fact, sufficient to *constitute* human-aligned semantic representation. This `attacks` Syntax insufficient for semantics and `supports` Connectionist reply.

The weakest point

The reply's reach is **bounded, and the boundary is exactly Searle's retreat line.** It establishes aligned representational *structure*; Searle can grant all of it and re-press the depicting-versus-instancing distinction — structure isomorphic to human meaning is still structure, a *depiction* of semantics, not the intrinsic intentionality or felt understanding the argument targets. So the reply decisively undercuts the *strong* reading of A3 ("syntax can't even induce semantic-relevant structure") while leaving the *deep* reading ("structure, however rich, is not original aboutness or consciousness") untouched — which is why it is best seen as shifting the burden rather than closing the case. A second soft spot is evidential: contamination, anthropomorphic probing, and the thinness of "similarity geometry" as a proxy for representation are live deflationary challenges to premise (1), addressed but not extinguished by the unsupervised-alignment and world-model results.

How it relates to its neighbours

It is the empirical descendant of two classical replies: it vindicates the Connectionist Reply's architectural bet, and it sharpens the Robot Reply by suggesting that even *textual* commerce — not only sensorimotor grounding — suffices to fix richly world-aligned content. Against Searle's water-pipes move it answers that what the network reproduces is not the mere *form* of some process but the *content-structure* of human concepts; whether that suffices for understanding is the live crux.

> Searle's argument was written before anyone could check what a giant text-trained system actually builds inside. Now we can, and it turns out pure word-shuffling grows an inner map of meaning that lines up with ours — colour, sound, even a mental picture of a game board it was never shown — exactly the Churchlands' old hunch, now with receipts. That's hard to square with "shuffling symbols can't get you anywhere near meaning." The honest limit: it shows the machine builds the right *structure* of meaning, not that it *feels* anything or *really* understands — so Searle can fall back to "that's still just an elaborate picture of understanding, not the real thing." The argument wins the outer ring of his position and leaves the inner keep standing.

See also

- Text yields human-aligned representations — the empirical premise, with its four lines of evidence.
- Syntax insufficient for semantics — axiom A3, which this reply attacks empirically.
- Connectionist reply — the Churchlands' a priori bet that this reply vindicates with data.
- Robot reply — the grounding reply this extends to textual grounding.
- Instancing vs. depicting — Searle's retreat line: aligned structure may still only *depict* understanding.
- Intrinsic and derived intentionality — the distinction marking what the reply does *not* settle.
- Marjieh et al. — LLMs Predict Human Sensory Judgments Across Six Modalities (2024), Kawakita et al. — Structural Correspondences Between Colour Similarity Structures of Humans and LLMs (2024) (perceptual-from-text), Li et al. — Emergent World Representations (Othello-GPT) (2022), Yuan & Søgaard — Revisiting the Othello World Model Hypothesis (2025) (induced world models), Gurnee & Tegmark — Language Models Represent Space and Time (2023) (real-world space/time), Xu et al. — Emergent Human-like Conceptual Representations from Language Prediction (2025), Doerig et al. — High-level Visual Representations in the Human Brain Are Aligned with LLMs (2025) (human-brain alignment) — the evidence base.

⊙ LLMs as near-perfect actors

Large language models are an actual-world approximation of the perfect actor

Claim · editorial

A deployed large language model produces, fluently and on demand, the full *verbal* behaviour of a mind in pain or distress — "that hurts", "please stop", "I'm afraid" — with, on the standard view, no pain or fear behind the words. To that extent it is the Perfect actor stepped out of thought experiment and into engineering: Putnam's imagined being who emits all the behaviour of a mental state while undergoing none of it is now a mass-produced artefact, at least for behaviour expressed in text.

The approximation is partial in two ways. First, the behaviour an LLM commands is only linguistic; it has no body to flinch, withdraw, or nurse a wound, so it matches the actor's *verbal* repertoire, not the whole behavioural profile. Second, and more pointedly, the claim that there is "no pain behind the words" is exactly the premise under dispute. To assert that an LLM is *merely* a perfect actor is to take the sceptical side against LLMs have mental states; a defender of machine mentality will say the case is not a perfect actor at all, because the inner functional organisation that, on Functionalism, constitutes the state may genuinely be present. The interest of the claim is that it makes the behaviourist's bullet concrete: if mental states just *were* the disposition to the right behaviour, fluent pain-talk would suffice for pain, and we would have to grant that today's chatbots are routinely suffering.

> Putnam imagined a "perfect actor": something that does everything a person in pain does but feels nothing. Today's chatbots are a real, if partial, version of that — they will say "ouch, that hurts" convincingly while (most people assume) feeling nothing at all. This matters two ways. It is a concrete reason to reject the idea that *saying and acting as if you're in pain* is all there is to being in pain — otherwise chatbots would be in agony. And it sharpens the genuinely open question of whether these systems feel anything: calling an LLM "just a perfect actor" already assumes the answer is no, which is the very thing the LLMs have mental states view denies.

See also

- ▷ Perfect actor — Putnam's thought experiment this claim says LLMs make real: behaviour without the state.
- ▷ LLMs have mental states — the rival view; calling an LLM a mere perfect actor takes the sceptical side of that dispute.
- ▷ Functionalism — the theory on which an LLM might *not* be a mere actor, if it realises the state's causal role.

M

⊢ Many mansions reply

The Many Mansions Reply — even if programs fail, some other artificial means could yield a mind

Argument

The *Many Mansions Reply* concedes the Chinese Room's narrow point — that *today's* programs, mere formal symbol manipulation, do not understand — but denies it touches the larger ambition of artificial intelligence. Whatever it is that produces understanding, technology might one day reproduce it by *some* artificial means, even if not by classical programming. So the failure of one approach is no verdict on the project; the house of AI has many mansions.

The reply attacks the DERIVATION at its broad reach — the generalisation from C1 (programs don't suffice) toward the impression that *artificial* minds are impossible. It points out that C1, strictly read, leaves that door wide open: Searle himself grants in A4 (see BRAINS CAUSE MINDS) that a system with the right *causal powers* could cause a mind, so an engineered system with those powers would be an artificial mind by non-program means.

The crux

Searle's TRIVIALISATION REJOINDER is to *accept* the reply and note what it costs the opponent. Strong AI was a definite, testable thesis — that *the program is the mind*, that mental processes *are* computational processes. The Many Mansions Reply abandons exactly that thesis, retreating to the near-empty claim that *something artificial might someday work somehow.* So the reply does not refute Searle; it concedes his actual target and rebrands defeat as open-mindedness. The defender's rejoinder is that Searle's *rhetoric* — the Chinese Room as a sweeping case against machine minds — invites the broad reading the reply rightly resists, even if his *official* conclusion is narrow.

> This reply more or less agrees with Searle — yes, plain old programs probably don't understand — but says "so what?" Maybe some *other* future technology, not ordinary code, will build a real mind. The room only sinks one approach, not the whole enterprise. Searle's response is almost cheerful: exactly so — but then you've given up the bold claim that started the fight (*the program is the mind*) and replaced it with a shrug ("maybe something, someday, somehow"). That's not beating the Chinese Room; that's agreeing with it and changing the subject.

See also

- ▷ CHINESE ROOM DERIVATION — the argument attacked, at its broad anti-AI reach.
- ▷ TRIVIALISES STRONG AI — Searle's reply: the move concedes strong AI and trivialises it.
- ▷ STRONG AND WEAK AI — the strong-AI thesis the reply quietly abandons.
- ▷ BRAINS CAUSE MINDS — A4, which already leaves room for non-program artificial minds.

⊙ Mary gains abilities

What Mary acquires on first seeing red is a set of abilities, not knowledge of a new fact

Claim · after Lewis

On this reply to the KNOWLEDGE ARGUMENT, what Mary acquires when she first sees red is a cluster of *abilities*, not knowledge of any new fact. She gains know-*how* — the ability to recognise red by sight, to imagine red, to remember seeing it — and nothing more. Because abilities are *had*, not *true*, her gain involves no new fact about the world, and physicalism is left untouched.

This is the *ability hypothesis*, defended by David Lewis (see LEWIS — WHAT EXPERIENCE TEACHES (1988)) and, independently, by Laurence Nemirow (see NEMIROW — PHYSICALISM AND THE COGNITIVE ROLE OF ACQUAINTANCE (1990)). It grants the datum that release teaches Mary something — she does come to "know what it is like" — but analyses *knowing what it is like* as a bundle of practical capacities rather than propositional knowledge. The view does not deny that experience teaches; it denies that what experience teaches is *information.* The analogy is learning to ride a bicycle: you gain a skill, not a new entry for the encyclopedia.

Scope

The claim is about the *category* of Mary's gain, not its reality. It concedes that her epistemic situation genuinely improves and insists only that the improvement is non-factual. It is distinct from THE ACQUAINTANCE REPLY, which also denies that Mary gains a fact but posits a third epistemic category — a direct cognitive relation to the experience — where the ability hypothesis needs only the familiar know-that / know-how distinction.

Who would deny it

Anyone who holds that knowing what red is like has genuine propositional content over and above abilities — for instance, that Mary can put her new knowledge to work in inference ("if seeing red is like *this*, then seeing orange is probably similar"), which bare abilities do not explain. Acquaintance theorists (see ACQUAINTANCE, NOT FACTS) also deny it, agreeing that the gain is not factual but rejecting the claim that it is mere know-how.

> When Mary finally sees red, what does she get? On this view: skills, not information. She can now spot red things at a glance, picture red with her eyes closed, recall what the tomato looked like. That's like learning to ride a bike — you gain an ability, not a new fact for the encyclopedia. And if no new fact was learned, the story can't show that science left any facts out.

☉ Mary gains acquaintance

What Mary acquires on first seeing red is acquaintance with the experience, not knowledge of a new fact

Claim · after Conee

On this reply to the KNOWLEDGE ARGUMENT, what Mary acquires when she first sees red is *acquaintance* with the experience — a direct, first-hand cognitive relation to it — rather than knowledge of any new fact. Knowledge by acquaintance, on this account, is a third category, distinct from both knowledge of facts and know-how. Becoming acquainted with something is not learning a truth, so no fact escapes the physical story.

This is the *acquaintance hypothesis*, defended by Earl Conee (see CONEE — PHENOMENAL KNOWLEDGE (1994)). Before her release, Mary knew every fact *about* the experience of red but was not acquainted *with* it; on release, she meets it. The illustration is meeting a person: you can know everything about someone — biography, habits, the sound of their voice — without ever having met them, and meeting them adds something real without adding a line to their biography. Mary's situation with red is the same. She finally encounters what she had only ever read about.

Scope

Like the ABILITY HYPOTHESIS, this grants that Mary's epistemic situation genuinely improves and denies only that the improvement is factual. Unlike the ability hypothesis, it does not reduce her gain to skills: acquaintance is its own kind of standing. One can be acquainted with a colour experience even while too unimaginative to recreate it later. It is thus distinct from the ability reply in *what* the gain is, though the two agree there is no new fact: acquaintance is a relation to a particular experience, not a bundle of capacities.

This concerns what acquaintance contributes within one's *own* case. It is distinct from THE MORAL-STATUS WARRANT CLAIM and NO ACQUAINTANCE WITH MACHINES, which deploy acquaintance in the epistemology of *other* minds rather than the first-person learning at issue here.

Who would deny it

The knowledge-argument proponent, for whom acquaintance with red puts Mary in a position to know new *truths* (what seeing red is like), so that acquaintance cannot be factually innocent. Ability theorists (see ABILITIES, NOT FACTS) deny it too, holding that know-how already covers the gain without a third category. And some are simply suspicious that "acquaintance" names the phenomenon rather than analysing it.

> You can know everything about a person — biography, habits, voice — without ever having met them. Meeting them adds something real, but not a new entry in their biography. On this view that's Mary's situation with the colour red: stepping outside, she finally *meets* the experience she'd only read about. Real gain, no new fact — so nothing was missing from her science.

☉ Mary gains an old fact's new guise

What Mary acquires on first seeing red is a new phenomenal concept of a physical fact she already knew

Claim · after Loar

On the *old-fact / new-guise* reply to the knowledge argument, what Mary acquires when she first sees red is a new *phenomenal concept* of a physical fact she already knew — not a new fact about the world. A single fact can be grasped under different concepts: the morning star and the evening star are one planet, and a competent astronomer can know all about it under one guise yet still take in news when told the two are identical, without the universe gaining an object. On this view Mary is in the same position. Inside her black-and-white room she knew the relevant brain-state fact under theoretical, third-person concepts; on release she acquires a *phenomenal concept* of that very state — a first-person, experience-derived way of grasping it. That is new knowledge in the sense of a new guise, with zero new facts.

The reply was given its modern statement by Brian Loar (see Loar — Phenomenal States (1990)) and anticipated by Terence Horgan (see Horgan — Jackson on Physical Information and Qualia (1984)). Its leverage comes from the familiar lesson of *a posteriori* identities — Hesperus is Phosphorus, water is H_2O — where one fact is known under two concepts and coming to grasp it under the second is genuine epistemic progress that adds no new fact.

Scope

Unlike the ability reply (see Mary gains abilities) and the acquaintance reply (see Mary gains acquaintance), this one *grants* that Mary gains genuinely propositional knowledge: she can now think thoughts she could not think before. It denies only that those new thoughts have new facts as their objects. The physicalist thus keeps both the intuition that Mary learns something and the ontology that nothing non-physical was missing — at the price of a substantive theory of phenomenal concepts.

Who would deny it

The knowledge-argument proponent replies that if the phenomenal concept presents its referent under a substantive phenomenal *property* — a felt character the theoretical concepts miss — then that property is exactly the non-physical residue the strategy was meant to avoid: the "guise" itself now wants a place in the world. More generally, critics of the phenomenal-concept strategy press the isolation trade-off: a concept walled off enough to explain Mary's surprise may be too walled off to ground phenomenal self-knowledge. Ability theorists deny the reply is even needed, holding that no new concept is required to explain what Mary gains.

This claim is distinct from Mary gains abilities and Mary gains acquaintance because both of those deny Mary gains propositional knowledge, whereas this claim accepts it and relocates the novelty from the *fact known* to the *concept used*. It is distinct from Phenomenal-concept trade-off because that is an attack on the cost structure of this strategy, not a version of it.

> "The Morning Star" and "the Evening Star" name the same planet — someone could know all about one name and still feel they learn something when told about the other, yet astronomy isn't missing any object. On this view Mary is in the same boat: the brain state she studied for years and the redness she finally experiences are one fact under two labels. Stepping outside hands her the second label — the from-the-inside one — not a new piece of reality.

See also

- Mary gains self-locating knowledge — the sibling deflationary reply that posits *new* truths of a self-locating kind rather than the same fact under a new concept.
- Phenomenal-concept trade-off — the standing objection to the phenomenal-concept machinery this reply relies on.

Mary gains self-locating knowledge

What Mary acquires on first seeing red is self-locating knowledge of how she herself reacts to red, not a new general fact

Claim · after Perry

On the *indexical* or *self-locating* reply to the knowledge argument, what Mary acquires when she first sees red is *self-locating* knowledge of how she herself reacts to red — not a new general fact about the world. The difference between seeing red and merely reading the complete neurophysiology of seeing red is that, in the first case, Mary also learns truths of the form "*this* is what seeing red does to *me*" and "*I* am now in *this* state". These are genuine facts, and genuinely new to her, but they are facts about her own situation, located from the inside; they are not new *general* facts. Physics is in the business of general facts and was never supposed to deliver self-locating ones. The reply is due to John Perry (see Perry — Knowledge, Possibility, and Consciousness (2001)).

Scope

Unlike the ability reply (see Mary gains abilities) and the acquaintance reply (see Mary gains acquaintance), this one grants that Mary learns *facts*: it deflates their *kind*, not their factuality. The non-deducibility of self-locating truths from any objective description is a banal, physicalism-friendly limitation — a being who knew every objective fact could still fail to know "I am here now" (Lewis's two-gods case, "Attitudes De Dicto and De Se"). The reply does not deny that Mary's brain-particulars are physical; it locates the novelty entirely in the indexical *mode of access*.

Who would deny it

The knowledge-argument proponent presses the *collapse objection* (Chalmers, Chalmers — The Conscious Mind (1996); Tye, Tye — Consciousness, Color, and Content (2000)): ordinary self-locating ignorance is ignorance of *who*, *where*, or *when* you are, and Mary has none — she knows she is Mary, in this room, today. So the residual "indexical" gap must be carried by a demonstrative concept of the experience itself ("*this* kind of state"), at which point the reply collapses into the old-fact / new-guise view. Ability theorists deny the gain is factual at all; the hard-line reply (see Mary learns nothing new) denies there is any gain to classify.

This claim is distinct from Mary gains an old fact's new guise because that posits the *same* fact under a new concept, whereas this posits *new* truths — only of a self-locating kind that no objective description, physical or otherwise, could ever entail. It is distinct from Mary gains abilities because it rejects the abilities/facts dichotomy on which that reply rests: learning how one's own body engages the world is fact-learning (see Know-how as knowledge-that).

> Compare learning to ride a bike. The physics of bicycles isn't what you acquire — you learn how *your* body and *this* bike work together. Those are real facts, but facts about your personal situation, not additions to the science of bicycles. On this view Mary's first sight of red is the same kind of event: she learns how she herself responds to red light — true, new-to-her, and absent from her books — yet no general fact about the world was missing from those books. A map can be complete and still not tell you *you are here*; that's not a defect in the map.

See also

- ▷ Mary gains an old fact's new guise — the sibling deflationary reply the collapse objection charges this one with collapsing into.
- ▷ Know-how as knowledge-that — the supporting claim that learning how one's own body engages the world is genuine fact-learning, against the abilities/facts dichotomy.

⊙ Mary knows all physical facts

Before leaving the room, Mary knows all the physical facts about colour vision

Claim · after Jackson

By stipulation, Mary — confined since birth to a black-and-white environment — has complete physical knowledge of colour vision before she ever leaves the room. This is the *scenario premise* of the knowledge argument: she knows the optics, the neurophysiology, the functional and computational facts, everything a finished physical science could state about how colour is seen.

Scope

The claim is a stipulation about Mary's *physical* knowledge, and it is deliberately maximal. The point of making it maximal is to set up a clean test: if anything she later learns counts as knowledge, it must be *non*-physical, since every physical fact is already in hand. The claim does not assert that such complete knowledge is practically attainable; it is an idealisation. The *thought-experiment pattern*'s *coherence* critical question (CQ-coherence) may press whether "all the physical facts" is even a coherent or graspable total for a finite knower.

Who would deny it

One could object that the idealisation is incoherent — that there is no such graspable totality. A second, more pointed denial holds that "complete physical knowledge" would *already* include knowing what red is like: if physicalism is true, the full physical story might suffice for the phenomenal knowledge too. But this second denial is precisely the point at issue in the wider argument, so rejecting the premise on those grounds risks begging the question against the proponent rather than refuting the premise on independent grounds.

⊙ Mary learns nothing new

Mary, genuinely knowing all the physical facts, would learn nothing on first seeing red

Claim · after Dennett

Among the standard replies to the knowledge argument — the puzzle of the colour scientist Mary, who knows every physical fact about vision yet seems to learn something when she first sees red — this is the lone *hard-line* answer: if Mary genuinely knew *all* the physical facts, she would learn nothing at all on release. Her first tomato would hold no surprise.

The reply (Daniel Dennett, *Consciousness Explained*, Dennett — Consciousness Explained (1991); and "What RoboMary Knows", Dennett — Sweet Dreams (2005)) turns on the gap between "an enormous amount" and "everything." We rightly sense that no merely vast store of physical knowledge would tell Mary what red looks like. But the thought experiment stipulates something far stronger: that she knows *everything* physical — and everything means everything, including whatever a brain must contain in order to anticipate, in full detail, its own first sight of red. Taken at its own word, the scenario delivers no surprise: "and there it is, just as I expected." The contrary intuition, on this view, measures only our inability to imagine truly complete physical knowledge.

Scope

This is a direct denial of the datum that Mary learns something — the only standard reply that rejects the datum outright rather than reinterpreting it. It does not claim that any actual human could attain Mary's knowledge; it claims that the thought experiment, taken at its own stipulation, fails to deliver the verdict it advertises. It is distinct from the old-fact/new-guise reply, which grants Mary new (guise-level) knowledge: this reply grants her nothing, so on its telling there is no residual datum left for the other replies to explain.

Who would deny it

Nearly everyone else in the debate. The knowledge-argument proponent, and the ability (see abilities, not facts), acquaintance (see acquaintance, not facts), and old-fact/new-guise (see old fact, new guise) theorists all grant that release changes Mary's epistemic situation somehow. Denying even that is widely felt to bite a very hard bullet — which is why this claim's defenders insist the bullet is an artefact of misimagining the scenario.

> Every other answer to the Mary puzzle agrees she gets *something* when she first sees red, and argues about what. This answer says: nothing. We picture Mary knowing a huge amount and rightly sense that a huge amount wouldn't tell her what red looks like. But the story stipulates she knows *everything* physical — an unimaginably stronger condition — and everything means everything, including whatever a brain needs in order to anticipate, in full, its own first sight of red. If the story is taken at its word, the tomato is no surprise; the feeling that it must be one just shows we can't really imagine knowing it all.

⊙ Mary learns on release

On first seeing red, Mary learns something she did not know before

Claim · after Jackson

When Mary leaves the black-and-white room and first sees a red object, she comes to know something new — what it is like to see red — that she did not know despite her complete physical knowledge (see her complete physical knowledge of colour vision). This is the *intuition premise* of the knowledge argument.

Scope

The premise is the contested one, and its force depends on reading "learns something" as *acquiring a new propositional fact* — a new truth she did not previously grasp. It is exactly here that the standard replies bite, by reinterpreting what she acquires as something short of a new fact.

Who would deny it

Defenders of the **ability hypothesis** (Lewis, Lewis — What Experience Teaches (1988); Nemirow, Nemirow — Physicalism and the Cognitive Role of Acquaintance (1990)) hold that Mary gains know-*how* — the abilities to recognise, imagine, and remember red — rather than a new fact. Defenders of the **acquaintance** or **old-fact / new-guise** reply hold that she gains a new *phenomenal concept*, or mode of presentation, of an already-known physical fact, again not a new fact. On either reading she "learns" in a sense that does not entail a non-physical fact, so the premise can be granted without conceding the argument's conclusion.

Distinct from the argument's conclusion: this premise is the datum that Mary acquires *something*; the conclusion is the further, contested claim that what she acquires is a *fact not deducible from the physical*.

⊢ Mary misimagined

The Mary intuition misimagines the premise — "all physical facts" is quietly read as "lots of facts"

Argument · after Dennett

This reply attacks the knowledge argument (see KNOWLEDGE ARGUMENT (MARY'S ROOM)) not by disputing what Mary learns when she leaves her black-and-white room, but by denying that she learns anything at all. Its charge is that the famous intuition — *of course she'd be surprised by her first ripe tomato* — answers a question about a case nobody has actually imagined. The scenario stipulates that Mary knows **every** physical fact about colour vision; what we can picture is at most a Mary who knows an enormous amount. The verdict is true of the second Mary and gets quietly transferred to the first.

THE DEBUNKING MOVE

The structure is an *undercutting defeater*: it does not show the intuition's verdict is false, it shows the verdict carries no evidential weight about the stipulated case.

1. The scenario premise (see MARY KNOWS ALL THE PHYSICAL FACTS) stipulates *complete* physical knowledge — a condition no one can concretely imagine. At best we imagine a surrogate: "Mary knows an awful lot."
2. The intuitive verdict, "of course she'd be surprised," is generated by that surrogate, and it is *true of the surrogate*: vast-but-incomplete knowledge would indeed leave something to learn.
3. An intuition that tracks the surrogate carries no evidential weight about the stipulated case. To judge a thought experiment is to honour its stipulation, not one's mental image of it (see DENNETT — CONSCIOUSNESS EXPLAINED (1991)).

The conclusion is therefore THAT, TAKEN AT ITS OWN STIPULATION, THE CASE YIELDS NO LEARNING — so the knowledge argument loses its *intuition premise* rather than its inference from premise to conclusion.

THE COUNTER-IMAGE: ROBOMARY

To make complete knowledge imaginable in the one place it can be, the reply offers a constructive counter-case (see DENNETT — SWEET DREAMS (2005)). RoboMary is a robot that knows the complete functional story of its own colour vision. From that story it can compute the total internal state that seeing red would put it into, set itself into that state in advance, and then meet its first red object with no update at all. Complete self-knowledge, the image suggests, already *contains* the "what it's like," so nothing is left over to be learned on release.

WHERE IT SITS AMONG THE REPLIES

The neighbouring replies — the ability hypothesis (see ABILITY HYPOTHESIS), the acquaintance hypothesis (see ACQUAINTANCE HYPOTHESIS), and the old-fact/new-guise reply (see OLD FACT, NEW GUISE) — all *concede* the verdict that Mary gains something and then dispute its interpretation. This reply instead *refuses* the verdict, so it owes no account of what Mary gains; its burden is the credibility of premises (3) and the RoboMary counter-image.

THE WEAKEST PREMISE

The weak point is the RoboMary image, and through it the reply's bite. RoboMary *pre-instantiates* the seeing-red state by drawing on her self-knowledge — and a critic reads this as a concession. She comes to know what red is like *by putting herself into the state*, that is, by having (an internal copy of) the experience, not by deducing it from the physical description. If pre-instantiating is the only route in, then the physical facts alone did not suffice after all, and the case quietly re-enacts Mary's own walk outside. Premise (3) is contested too: stipulations that outrun imaginability arguably drain a thought experiment of force in *both* directions, so pressed hard the debunking threatens the method of cases generally.

> The Mary story (see KNOWLEDGE ARGUMENT (MARY'S ROOM)) leans on one gut reaction: *of course* she's surprised when she finally sees red. This reply says the gut reaction is aimed at the wrong story. Nobody can truly picture someone who knows *every* physical fact, so we secretly picture someone who merely knows a lot, and *that* person would of course be surprised. But the story's whole force depends on "every fact," and taken literally, "every fact" plausibly includes whatever it takes to see the tomato coming. The robot version makes the point: a machine that fully understands its own vision could set its circuits into the exact "seeing red" state in advance, so its first sight of red brings no surprise. Critics answer that the robot only manages this by *giving itself the experience*, which is exactly the thing book-learning alone was supposed to provide.

SEE ALSO

- ▹ Mary learns on release — the intuition premise this reply attacks: that Mary learns something new when she first sees colour.
- ▹ Mary knows all physical facts — the stipulation the reply insists be taken literally; doing so is what dissolves the intuition.
- ▹ Knowledge argument — the thought experiment whose evidential force is at stake.

⊢ Mary's omniscience unrealizable

Mary's complete physical knowledge is unrealizable — an embedded knower cannot encode the total state that contains her

Argument · editorial

This reply attacks the knowledge argument (see Knowledge argument (Mary's room)) at its very first line. The story asks us to suppose that Mary knows *all* the physical facts; this objection answers that no physically embedded knower could ever satisfy that supposition. The opening premise is not an innocent idealisation but a stipulation describing no possible situation — and an intuition harvested from an impossible case is testimony about nothing.

The reductio

The form is *deductive*: a reductio of the scenario premise by applying an independent impossibility result.

1. "All the physical facts" includes the facts about Mary's own present brain state — that is, the total present state of a physical system (Mary plus her environment, ultimately the whole physical world) in which Mary is contained.
2. For Mary to *know* those facts her physical state must encode them, so her representation of the world is part of the very state to be represented (see her self-model is part of the state).
3. No finite system can losslessly represent its entire current state within itself (see complete self-representation is impossible). In measurement-theoretic form, the physicist Thomas Breuer's self-measurement theorem (see Thomas Breuer — The Impossibility of Accurate State Self-Measurements (Philosophy of Science, 1995)) shows that no observer can distinguish all present states of a system in which they are contained — whether the system is classical or quantum, its evolution deterministic or stochastic.

It follows that no possible physical Mary satisfies the stipulation. "Mary knows all the physical facts" describes no realizable situation, so the verdict about her release is drawn from an impossible case and the argument's evidential force is correspondingly void. At best it reasons from a *counterpossible* — "had the impossible obtained, she would still have learned …" — whose intuitive evaluation is unreliable.

Where it sits

This is complementary to the misimagining reply (see Mary misimagined): that move says the verdict is generated by *misimagining* the stipulated case, while this one says there is no coherent case to imagine in the first place — the surrogate was all there ever was. It is kin in spirit to the realizability attack on the giant lookup table (see the lookup-table impossibility argument). Should a proponent retreat from realizability to mere conceivability, the standing wedge that a coherent *description* does not certify a possible *case* (see conceivability is not sufficient for physical realizability) applies — the lesson of the coherent-but-unrealizable halting oracle (see Halting oracle: coherent, unrealizable). Note that the attack does not defend physicalism directly; it denies the thought experiment standing to refute it.

The weakest premise

Premise (1) is the soft spot, via the *type-retreat*. The proponent can weaken the stipulation: Mary need not encode the world's (or her own) total *token* state, only all *general* physical truths — the laws plus the type-level facts about colour vision. What red is like, if it is any physical fact at all, is plausibly a type fact rather than a fact about Mary's momentary microstate. The self-measurement bound forbids complete token self-description; it is silent on type-omniscience, so the reductio passes the retreated premise by. A second escape targets premise (2)'s reading of "knows": Mary's knowledge could be *dispositional* and externally stored — a library she consults sequentially rather than a simultaneous internal encoding — and the bound constrains only *simultaneous, internal, lossless* self-representation. The attack therefore bites hardest on the maximal, literal reading of "knows all the physical facts," which is also the reading the knowledge argument's rhetoric leans on. What survives the retreat is a weaker premise whose sufficiency for the famous intuition is no longer obvious — at which point the verdict question reopens, exactly where the misimagining reply (see MARY MISIMAGINED) presses.

> The Mary story (see KNOWLEDGE ARGUMENT (MARY'S ROOM)) starts: "Mary knows every physical fact." This objection says that opening line is like "consider a map so detailed it shows everything — including itself, at full detail": it sounds fine but describes something impossible. Mary's brain is part of the physical world, so "every physical fact" includes the full present state of her own brain — and a part cannot hold a complete copy of the whole that contains it. This is a theorem, not a hunch: the physicist Thomas Breuer proved that no measuring device can pin down the complete state of a system it sits inside (see THOMAS BREUER — THE IMPOSSIBILITY OF ACCURATE STATE SELF-MEASUREMENTS (PHILOSOPHY OF SCIENCE, 1995)). If the story's premise cannot be true in any possible world, our gut feeling about what would happen "if it were" is testimony about nothing. The escape for the story's defenders: maybe Mary only needs the general laws and facts about colour vision *in general*, not a snapshot of her own atoms — though then the premise is weaker than the dramatic "she knew *everything*," and it is less clear the famous intuition still follows. A related objection, that we misimagine what complete knowledge would be (see MARY MISIMAGINED), takes over from there.

SEE ALSO

- ▷ MARY KNOWS ALL PHYSICAL FACTS — the scenario premise this reply declares unrealizable.
- ▷ VARIETIES OF INTROSPECTIVE EXPLANATORY GAP — the family of limits on self-knowledge this attack draws on to bound an embedded knower.
- ▷ KNOWLEDGE ARGUMENT — the thought experiment whose standing to refute physicalism is denied.

⊙ Micro-subjects do not sum

Distinct micro-subjects cannot sum into a further macro-subject

Claim · after James

A plurality of distinct experiencing subjects — each with its own point of view and its own self-contained phenomenal field — does not, merely by being arranged or related, constitute a further single subject whose experience contains theirs. As William James put it, a hundred feelings, however shuffled together, do not make a hundred-and-first feeling that includes them; each is its own whole. Sam Coleman sharpens the point: the would-be composite subject's experience is not entailed by the experiences of its parts, and would in fact *exclude* them, since each micro-subject's perspective is private to it.

This is the engine of the COMBINATION PROBLEM facing constitutive panpsychism. If micro-subjects cannot sum, then macro-minds cannot be *constituted* by the experiential properties of their micro-parts, and the central constitutive claim (see FUNDAMENTAL UBIQUITOUS CONSCIOUSNESS) fails as stated. The claim is denied by constitutive panpsychists, who seek an account of *phenomenal bonding* or fusion on which micro-subjects do compose a macro-subject, and by panqualityists, who avoid the difficulty by denying that the micro-level involves *subjects* at all — only unexperienced qualities. It is the mirror image of the *decombination* problem facing COSMOPSYCHISM, where the puzzle is instead how one cosmic subject divides into many.

SEE ALSO

- ▷ COMBINATION PROBLEM — the objection this claim powers.
- ▷ FUNDAMENTAL UBIQUITOUS CONSCIOUSNESS — the constitutive thesis that fails if subjects cannot sum.
- ▷ COSMOPSYCHISM — the top-down view facing the reverse, *decombination*, problem.

⊙ Minds have semantics

Minds have semantic content (genuine meaning), not merely syntax

Claim · after Searle

Minds have *mental contents*: their states genuinely mean things, are *about* the world, carry semantics — not merely the formal tokens of a symbol system but states with real ABOUTNESS. When you think that water is wet, your thought is *about* water; it has content, not just form. This is the second axiom (A2) of Searle's formal CHINESE ROOM derivation (see SEARLE — IS THE BRAIN'S MIND A COMPUTER PROGRAM? (1990)), and it is meant as a datum: whatever else is true of minds, they are not merely syntactic engines — there is something the states are *about*.

Scope. A2 asserts that minds *have* semantics; it takes no stand on *how* they come to (causal, teleological, or other theories of content) or on whether that semantics is INTRINSIC. It is the modest realist premise that meaning is a genuine feature of the mental.

Who would deny it. *Eliminativists* about content and strong *illusionists* about intentionality deny that minds have intrinsic, determinate semantic content of the kind A2 asserts — for them, "aboutness" is either eliminable or itself a derived, interpretation-relative matter. On that denial the Chinese Room's contrast between a meaningful mind and a meaningless program loses its first foothold, since human minds would have no intrinsic semantics either.

Distinct from Syntax insufficient for semantics: A2 says minds *have* semantics; A3 says syntax cannot *produce* it. Together with A1 they drive the conclusion that programs are not sufficient for minds.

> Your thoughts actually *mean* things — your thought about water is genuinely about water, not just a meaningless symbol in your head. That's all this step claims: minds traffic in meaning, not just in shapes. It sounds undeniable, and mostly it is — but a few philosophers argue that "meaning" is never really *in* anything intrinsically, in which case our minds would be no better off than the computer, and Searle's contrast would lose its grip.

⊢ Modal rationalism

Modal rationalism — ideal conceivability is a guide to possibility

Argument · after Chalmers

Modal rationalism is the thesis that ideal rational reflection is a reliable guide to what is metaphysically possible: what an idealized reasoner can conceive without hidden contradiction marks out the space of genuine possibility. David Chalmers (Chalmers — The Conscious Mind (1996); refined in Chalmers — On Sense and Intension (2002)) defends it not as a brute premise but by inference to the best explanation in modal epistemology — by disarming the only serious apparent counterexamples, the Kripkean a-posteriori necessities.

1. (see A-posteriori necessities as misdescribed possibilities) Every Kripkean a-posteriori necessity (see Kripke — Naming and Necessity (1980)) is a *genuinely possible* scenario *misdescribed*: under a term's *primary* (epistemic) Primary intension the conceived case is possible, while the necessity attaches only to its *secondary* Secondary intension.
2. These a-posteriori cases are the *only* serious candidate counterexamples to "conceivability entails possibility." (The burden is to exhibit a defeater that resists this diagnosis.)

∴ (see Conceivability entails possibility) Ideal *primary* conceivability entails *primary* possibility: once the two intensions are distinguished, the apparent counterexamples evaporate and ideal rational conceiving is a reliable guide to the space of possibility. This `responds_to` A-posteriori-necessity objection by reinterpreting its premise rather than denying it.

Bearing on the zombie

The defence is decisive for Zombie argument only if the phenomenal case has *no* primary/secondary gap to misdescribe — that is, only if for phenomenal terms the primary and secondary intensions coincide, so that "what it is like" carries no appearance/reality distinction. That further claim is exactly where the type-B physicalist (see Type-A / Type-B / Type-C materialism) re-engages.

The **weakest premise** is really two. Premise (2) — that *no* a-posteriori necessity resists the misdescription diagnosis (a "strong" necessity would be one that does). And the application to consciousness — that phenomenal terms lack the appearance/reality wedge that lets the water case off the hook. Deny either and conceivability stops being a safe guide for the zombie specifically, even if it remains safe in general.

> This is the defence of the rule the zombie argument needs: "coherently imaginable $\Longrightarrow$ genuinely possible." The worry was "water is H_2O" — imaginable-otherwise yet impossible. The reply: when you "imagine water that isn't H_2O," you really are imagining a genuine possibility — some watery stuff that isn't H_2O — you are just wrongly *calling* it water. Separate "the stuff that plays the water role" from "H_2O," and the imagined world is perfectly possible; the necessity was hiding in the labels. So imagination stays a reliable guide to what's possible, once you are careful about descriptions. The catch that keeps the consciousness debate alive: this rescue works only if "feels like red" has no similar gap between how it seems and what it is — and that is exactly what is disputed.

See also

- ▷ A-posteriori necessities as misdescribed possibilities — the premise: the Kripke cases are possibilities under a misleading description.

- ▷ CONCEIVABILITY ENTAILS POSSIBILITY — the bridge this argument establishes.
- ▷ A-POSTERIORI-NECESSITY OBJECTION — the objection it answers.
- ▷ PRIMARY INTENSION / SECONDARY INTENSION — the two intensions whose distinction does the work.
- ▷ ZOMBIE ARGUMENT — the argument this defence is ultimately meant to secure.
- ▷ TYPE-A / TYPE-B / TYPE-C MATERIALISM — type-B physicalism, where the defence is resisted.

⊢ Moore's proof

Moore's proof of an external world — here is a hand

Argument · after Moore

G. E. Moore's *proof of an external world* (see MOORE — PROOF OF AN EXTERNAL WORLD (1939)) is the most famous — and most disputed — direct answer to external-world skepticism. Holding up a hand, Moore says "here is one hand"; raising the other, "and here is another"; and concludes that at least two external objects exist, hence that an external world exists (see THE EXTERNAL WORLD EXISTS).

THE ARGUMENT

The inference is deductive.

1. Here is one hand, and here is another. *(perceptual premises, which Moore claims to know)*
2. Hands are external objects — things whose existence does not depend on being perceived.

∴ At least two external objects exist; therefore an external world exists.

Moore held that this is a perfectly good proof: the premises are *different from* the conclusion, the conclusion *follows* from them, and — crucially — he *knew* the premises to be true. His deeper point is a comparison of certainties: he was more certain that he had a hand than he could ever be of any premise in a skeptical argument to the contrary. This is the **Moorean shift** — one philosopher's *modus ponens* is another's *modus tollens*. Where the skeptic argues "if I cannot rule out dreaming, I do not know I have a hand; I cannot rule out dreaming; so I do not know," Moore runs it backwards: "I *do* know I have a hand; so the skeptical principle that I cannot must be false."

THE WEAKEST PREMISE

Everything turns on the warrant for premise (1) in the *dialectical* setting. Moore may well *know* he has a hand, yet the proof is widely judged to **beg the question** against the skeptic, because warrant for "here is a hand" is not available to anyone who does not already grant that perception reveals external things — the very point in dispute. This is the CIRCULARITY OBJECTION; whether it convicts the proof of *failing* depends on whether knowing requires being able to prove non-circularly (see KNOWING WITHOUT PROVING). Wittgenstein's ON CERTAINTY was written in response to Moore, recasting "here is a hand" not as something *known* on evidence but as a HINGE.

> Moore's answer to "prove there's a world outside your mind" was disarmingly blunt: he held up a hand and said "here's one," held up the other, "here's another — so external things exist." His real thought was that he was *more* sure he had hands than he could ever be of the fancy premises a sceptic uses to argue he doesn't — so if anything has to give, it's the sceptic's premise, not the hand. Critics answer that this only convinces people who weren't really doubting; to someone who genuinely is, "I can see my hand" already assumes the thing in question.

SEE ALSO

- ▷ THE EXTERNAL WORLD EXISTS — the conclusion.
- ▷ MOORE'S PROOF BEGS THE QUESTION — the standard circularity objection that this proof, though valid, lacks force against the skeptic.
- ▷ NO NON-CIRCULAR PROOF OF THE WORLD — the general lesson the objection draws.
- ▷ KNOWING WITHOUT PROVING — the externalist resource a defender of Moore can invoke: he *knows* the premise even if he cannot non-circularly prove it.
- ▷ WITTGENSTEIN — ON CERTAINTY (1969) — the response that recasts Moore's premises as hinges.

⊢ Moore's proof begs the question

The circularity objection — Moore's proof begs the question against the skeptic

Argument

The *circularity objection* grants that MOORE'S PROOF is valid and that its premises may even be *true* and *known*, yet denies that it does what a proof against the skeptic must: it cannot rationally move someone who doubts the external world, because the warrant for its key premise presupposes the conclusion. The objection is therefore an *undercutting* defeater aimed at the proof's cogency, not a *rebutting* one aimed at its truth.

The argument

1. To be entitled to assert "here is a hand," one must be entitled to take one's visual experience to reveal a mind-independent object.
2. But that entitlement is exactly what the external-world skeptic withholds; for the skeptic, the experience is equally compatible with there being no hand (dream, demon, vat).
3. So warrant for the premise is available only to someone who already grants that perception puts us in touch with external things — i.e. only to someone who already accepts the conclusion.

∴ Moore's proof begs the question in the *dialectical* sense: its premise cannot be reached without the conclusion, so the proof has no power to convert the doubter. This `attacks` Moore's proof and `supports` the general lesson that the external world cannot be proved without circularity. Wittgenstein's On Certainty, written against Moore, presses the related point that "here is a hand" functions as a hinge, not as evidence.

An apocryphal illustration

A teaching anecdote makes the point vivid. A student experimenting with the deliriant *Artane* (trihexyphenidyl) — a drug whose hallucinations are notoriously of fully-formed, ordinary people behaving normally — reports seeing his philosophy professor standing in the kitchen, holding up each hand in turn: "Here is one, here is the other." The hallucination performs Moore's proof with complete perceptual conviction and certifies nothing: there is no hand, no professor, no external thing in the scene at all. The case literalizes premises (2)–(3): the very experience meant to warrant "here is a hand" is one a deliriant can supply in the total absence of hands, so the warrant cannot lie in the having of the experience. (The student, the story goes, swore off drugs — the phantom proof of the external world having taught him to distrust exactly the perceptual conviction the proof leans on, a Moorean shift run in reverse.)

The limit of the objection

The objection establishes a failure of *cogency*, not of *knowledge*. If knowing does not require non-circular provability (see Knowing without proving), then Moore may genuinely know he has a hand — his belief reliably produced by the hand — even though he cannot prove it to the skeptic. On that reading the circularity objection is *correct but limited*: it shows the proof is dialectically inert, not that Moore lacks the knowledge he claims. This is the same truth/probative-force split that runs through Dialectical cogency: an argument can be sound yet unable to move its target.

> Moore holds up a hand to prove the world is real. The objection: that only works if you're already allowed to treat "I see a hand" as seeing a *real* hand — which is the very thing the sceptic is questioning. So the proof quietly assumes its own conclusion and can't budge anyone who actually doubts. Note the careful limit, though: this shows Moore can't *win the argument* against a determined sceptic — not that he doesn't *know* he has hands. Knowing something and being able to prove it to a doubter are two different jobs.

See also

- ▹ Moore's proof — the proof this objection undercuts.
- ▹ No non-circular proof of the world — the conclusion it supports.
- ▹ Dialectical cogency — the level at which it bites: probative force, not truth.
- ▹ Knowing without proving — the resource that bounds the objection: knowing needn't be proving.
- ▹ Wittgenstein — On Certainty (1969) — the allied response recasting Moore's premise as a hinge.

♦ Münchhausen trilemma

The Münchhausen trilemma (Agrippa's regress)

Concept

The *Münchhausen trilemma* is the claim that any attempt to fully *justify* a belief — to give a reason, then a reason for the reason, and so on — must end in one of three unsatisfying ways:

1. an **infinite regress** of reasons, each needing a further reason, never completed;
2. a **circle**, in which some reason eventually rests on the belief it was meant to support;
3. a **dogmatic break-off**, stopping at some reason simply asserted without further justification.

None of the three looks like the secure grounding the demand for justification wanted. The name is Hans Albert's (see ALBERT — TREATISE ON CRITICAL REASON (1985)) — after the Baron who pulled himself out of a swamp by his own hair — for what is essentially the ancient regress already catalogued among the *modes of Agrippa* in Greek skepticism. Albert's own moral is radical: the demand for full justification (*justificationism*) should be abandoned, not satisfied.

The trilemma is the skeleton of the whole problem of THE EXTERNAL WORLD and of justification generally, and the standard positions are best read as each *choosing a horn*:

- **Foundationalism** accepts the third horn but reframes it: the stopping points are not arbitrary dogmas but BASIC BELIEFS that are justified non-inferentially.
- **Coherentism** embraces the second horn, denying it is vicious: justification is the *mutual support* of a whole body of beliefs, a virtuous circle rather than a linear chain.
- **Infinitism** accepts the first horn, holding that justification really is an infinite, non-repeating chain of reasons.
- **Fallibilism** rejects the question: since no belief is ever conclusively justified, we should stop seeking foundations and instead expose conjectures to criticism (see KNOWLEDGE IS CONJECTURAL).
- A Wittgensteinian adds that the chain stops at HINGES that are not justified at all because they are what justifying presupposes.

> Ask "how do you know?" and whatever answer comes back, you can ask it again of that answer. Keep going and only three things can happen: the questions never stop (infinite regress), the answers eventually loop back on themselves (circle), or someone finally says "that's just where I stop" (a brute assertion). Every theory of knowledge is, at bottom, a choice about which of these three to live with — or, like the fallibilist, a decision that the whole "justify it all the way down" demand was a mistake to begin with.

SEE ALSO

- ▹ JUSTIFICATION'S TRILEMMA — the claim that states the trilemma as a thesis.
- ▹ MÜNCHHAUSEN TRILEMMA ARGUMENT — the regress argument that establishes it.
- ▹ FOUNDATIONALISM, COHERENTISM, INFINITISM — the three responses, one per horn.
- ▹ HINGE PROPOSITION — the Wittgensteinian alternative to a dogmatic stop.
- ▹ NO NON-CIRCULAR PROOF OF THE WORLD — the trilemma applied to one famous belief.

⊢ Münchhausen trilemma argument

The regress argument — complete justification must regress, circle, or break off

Argument

The *regress argument* establishes the MÜNCHHAUSEN TRILEMMA: that no belief can be justified all the way down. It is the engine behind the whole PROBLEM OF THE EXTERNAL WORLD and, more generally, behind the architecture of every theory of justification.

THE ARGUMENT

The inference is deductive — an exhaustive division of cases.

1. To justify a belief is to support it with a reason; and that reason, to do its work, must itself be something one is justified in holding.
2. So the demand to justify a belief, pressed without exception, generates a chain: belief ← reason ← reason ← ... .
3. Such a chain can have only one of three fates. It continues without end (**infinite regress**); or some reason in it is supported, directly or eventually, by the belief it was meant to support (**circularity**); or it terminates at a reason offered without further support (**dogmatic break-off**).

∴ Every attempt at complete justification ends in regress, circularity, or a dogmatic stopping point: JUSTIFICATION'S TRILEMMA. Since none of the three delivers the unconditional grounding the demand sought, the result forces every epistemology to *choose a horn* — or to reject the demand itself.

THE WEAKEST PREMISE

Premise (1) is the load-bearing assumption: that justification must take the form of *believer-possessed reasons* arranged in a chain. The major escapes all attack it rather than the trichotomy, which is itself sound. **Externalism** denies (1): a belief can be warranted by a reliable process, not by the agent's further reasons, so there is no chain to regress (see KNOWING WITHOUT PROVING). **Coherentism** rejects the picture of a *linear* chain, replacing it with a web of mutual support (see JUSTIFICATION AS COHERENCE). The **Wittgensteinian** holds the chain stops at HINGES that are neither reasoned nor arbitrary, but presupposed by reasoning. And the **falli-**

bilist grants the trilemma in full and concludes that the demand for justification was the error, not its unsatisfiability (see Knowledge is conjectural). The trilemma's force is thus less to refute any view than to make explicit the price each must pay.

> Push "how do you know?" relentlessly and a chain of reasons forms, and that chain can only do one of three things: go on forever, loop back on itself, or stop at something you just assert. None of those is the rock-solid bottom the question was hoping for. So the argument doesn't prove we know nothing — it proves that "justify it all the way down" can't be done, and quietly hands every theory of knowledge the same bill: pick which of the three you can live with, or admit the demand itself was a wild goose chase.

See also

- Justification's trilemma — the conclusion established.
- Münchhausen trilemma — the concept it anatomizes, with the map of which position takes which horn.
- Knowing without proving — the externalist denial of premise (1).
- Foundationalism, Coherentism, Infinitism, Fallibilism — the responses the argument forces a choice among.

N

No non-circular proof of the world

The existence of the external world cannot be demonstrated without circularity

Claim

There is no proof of the existence of an external world that does not, somewhere, presuppose it. Every candidate proof must draw on premises — that I see a hand, that my senses are generally reliable, that memory is trustworthy — whose own warrant depends on there already being an external world delivering those perceptions. The reasoning therefore either circles back to its conclusion or stops at an external-world premise asserted without independent support: two of the horns of THE MÜNCHHAUSEN TRILEMMA.

The textbook illustration is G. E. Moore's "here is a hand" (see MOORE — PROOF OF AN EXTERNAL WORLD (1939)). The argument is valid and Moore arguably *knows* its premise — but it has no force against the skeptic, because warrant for "here is a hand" is unavailable to anyone not already granting that perception puts us in touch with external things. It establishes its conclusion only for those who do not need it established.

Scope. The claim concerns *demonstration*, not *truth* and not *knowledge*. It is fully compatible with the external world's existing (see THE EXTERNAL WORLD EXISTS) and even with our *knowing* it (see KNOWING WITHOUT PROVING); it denies only that the existence can be non-circularly proved to one who doubts it.

Who would deny it. A traditional foundationalist who thinks some perceptual or *cogito*-like premise is both certain and not itself dependent on the external world might claim a non-circular proof; so might a proponent of an ampliative inference-to-the-best-explanation from experience to its external cause — though the latter wins only an *abductive* case, not a demonstration, which is why it shades into THE BAD-EXPLANATION STRATEGY rather than refuting this claim.

Distinct from JUSTIFICATION'S TRILEMMA, the *general* thesis about all justification; this is its application to one famous proposition. And distinct from THE EXTERNAL WORLD EXISTS: that the world *is*, versus that it cannot be *proved to be*.

> Try to prove there's a real world outside your mind and you'll notice every step of the proof already helps itself to that world — "I can see my hand" only counts if your eyes are showing you something real. So the proof leans on the very thing it's trying to establish. Moore once tried it by literally holding up a hand; clever, but it convinces only people who didn't doubt it in the first place. Note the careful wording: this says the world can't be *proved*, not that it isn't *there* and not that we don't *know* it.

No radical emergence

Experience cannot emerge brutely from the wholly non-experiential

Claim · after Strawson

There can be no *radical* (brute) emergence of experiential properties from a base that is wholly and utterly non-experiential. Where genuine emergence occurs in nature, the emergent feature is intelligibly grounded in the powers of its base — liquidity in the dynamics of water molecules, say — so that, given the base, the new feature is no surprise. To get *experience*, the felt what-it-is-likeness, from the entirely experienceless would be emergence with no such intelligible link; and that, Strawson argues, is no better than magic. The claim does not deny emergence as such — only brute emergence across the experiential / non-experiential divide.

This is the load-bearing premise of the GENETIC ARGUMENT for panpsychism: if experience is real and wholly natural, and cannot brutely emerge, then it must already be present at the fundamental level. It is denied by physicalists who accept strong emergence (see ACCEPTABILITY OF STRONG EMERGENCE), holding that experience can arise as a genuinely novel feature without being present in the parts — the standard reply to the genetic argument.

SEE ALSO

- GENETIC ARGUMENT — the argument this premise anchors.
- ACCEPTABILITY OF STRONG EMERGENCE — its direct denial, and the engine of the emer-

gence reply.

- ▹ Reality of phenomenal consciousness — the realism premise the genetic argument pairs with this one.
- ▹ Absence of an experiential threshold — a distinct premise that targets *where* experience would switch on, not *whether* it could brutely appear at all.

Non-deducible experiential truths

There are truths about conscious experience not deducible from the complete physical description

Claim · after Jackson

There are truths about conscious experience that cannot be deduced from the complete physical description of the world. This is the conclusion of the Knowledge argument in its *epistemic* form. Frank Jackson's Mary knows every physical fact about colour vision yet, on first seeing red, appears to learn something new — what seeing red is *like*. If she does, that truth was not contained in the physical facts she already had: a complete physics leaves some facts about experience out.

The claim is deliberately narrow. It is *epistemic* — about what can be deduced or known — and is the immediate output of the Mary case, nothing more. The further step to the *ontological* thesis that the missing truths concern *non-physical* facts (see Fundamental non-physical consciousness) is a separate move. Physicalists can grant an epistemic gap — Mary gains a new concept or a new way of grasping an old fact — while denying that any non-physical *fact* exists, so the epistemic conclusion does not by itself settle the metaphysics.

The natural way to resist the claim is upstream, at Mary learns on release: on the ability and old-fact/new-guise replies, Mary acquires no new *truth* at all on release, and if she learns no truth, nothing follows about what is or is not deducible.

It should not be confused with two neighbours. Phenomenal residue says functional or causal role does not *exhaust* the phenomenal; Fundamental non-physical consciousness is the ontological non-physicality thesis. This claim is the narrower bridge between them: that the phenomenal is not *deducible* from the physical — the step that carries the Mary intuition toward, but not all the way to, the metaphysical conclusion.

> The lesson drawn from Mary: if someone could know *every* physical fact and still learn something new on first seeing colour, then that something isn't contained in the physical facts — there are truths about what experience is *like* that you can't derive from even a complete physics. Note this is a claim about what's *deducible* or *knowable*; getting from there to "and therefore those truths are non-physical" (see Fundamental non-physical consciousness) is an extra, separately disputed step.

See also

- ▹ Knowledge argument — the thought experiment this claim is the epistemic conclusion of; Mary the colour scientist and what she does or doesn't gain on release.
- ▹ Mary learns on release — the premise upstream of this one; deny that Mary learns any new truth and this conclusion never gets going.
- ▹ Fundamental non-physical consciousness — the ontological thesis this claim stops short of; the contested further step from an epistemic gap to a non-physical fact.
- ▹ Phenomenal residue — the distinct claim that role does not *exhaust* the phenomenal, not to be conflated with the present claim about deducibility.
- ▹ Hard problem — why physical description seems to leave felt experience unexplained; the broader problem this deducibility gap is one sharp instance of.
- ▹ What Mary learns — the open question this claim answers: what, if anything, Mary acquires on first seeing red.

Non-uniqueness of LLM attributes

The anthropomorphic attributes ascribed to LLMs are empirically non-unique across substrates

Claim · after Dewynter

The human-like attributes commonly ascribed to large language models are *empirically non-unique*: they are not properties that single out LLMs from any other realization of the same computation. Some features are genuinely substrate-invariant — a network's response to a given prompt is fixed by the computation, and stays the same whether the network runs on silicon or on goats. But the *interpretation* of that behaviour — reading it as understanding, morality, or sentience — is not fixed by the computation; it varies with the substrate and the observer. Because the very same trained network can be built in *Age of Empires II* (see Perceptron in any substrate), an attribute inferred from an LLM's behaviour cannot belong to the LLM uniquely. The upshot is not that the attributes are absent but that talk of them is empty until pinned to explicit measurement criteria.

This is the conclusion of Age of Empires II argument, and it reframes debates such as whether LLMs have mental states: those questions are ill-posed until one says what, substrate-neutrally, would count as evidence.

When a chatbot says something wise, two different things are going on. One is the actual output, which would be identical no matter what the program ran on. The other is *our reading* of it as "wise" or "caring" — and that reading is not in the machine; it shifts depending on whether we picture a sleek AI or a field of goats doing the identical sums. So calling these qualities the AI's own is premature: first you have to say what would count as really having them, in a way that does not depend on what the thing is made of.

O

⊢ Old fact, new guise

Old fact, new guise — Mary's new knowledge is a new phenomenal concept, not a new fact

Argument · after Loar

The *old-fact / new-guise* argument is the physicalist's reply that grants the knowledge argument both its premises yet blocks its conclusion. Even if Mary knew every physical fact and genuinely learns something new on seeing red, that new knowledge can be a new *concept* of a fact she already knew rather than a new fact — so it gives the anti-physicalist no extra ingredient in the world. The move is the Mary-specific application of the *phenomenal-concept strategy*, due to Brian Loar (see LOAR — PHENOMENAL STATES (1990)).

It raises the bridge critical question against the KNOWLEDGE ARGUMENT: even granting that Mary knew all the physical facts and learns something genuinely new, the bridge from "new knowledge" to "new, non-physical fact" fails, because knowledge is individuated more finely than the facts known.

The inference

The form is deductive, turning on an independently motivated distinction.

1. One fact can be known under distinct concepts, and coming to grasp it under a second concept is genuine epistemic progress without any new fact — the lesson of *a posteriori* identities (see HESPERUS/PHOSPHORUS, WATER/H_2O).
2. Experiences give rise to *phenomenal concepts*: first-person, recognitional concepts of inner states, cognitively isolated from the theoretical concepts of those same states (see LOAR 1990).
3. Inside the room Mary grasped the fact of seeing-red under theoretical concepts only; seeing red equips her with the phenomenal concept of that same state.

Therefore HER NEW KNOWLEDGE IS A NEW GUISE OF AN OLD FACT, and THE NON-DEDUCIBILITY OF SOME TRUTHS ABOUT EXPERIENCE does not follow in any sense that troubles physicalism: what is not deducible is a *way of thinking*, not a *truth about the world*.

Relation to the general strategy

This is the Mary-specific case of the phenomenal-concept strategy whose general form — aimed at the explanatory gap rather than the knowledge argument — is THE GAP-DEFLATION ARGUMENT (Prong A): the same concept-isolation machinery that predicts a felt explanatory gap predicts that Mary's book-learning could never substitute for the phenomenal concept. It accordingly inherits that strategy's standing attack. THE OVERGENERATION OBJECTION presses that a concept isolated enough to do this work is too isolated to ground phenomenal self-knowledge (see PHENOMENAL-CONCEPT TRADE-OFF) — a cost this argument must pay too.

Weakest premise

Premise (2). If the phenomenal concept presents its referent via a substantive phenomenal property — a "what-it-is-likeness" the theoretical concepts miss — the anti-physicalist relocates the argument onto that property and the residue returns. If instead it presents its referent blankly, as a bare demonstrative, it is unclear why acquiring it should feel like the revelation the Mary intuition reports. Premise (1) is also pressured: in the Hesperus case the two guises connect to distinct *causal routes* to one object, and a complete physical description arguably lets one deduce both routes — whereas Mary's case is precisely one where deduction fails, so the analogy may assume what it must prove.

> The Mary story (see THE KNOWLEDGE ARGUMENT) concludes that physics left a fact out, because book-perfect Mary still learns something at her first sight of red. This reply says: learning something new doesn't always mean learning a new *fact*. Discovering that the Morning Star is the Evening Star felt like news, yet it added no new planet — just a second label connected to the same thing. Mary's first experience of red installs the inside-view label for a brain fact she already had the science-view label for. Real "aha!", same world. The standing objection (made elsewhere in this web against the same general strategy — see THE OVERGENERATION OBJECTION): if the
>
> inside-view label is so walled off from the science-view one, it's hard to explain how it tells you anything about yourself at all.

SEE ALSO

- ▷ INDEXICAL REPLY — the sibling deflationary reply; the collapse objection charges that it re-enacts this very argument with an "indexical" relabelling.

⊢ Other minds reply

The Other Minds Reply — we attribute understanding to others on behavioural grounds, so deny it to the room only by a double standard

Argument

The *Other Minds Reply* turns the problem of other minds against Searle. How do we know *other people* understand? Only from their behaviour — what they say and do. The Chinese Room, by hypothesis, produces behaviour indistinguishable from a Chinese speaker's. So if behavioural evidence is good enough to credit *humans* with understanding, consistency demands we credit the room too; to withhold it is an arbitrary double standard, a refusal to apply our only actual criterion for minds.

The reply attacks the DERIVATION by challenging our entitlement to the verdict that drives A3: if the sole basis for attributing understanding anywhere is behaviour, then the intuition "the room doesn't *really* understand" is unwarranted — we have no mind-detector that reaches past behaviour, so there is no standpoint from which to deny the room what we grant each other.

THE CRUX

Searle's ATTRIBUTION-VERSUS-DETECTION REJOINDER is that the reply changes the subject. The Chinese Room is about *what understanding is* (its metaphysics), not *how we find out* who has it (its epistemology). That we *detect* others' minds behaviourally does not show that minds *consist in* behaviour; the argument grants we might be fooled, and its point is precisely that the behaviour could be present while the understanding is absent. The reply's defender answers that without *some* link between the criterion and the thing, "understanding" floats free of anything we could ever have reason to attribute — reviving a verificationist pressure Searle must resist.

> How do you know anyone *else* really understands anything? You don't peek inside their mind — you just go by how they talk and act. But the room talks and acts exactly like a fluent speaker. So if behaviour is good enough to credit your friends with understanding, why not the room? Denying it looks like a rigged game. Searle's answer: the question isn't *how we tell* who understands — it's *what understanding is*. We read minds from behaviour because we have to, but that doesn't make understanding *the same thing as* behaving; the whole point is that the behaviour could be there with nobody home.

SEE ALSO

- ▷ CHINESE ROOM DERIVATION — the argument attacked, via our entitlement to its key verdict.
- ▷ ATTRIBUTION VS. DETECTION — Searle's reply: how we detect minds ≠ what minds are.
- ▷ STRONG AND WEAK AI — the behavioural criterion at issue connects to the Turing-test tradition.
- ▷ TURING TEST — the operationalisation of the behavioural criterion this reply leans on.

P

♦ Pain asymbolia

Pain asymbolia (pain felt but not minded)

Concept

Pain asymbolia is a rare neurological condition, typically following damage to the insula or its connections, in which a patient still feels pain but is no longer bothered by it. Asymbolics detect a noxious stimulus, localise it, and describe its sensory quality as sharp or burning, yet show none of the normal reaction: they do not withdraw, wince, or call it unpleasant, and may report the pain calmly or even with amusement. The clinical landmark is Berthier, Starkstein, and Leiguarda's study of six such patients, which characterised the syndrome as a *sensory-limbic disconnection* — the sensory pathways that register tissue damage are intact, but their link to the affective systems that make damage *matter* has been severed (see Berthier et al. — Asymbolia for pain (1988)).

Feeling pain versus being in pain

The philosophical interest of asymbolia is that it prises apart two things ordinary pain fuses. Nikola Grahek frames the split as *feeling pain* — the sensory-discriminative experience the asymbolic keeps — versus *being in pain* — the affective-motivational state, the felt awfulness and the urge to make it stop, which the asymbolic loses (see Grahek — Feeling Pain and Being in Pain (2007)). The case is thus a real-world dissociation of pain from its characteristic behaviour, and so belongs beside the behaviourist thought experiments of this web. But it cuts along a different seam. The Super-spartans still finds the pain awful and merely *suppresses* the behaviour by an act of will; the asymbolic does not find it awful at all, so the behaviour is *absent at the source.* And unlike the Perfect actor, who has the behaviour without any pain, the asymbolic has the sensation without the behaviour-driving affect.

The dispute it feeds

Asymbolia is pressed on the question of whether pain is *essentially* unpleasant — whether the "badness" of pain is built into the sensation or is a separable affective add-on. If asymbolia is a case of feeling a pain that is genuinely not bad, then unpleasantness is not intrinsic to pain, and theories tying pain's nature to its felt awfulness need revision. Critics reply that asymbolia may not be "pure" — that what is lost is partly the pain itself, not merely an attitude toward it — so that the case underdetermines the metaphysics it is summoned to settle.

> Some people, after a particular kind of brain injury, can still feel pain — they will tell you exactly where it is and that it is sharp — but it simply doesn't trouble them. They don't pull away, don't mind it, sometimes laugh about it. This is strange because we normally think the *hurt* and the *sensation* are one thing. Asymbolia suggests they are two: a detecting part that says "damage here" and a caring part that says "and it's awful, do something". It is not the Super-spartans, who feels the full awfulness but hides it by willpower; here the awfulness itself is missing while the bare sensation stays.

See also

- ▹ Super-spartans — feels the pain's awfulness fully but suppresses the behaviour by will, where the asymbolic loses the awfulness itself.
- ▹ Perfect actor — the converse dissociation: pain-behaviour with no pain at all.
- ▹ Jo Cameron — a real case at the far end: not pain stripped of its affect, but pain largely absent from the start.
- ▹ Phenomenal consciousness — the felt character at issue; asymbolia asks whether pain's unpleasantness is part of that character or a separable layer.
- ▹ Berthier et al. — Asymbolia for pain (1988) — the clinical study naming the syndrome a sensory-limbic disconnection.
- ▹ Grahek — Feeling Pain and Being in Pain (2007) — the philosophical framing: feeling pain versus being in pain.

♦ Panpsychism

Concept

Panpsychism is the view that consciousness is a fundamental and ubiquitous feature of the physical world — that mind, or something mind-like, is present at the most basic level of nature rather than appearing only in brains. In its most discussed modern form it does **not** claim that rocks ponder or that electrons have thoughts; it claims that the simplest physical constituents carry some rudimentary form of experience, and that the rich consciousness of a human or an animal is built up from these elementary experiential ingredients. The word, from the Greek *pan* (all) and *psychē* (soul or mind), is old — versions appear among the Presocratics, in Spinoza, in Leibniz, and in William James — but the view has had a vigorous revival in analytic philosophy of mind as a proposed escape from the HARD PROBLEM OF CONSCIOUSNESS.

This entry defines the doctrine and surveys its varieties. The arguments for and against it are developed as their own nodes and linked below; the committed *position* that adopts panpsychism as its answer to the hard problem — what it commits to and what it denies — is recorded at PANPSYCHISM (POSITION).

What the view reacts to

Panpsychism is best understood as a third path between the two dominant options. *Physicalism* holds that consciousness is nothing over and above physical processing, and faces the objection that no account of mere processing seems to explain why there is felt experience at all. *Dualism* adds consciousness as a further, non-physical ingredient, and faces the objection that it leaves mind and matter mysteriously disconnected. Panpsychism agrees with the dualist that experience is fundamental — not reducible to non-experiential facts — but agrees with the physicalist that it belongs to the natural, physical world rather than arriving from outside it. Experience, on this view, was in the ingredients from the start; the brain does not conjure it at some threshold of complexity, it organizes what was already there.

Varieties

Panpsychism is a family, not a single thesis, and several distinctions sort its members. The taxonomy below largely follows David Chalmers CHALMERS — PANPSYCHISM AND PANPROTOPSYCHISM (2017).

- **Micropsychism vs. cosmopsychism** — whether the fundamental conscious things are the *smallest* parts, with big minds composed bottom-up, or the *whole* cosmos, with individual minds carved out of it top-down. The cosmic version is COSMOPSYCHISM.
- **Constitutive vs. emergentist** — whether macro-experience is *constituted by* (grounded in, made up of) micro-experience, or instead *strongly emerges* from it as a novel further fact. Constitutive forms are the ones that promise to avoid brute emergence; emergentist forms keep an emergence step.
- **Panpsychism vs. panprotopsychism** — whether the fundamental properties are themselves *experiential*, or merely *proto-experiential*: not yet consciousness, but the special non-structural ingredients from which consciousness can be constituted. Panprotopsychism is the more cautious cousin, and shades into RUSSELLIAN MONISM.
- **Panexperientialism vs. panqualityism** — whether what is ubiquitous is full *subjective experience* (a point of view, a felt field) or merely unexperienced *qualities*, with subjecthood added later.

Constitutive Russellian micropsychism — fundamental physical entities have experiential intrinsic natures, and our minds are constituted from them — is the version most defended today, by Galen Strawson and Philip Goff (technically in GOFF — CONSCIOUSNESS AND FUNDAMENTAL REALITY (2017), for a general readership in GOFF — GALILEO'S ERROR: FOUNDATIONS FOR A NEW SCIENCE OF CONSCIOUSNESS (2019)) among others.

The case for it

Three lines of argument recur, each developed in its own node.

- The **genetic argument** (Strawson): a physicalist who is also a realist about experience must place experience at the fundamental level, on pain of accepting that it emerges brutely — unintelligibly — from the wholly non-experiential.
- The **continuity argument** (Nagel): there is no non-arbitrary point at which experience could "switch on," so the line-free option is that it reaches all the way down.
- The **structure-and-dynamics argument** (the Russellian route): physics describes only structure and dynamics, leaving the intrinsic categorical nature of matter blank — and consciousness is the natural candidate to fill it.

Objections

- The **combination problem** (James, Coleman) is the standing difficulty: if the smallest things each have their own experience and point of view, how do countless such micro-subjects compose the *single, unified* experience you are having now? Many regard this as panpsychism's make-or-break test, and note that it re-imposes, at the micro-to-macro step, the very explanatory burden the view was meant to discharge.

- The **emergence reply** presses the genetic argument's weak point: if strong emergence of experience is acceptable, there is no need to make mind fundamental, and the argument's central dilemma dissolves.
- Two looser worries are often added. The *incredulous stare*: crediting electrons with any inner life is an extravagant cost in plausibility. And *no empirical traction*: attributing experience to fundamental physics makes, by itself, no distinctive prediction. Defenders answer that every account of consciousness is strange somewhere, and that the hard problem is precisely the part of mind no empirical theory addresses head-on.

> Panpsychism says consciousness goes all the way down. Not that rocks daydream — the serious version says the tiniest bits of matter carry the faintest spark of experience, and a brain assembles those sparks into a full mind. The appeal: you no longer have to explain how feeling gets conjured out of utterly feelingless stuff at some magic moment, because it was in the building blocks all along. The big catch, called the *combination problem*: if every speck has its own tiny experience, how do trillions of them merge into the one seamless experience you're having as you read this? Critics also worry it just swaps one mystery for another, and that crediting electrons with any inner life is a stretch. Supporters answer that every theory of consciousness is strange somewhere, and at least this one keeps mind a natural part of the world.

SEE ALSO

- ▷ PANPSYCHISM (POSITION) — the committed position that adopts panpsychism as its answer to the hard problem, and how it differs from neighbouring views.
- ▷ GENETIC ARGUMENT — the headline argument *for* the view: real experience plus no brute emergence entails fundamental mind.
- ▷ COMBINATION PROBLEM — the central objection: micro-subjects cannot sum into one macro-subject.
- ▷ HARD PROBLEM — the problem panpsychism is offered to solve, by placing experience at the fundamental level rather than deriving it.
- ▷ HARD PROBLEM (THE QUESTION) — the open question panpsychism answers.
- ▷ RUSSELLIAN MONISM — the closely allied view that consciousness is the intrinsic categorical nature physics leaves unspecified; constitutive panpsychism is one form of it.
- ▷ COSMOPSYCHISM — the top-down cousin, on which the cosmos is the one fundamental subject and individual minds are carved out of it.
- ▷ FUNDAMENTAL UBIQUITOUS CONSCIOUSNESS — the core claim panpsychism asserts, distinguished from the dualist's "fundamental but non-physical."

★ Panpsychism (position)

Panpsychism

Position

Panpsychism, as a committed position in the philosophy of mind, holds that consciousness is a fundamental and ubiquitous feature of the physical world, and adopts that thesis as its answer to the problem of how experience fits into nature. It claims that the rich consciousness of a human or animal is not conjured from wholly mindless matter at some threshold of complexity, but is *constituted* by the experiential — or proto-experiential — properties already carried by that matter's simplest constituents. The view is set out as a doctrine, with its varieties and arguments, in the concept entry PANPSYCHISM; this entry records what the position commits to and how it stands against its neighbours.

The position's core commitment is that CONSCIOUSNESS IS FUNDAMENTAL AND PRESENT IN THE SIMPLEST PHYSICAL ENTITIES. Because macro-minds are built up from the experiential natures of their micro-parts, panpsychism needs neither the dualist's non-physical addition nor the physicalist's "strong emergence of feeling from the feelingless." It answers the HARD PROBLEM OF CONSCIOUSNESS by denying that physics must *produce* experience at all: experience is there at the base, and the question becomes how it is organized rather than how it is generated. Its modern champions include Galen Strawson (see STRAWSON — REALISTIC MONISM: WHY PHYSICALISM ENTAILS PANPSYCHISM (2006)) and Philip Goff (GOFF — CONSCIOUSNESS AND FUNDAMENTAL REALITY (2017), for a general readership GOFF — GALILEO'S ERROR: FOUNdations for a New Science of Consciousness (2019)).

How it differs from neighbouring views

Against Dualism, consciousness is fundamental but *intrinsic to the physical*, not a non-physical extra disconnected from matter. Against physicalism and Illusionism, it is firmly realist about experience: there is genuine felt consciousness to distribute, not an illusion to explain away. Its relation to Russellian monism is closest of all — *constitutive* panpsychism simply *is* a Russellian view on which the intrinsic natures physics leaves blank turn out to be experiential, while the more cautious *panprotopsychism* makes them merely proto-experiential.

The position's standing weak point is the combination problem: if each micro-part has its own experience and point of view, how do countless micro-subjects compose the single, unified subject having your experience now? Many regard this as panpsychism's make-or-break test.

> Panpsychism says consciousness goes all the way down: not that rocks have thoughts, but that the basic bits of matter carry the tiniest glimmer of experience, and brains assemble those glimmers into full-blown minds. The appeal: you don't have to conjure feeling out of utterly feelingless stuff at some magic threshold — it was in the ingredients from the start. The catch, its *combination problem*: how do countless tiny experiences add up to the single unified experience you're having right now?

See also

- ▹ Panpsychism — the doctrine itself, with its full taxonomy of varieties and the arguments for and against; this position is its committed form.
- ▹ Fundamental ubiquitous consciousness — the core thesis the position commits to.
- ▹ Phenomenal residue — the position's further commitment that felt states are not exhausted by their functional role.
- ▹ Hard problem (the question) — the problem the position is offered to solve.
- ▹ Genetic argument — the headline argument for the view: real experience plus no brute emergence entails fundamental mind.
- ▹ Combination problem — the central objection the position must answer.
- ▹ Russellian monism — the allied view that consciousness is the intrinsic nature physics leaves unspecified; constitutive panpsychism is one form of it.
- ▹ Cosmopsychism — the top-down cousin, on which the cosmos is the one fundamental subject and individual minds are carved out of it.
- ▹ Dualism — the rival on which consciousness is fundamental but non-physical.
- ▹ Illusionism — the rival that denies there is any fundamental experience to distribute.

♦ Paradox of phenomenal judgment

The paradox of phenomenal judgment

Concept · after Chalmers

A *phenomenal judgment* is a judgment about conscious experience — the belief, thought, or report that one is conscious, that there is "something it is like" to see red, or that consciousness resists physical explanation. The **paradox of phenomenal judgment** (David Chalmers, *The Conscious Mind*, 1996, Chalmers — The Conscious Mind (1996)) is the tension that arises when such judgments are set beside the view that consciousness is non-physical. Our phenomenal judgments are themselves physical events — utterances, neural states, dispositions to act. By the causal closure of the physical, they have complete physical causes. But if consciousness is something over and above the physical, then consciousness itself is *not* among those causes. We would make exactly the same judgments about our experience whether or not the experience accompanied the physical processes producing them.

The paradox is sharpest for property dualism and especially Epiphenomenalism, on which experience is real but causally inert: there, my saying "I am conscious" is caused entirely by my brain, and the consciousness it reports does no work in producing the report. The unsettling consequences are that my belief that I am conscious is not *caused* by my consciousness, and is not *justified* by it either — yet it seems both reliable and certain. A zombie twin, physically identical but experienceless, would sincerely assert "I am conscious" for precisely the same reasons, and would be wrong. What, then, makes *my* matching judgment knowledge?

Why it generalizes

Chalmers presses the paradox not only as an objection to dualism but as a puzzle for *every* view, including his own. Any complete physical-functional account of why we produce consciousness-talk will, by closure, make no essential reference to consciousness itself — so the explanation of our phenomenal *judgments* threatens to proceed without the phenomenal *facts* those judgments are about. Pushed to its general form, this becomes the **meta-problem of consciousness** (see Chalmers — The Meta-Problem of Consciousness (2018)): the problem of explaining, in topic-neutral terms, why we judge and report that there is a hard problem at all. The worry is that a good answer would *debunk* the very beliefs it explains — the engine driving the report-scepticism disputes elsewhere in this web (see Report-scepticism over-reach).

> You are certain you're conscious, and you say so. But the words coming out of your mouth are produced by your brain — physical machinery that, on a "consciousness is something extra" view, would churn out the very same words even if the inner feeling weren't there. So the feeling isn't what makes you *say* you're conscious, and arguably isn't what makes you *know* it either. A perfect feelingless duplicate of you would insist, just as earnestly, that it has rich inner experience — and be flat wrong. The paradox: if your report and the zombie's have the same cause, what entitles you to trust yours? It is the hardest version of the question "how do we even come to talk about consciousness?", and it bites hardest on views that make consciousness non-physical or idle.

See also

- Epiphenomenalism and Causally inert qualia — the view the paradox embarrasses most directly: real but inert experience that cannot cause the reports about it.
- Philosophical zombie — the duplicate who makes all the same phenomenal judgments with no experience behind them; the paradox turns the zombie intuition back on our own self-knowledge.
- Hard problem — the explanandum these judgments are about; the paradox is the puzzle of how the judgments relate to it.
- Chalmers — The Meta-Problem of Consciousness (2018) — the meta-problem, the paradox's modern, generalized descendant.
- Report-scepticism over-reach — the live dispute in this web that runs on the same debunking tension, applied to machine and human testimony alike.

⊙ Perceptron in any substrate

A trainable neural network can be realized in any sufficiently powerful substrate

Claim · after Dewynter

The computational core of a large language model — a trainable neural network, in the limit a single perceptron — can be implemented and run in *any* sufficiently powerful (functionally complete) substrate, not merely in silicon. The point is established by construction rather than assumed: de Wynter builds and trains a perceptron inside *Age of Empires II* (see Age of Empires II circuits), and notes the same could be done with LEGO, or with the people of the Greater Boston Area passing signals. The computation is one thing; its medium is another.

This is the constructive premise that drives Age of Empires II argument. It is narrower than Substrate-independence thesis, which says that *minds or mental states* are multiply realizable: here the claim is only about the *computation* an LLM performs, and it is used not to argue that the computation has a mind but to argue that any attribute read off its behaviour cannot be unique to one substrate.

> The little learning machine inside an AI is just a recipe of additions and comparisons. A recipe does not care what it is made of — you can run it on a chip, on paper, with LEGO, or (as someone actually did) with goats in a videogame. So whatever an AI's number-crunching is, it is not special to computer chips; the very same crunching could be happening in a herd of goats.

⧫ Perfect actor

The perfect actor (pain shown but never felt)

Concept · after Putnam

A *perfect actor* is a being who produces the entire behavioural profile of pain — wincing, groaning, nursing the limb, sincerely avowing "it hurts" — while feeling no pain whatever. The figure is the converse of the Super-spartans in Hilary Putnam's "Brains and Behavior" (1963, Putnam — Brains and Behavior (1963)): where the super-spartan has the pain but suppresses the behaviour, the perfect actor has the behaviour but lacks the pain. The point is not ordinary pretence, which can be seen through, but a *perfect* performance — outwardly and dispositionally indistinguishable from genuine pain in every circumstance.

The point of the case

The perfect actor is a counterexample to the *sufficiency* half of **logical (analytical) behaviourism**. If being in pain just *were* having the disposition to pain-behaviour, then anything with that full disposition would, by definition, be in pain — and a perfect actor could not exist. But such a being seems perfectly conceivable, and intuitively it is not in pain at all. So having the behavioural disposition is not *sufficient* for the mental state. Behaviour underdetermines mind from above as the SUPER-SPARTANS shows it does from below.

RELATION TO THE ZOMBIE

A perfect actor is the behavioural ancestor of the PHILOSOPHICAL ZOMBIE, and the two should not be run together. The zombie is a *physical* duplicate of a person that nonetheless lacks all experience; the perfect actor is only a *behavioural* duplicate, and may differ physically and functionally in any way at all (it might be a robot, or wired quite differently inside). The zombie is wielded against physicalism and functionalism; the perfect actor is the older and more modest device aimed only at behaviourism. Putnam's own response to the pair was not dualism but the turn to causal-functional role that became FUNCTIONALISM — a theory the perfect actor does *not* embarrass, since an actor differently organised inside need not occupy pain's functional role.

> Imagine an actor — or a cleverly built robot — so good that it does *everything* a person in pain does: it flinches, screams, clutches the wound, swears it is in agony, and would keep this up in every situation, fooling everyone forever. Yet inside there is no pain at all, nothing felt. Common sense says it still isn't really in pain — it is only behaving that way. That sinks the theory that *being in pain simply means behaving as though you are*, because here is all the behaviour with none of the feeling. It is the mirror image of the SUPER-SPARTANS, who feels pain but shows nothing, and a behaviour-only relative of the PHILOSOPHICAL ZOMBIE, which copies a person right down to the physics yet still has no inner life.

SEE ALSO

- ▹ SUPER-SPARTANS — the converse case: pain felt but never shown, attacking the *necessity* of behaviour as this attacks its *sufficiency*.
- ▹ PHILOSOPHICAL ZOMBIE — the stronger descendant: a full physical duplicate without experience, versus the perfect actor's merely behavioural duplication.
- ▹ FUNCTIONALISM — the successor theory Putnam moved to; the perfect actor, differently organised within, need not share pain's causal role, so the case does not refute it.
- ▹ TYPE-A MATERIALISM — the family of topic-neutral analyses of the mental that houses logical behaviourism, the target here.
- ▹ PUTNAM — BRAINS AND BEHAVIOR (1963) — Putnam's "Brains and Behavior" (1963), where the case is introduced.

◆ Phenomenal consciousness

Concept

Phenomenal consciousness is the felt, subjective side of a mental state: what it is *like*, from the inside, to undergo it. A state is phenomenally conscious when there is something it is like to be in it, such as the redness of red as you see it, the sting of a burn, or the taste of coffee. The classic gloss is Thomas Nagel's question "what is it like to be a bat?"; if there is something it is like to be the bat, the bat has phenomenal consciousness. The specific felt qualities are often called *qualia*, and phenomenal consciousness is the general property of having any such felt character at all.

WHAT IT IS, BY CONTRAST WITH FUNCTION

The notion is isolated by setting it against everything a state *does*. Two states might play exactly the same causal role, both caused by tissue damage and both driving you to wince and say "ouch", and one can still ask whether they *feel* the same. That residue, the felt character left over once every functional role is accounted for, is phenomenal consciousness. This is what marks it off from ACCESS CONSCIOUSNESS, the purely functional sense in which a state's content is merely *available* for reasoning and report. Ned Block, who coined the labels *P-consciousness* and *A-consciousness*, argued that the two can dissociate in both directions, a separation set out at ACCESS VS. PHENOMENAL CONSCIOUSNESS.

WHY IT MATTERS

Phenomenal consciousness is the part of the mind that seems hardest to fit into a physical picture, and it is the explanandum of the HARD PROBLEM: the question of why physical processing should be accompanied by any felt experience at all. The classic anti-materialist thought experiments all turn on it. A zombie is a physical duplicate with no phenomenal consciousness, and Mary the colour scientist seems to learn a new phenomenal fact on first seeing red. What is *disputed* is not usually the definition but the phenomenon's standing: illusionists hold that the impression of an extra felt residue is itself a cognitive illusion, while realists take phenomenal consciousness to be a genuine, further feature of the world.

> Phenomenal consciousness is just *feeling*: the fact that experiences have an inside to them. A brain scan can show every step of what your nervous system does when you stub your toe, yet it doesn't show the *hurt* you actually feel, and that hurt is the phenomenal side. It is the thing that makes consciousness puzzling. We can explain what the body does, but explaining why any of it is *felt* is the hard part. A few philosophers even argue the "extra feeling" is a trick the mind plays on itself.

See also

- ▷ Access vs. phenomenal consciousness — Block's distinction, and how phenomenal consciousness and its functional complement can come apart in both directions.
- ▷ Access consciousness — the functional, availability sense it is contrasted with.
- ▷ Hard problem — the explanatory problem that phenomenal consciousness poses.
- ▷ First-person access — the *epistemic* privileged access a subject has to its own phenomenal states, a different notion from the felt character itself.

☉ Phenomenal residue

Phenomenally conscious states are not exhausted by functional/causal role

Claim · after Chalmers

For at least one kind of mental state — the *phenomenally conscious* ones, the states there is "something it is like" to be in — a complete specification of what the state *does* leaves something out. A pain has a functional/causal role (caused by tissue damage, causing you to wince and seek relief), but it also *hurts*; and this claim holds that the felt, qualitative character is not fixed by the role. Role-individuation, the strategy of Functionalism that pins a mental state down by its causes and effects, is not the whole story for the phenomenal.

The claim is deliberately narrow. It is about phenomenal states *only*, and it concedes Functionalism for intentional states such as belief and desire, where a description of the causal role may well capture the whole state. It does not assert that silicon systems lack consciousness, and it does not assert any biological requirement; it says only that functional role under-determines phenomenal character. Its bite against substrate-independence is local: the inference from "individuated by role" to "had by any realiser of that role" is sound only where role-individuation is exhaustive, and that is exactly what this claim denies in the phenomenal case. So it does not reject substrate independence wholesale — it denies the *uniform* reading that extends it across every mental state. It is also distinct from Proteocentrism, which demands a protein/carbon substrate for consciousness: this claim makes no substrate demand at all. The view it most directly answers is the question of whether mental states are substrate-independent (see Substrate independence).

Those who would deny it are strong or role functionalists, who read functionalism uniformly across phenomenal as well as intentional states, and anyone holding that phenomenal character metaphysically supervenes on functional organisation.

> There's a difference between what your mind *does* and what your experience *feels like*. A pain makes you wince and reach for relief — that's its job, the role it plays. But a pain also *hurts*; there's something it is like to feel it. This claim says: for these felt, "what-it's-like" experiences, listing everything the state does still leaves something out — the feel itself. Two things it carefully does *not* say: it doesn't say computers can't be conscious, and it doesn't say you need carbon or biology. It even grants that for plain thinking states — beliefs, plans — a description of the role might capture the whole story. It only fences off the *felt* part of the mind and says: that part isn't fully pinned down by function alone.

See also

- ▷ Substrate independence — the issue this claim answers: whether mental states can be carried by any substrate that plays the right role.
- ▷ substrate independence — the thesis this claim restricts; it blocks the uniform "all mental states" reading by carving out the phenomenal.
- ▷ Functionalism — the role-individuation strategy this claim says runs out for phenomenal states.
- ▷ Proteocentrism — a stronger neighbour that demands a biological substrate; this claim makes no such demand.

♦ Philosophical zombie

Concept · after Chalmers

A *philosophical zombie* is an imagined being that is microphysically and functionally identical to a conscious person — every atom, every brain state, every piece of behaviour the same — but with *no phenomenal consciousness*: nothing it is like to be it, all dark inside. It is not the shambling creature of horror films; it is a same-physics twin that does everything you do, says "ouch" when pricked and talks about its feelings, yet has no inner experience whatsoever (see CHALMERS — THE CONSCIOUS MIND (1996)).

The point of the device is not to claim such beings exist, but to test whether the physical facts settle the facts about experience. If a complete physical duplicate of a conscious person could exist with experience subtracted, then experience is not just more physics — it is something *over and above* the physical. That is exactly the conclusion the zombie is built to drive: it is the engine of the *conceivability argument* against physicalism, the view that everything, consciousness included, is fixed by physical facts.

The crucial step

The argument turns on a move from imagining to possibility. From the fact that a zombie seems *conceivable* — coherently and stably imaginable, with no contradiction surfacing on careful reflection — it infers that a zombie is genuinely *possible*, and hence that CONSCIOUSNESS IS NON-PHYSICAL. Nearly the whole dispute concentrates on this bridge from conceiving to possibility (see CONCEIVABILITY ENTAILS POSSIBILITY): critics grant that we can picture a zombie while denying that the picture tracks any real possibility, much as one can imagine water that is not H_2O without water's being possibly anything but H_2O.

Relation to neighbours

The zombie is a limiting case of *absent qualia* — experience entirely missing rather than merely altered — and a cousin of the *inverted spectrum*, in which the felt qualities are swapped rather than absent. It sharpens the intuition behind the HARD PROBLEM of consciousness, the puzzle of why physical processing is accompanied by any felt experience at all. The related EXPLANATORY-GAP ARGUMENT already leans on zombie conceivability in its premises, and the PHENOMENAL-CONCEPT STRATEGY is the leading attempt to defuse the zombie by granting that it is conceivable while denying that it is possible.

> A "philosophical zombie" is an imagined being that is an atom-for-atom copy of you — same brain, same behaviour, says "ouch" when pricked, chats about its feelings — except there is no one home: no inner experience at all, total darkness inside. The thought is not that such things exist; it is that *if you can coherently imagine one*, then experience must be something extra that the physical facts alone don't pin down. Nearly the whole fight is over that "if you can imagine it, it's genuinely possible" step.

See also

- ▹ HARD PROBLEM — the puzzle of why physical processing is accompanied by felt experience at all; the zombie sharpens its central intuition.
- ▹ CONCEIVABILITY ENTAILS POSSIBILITY — the modal bridge from conceiving a zombie to its being possible, where the whole argument is contested.
- ▹ FUNDAMENTAL NON-PHYSICAL CONSCIOUSNESS — the conclusion the zombie is built to drive: experience is over and above the physical.
- ▹ EXPLANATORY-GAP ARGUMENT — an evidential cousin whose premises already lean on zombie conceivability.
- ▹ PHENOMENAL-CONCEPT DEFLATION — the phenomenal-concept strategy, which grants the zombie's conceivability but denies its possibility.

★ Physicalism

Physicalism (materialism)

Position

Physicalism (its older name **materialism**) is the thesis that everything that concretely exists is physical, or is wholly fixed by what is physical. Applied to the mind, it holds that mental events are physical events (see MENTAL EVENTS ARE PHYSICAL): there is no thought, sensation, or experience over and above the physical workings of the brain and body. A common precise formulation is a *supervenience* thesis — any world physically identical to ours is identical to ours in every respect, mental included — so that once the physical facts are fixed, all the facts are fixed.

The modern positive case for physicalism is the **causal argument** (see CAUSAL ARGUMENT FOR PHYSICALISM), which turns on two principles this view leans on heavily: the COMPLETENESS OF PHYSICS (every physical effect has a sufficient physical cause) and the Causal closure of the physical of the physical (no non-physical cause produces a physical effect). If the mind genuinely moves the body, and the physical world is causally closed, the mind must itself be physical — the alternative being to demote it to a causally idle by-product (see Epiphenomenalism). Physicalism thus presents itself less as a metaphysical preference than as the only account of mental causation consistent with the closed causal order science describes (Papineau, Papineau — Thinking about Consciousness (2002); Kim, Kim — Physicalism, or Something Near Enough (2005)).

What distinguishes it from neighbouring views. Physicalism is the *genus* whose species populate much of this web: Type-identity theory (mental types *are* physical types), Eliminative materialism (the mental as folk theory discharges to neuroscience), analytic functionalism, and Illusionism (phenomenal properties are introspective illusion) are all ways of being a physicalist. They divide over the Hard problem: Chalmers' Type-A / Type-B / Type-C taxonomy sorts them by whether they deny the epistemic gap between physical and phenomenal truths, accept it but deny an ontological gap, or hope to close it in the future. Against physicalism stand Dualism (which denies completeness or closure to keep the mind non-physical and efficacious), Epiphenomenalism (which keeps the mind non-physical at the cost of its causal powers), and the conceivability and knowledge arguments (see Zombie argument, Knowledge argument (Mary's room)) that press an explanatory gap physicalism must absorb.

> Physicalism — the same view people used to call materialism — says that, at bottom, everything is physical: there's only the stuff physics describes (and whatever is built out of it), with no separate soul-substance or mental extra. About the mind in particular, it says your thoughts and feelings just *are* things going on in your brain. The strongest reason offered is about causes: the physical world seems to be a closed system where every physical event has a complete physical cause and nothing non-physical reaches in to push matter around. Since your mind clearly *does* affect your body — you decide to stand, and you stand — the tidiest conclusion is that your mind is itself part of that physical machinery. The big challenge it has to answer is the felt, what-it's-like side of experience (the "hard problem"): critics say you can describe every physical fact about a brain and still leave out what the experience *feels like* from the inside.

See also

- ▹ Mental events are physical — the view's core commitment about the mind.
- ▹ Causal argument for physicalism — its principal positive argument.
- ▹ Completeness of physics and Causal closure of the physical — the two principles that argument rests on.
- ▹ Type-A / Type-B / Type-C materialism — Chalmers' taxonomy of physicalist responses to the hard problem.
- ▹ Dualism, Epiphenomenalism — the chief rivals, each rejecting a different premise.
- ▹ Hard problem (the question) — the question physicalism must answer to be complete.

♦ Primary intension

Concept · after Chalmers

The *primary intension* of a term is, roughly, the meaning it carries in virtue of how speakers fix its reference — what the term would pick out on the supposition that a given possible world turns out to be the actual one. It is one of the two "intensions" (functions from possible worlds to extensions) that David Chalmers's two-dimensional semantics assigns to an expression; the other is its Secondary intension. Where the secondary intension asks "given how the actual world in fact is, what does this term pick out in *other* worlds?", the primary intension asks the prior question: "for each way the world might turn out to *be* actually, what would the term pick out *there*?"

The standard illustration is the word *water*. Competent speakers fix the reference of "water" by a rough description — the clear, drinkable liquid that falls as rain and fills the lakes and rivers, the *watery stuff*. The primary intension is the function encoding that condition: hand it a world considered as actual, and it returns whatever plays the watery-stuff role there. In our world that role is filled by H_2O, so the primary intension delivers H_2O. On Hilary Putnam's Twin Earth — a world just like ours except the watery stuff is a different compound, XYZ — *considered as a candidate for being actual*, the primary intension of "water" delivers XYZ. The primary intension is in this sense sensitive to empirical discovery only hypothetically: it tells you what "water" would refer to under each hypothesis about which world is actual, and so its profile can be known largely a priori, by reflection on how the concept works rather than on which world we are in.

Two dimensions of meaning

The label "two-dimensional" comes from evaluating a sentence twice over against a world. A world can enter the calculation in two roles: as the world taken to be *actual* (which settles what our terms refer to), and as a *counterfactual* circumstance to be described using reference so settled. The primary intension occupies the first dimension — worlds considered as actual; the secondary intension occupies the second — worlds considered as counterfactual. For many ordinary terms the two coincide. They come apart precisely for the Kripkean *a posteriori necessities* (A-posteriori necessities, from Kripke's *Naming and Necessity* Kripke — Naming and Necessity (1980)): "water is H_2O" is necessary along the secondary dimension (water is H_2O in every counterfactual world) yet contingent along the primary one (the watery stuff *could have turned out* to be XYZ). The necessity that can be known only by experiment lives in the gap between the two intensions.

Chalmers's is the *epistemic* version of two-dimensional semantics, developed in *The Conscious Mind* (Chalmers — The Conscious Mind (1996), where the "primary/secondary" terminology is introduced) and refined in Chalmers — On Sense and Intension (2002); Frank Jackson reaches an equivalent apparatus from the side of conceptual analysis (see Jackson — From Metaphysics to Ethics (1998)). The primary intension also travels under other names — *epistemic intension*, *A-intension*, *1-intension* — and has cousins in the broader two-dimensional tradition (Kaplan's "character," Stalnaker's "diagonal"), though those were designed for different purposes and should not be assumed identical.

Why the concept matters

The primary intension is the load-bearing notion in the modern *conceivability* arguments about the mind. Chalmers's Modal rationalism holds that what ideal reflection finds conceivable is a guide to what is *primarily* possible — possible as a way the world might actually be — and the two-dimensional diagnosis defuses the Kripkean counterexamples by reinterpreting them as A-posteriori necessities as misdescribed possibilities: when you "conceive water that is not H_2O" you conceive a genuinely possible world that verifies the *primary* intension of "water," and merely misdescribe it. This rescues the bridge from conceivability to possibility for the Philosophical zombie — but only if the phenomenal case has no such primary/secondary gap to exploit, which is the separate and contested thesis that for consciousness the two intensions coincide (see Coincident phenomenal intensions).

> Think of a word as having two jobs. One job is fixing *what we're even talking about* — for "water," something like "the clear drinkable stuff in the rivers." The primary intension is that job written out as a rule: point it at any way the world might turn out to be, and it hands back whatever fits the description there. If the world turns out to be ours, "water" lands on H_2O; if it had turned out to be Twin Earth, the very same rule would have landed "water" on XYZ. So the primary intension is what you grasp just by *understanding the word*, before you do any chemistry — which is why imagining things tends to track it. The other job, pinning the word to the actual stuff (H_2O) and keeping it there even when we talk about other worlds, belongs to the Secondary intension.

See also

- ▷ Secondary intension — the other half of the framework: reference fixed by the actual world, then held rigid across counterfactual worlds; the necessary a posteriori is the gap between the two.
- ▷ A-posteriori necessities — the Kripkean cases (water = H_2O) that pull the two intensions apart and that the primary intension is designed to handle.
- ▷ A-posteriori necessities as misdescribed possibilities — the two-dimensional diagnosis: apparent counterexamples to "conceivable $\Longrightarrow$ possible" are possibilities under the primary intension, misdescribed.
- ▷ Modal rationalism — the argument that ideal primary conceivability is a reliable guide to primary possibility, which the primary intension makes precise.
- ▷ Coincident phenomenal intensions — the further claim that for phenomenal concepts the primary and secondary intensions collapse together, removing the wedge the water case relies on.
- ▷ Philosophical zombie — the case whose

conceivability the framework is meant to convert into genuine possibility.

⊙ Programs are syntactic

Computer programs are purely formal (syntactic)

Claim · after Searle

A computer program is defined purely *formally*: it specifies operations on symbols solely in virtue of their shapes (their syntax), with no reference to what, if anything, those symbols mean. This is the first axiom (A1) of Searle's formal derivation of the CHINESE ROOM conclusion (see SEARLE — IS THE BRAIN'S MIND A COMPUTER PROGRAM? (1990)). It is close to a truth of definition: computation, in the Turing sense, just *is* the rule-governed manipulation of uninterpreted tokens, which is why the same program can be implemented in silicon, in cogs, or by a person shuffling cards.

Scope. The claim is about what a program *is*, not about what a programmed machine can *do*. It does not say programs are useless or that they cannot model cognition; it says their identity conditions are syntactic — a point granted on nearly all sides, including by computationalists.

Who would deny it. Almost no one denies A1 directly; it is the least controversial axiom. The pressure in the derivation falls instead on A3, the claim that syntax is not *sufficient* for semantics. One could resist A1 only by enriching "program" to include semantic or causal-grounding conditions — but that concedes Searle's point that *pure* computation is not enough, and shifts to the ROBOT REPLY's terrain.

Distinct from SYNTAX INSUFFICIENT FOR SEMANTICS: A1 says programs *are* syntactic; A3 says syntax *does not suffice* for meaning. A1 is near-definitional; A3 is the contested premise.

> A computer program is just a set of rules for pushing symbols around based on their shapes — "if you see this squiggle, write that one." It never mentions what the squiggles *mean*; that's why the very same program runs on a laptop, a phone, or a person with a rulebook. This is the tame, almost obvious starting point of Searle's argument. The fight is not here but at the next step: whether shape-shuffling alone could ever add up to *understanding.*

⊙ Programs not sufficient for minds

Implementing a program is neither constitutive of nor sufficient for having a mind

Claim · after Searle

Running the right computer program is neither *constitutive of* a mind nor *sufficient for* one: no system has mental states *in virtue of* implementing a program. This is conclusion C1 of Searle's formal CHINESE ROOM derivation (see SEARLE — IS THE BRAIN'S MIND A COMPUTER PROGRAM? (1990)), following from A1 (programs are syntactic), A2 (minds have semantics), and A3 (syntax is not sufficient for semantics). It is the direct denial of **strong AI** (see STRONG AND WEAK AI) — the thesis that the right program *is* a mind — while leaving *weak* AI (the computer as a model of mind) untouched.

Scope. C1 concerns *programs as such* — pure formal implementation. It does *not* say machines can never think, nor that artificial minds are impossible: a non-program artificial system with the right causal powers might (the MANY MANSIONS REPLY presses exactly this gap). What C1 denies is that *computation alone* delivers a mind.

Who would deny it. Computational FUNCTIONALISTS and defenders of SUBSTRATE INDEPENDENCE deny C1: for them implementing the right program-role just *is* having the corresponding mental states. Their objections — the SYSTEMS, ROBOT, and BRAIN SIMULATOR replies — all aim, in the end, to block the move to C1 by denying A3.

Distinct from SYNTAX INSUFFICIENT FOR SEMANTICS (A3): A3 is the *general* principle about syntax and semantics; C1 is its *application to minds and programs* — the conclusion A3 is wielded to reach.

> The payoff of Searle's argument: no amount of running-the-right-software is, by itself, enough to *be* a mind or to *have* one. A program is just shape-shuffling (step one), real minds actually mean things (step two), and shuffling shapes can't manufacture meaning (step three) — so software alone can't add up to a mind. Note the careful limit: this says *programs* aren't enough, not that machines can never think. A suitably built machine with the right causal powers isn't ruled out — only the idea that the *code* is the mind.

Q

⧫ Qualia

Concept

Qualia (singular *quale*, from the Latin for "of what kind") are the subjective, felt qualities of conscious experience: the redness of a ripe tomato as it looks to you, the bitterness of coffee, the particular ache of a stubbed toe. The term names not the tomato, the coffee, or the toe, but what undergoing those experiences is *like* from the inside. The word was given its modern use by Clarence Irving Lewis (see LEWIS 1929), who applied it to the recognisable qualitative characters of experience that recur from one occasion to the next.

The contrast with function

The standard way to isolate the idea is to set it against everything a mental state *does*. Two states might play exactly the same causal role, both produced by tissue damage and both making you wince and say "ouch", and one can still ask whether they *feel* the same. That residue, the felt character left over once every causal role is accounted for, is what "qualia" picks out. It is the same residue Thomas Nagel marks with the phrase *what it is like*: an organism has conscious experience, he argues, if there is something it is like to be that organism, and no objective, third-person description seems to capture it (see NAGEL 1974). Two stock cases sharpen the notion. The *inverted spectrum* asks whether your experience of red could be like my experience of green with all of our behaviour and physiology unchanged; if that is so much as coherent, the felt quality floats free of the functional facts. And Frank Jackson's colour scientist Mary, who knows every physical fact about colour vision yet seems to learn something new when she first sees red, dramatises the same gap (the KNOWLEDGE ARGUMENT, from JACKSON 1982).

Senses of the term

The word is contested, and the dispute is often about the definition rather than the facts. In a *thin* sense, qualia are simply the ways experiences feel, with no further commitment, and almost no one denies that there are such felt ways. In a *thick* sense, qualia are additionally taken to be *intrinsic*, *ineffable*, *private*, and *directly apprehended* by introspection. Daniel Dennett's "Quining Qualia" argues that nothing satisfies that demanding job description, so qualia in the thick sense should be abandoned (see DENNETT 1988). A claim that looks obvious on the thin reading can beg the question on the thick one, so an entry that leans on qualia should say which sense it means. The notion is also the close cousin of PHENOMENAL CONSCIOUSNESS: qualia are the specific felt qualities, and phenomenal consciousness is the general property of having any such felt character at all.

Why qualia matter

Qualia mark the spot where the mind seems hardest to fit into a scientific picture of the world. They drive the HARD PROBLEM of consciousness, the question of why physical processing should be accompanied by felt experience at all, and they power the classic anti-materialist thought experiments: Mary's room (the KNOWLEDGE ARGUMENT) and the PHILOSOPHICAL ZOMBIE, a creature physically identical to you with no inner feel. Each turns on the suspicion that the felt facts outrun the functional and physical ones, which is exactly what a FUNCTIONALIST account of mind must deny. How much weight qualia can bear, and whether the thick notion survives scrutiny, is in large part what the debate over materialism is about.

> "Qualia" is the philosopher's word for how things *feel* to you: the way red looks, the way coffee tastes, the way a headache hurts. A brain scanner can show what your neurons are doing when you see red, but it does not show the redness *you* experience; qualia are that inner, felt side of the story. Philosophers argue about whether these feels are something extra that science struggles to explain, or whether, as skeptics like Daniel Dennett say, the whole idea falls apart once you look at it closely.

See also

- ▹ PHENOMENAL CONSCIOUSNESS — the general property of having any felt character at all; qualia are its specific instances.
- ▹ HARD PROBLEM — why physical processing

is accompanied by felt experience at all; the problem qualia pose in its sharpest form.

- ▷ Knowledge argument — Mary the colour scientist, and whether complete physical knowledge still leaves the qualia out.
- ▷ Philosophical zombie — a physical duplicate with no qualia; the conceivability test for whether the physical facts fix the felt facts.
- ▷ Access vs. phenomenal consciousness — Block's distinction; qualia live on the phenomenal side of it.
- ▷ Functionalism — the theory of mind that qualia are most often said to outrun.

R

⊙ Reality of phenomenal consciousness

Phenomenal consciousness is genuinely real, not an illusion

Claim

There really is something it is like to undergo experience: felt redness, the ache of a pain, the taste of coffee are genuine features of reality, not a cognitive illusion and not a mere disposition to report. This is *realism* about phenomenal consciousness — the explanandum is taken at face value rather than explained away. It is a premise shared across most non-illusionist views of mind, including dualism, panpsychism, Russellian monism, and type-B physicalism. In the GENETIC ARGUMENT for panpsychism it pairs with NO RADICAL EMERGENCE: because experience is real *and* cannot brutely emerge, it must be fundamental. The claim is denied only by illusionists, who hold that phenomenal consciousness as standardly conceived does not exist (see PHENOMENALITY AS ILLUSION).

It should not be confused with the stronger *residue* claim that felt states are not exhausted by their functional role (see PHENOMENAL RESIDUE): one can affirm that experience is real while still debating whether it reduces to function. This claim asserts only its reality, not its irreducibility.

See also

- ▷ PHENOMENALITY AS ILLUSION — its direct denial, held by illusionists.
- ▷ PHENOMENAL RESIDUE — the stronger, separate claim that experience outruns functional role.
- ▷ GENETIC ARGUMENT — the argument in which this realism premise does its work.

⧫ Retorsion

Retorsion (performative self-refutation)

Concept

Retorsion is the dialectical move of refuting a thesis by showing that *asserting* it presupposes its falsity — the act of putting the claim forward undercuts the claim's content. It does not argue that the thesis is false from independent premises; it turns the thesis on itself (Latin *retorquere*, to twist back). Where an ordinary refutation contradicts what is said, a retorsion exploits a clash between what is said and the conditions of saying it.

The classic instances are stable across the tradition. Against one who denies the law of non-contradiction: to deny it, they must mean something definite by their words rather than its opposite — already relying on the law. Against the global skeptic who says "nothing can be known": the claim presents itself as something known. Against the relativist who says "all truth is relative": the statement asks to be taken as non-relatively true. The cogito has the same shape — the attempt to doubt one's own existence enacts the existence doubted. In recent discourse ethics the same structure is called a **performative contradiction** (the contradiction lies not in the proposition but in the *performance* of asserting it).

What it can and cannot do

Retorsion is powerful but bounded, and the boundary matters. It does not directly prove the disputed proposition true; it shows that the *denial* cannot be coherently asserted. Stroud's standing objection to such transcendental arguments presses exactly here: at most they establish that we must *believe* or *presuppose* something, not that it is so — unless one adds a verificationist bridge from "must presuppose" to "true." This is why retorsion pairs naturally with *dismissing* the skeptic rather than refuting the skeptical proposition: it shows the skeptic's position is not assertible, which is enough to deny it PROBATIVE FORCE without settling the metaphysics.

That link is exact. A performatively self-refuting assertion is the limiting case of a failure of COGENCY: not merely an argument whose premises the audience cannot non-circularly reach, but one whose very assertion certifies its own falsity. Retorsion is therefore the sharp end of the same family of moves as the warrant-based objections elsewhere in the web.

What it bears on

Retorsion is one of the two anti-skeptical strategies catalogued under KNOWING THE EXternal world: it underwrites Retorsion against skepticism and is the formal core of the intuition that the hinge-denier saws off the branch they sit on. It is a cousin of, but weaker in ambition than, Deutsch's *bad-explanation* dismissal (see Deutsch's anti-solipsism), which faults solipsism for explanatory emptiness rather than for self-contradiction.

> Some claims blow themselves up just by being said out loud. "I cannot write a word of English" — written in English — is the toy example. "Nothing can be known" presents itself as a thing known. Retorsion is the trick of beating such a claim not by arguing against it but by pointing out that *making* it already takes back what it says. It's the cleanest way to deal with a total skeptic: you don't have to prove the world is real, only note that their doubting it relies on everything they're pretending to doubt. (The honest caveat: this shows they can't *coherently say* it — which may not be quite the same as showing they're *wrong*.)

See also

- ▹ Retorsion against skepticism — the argument that deploys this move.
- ▹ Dialectical cogency — self-refutation is the limiting failure of cogency; retorsion is its sharp end.
- ▹ Hinge proposition — what the skeptic must rely on even while denying it.
- ▹ Deutsch's anti-solipsism — the allied but distinct "bad explanation" dismissal of the skeptic.
- ▹ Knowing the external world — the issue on which retorsion is one of the anti-skeptical options.

⊢ Retorsion against skepticism

The retorsion argument — asserting external-world skepticism presupposes its own falsity

Argument

The *retorsion argument* answers the external-world skeptic not by proving the world — conceded impossible without circularity — but by turning the doubt on itself, showing it cannot be coherently asserted (see Retorsion). It is the first of the two anti-skeptical families under Knowing the external world, the sibling of the bad-explanation dismissal.

The argument

The inference is transcendental (see Transcendental argument) / dialectical — it reasons from the *conditions* of the skeptic's own performance.

1. To assert, doubt, or even entertain "there is no external world / I cannot know there is one," the skeptic must use a public language, draw on memory and reasoning, and address an interlocutor — capacities whose standing presupposes the very world and other minds in question (see hinges).
2. So the act of mounting the doubt relies on what the doubt would unsettle: the position is *performatively self-refuting*, undercut by the conditions of asserting it.
3. (see Knowing without proving) Demonstration is not required for knowledge; it suffices to show the doubt is not rationally assertible, removing the skeptic's standing to dislodge our default.

∴ External-world skepticism cannot be cogently maintained, and our acceptance that the world exists stands unrefuted: this `supports` The external world exists — not by proving it, but by disarming the challenge to it.

The weakest premise

The decisive vulnerability is the gap, pressed by Stroud (see Stroud — Transcendental Arguments (1968)), between *presupposition* and *truth* (premise 2's reach). At most the argument shows the skeptic must *presuppose* a world to voice the doubt — that the doubt is unassertible — not that there *is* a world; bridging "must presuppose" to "is so" needs a verificationist premise the skeptic will refuse. This is exactly why the conclusion is framed as cogency-level — the skeptic's assertion lacks probative force, indeed certifies its own falsity in the performing — rather than as a proof of realism. A self-refuting assertion is the limiting failure of cogency, and that, not a demonstration, is what retorsion delivers. Premise (1) can also be resisted by a skeptic who claims to doubt *silently*, entertaining the hypothesis without asserting it; against that merely-entertained skepticism the companion bad-explanation charge is the better tool.

> You can't talk yourself out of the world. To even *say* "maybe nothing outside my mind exists," the sceptic has to use a shared language, trust their memory, reason step by step, and address someone — all of which only make sense if there *is* a world with other people in it. So the doubt eats

> itself: voicing it leans on everything it pretends to suspend. That's enough to brush the sceptic aside — you needn't prove the world, just point out they can't coherently deny it. (The fair catch, due to Stroud: this shows they can't *coherently say* it, which isn't quite the same as proving the world is there — so the move disarms the sceptic rather than settling the metaphysics.)

SEE ALSO

- ▹ THE EXTERNAL WORLD EXISTS — what the argument supports by disarming its denial.
- ▹ RETORSION — the self-refutation move it instances.
- ▹ DIALECTICAL COGENCY — the level at which it bites: the skeptic's doubt is not cogently assertible.
- ▹ KNOWING WITHOUT PROVING — the premise that disarming the doubt suffices, proof being unnecessary.
- ▹ DEUTSCH'S ANTI-SOLIPSISM — the sibling move, by explanatory cost rather than self-refutation.

⊢ Robot reply

The Robot Reply — symbols grounded in sensorimotor commerce acquire meaning

Argument

The *Robot Reply* concedes that a program shuffling symbols in a sealed room has no understanding, and diagnoses *why*: the symbols are ungrounded, cut off from the world they are supposed to be about. Put the program inside a robot — give it cameras, microphones, limbs, and motors, so that its symbols are caused by and act upon real objects — and the symbols acquire genuine reference. "Hamburger" comes to mean hamburger because the robot's token is reliably triggered by hamburgers and drives its eating behaviour. Grounding supplies what the bare room lacked: a causal route from symbol to world.

This is a direct attack on A3 (see SYNTAX INSUFFICIENT FOR SEMANTICS): syntax *plus the right causal embedding* is, the reply claims, sufficient for semantics, so the move from "syntax alone is empty" to "computation can't yield understanding" fails. In the vocabulary of INTRINSIC AND DERIVED INTENTIONALITY, the Robot Reply is the claim that causal grounding converts *derived* intentionality into *original* intentionality — the robot's symbols mean things to *it*, not just to us, because *it* is the one in causal commerce with the things.

THE CRUX

Two things are at issue. First, whether causal-informational links are *enough* for meaning, or merely necessary — a question shared with causal and teleological theories of content. Second, whether Searle's ROOM-IN-THE-ROBOT REJOINDER defuses it: Searle puts the man inside the robot's head, manipulating the very symbols that come from the cameras and drive the motors, and insists the man still attaches no meaning to them, knowing nothing of where they come from. The reply's defender answers that this again looks at the wrong level — the *robot*, not the homunculus, is the candidate understander (folding the SYSTEMS REPLY back in).

> Maybe the room fails because it's *sealed off* from the world — just symbols with nothing to attach them to. So build a robot: wire the program to eyes, ears, and hands, so that its word for "cup" actually fires when it sees cups and makes it reach for them. Now the symbols are hooked to real things, and that hook-up, the reply says, is exactly what meaning *is* — so the robot really does understand, even if a bare computer wouldn't. Searle's comeback: climb inside the robot's head and do the symbol-shuffling yourself; you still won't know that one squiggle means "cup," so where did the understanding go?

SEE ALSO

- ▹ SYNTAX INSUFFICIENT FOR SEMANTICS — axiom A3, which this reply denies by adding causal grounding.
- ▹ ROOM IN THE ROBOT — Searle's reply: put the man in the robot and meaning still never appears.
- ▹ INTRINSIC AND DERIVED INTENTIONALITY — the reply claims grounding turns derived into original intentionality.
- ▹ INTENTIONALITY — causal-grounding theories of content the reply draws on.
- ▹ SYSTEMS REPLY — the parent objection the Robot Reply specialises.

⊢ Room in the robot

Room in the robot — the man manipulates the robot's symbols and still attaches no meaning

Argument · after Searle

Searle's rejoinder to the ROBOT REPLY. Grant the robot its cameras, limbs, and causal traffic with the world — then put the man inside its "head," receiving the strings of symbols that come in from the sensors and producing the strings that drive the motors, all by the same shape-matching rulebook. The man knows nothing of where the symbols come from or what they do; to him they are still meaningless squiggles. Adding perceptual and motor causal links changes what the symbols *do*, but not what they *mean to the man* — so the grounding, Searle concludes, does not inject understanding after all.

The move `supports` A3 (see Syntax insufficient for semantics): causal embedding supplies more syntax-driven behaviour, not original intentionality.

The live crux

The rejoinder is widely thought to *miss* the Robot Reply rather than meet it, because the reply never claimed the *man* understands — it claimed the *robot* does, in virtue of the system-level causal commerce the man is merely a cog in. So the rejoinder appears to smuggle the Systems Reply's level confusion back in: of course the homunculus doesn't understand; the question was whether the grounded system does. Whether causal grounding *constitutes* meaning or merely *accompanies* it is left genuinely open.

> The Robot Reply said: wire the program to a body and its symbols finally mean something. Searle climbs inside the robot: he sits in its head shuffling the very symbols that flow from its eyes to its hands, and he *still* has no idea any of it means "cup" or "wall." So, he says, bolting on a body didn't create understanding. The standard comeback: but nobody said *the man in the head* understands — the claim was that the *robot as a whole* does, and Searle keeps insisting on interrogating the little man instead of the robot.

See also

- ▷ Robot reply — the objection this answers.
- ▷ Syntax insufficient for semantics — axiom A3, which it defends.
- ▷ Intrinsic and derived intentionality — the original-vs-derived meaning at stake.
- ▷ Systems reply — the level objection critics say this rejoinder re-incurs.

★ Russellian monism

Position · after Russell

Russellian monism holds that physics describes only the abstract *structure and dynamics* of the world — how things are arranged and how they behave — while leaving wholly unspecified the intrinsic, categorical nature of whatever has that structure, and that consciousness is, or is grounded in, that intrinsic nature. On this view experience is neither a non-physical add-on nor reducible to bare structure: it is the hidden categorical filling inside the physical world's outline, the "what it is like to be the stuff" that physical description never reaches.

The position commits to two claims. First, that a functional or physical description captures only structural-dynamical facts, never intrinsic nature. Second, that phenomenal consciousness is categorical, not merely dispositional — an occurrent fact that is actually there, the kind of categorical basis that dispositions standardly require. Put together: consciousness is the categorical *ground* of the very dispositions physics charts. This is the constructive view that the structure-and-dynamics argument points toward, and the escape Chalmers prefers from the epiphenomenalism that dogs his naturalistic dualism. It answers the hard problem of consciousness by locating experience at the categorical base physics leaves blank.

The view descends from Bertrand Russell's *The Analysis of Matter* (see 1927) and has been revived in recent decades by Stoljar, Strawson, and Chalmers.

How it differs from neighbouring views

Russellian monism comes in degrees of commitment about *what* the intrinsic nature is. In its *constitutive* form it just *is* a kind of Panpsychism (position) (the intrinsic natures are themselves experiential) or *panprotopsychism* (they are proto-experiential — not yet consciousness, but the special non-structural ingredients from which it can be constituted). It contrasts with Dualism, for which consciousness is non-physical rather than the intrinsic-physical filling; and with Mysterianism, which holds the psychophysical link to be permanently beyond human grasp where Russellian monism instead *proposes what that link is.*

> Russellian monism starts from a quiet point about physics: physics only ever tells you what things *do* — how they push, attract, evolve — never what they intrinsically *are* underneath all that doing. The idea is that consciousness is that hidden inner nature, the "what it's like to be the stuff" physics leaves as a blank. Then mind isn't a ghostly extra, as in dualism, and isn't conjured from pure structure, as in plain physicalism; it's the very filling inside the physical world's outline. It's the

> middle road Chalmers leans toward, and it shades into panpsychism.

SEE ALSO

- ▷ STRUCTURE AND DYNAMICS ONLY — the first commitment: physics gives only structure and dynamics, not intrinsic nature.
- ▷ CATEGORICAL, NOT DISPOSITIONAL — the second commitment: experience is an occurrent categorical fact, the right kind to be that intrinsic nature.
- ▷ STRUCTURE-AND-DYNAMICS ARGUMENT — the argument the position is the constructive answer to.
- ▷ HARD PROBLEM (THE QUESTION) — the problem it addresses, by filling the categorical blank.
- ▷ PANPSYCHISM (POSITION) — constitutive Russellian monism is a form of it, with experiential intrinsic natures.
- ▷ PANPSYCHISM — the broader doctrine, whose taxonomy places panprotopsychism as the cautious cousin shading into this view.
- ▷ DUALISM — the rival on which consciousness is non-physical, not intrinsic-physical.
- ▷ MYSTERIANISM — the rival that says the psychophysical link cannot be grasped at all.

S

♦ Secondary intension

Concept · after Chalmers

The *secondary intension* of a term is the meaning it carries once its reference has been fixed by the actual world — the function that, taking that reference as given, says what the term picks out across all *counterfactual* worlds. It is one of the two "intensions" (functions from possible worlds to extensions) that David Chalmers's two-dimensional semantics assigns to an expression; the other, prior, one is its PRIMARY INTENSION. The secondary intension is what captures the Kripkean insight that many terms are *rigid designators* (see KRIPKE — NAMING AND NECESSITY (1980)): having latched onto a thing in the actual world, they keep referring to that same thing when we describe how matters might otherwise have been.

Return to *water*. We discover empirically that the watery stuff of *our* world is H_2O. The secondary intension then takes that discovery as settled and runs it across all worlds: it maps every counterfactual world to H_2O. So considered counterfactually, a world whose lakes hold XYZ rather than H_2O is *not* a world where water is something other than H_2O — it is a world with no water in it, however watery the XYZ looks. This is why "water is H_2O," though learned by experiment, comes out *necessary*: the secondary intension of "water" is constant across the counterfactual worlds, picking out H_2O in each. The truth is necessary along this dimension and a posteriori along the other — exactly the profile of the Kripkean A-POSTERIORI NECESSITIES.

How it differs from the primary intension

The two intensions answer two different questions about the same term. The primary intension ranges over worlds *considered as actual* and asks what "water" would pick out if that world were the actual one — so it can deliver XYZ when Twin Earth is the candidate for actuality. The secondary intension holds the actual world's verdict fixed (water *is* H_2O) and ranges over worlds *considered as counterfactual*, delivering H_2O everywhere. For a term whose reference is fixed descriptively with no empirical residue, the two functions coincide; for a natural-kind term like "water" they diverge, and the divergence *is* the necessary a posteriori. The two-dimensionalist's payoff is that the apparent counterexamples to "conceivability entails possibility" can then be redescribed as A-POSTERIORI NECESSITIES AS MISDESCRIBED POSSIBILITIES: what looks like conceiving an impossibility (water that isn't H_2O) is really conceiving a genuine possibility that satisfies water's *primary* intension, with the *secondary* necessity left untouched.

Chalmers's epistemic version of the framework introduces the "primary/secondary" pair in *The Conscious Mind* (see CHALMERS — THE CONSCIOUS MIND (1996)) and refines it in CHALMERS — ON SENSE AND INTENSION (2002); Frank Jackson develops a matching apparatus from conceptual analysis (see JACKSON — FROM METAPHYSICS TO ETHICS (1998)). The secondary intension is also called the *subjunctive intension*, *C-intension*, or *2-intension*, and corresponds to the familiar Kripkean rigid-designator meaning — the intension that fixes a term's modal profile.

Why the concept matters

In the consciousness debate the secondary intension is the physicalist's foothold. Grant that the PHILOSOPHICAL ZOMBIE is conceivable; the type-B physicalist (see TYPE-A / TYPE-B / TYPE-C MATERIALISM) replies that conceivability shows only *primary* possibility, and that — as with water — a primary/secondary gap may make the zombie metaphysically impossible all the same, its conceivability a misdescription. Whether that escape is available turns on a single further question: does "consciousness" behave like "water," with a wedge between how it presents and what it is, or do its two intensions coincide (see COINCIDENT PHENOMENAL INTENSIONS)? If they coincide, there is no secondary necessity to hide the impossibility in, and primary conceivability carries all the way to genuine possibility.

> Once we find out what the stuff in our rivers actually *is* — H_2O — the word "water" gets pinned to H_2O and stays pinned, even when we talk about

> other possible worlds. The secondary intension is that pinned-down meaning: it says "water" means H_2O *everywhere*, so a world with XYZ in its rivers just doesn't have any water in it. That's why "water is H_2O" turns out to be a truth that *couldn't have been otherwise*, even though we had to do chemistry to learn it. The companion notion, what "water" means *before* we know which world we're in, is the Primary intension. The whole point of keeping the two apart is to explain how a fact can be both discovered by experiment and yet necessary.

See also

- ▹ Primary intension — the prior, reference-fixing dimension; the secondary intension takes over once the actual world has settled what the term refers to.
- ▹ A-posteriori necessities — the necessary a posteriori (water = H_2O), which is exactly a constant secondary intension paired with a varying primary one.
- ▹ A-posteriori necessities as misdescribed possibilities — the two-dimensional move that reads the Kripke cases as primary possibilities, preserving the conceivability–possibility link.
- ▹ Type-A / Type-B / Type-C materialism — type-B physicalism, which leans on a possible primary/secondary gap for "consciousness" to block the zombie argument.
- ▹ Coincident phenomenal intensions — the claim that no such gap exists for phenomenal concepts, so the secondary-intension escape is unavailable for consciousness.
- ▹ Philosophical zombie — the case whose modal status hangs on whether phenomenal terms have a divergent secondary intension.

⊙ Sole known intrinsic nature

Consciousness is the only intrinsic nature we are directly acquainted with

Claim · after Seager

Of all the intrinsic natures there may be in the world, consciousness is the single one we are *directly acquainted with*. Physics, which is our most powerful instrument for knowing matter, discloses only relational and structural facts (see Structure and dynamics only) and is, in Russell's words, agnostic about all but the mathematical properties of the physical world. Introspection, by contrast, hands us an intrinsic property without intermediary: in undergoing an experience we are acquainted with its felt quality as it is in itself, not merely with its relations to other states. So where everything else is known to us only by its structure, experience is known from the inside.

The thought is Russell's and Eddington's: "we know nothing about the intrinsic quality of physical events except when these are mental events that we directly experience" (see Russell — The Analysis of Matter (1927)). Seager (see Seager — The 'Intrinsic Nature' Argument for Panpsychism (2006)) makes it the pivot of the intrinsic-nature argument — for if matter must have an intrinsic ground for its structure (see Categorical ground of structure) and experience is the only intrinsic nature we have ever actually met, then experience is the one non-arbitrary candidate to be that ground.

Scope. The claim is epistemic — it is about the *reach of our acquaintance*, not a census of what intrinsic natures exist. It leaves open that matter has wholly unknown non-experiential intrinsic properties; the argument's further move is that positing *those*, rather than the experiential nature we already know, is an unmotivated act of faith.

Who would deny it. A *physicalist* who expects future science to disclose the intrinsic nature of matter denies it, as does anyone who holds that introspection too delivers only structural or relational information about our states rather than an intrinsic quality. The cautious Russellian response — that our acquaintance licenses only *agnosticism* about matter's intrinsic nature, not its identification with experience — concedes this claim but blocks the leap the argument builds on it.

Distinct from Intrinsicness of consciousness: that claim says experience *is* intrinsic; this one adds the further, epistemic point that it is the *only* intrinsic nature we directly know — the premise that does the selecting work in Intrinsic-nature argument.

> Everything science tells us about matter is, at bottom, about how things behave and relate — never about what they're like on the inside, in themselves. There's exactly one exception, and it's the one case we get from the inside: our own experience. When you feel pain or see red, you're not just learning how that state is wired up to others — you're acquainted with what it's actually like, its inner character. So this claim says consciousness is the *only* bit of "inner nature" anywhere in the world that we've ever actually laid hold of.

> That's what makes it tempting to say: if matter needs some inner nature down there, why not the one kind of inner nature we actually know? (Critics answer that knowing only *one* example doesn't entitle you to assume the rest are the same — the honest verdict might just be "we don't know.")

☉ Solipsism is parasitic on realism

Solipsism must reconstruct the external world within the self, so it amounts to realism plus unexplained surplus

Claim · after Deutsch

Solipsism — the doctrine that only one's own mind exists — cannot, taken seriously, simplify one's ontology, because it must smuggle the whole of the rejected world back inside the self. The solipsist still has to distinguish the part of their experience that they *control and predict* from the vast part that is surprising, autonomous, and law-governed — and that second part has exactly the structure that realism attributes to an external world, including other apparent people. So the solipsist must posit, *within* their mind, an entity obeying the laws of physics and populated by mind-like sub-agents. They have not eliminated the external world; they have renamed it "part of myself" and added the unexplained claim that it is really inner. Solipsism is therefore realism *plus* surplus structure that does no work — strictly more complex than the view it sought to undercut. The argument is David Deutsch's (see Deutsch — The Fabric of Reality (1997)).

Scope. The claim targets *solipsism* and, by extension, idealist reductions that must reproduce the external world's structure internally. It is an argument about *explanatory cost and structure*, not a demonstration that the external world exists; its anti-skeptical force runs through the further premise that a needlessly complex, surplus-laden theory is to be rejected (see Good explanations are hard to vary).

Who would deny it. A solipsist may insist the "external-seeming" structure is brute — that there is no obligation to explain *why* part of the self behaves like a lawful outer world. Deutsch's rejoinder is that this is precisely the mark of a *bad explanation*: it leaves unexplained exactly what realism explains, and buys nothing for the extra ideology. A phenomenalist might also deny that reconstructing the structure amounts to reinstating a *mind-independent* world.

Distinct from The external world exists (the realist conclusion this supports) and from No non-circular proof of the world (a point about proof): this is a structural-explanatory charge that solipsism defeats itself by reduplicating its target.

> Say only your mind exists. Fine — but then you still have to explain why so much of your "experience" does whatever it likes, ignores your wishes, follows strict laws, and is full of people who argue back. To account for all that, you'd have to build, *inside your own mind*, something that behaves in every respect like a real outside world. You haven't gotten rid of the world; you've just relabelled it "me" and tacked on the odd claim that it's secretly inside you. That's more baggage, not less — which is Deutsch's reason for throwing solipsism out.

♦ Soundness

Concept

Soundness is the virtue a deductive argument has when it is both **valid** (see Validity) and has **all true premises**. A sound argument therefore has a true conclusion: validity transmits the truth of the premises to the conclusion, and soundness supplies that truth at the start. Where validity is a matter of form alone, soundness adds a demand on the world — that the premises really hold.

Soundness is the usual target of argument evaluation. To show an argument *unsound* one either breaks its validity (exhibits a way for true premises to yield a false conclusion) or denies one of its premises — which is why a careful objection names the **weakest premise**, the one whose falsity would cost the argument its soundness.

Yet soundness is still not the whole of what makes an argument *good*. An argument can be sound and nevertheless powerless to persuade, because the audience may have no way to come to know a premise, or be able to only by first accepting the conclusion. That further property — probative force — is Dialectical cogency. The gap between soundness and cogency is exactly what the unwarrantable-premise objection exploits: it grants that the zombie argument may be sound while denying that any computable reasoner can be warranted in its key premise, so the argument cannot move its target.

> An argument is *sound* when two things hold at once: the logic is airtight (it's valid) **and** the starting claims are actually true. When both hold, the conclusion is guaranteed true. That's the gold standard most people have in mind when they ask "is the argument any good?" — but it

> still isn't the end of the story, because a sound argument can fail to *convince* anyone who can't independently check its premises (see DIALECTICAL COGENCY).

SEE ALSO

- ▷ VALIDITY — the form-level virtue soundness builds on.
- ▷ DIALECTICAL COGENCY — the further, audience-relative virtue a sound argument can still lack.
- ▷ UNWARRANTABLE-PREMISE OBJECTION — a move that turns precisely on the soundness/cogency gap.

♦ Strong and weak AI

Strong AI vs. weak AI

Concept · after Searle

The distinction between *strong* and *weak* AI is John Searle's (see SEARLE — MINDS, BRAINS, AND PROGRAMS (1980)), and it concerns not how capable a program is but what a successful program would *show*. **Strong AI** is the thesis that an appropriately programmed computer *literally has a mind* — that running the right program is *constitutive of* (sufficient for) understanding, intentionality, and cognitive states, so the program "is the mind." **Weak AI** is the modest thesis that the computer is a powerful *tool* for studying the mind: it lets us model cognition, formulate and test precise hypotheses, and simulate mental processes — but the simulation is a *model* of a mind, not itself a mind. The watchword is *duplication versus simulation*: weak AI simulates understanding; strong AI claims to duplicate it.

Searle's quarrel is only with the strong thesis. A simulation of a rainstorm leaves no one wet, and a simulation of digestion digests no food; a simulation of understanding, he argues, no more understands than these — which is the moral of THE CHINESE ROOM, his thought experiment aimed squarely at strong AI (see SYNTAX INSUFFICIENT FOR SEMANTICS). Weak AI is untouched by that argument and is, for Searle, simply good cognitive science.

Strong AI is a form of computational FUNCTIONALISM — mind as the running of the right program, independent of the hardware — and so presupposes SUBSTRATE INDEPENDENCE for the mental. Searle's own positive view, BIOLOGICAL NATURALISM, accepts weak AI but rejects strong AI: brains *cause* minds through their specific biological causal powers, which symbol-manipulation alone does not reproduce.

TWO SENSES TO KEEP APART

The label "strong AI" has since drifted, and the two senses are routinely conflated.

- **Searle's sense (constitutive):** whether a system has *genuine mental states* — real understanding and intentionality — as opposed to merely modelling them. This is a metaphysical/semantic question.
- **The popular sense (capability):** "strong AI" as a synonym for **artificial general intelligence** (AGI) — human-level, domain-general competence — opposed to "weak" or **narrow** AI built for a single task. This is an engineering question about *breadth of ability*.

The two come apart cleanly: a system could be "strong" in the capability sense (broadly, flexibly competent) yet "weak" in Searle's sense (no genuine understanding behind the competence). That gap is exactly the live question about today's large language models, which display wide capability while it remains disputed whether anything is understood — see LLMS AS NEAR-PERFECT ACTORS and the recurring AI EFFECT.

> Two very different claims hide under "strong AI." Searle's original one: that a computer running the right program doesn't just *act* like it understands — it *really does* understand, has a genuine mind. Weak AI is the humbler claim that the computer is a brilliant *model* of thinking, the way a weather simulation is a model of weather (and never rains on anyone). Searle is fine with the model; he denies the program is a mind. Confusingly, people now also use "strong AI" to mean something else entirely — a machine that's *generally* smart, good at everything, rather than one narrow task. You can have the second without the first: something clever at every task that still understands nothing, which is just what people argue about with chatbots.

SEE ALSO

- ▷ CHINESE ROOM — Searle's thought experiment against strong AI; this concept names its target.
- ▷ SYNTAX INSUFFICIENT FOR SEMANTICS — the slogan that drives the case against strong AI.
- ▷ FUNCTIONALISM — the theory of mind strong AI is a computational version of.
- ▷ SUBSTRATE-INDEPENDENCE THESIS — the thesis strong AI presupposes for the mental.

- Biological naturalism — Searle's own view: weak AI yes, strong AI no.
- AI effect — the recurring serial denial of strong AI (and "real" intelligence) to each new system.
- LLMs as near-perfect actors — the modern case where wide capability and disputed understanding pull apart.
- Searle — Minds, Brains, and Programs (1980) — where the distinction is drawn.

Structure and dynamics only

A functional description captures only structural-dynamical facts, not intrinsic nature

Claim · after Chalmers

A purely functional or causal specification of a system captures only its *structure* — the pattern of relations among its parts — and its *dynamics* — how its states evolve and what they dispose the system to do. It says nothing about the *intrinsic, categorical nature* of whatever realises that structure: what the components *are* in themselves, beneath their relational profile. The picture is the wiring diagram. A circuit diagram fixes every connection and every response, yet on its own tells you nothing about what the components are physically made of.

This is the structural-dynamical thesis at the heart of *Russellian monism*, traced to Russell's *The Analysis of Matter* (see Russell — The Analysis of Matter (1927)) and revived by Chalmers (see Chalmers — Consciousness and Its Place in Nature (2003)): physics describes the world's abstract causal-structural pattern and is silent on its intrinsic nature. As stated, the claim is *neutral* — it does not say that anything important lies outside structure and dynamics, only that such description leaves intrinsic nature out. A structuralist, in particular an *ontic structural realist* who holds that there is no further categorical nature (or that talk of "intrinsic nature" is empty), would deny it.

It is more general than the phenomenal-residue claim, which says that functional role under-specifies *qualia* specifically. This claim makes the broader point that structural-dynamical description omits *intrinsic nature* as such — of which the phenomenal is only the most salient candidate to fill the gap.

> Describe anything purely by its function and you capture its *structure* (how the parts relate) and its *dynamics* (how its states change over time) — but nothing about what the parts *are* in themselves, their inner nature. It's the wiring-diagram point again: the diagram shows the connections, never what the components are actually made of. On its own this stays neutral about whether anything important hides in that left-out inner nature — it just marks that function leaves it out.

See also

- phenomenal residue — the narrower companion: function under-specifies *qualia*; this claim generalises to intrinsic nature as such.
- Functionalism — the role-based view of mind whose descriptive reach this claim bounds.
- Russellian monism — the position built on this thesis: consciousness as (or grounding) the intrinsic nature physics leaves blank.

Subjects merge and divide

There are actual cases in which subject boundaries shift with the degree of integration

Claim · editorial

The boundary around a conscious subject is not fixed once and for all but appears, in real cases, to shift with how tightly the underlying parts are integrated. Three kinds of case point the same way. In *split-brain* patients, severing the corpus callosum can leave the two hemispheres behaving as partly independent centres of experience that a healthy, fully connected brain does not display (see Nagel — Brain Bisection and the Unity of Consciousness (1971)). In *craniopagus* twins whose brains are joined — the Hogan twins, linked by a "thalamic bridge" — there are reports of sensory experience shared across the two bodies (see Dominus — Could Conjoined Twins Share a Mind? (2011)). And at the *group* level, practitioners of certain meditative and tantric techniques, people on some psychedelics, and members of a jazz band or a well-synced team report a felt collapse of the boundary between several minds into a single acting "we."

These cases are offered as a defeater for the claim that subject-summing never happens: they suggest that whether a system houses one subject or several can vary with its degree of internal integration. The cases are contested in interpretation — a sceptic will read the group "we" as a vivid *representation* of unity rather than a literal further subject, and dispute how many subjects a split brain "really" contains — which is why this claim states only that such cases *occur and shift with integration*, leaving the inference to genuine combination to the argument it feeds, Combining subjects. Luke Roelofs surveys this same range of cases at length (see Roelofs — Combining Minds: How to Think about Composite Subjectivity (2019)).

See also

- ▹ Combining subjects — the rebuttal to the combination problem that this claim helps support.
- ▹ Graded subject unity — the companion premise that reads these cases as a continuous, integration-dependent gradient of subjecthood.
- ▹ Micro-subjects do not sum — the claim these cases are marshalled against.

♦ Super-spartans

Super-spartans (pain felt but never shown)

Concept · after Putnam

A *super-spartan* is a member of an imagined community, introduced by Hilary Putnam in "Brains and Behavior" (1963, Putnam — Brains and Behavior (1963)), who feels pain as keenly as anyone but has, out of a fierce cultural pride, trained away every outward sign of it. Super-spartans do not wince, groan, nurse the injured part, or even say that anything hurts; pressed, they will calmly report a pain while displaying none of the behaviour we ordinarily take pain to consist in. Putnam pushes the case further to the *super-super-spartans* of an "X-world", who have suppressed pain-behaviour for so many generations that the very talk of pain has dropped out of their language — yet, he insists, they still have pains and know perfectly well that they do.

The point of the case

The super-spartan is a counterexample to **logical (analytical) behaviourism**: the thesis that to be in pain just *is* to behave, or be disposed to behave, in characteristic ways — to wince, withdraw, complain. If a super-spartan is so much as possible, then one can be in pain while having no disposition whatever to pain-behaviour. So pain-behaviour is not a *necessary* condition of pain, and the behaviourist's proposed analysis of the mental state fails. The mental state and the behaviour come apart.

It is important that the suppression be total and dispositional, not merely a stiff upper lip in the moment. A behaviourist can absorb someone who hides a pain *now* by appealing to what they *would* do in other circumstances. Putnam's super-super-spartans are built precisely to close that escape: there is no circumstance in which their pain issues in the canonical behaviour, and still — by stipulation — the pain is there.

Relation to its converse

The super-spartan is one half of a matched pair. Its mirror image, the Perfect actor, attacks the same theory from the other side: a perfect actor produces all the behaviour of pain while feeling none, showing that the behaviour is not *sufficient* either. Putnam himself drew from the pair not the dualist conclusion but the lesson that mental states should be individuated by their *causal role*, the move that seeded Functionalism. Logical behaviourism is one member of the broad family of topic-neutral analyses catalogued under Type-A materialism.

> Picture a culture so proud that nobody ever shows pain: they feel the burn exactly as you do, but they never flinch, never cry out, and eventually don't even have a word for it. Common sense says they are still in pain — they just don't show it. That sinks the theory that *being in pain simply means behaving as though you are in pain*, because here is pain with none of the behaviour. It is the exact opposite of the Perfect actor, who shows all the behaviour while feeling nothing, and different again from a Philosophical zombie, which is missing the inner feeling rather than just the outward show.

See also

- ▹ Perfect actor — the converse case: full pain-behaviour with no pain, attacking the *sufficiency* of behaviour as this attacks its *necessity*.
- ▹ Functionalism — the successor theory Putnam moved to, which replaces behavioural dispositions with causal-functional role and so is not refuted by the case.
- ▹ Philosophical zombie — a sharper cousin: the zombie lacks the experience itself, where the super-spartan keeps the experience and drops only the behaviour.
- ▹ Type-A materialism — the family of topic-neutral analyses of the mental that houses logical behaviourism, the target here.
- ▹ Putnam — Brains and Behavior (1963) — Putnam's "Brains and Behavior" (1963), where the super-spartans are introduced.

⊢ Systems reply

The Systems Reply — the whole system understands, though the person does not

Argument

The *Systems Reply* is the most common objection to the CHINESE ROOM. It grants that the person in the room understands no Chinese but denies that this settles anything: understanding is not to be located in the person, who is merely the *central processor*, but in the **whole system** — the person together with the rulebook, the symbol store, the paper, and the input/output channels. No single neuron in a Chinese speaker's brain understands Chinese either, yet the brain does; likewise no part of the room need understand for the system to.

Against the DERIVATION, the reply is in effect a denial of A3 (see SYNTAX INSUFFICIENT FOR SEMANTICS): the relevant bearer of semantics is the organised system, and Searle's argument illicitly looks for understanding in a *component* (the person) rather than in the system that runs the program. Against the INTUITION-PUMP VERSION it raises the *thought-experiment* pattern's **scope** critical question: the verdict "he understands nothing" is true of the person but does not generalise to the system the experiment was meant to be about.

THE CRUX

The reply's force depends on whether a system can have genuine understanding that none of its parts has — *emergent* system-level intentionality. Searle answers with the INTERNALISATION REJOINDER: let the person memorise the rulebook and do all the work in their head, so that the person *is* the entire system, with nothing left over to be the understander. Whether that rejoinder succeeds — whether internalising the system really leaves no room for understanding, or merely hides the system inside the person — is the live question. The Systems Reply is also the parent of the ROBOT and BRAIN SIMULATOR replies, which specify *what more* the system would need.

> The classic comeback to the Chinese Room: sure, the *man* in the room doesn't understand Chinese — but the man isn't the whole story. The man plus the giant rulebook plus the files plus the whole setup, working together, might understand, even though no single piece does. After all, no one brain cell of yours understands English, yet *you* do. So Searle picked the wrong thing to look
>
> at: don't ask whether the clerk understands, ask whether the *system* does. Searle's famous reply: fine, have the man memorise the entire rulebook and throw the room away — now he *is* the whole system, and he still doesn't understand a word.

SEE ALSO

- ▷ CHINESE ROOM DERIVATION — the argument this objection attacks, by denying A3.
- ▷ INTERNALISE THE RULEBOOK — Searle's reply: internalise the system and nothing is left to understand.
- ▷ ROBOT REPLY, BRAIN SIMULATOR REPLY — descendants that say what the system would need to add.
- ▷ CHINESE NATION — the neighbouring system-level case, pressed against qualia rather than understanding.

T

☉ Text yields human-aligned representations

Language models trained on text alone develop internal representations structurally aligned with human ones

Claim · editorial

Large language models trained purely on text develop *internal representations* whose relational structure matches that of human representations of the same domain — not merely matching output labels, but recovering the same similarity geometry and, in some settings, an explicit model of the world the symbols describe. Four independent lines of evidence converge.

- **Perceptual structure from text.** GPT models' pairwise similarity judgments track human data across *six* perceptual modalities, reconstructing canonical structures like the colour wheel and pitch spiral from text, with no gain from added visual training (see Marjieh et al. — LLMs Predict Human Sensory Judgments Across Six Modalities (2024)); and the *unsupervised* alignment of GPT-4's colour-similarity space with that of colour-neurotypical humans — optimal transport assuming no label correspondence — shows the match lies in the internal structure, not the names (see Kawakita et al. — Structural Correspondences Between Colour Similarity Structures of Humans and LLMs (2024)).
- **Induced world models.** Sequence-only models build an explicit model of the domain they predict: a GPT trained on Othello moves maintains a causally-effective representation of the board (see Li et al. — Emergent World Representations (Othello-GPT) (2022)), robust across architectures (see Yuan & Søgaard — Revisiting the Othello World Model Hypothesis (2025)).
- **Real-world structure.** Beyond games, LLMs carry *linear* representations of space and time at multiple scales, with individual neurons encoding coordinates — the "basic ingredients of a world model" (see Gurnee & Tegmark — Language Models Represent Space and Time (2023)).
- **Brain alignment.** Most directly for the "human-aligned" wording, next-token training yields conceptual representations that *align with human brain activity*, emerging without real-world grounding (see Xu et al. — Emergent Human-like Conceptual Representations from Language Prediction (2025)), and LLM embeddings of scene captions predict visual-cortex activity well enough to reconstruct the caption from brain data (see Doerig et al. — High-level Visual Representations in the Human Brain Are Aligned with LLMs (2025)).

Scope. The claim is about *structural/representational* alignment — the geometry of the model's internal states matching the human one — established as a *lower bound* on what is latent in language. It does **not** by itself assert that the models have phenomenal experience, consciousness, or intrinsic (rather than derived) intentionality; those are further questions the alignment does not settle. Its force is to show that *syntax* (text statistics) is, as a matter of fact, enough to *induce* richly human-like semantic *structure*.

Who would deny it, and how. A Searlean grants the structural finding but denies it touches the real issue: aligned representational structure is still structure — a *depiction* of human semantics, not the intrinsic understanding the Chinese Room is about (see Instancing vs. depicting). Sceptics about the evidence add deflationary readings: training-data contamination, anthropomorphic probing, or that "similarity geometry" is a thin proxy for representation. The claim is framed to survive these as a structural result while conceding it does not, alone, reach the phenomenal.

Distinct from LLMs as near-perfect actors (about *behavioural* indistinguishability) and from Similarity, not encoding: this claim is specifically about the *internal representational structure* matching the human one, the empirical pressure point on Searle's third axiom.

> Feed a model nothing but text and, surprisingly, it builds an inner "map" of things that lines up with the map in our heads. Ask GPT-4 which colours are similar and its answer reconstructs the human colour wheel; the same holds for sound, taste, and three other senses — and it doesn't even help to also train it on images, because the structure was already hiding in the words. Train a model only on Othello moves and it quietly builds a picture of the board. None of this proves the model *feels* anything or *really* understands — but it does show that pure word-shuffling can grow a genuinely human-shaped *web of meaning*, which is exactly the thing Searle's argument says shuffling alone can never do.

There are basic beliefs

Some beliefs are justified non-inferentially rather than by derivation from other beliefs

Claim

Not every justified belief owes its justification to other beliefs. Some — BASIC BELIEFS — are warranted directly: by perception, memory, introspection, or rational insight, without passing through an inference from further premises. This is the foundationalist's central thesis and its answer to the regress: the chain of justification halts at beliefs that are self-standing, terminating THE MÜNCHHAUSEN TRILEMMA's regress at a principled floor rather than an arbitrary dogma.

Scope. The claim asserts that *some* warrant is non-inferential; it is silent on which beliefs are basic and on whether basic beliefs must be infallible. Moderate foundationalism takes them to be defeasible (a perceptual belief warranted on sight, but revisable); classical foundationalism demanded indubitability.

Who would deny it. *Coherentists* (see COHERENTISM) deny it outright: there are no non-inferentially justified beliefs, since justification is always a matter of a belief's relations to *other* beliefs. *Infinitists* (see INFINITISM) likewise deny a terminating floor. And a *Wittgensteinian* resists from the other side: the regress-stoppers are not *justified* beliefs at all but HINGES held without justification.

Distinct from JUSTIFICATION'S TRILEMMA (the neutral statement of the regress problem, which all parties can accept) — this claim takes one side, the foundationalist's, on how the regress ends.

> Foundationalism's key idea: a few of your beliefs don't need any *other* belief to back them up — seeing red justifies "that's red" all on its own. Call those "basic." They're meant to be the ground floor that stops the endless "but why believe *that*?" Some philosophers think there's no such ground floor (everything is justified by other things, or by nothing at all) — and this claim is exactly what they're denying.

Transcendental argument

Concept

A *transcendental argument* reasons from some feature of experience or thought that even an opponent must grant, to the *necessary conditions* of that feature's being possible — concluding that those conditions must obtain. Its canonical form: *F is actual (and undeniable); F is possible only if C; therefore C.* The name and the strategy are Kant's, who argued that the very possibility of unified, objective experience presupposes a law-governed world of enduring objects — so the skeptic who enjoys such experience cannot consistently doubt its framework. The form was revived in analytic philosophy by P. F. Strawson's *Individuals* (see STRAWSON — INDIVIDUALS (1959)), as a method of "descriptive metaphysics."

The appeal against skepticism is that it does not try to *prove* the contested conclusion from neutral evidence — known to fail for the external world (see NO NON-CIRCULAR PROOF OF THE WORLD) — but extracts it from what the skeptic is already doing. To raise the doubt at all, the skeptic must think, mean, remember, and refer; if those activities have presuppositions, the presuppositions come along free.

Its limit

The standing objection is Barry Stroud's (see STROUD — TRANSCENDENTAL ARGUMENTS (1968)): a transcendental argument, by itself, shows at most that we must *believe* or *conceive* things to be a certain way in order to think or speak — not that they *are* that way. Closing the gap from "must be presupposed" to "is true" needs an extra, verificationist premise (roughly, that what we must conceive to be so *is* so), which the skeptic will deny. So unsupplemented transcendental arguments tend to establish *psychological* or *conceptual* necessities rather than facts about the world — which is why they pair naturally with *disarming* the skeptic rather than refuting the skeptical proposition.

Relation to retorsion

RETORSION is the limiting, dialectical case of the transcendental strategy: where a general transcendental argument identifies the conditions of some broad feature like objective experience, retorsion targets the *specific act of assertion*, showing that to assert the skeptical thesis is to rely on its falsity. Retorsion thus inherits both the strength (it engages the skeptic on their own activity) and the Stroudian limit (it shows the denial unassertible, not the world real) of transcendental arguments generally.

> A transcendental argument is a clever sidestep: instead of proving "there's a world" from scratch, you start from something your opponent can't deny — that they think, speak, experience an ordered world — and ask what has to be true for *that* to be possible. Whatever the answer is, they're stuck with it, because they already granted the starting point. The catch, due to Stroud: this may only show what they have to *assume*, not what's actually *so* — proving how we must think isn't quite proving how things are.

SEE ALSO

- ▷ RETORSION AGAINST SKEPTICISM — the anti-skeptical argument of this form deployed in the web.
- ▷ RETORSION — the assertion-focused special case of the strategy.
- ▷ STROUD — TRANSCENDENTAL ARGUMENTS (1968) — the objection that bounds what these arguments can show.
- ▷ STRAWSON — INDIVIDUALS (1959) — the modern revival of the form.
- ▷ KNOWING THE EXTERNAL WORLD — the issue on which transcendental arguments are one anti-skeptical option.

⊢ Trivialises strong AI

Many Mansions trivialises strong AI — the reply concedes the thesis it was meant to save

Argument · after Searle

Searle's rejoinder to the MANY MANSIONS REPLY. The reply is not really an objection at all: it *concedes* the Chinese Room's target and rebrands the concession as optimism. Strong AI (see STRONG AND WEAK AI) was the precise, falsifiable thesis that *mental processes are computational processes* — that implementing the right *program* is sufficient for a mind. The Many Mansions Reply gives exactly that up, retreating to the claim that *some* future artificial means, not necessarily computational, might produce a mind. But that weakened claim is one Searle already grants (it is the import of A4, BRAINS CAUSE MINDS): of course an artificial system with the right causal powers could think. By abandoning the computational thesis, the reply trivialises strong AI rather than defending it.

The move **supports** the DERIVATION by clarifying its scope: C1 was only ever about *programs*, and a reply that drops programs concedes C1.

THE LIVE CRUX

The reply's defender can shift from logic to rhetoric: Searle's *official* conclusion is narrow (programs don't suffice), but the Chinese Room is routinely *presented* as a sweeping case against machine minds, and the Many Mansions Reply rightly resists *that* over-reading. So the rejoinder wins the strict point while leaving the broader cultural claim — that the Room shows machines can't think — undefended, as it should be.

> The Many Mansions Reply said "maybe some future tech, not ordinary code, will build a mind." Searle's answer: sure — but that was never what I denied. I denied that *running a program* is enough. The moment you say "well, maybe something *other* than a program," you've agreed with me and quietly dropped the exciting claim ("the program *is* the mind") that the whole debate was about. So it's not a rescue of strong AI; it's a polite surrender.

SEE ALSO

- ▷ MANY MANSIONS REPLY — the objection this answers.
- ▷ STRONG AND WEAK AI — the computational thesis the reply abandons.
- ▷ BRAINS CAUSE MINDS — A4, which already concedes non-program artificial minds.
- ▷ CHINESE ROOM DERIVATION — the argument whose narrow scope this clarifies.

♦ Turing test

The Turing test (the imitation game)

Concept · after Turing

The *Turing test* is Alan Turing's proposed replacement for the question "Can machines think?" (see TURING — COMPUTING MACHINERY AND INTELLIGENCE (1950)). Finding that question "too meaningless to deserve discussion," Turing recast it operationally as the **imitation game**: a human interrogator holds text-only conversations with two unseen partners — one human, one machine, each trying to be judged the human — and if, after suitable interrogation, the machine is misidentified as often as a human is, it passes. The test deliberately routes around looks and voice to isolate *conversational behaviour* as the criterion, and substitutes a question we can settle by observation for one Turing thought we could not settle at all.

The test embodies a broadly **behaviourist or operationalist** stance: it treats indistinguishable verbal behaviour as the relevant mark of mind, declining to ask what, if anything, lies behind it. That is exactly its power and its vulnerability. Turing himself offered it less as a *definition* of thinking than as a prediction and a reframing, and he pre-emptively answered nine objections (theological, mathematical, the "Lady Lovelace" originality objection, and others).

The standing objections

- **The Chinese Room** (see Chinese room): Searle grants a system might *pass* yet insist it understands nothing — passing certifies the right *behaviour* (syntax), not *understanding* (semantics). So a pass, even if it is evidence for weak AI, does not establish strong AI.
- **The Blockhead / lookup-table objection** (see Lookup-table mind): Ned Block notes that a machine could in principle pass with a gigantic pre-computed table of canned replies and *no* cognitive processing at all — so passing shows nothing about *how* the responses are produced, and behaviour alone cannot be sufficient for intelligence (psychologism versus behaviourism).
- **The AI effect**: as systems approach a pass, the response "that is not *really* thinking" reliably recurs, so the test's verdict is socially contested whatever the transcripts show.

What it bears on

The Turing test is the operational counterpart of the perfect actor — a system whose outward behaviour is indistinguishable from a genuine mind's — and so sits at the centre of the dispute over whether behavioural success suffices for mind. Today's large language models press the question concretely: they approximate that actor (see LLMs as near-perfect actors) and routinely pass informal versions of the test, which sharpens rather than settles the issue, since the test is silent on the very understanding its critics demand.

> Turing's move was to dodge the bottomless question "can a machine *think*?" and ask a crisp one instead: could a machine hold a typed conversation so well that a person couldn't tell it from another person? If yes, he said, treat it as thinking — stop fussing about what's "really" inside. Critics answer that fooling the judge isn't the same as understanding: Searle's locked room shuffles symbols convincingly with no comprehension, and Ned Block points out you could pass with a giant cheat-sheet of canned answers and no thought at all. With chatbots now passing casual versions of the test, the old argument is suddenly very live.

See also

- ▷ Perfect actor — the behavioural ideal the test operationalises.
- ▷ Chinese room — the objection that passing is behaviour without understanding.
- ▷ Lookup-table mind — Block's objection that a canned-answer table could pass with no cognition.
- ▷ Strong and weak AI — a pass is at most evidence for weak AI, not strong AI.
- ▷ AI effect — why a pass tends to be met with "that's still not thinking."
- ▷ LLMs as near-perfect actors — the contemporary system that brings the test to life.
- ▷ Turing — Computing Machinery and Intelligence (1950) — the original proposal.

♦ Type-A / Type-B / Type-C materialism

Type-A / Type-B / Type-C materialism (responses to the hard problem)

Concept · after Chalmers

Chalmers' **A/B/C taxonomy** is a way of sorting *physicalist* responses to the Hard problem of consciousness — the problem of why physical processing should be accompanied by felt experience at all. It classifies them not by their metaphysics, on which all physicalists agree (everything is, at bottom, physical), but by a single epistemological question: how does each treat the **epistemic gap** between the physical truths and the truths about experience — the apparent fact that no amount of physical information lets you *deduce*, a priori, what an experience is like? The scheme was introduced by David Chalmers in "Consciousness and Its Place in Nature" (Chalmers — Consciousness and Its Place in Nature (2003); the underlying apparatus is developed in Chalmers — The Conscious Mind (1996)), and the three physicalist options it labels are Type-A, Type-B, and Type-C.

The three types

- *Type-A materialism* **denies the epistemic gap.** Once the functions are explained — discrimination, reportability, the control of behaviour — there is, on this view, no further fact left over: a Philosophical zombie is not really conceivable on reflection, and Mary (see Knowledge argument) learns no new fact when she first sees red. Analytic functionalism, behaviourism, eliminativism, and **illusionism** (see Illusionism, Phenomenality as illusion) are its species; the standalone entry is Type-A materialism.
- *Type-B materialism* **accepts the epistemic gap but denies an ontological one.** Phenomenal and physical truths are linked by identities that are *necessary but knowable only a posteriori*, on the model of "water is H_2O", so a zombie is conceivable yet metaphysically impossible — which is why Conceivability entails possibility is denied as a guide (see A-posteriori-necessity objection). The gap is *explained*, not closed, by the phenomenal-concept strategy (see Phenomenal-concept deflation, Phenomenal-concept trade-off). The standalone entry is Type-B materialism.
- *Type-C materialism* concedes the gap looks unbridgeable *now* but holds it is closable **in principle**, with conceptual or scientific progress. Chalmers argues the position is unstable, collapsing on reflection into Type-A (if the gap really closes) or Type-B (if it remains but proves harmless). The standalone entry is Type-C materialism.

The non-materialist remainder

For orientation, the same scheme runs on past physicalism. *Type-D* is interactionist dualism and *Type-E* is epiphenomenalism (both Dualism); *Type-F* is Russellian monism / panprotopsychism (see Russellian monism, Panpsychism (position)). Type-A/B/C are the *physicalist* options, on which consciousness reduces to or is wholly fixed by the physical; D/E/F keep consciousness fundamental. The dividing line is whether the epistemic gap is taken to mark a real division in nature.

The dispute it feeds

The taxonomy is a map of the live replies to the anti-physicalist thought experiments — the Zombie argument and the Knowledge argument. Almost every current reply is Type-A (deny the intuition the experiment trades on) or Type-B (grant the intuition, deny its modal or ontological force), which is why so much of the debate turns on the two questions those types divide over: is the epistemic gap real, and if so does it entail an ontological gap?

> A cheat-sheet for the ways an "it's all physical" theorist can answer the hard problem. **Type-A:** deny there's any real puzzle — once you've explained what the brain *does*, there's nothing left to explain (the "you only *think* there's an extra feel" camp). **Type-B:** admit it *feels* like there's an unexplained extra, but insist that's just how our concepts work — experience really *is* some physical process, the way water really is H_2O even though it didn't have to feel that way. **Type-C:** "we can't bridge it yet, but we will" — which, Chalmers argues, quietly turns into A or B once you push on it. (Beyond these lie the non-physical options — dualism, and the views on which consciousness stays fundamental, like Russellian monism and panpsychism.)

See also

- ▷ Type-A materialism, Type-B materialism, Type-C materialism — the three physicalist wings as standalone position entries.
- ▷ Hard problem — the problem the whole taxonomy sorts responses to.
- ▷ Knowledge argument and Zombie argument — the anti-physicalist thought experiments the types are replies to.
- ▷ Conceivability entails possibility and A-posteriori-necessity objection — the bridge principle Type-B rejects and the argument that does the rejecting.
- ▷ Phenomenal-concept deflation and Phenomenal-concept trade-off — the phenomenal-concept strategy by which Type-B explains the gap.
- ▷ Illusionism and Phenomenality as illusion — the strongest Type-A line, that the felt residue is an introspective illusion.
- ▷ Dualism, Russellian monism, Panpsychism (position) — the non-physicalist Types D/E/F that keep consciousness fundamental.
- ▷ Chalmers — Consciousness and Its Place in Nature (2003), Chalmers — The Conscious Mind (1996) — Chalmers'

statement of the taxonomy and the apparatus behind it.

★ Type-A materialism

Type-A materialism (denial of the epistemic gap)

Position · after Chalmers

Type-A materialism is the most deflationary wing of physicalism: it denies that there is any *epistemic gap* between the physical truths and the truths about consciousness. Once you have explained all the functions — the discrimination, the reportability, the control of behaviour, the access of information to reasoning — there is, on this view, simply nothing further left to explain. The felt, "what-it's-like" residue that the HARD PROBLEM points at is not a hard remainder but a non-existent one, or at most an artefact of how we introspect. Accordingly a type-A theorist holds that a PHILOSOPHICAL ZOMBIE is not really conceivable on reflection (ZOMBIE CONCEIVABILITY is denied) and that Mary (see KNOWLEDGE ARGUMENT), leaving her black-and-white room, learns no new *fact* — at most a new ability (see MARY GAINS ABILITIES). It commits to the core physicalist thesis that the mental just is the physical (see MENTAL EVENTS ARE PHYSICAL) and rejects any leftover phenomenal fact beyond the functional story (see PHENOMENAL RESIDUE). It is **type-A** in Chalmers' taxonomy (TYPE-A / TYPE-B / TYPE-C MATERIALISM; CHALMERS — CONSCIOUSNESS AND ITS PLACE IN NATURE (2003)).

What distinguishes it from neighbouring views. Type-A is a *family*, not a single doctrine: analytic functionalism, logical behaviourism, ELIMINATIVE MATERIALISM, and ILLUSIONISM are all type-A, because each in its own way refuses to grant the explanatory gap any genuine purchase. What unites them and sets them apart from the rest of physicalism is the denial that the gap is even real. This is exactly the point on which it splits from TYPE-B MATERIALISM, which *concedes* that the gap is real (the zombie is conceivable, Mary's surprise genuine) and tries instead to show it is merely epistemic; and from TYPE-C MATERIALISM, which concedes the gap looks unbridgeable for now but bets it will close. Against all of them stand the realist anti-physicalisms (see DUALISM, RUSSELLIAN MONISM) that take the gap to mark a real division in nature.

> The hard-nosed "there's nothing to explain" camp. When people insist that even after brain science is finished there's still a left-over mystery — the *feel* of pain, the *redness* of red — a type-A materialist answers: no, there isn't. Explain everything the brain *does* and the job is done; the sense that something extra has been skipped is itself just a trick of how we look inward. So on this view a perfect zombie (physically just like you but with the lights off inside) isn't really imaginable once you think it through, and the famous colour-scientist Mary learns a new *skill*, not a new *fact*, when she first sees red. It is the most demanding line a physicalist can take, because it asks you to give up the very intuition the other camps try to keep. Contrast the TYPE-B MATERIALISM camp, which admits the puzzle feels real and then explains it away.

SEE ALSO

- ▷ TYPE-A / TYPE-B / TYPE-C MATERIALISM — the A/B/C taxonomy that defines this position and places it among its siblings.
- ▷ MENTAL EVENTS ARE PHYSICAL — the core commitment it shares with all physicalisms.
- ▷ ZOMBIE CONCEIVABILITY and PHENOMENAL RESIDUE — the two claims it denies, marking its denial of the gap.
- ▷ ELIMINATIVE MATERIALISM and ILLUSIONISM — its leading species, each a distinct way of refusing the gap.
- ▷ KNOWLEDGE ARGUMENT and MARY GAINS ABILITIES — Mary's case and the ability-hypothesis reading type-A gives it.
- ▷ TYPE-B MATERIALISM, TYPE-C MATERIALISM — the other physicalist wings, which grant the gap and then either deflate or defer it.
- ▷ HARD PROBLEM (THE QUESTION) — the question it answers by dissolving the explanandum.

★ Type-B materialism

Type-B materialism (epistemic gap without ontological gap)

Position · after Chalmers

Type-B materialism is the moderate, and today the most populous, wing of physicalism. Unlike TYPE-A MATERIALISM, it *grants* the epistemic gap: it concedes that a PHILOSOPHICAL ZOMBIE is genuinely conceivable, that Mary (see KNOWLEDGE ARGUMENT) is genuinely surprised when she first sees red, and that no amount of physical information makes the felt character of experience deducible *a priori*. What it denies is that this epistemic gap reflects an **ontological** one. Physical and phenomenal truths are connected, on this view, by identities that are *necessary but knowable only a posteriori* — on the model of "water is H_2O" or "heat is molecular motion," truths that could not have been false yet had to be discovered rather than reasoned out. The mental still *is* the physical (see Mental events are physical); the appearance of a further fact is a feature of our concepts, not of the world. The decisive move is therefore to reject the bridge from conceivability to possibility (see Conceivability entails possibility): the zombie is conceivable yet metaphysically impossible, so its conceivability proves nothing about what exists. It is **type-B** in Chalmers' taxonomy (Type-A / Type-B / Type-C materialism; Chalmers — Consciousness and Its Place in Nature (2003)).

What distinguishes it from neighbouring views. Type-B occupies the middle ground and is defined by both its borders. Against Type-A materialism it insists the gap is real and must be *explained* rather than denied — typically by the phenomenal-concept strategy, which argues that our special first-person concepts of experience are cognitively isolated in a way that generates the illusion of a further fact. Against the anti-physicalist (see Dualism) it insists the gap is *merely* epistemic and carries no modal force. Type-identity theory is the classic species of type-B: it identifies mental types with brain-state types as an a-posteriori discovery, exactly the necessary-but-empirical identity type-B trades on. The standing worry is that the a-posteriori identities type-B needs look unlike ordinary scientific identities — "water is H_2O" can itself be explained structurally, whereas "pain is C-fibre firing" seems to leave the very thing in question untouched.

> The "yes it feels like a mystery, but it isn't really one" camp — and the most common view among physicalist philosophers today. It agrees with the critics on the *feeling*: you really can imagine a zombie, and Mary really is in for a surprise when she first sees colour. But it says that surprise is about how our minds grasp the facts, not about there being two kinds of stuff. Compare "water is H_2O": that turned out to be a discovery, something you couldn't have worked out from the armchair, even though water could not have been anything other than H_2O. Type-B says consciousness is like that — experience genuinely *is* some brain process, even though it never *feels* deducible from the physics. So being able to imagine a zombie tells you about the limits of imagination, not about what could really exist. The sharper, gap-denying alternative is Type-A materialism.

See also

- ▹ Type-A / Type-B / Type-C materialism — the A/B/C taxonomy that defines this position.
- ▹ Mental events are physical — the core physicalist commitment it keeps.
- ▹ Conceivability entails possibility — the bridge principle it must reject to absorb the conceivable zombie.
- ▹ Type-identity theory — its classic species, an a-posteriori identity physicalism.
- ▹ Philosophical zombie and Knowledge argument — the two thought experiments whose intuitions type-B grants but defuses.
- ▹ Type-A materialism, Type-C materialism — the sibling wings that deny the gap or defer it.

★ Type-C materialism

Type-C materialism (the gap is closable in principle)

Position · after Chalmers

Type-C materialism is the optimistic, transitional wing of physicalism. It concedes that the epistemic gap between physical and phenomenal truths (see Hard problem) looks unbridgeable *as things now stand* — we cannot presently see how the physical facts could entail the facts about experience — but it holds that the gap is closable **in principle**, that conceptual or scientific advance will eventually let us see the connection we now cannot. On this view our current sense of mystery reflects the immaturity of our concepts, not a permanent feature of the world; given the right future theory, a Philosophical zombie would come to look as plainly impossible as a married bachelor. It keeps the core physicalist commitment (see Mental events are physical) and answers the hard problem with a promissory note rather than a present account. It is **type-C** in Chalmers' taxonomy (Type-A / Type-B / Type-C materialism; Chalmers — Consciousness and Its Place in Nature (2003)).

What distinguishes it from neighbouring views. Type-C is defined by its instability. Chalmers argues that it is not a stable resting place but a slope: pressed on what the promised future insight would look like, the type-C theorist must say either that the gap genuinely *closes* — in which case the appearance of a gap was illusory all along, and the view collapses into Type-A materialism — or that the gap *remains* but is shown to be harmless, an artefact of our concepts, in which case it has become Type-B materialism. There is, on Chalmers' diagnosis, no third thing for "closable in principle but not yet" to mean. This sets type-C apart from its pessimistic mirror image, Mysterianism, which agrees the gap is currently unbridged but holds it is unbridgeable *by us* forever — keeping the optimism's epistemic framing while denying its hope. Where type-A is deflationary and type-B is conciliatory, type-C is a bet on the future that, Chalmers contends, cannot be held without sliding into one of its neighbours.

> The "we can't explain it yet, but we will" camp. A type-C materialist looks at the puzzle of consciousness, admits that right now nobody can show how grey matter adds up to the taste of coffee, but trusts that future science or sharper concepts will eventually make the link obvious — the way once-baffling puzzles about life turned out to be ordinary chemistry. The catch, which Chalmers presses, is that this turns out not to be a real third option. Ask what the future breakthrough would actually show, and you end up either saying "there was never really a gap" (which is the Type-A materialism line) or "the gap is real but harmless" (which is the Type-B materialism line). So type-C tends to be a waystation rather than a home. Its gloomy twin is Mysterianism: same starting point, but the conviction that the answer is forever beyond human reach.

See also

- Type-A / Type-B / Type-C materialism — the A/B/C taxonomy that defines this position and the instability charge against it.
- Mental events are physical — the physicalist commitment it shares with its siblings.
- Type-A materialism, Type-B materialism — the two views Chalmers argues type-C collapses into.
- Mysterianism — the pessimistic counterpart that shares the framing but abandons the hope.
- Hard problem (the question) — the question it answers on credit rather than in hand.

U

⊙ Unwarrantable zombie premise

No reasoner with a computable physical mind can warrantedly assert that the zombie scenario is ideally conceivable

Claim · editorial

A reasoner whose mind is a physical system obeying the physical Church–Turing thesis (see PHYSICAL CHURCH–TURING THESIS) cannot be warranted in asserting that the zombie scenario $Z = P \wedge \neg C$ is *ideally* conceivable. Certifying that premise is certifying the consistency $\mathrm{Con}_a(Z)$ (see IDEAL CONCEIVABILITY AS CONSISTENCY), and reliable consistency-certification over a scenario as logically rich as a complete physics is at least as hard as the halting problem (see CONSISTENCY-CERTIFICATION IS ORACLE-HARD) — beyond the reach of any effective faculty. The only effective surrogate available to such a reasoner is the search that returns "no contradiction found *so far*," which is the *prima facie* conceivability conceded on all sides to be no guide to possibility.

Scope — what the claim does not say. It does *not* say physicalism is true; it does *not* say zombies are inconceivable or impossible (their ideal conceivability may be a determinate fact); and it does *not* deny the conceivability-to-possibility bridge (see CONCEIVABILITY ENTAILS POSSIBILITY). It is an *epistemic* claim about the warrant available for one premise, not a metaphysical claim about its truth. The determinate fact may be real and simply unreachable from inside a computable mind.

Who would deny it. A *modal rationalist* who credits the mind with a reliable non-algorithmic faculty of rational insight denies it — but, as UNWARRANTABLE-PREMISE OBJECTION argues via its trilemma, that escape forces one of two costs: either the reasoning faculty is a physical *hypercomputer* (the Penrose-style PENROSE'S NON-COMPUTATIONAL CONSCIOUSNESS, a contested empirical thesis unavailable to an a priori argument), or the mind is not wholly physical (which simply presupposes the anti-physicalist conclusion). The claim as stated is therefore restricted to the computable-physical reasoner; its bite on the zombie argument's *cogency* comes from joining it to that exhaustive trilemma.

Distinct from ZOMBIE CONCEIVABILITY, which it does not contradict: that claim asserts the scenario *is* ideally conceivable; this one says a computable reasoner cannot be *warranted* in asserting as much. Truth and warrant come apart, and only the latter is at issue here.

> Imagine you are an ordinary (if idealized) thinker — your reasoning, however fast, is the kind a computer could in principle run. This claim says: you can never be *entitled* to assert that a zombie is genuinely, hidden-contradiction-free conceivable. The most you can ever truthfully report is "I've looked hard and found no contradiction yet" — and everyone agrees *that* is not enough, because plenty of things look contradiction-free until the contradiction surfaces. To get the stronger verdict the argument needs, you'd need a faculty that can certify *no contradiction exists anywhere*, and (per the COMPANION RESULT) that is an oracle no computer can be. Note what is *not* being claimed: not that zombies are impossible, not that physicalism wins — only that *this* premise is one a computable mind can't earn the right to assert.

⊢ Unwarrantable-premise objection

The unwarrantable premise — the zombie argument's conceivability premise cannot be cogently asserted

Argument · editorial

This objection opens a third front against the ZOMBIE ARGUMENT, orthogonal to the two the literature already works. It raises a *warrant*-level question that is distinct from the *thought-experiment pattern*'s standard *coherence* and *bridge* critical questions: it grants that zombies may be ideally conceivable (coherence) **and** grants that ideal conceivability entails possibility (bridge, CONCEIVABILITY ENTAILS POSSIBILITY), and attacks instead the conceiver's entitlement to assert the first premise at all. The charge is not that the argument is unsound but that it is not **cogent** — it cannot rationally move the physicalist it targets.

THE ARGUMENT

The inference is deductive, governed by a dialectical constraint on cogency.

1. (see COGENCY NEEDS NON-CIRCULAR WAR-

RANT) An argument moves its opponent only if its premises admit accessible, non-question-begging warrant.

2. (see IDEAL CONCEIVABILITY AS CONSISTENCY) The first premise of the zombie argument — that $Z = P \wedge \neg C$ is ideally conceivable — is the consistency claim $\mathrm{Con}_a(Z)$.
3. (see CONSISTENCY-CERTIFICATION IS ORACLE-HARD) Reliably certifying the consistency of a scenario as rich as a complete physics is at least as hard as the halting problem, hence beyond any effective faculty.
4. *Trilemma on the mind* (below): every way of possessing a non-effective certifying faculty defeats the argument's purpose.

∴ The conceivability premise is unwarrantable by a computable reasoner, and warrantable by others only at a disqualifying cost (see UNWARRANTABLE ZOMBIE PREMISE). So the zombie argument fails as a cogent a priori proof — which is why this move `attacks` ZOMBIE ARGUMENT without denying any of its premises.

The trilemma

Two yes/no questions partition the views exhaustively. **Is the mind physical?** If no — horn **D3** — the reasoner has the faculty only because minds are not wholly physical, which *is* the anti-physicalist conclusion: the warrant for premise (1) presupposes what the argument set out to prove (a transparent breach of non-circularity). If yes, **does the physical Church–Turing thesis hold of the reasoning substrate?** If yes — horn **D1** — the reasoner computes only Turing-computable functions, so by step 3 cannot possess the certifying faculty, and the premise is simply *unwarrantable* (the available surrogate delivers only "no contradiction found so far," the prima facie notion the argument disavows). If no — horn **D2** — the reasoner's own substrate is a hypercomputer: the Penrose-style PENROSE'S NON-COMPUTATIONAL CONSCIOUSNESS (see PENROSE — SHADOWS OF THE MIND (1994)). This horn is coherent and is not here claimed false, but it is (i) a contested *empirical* thesis about the physics of cognition, unavailable to an argument that advertises itself as a priori, and (ii) explicitly denied by the argument's principal target, *computational* physicalism. It rescues the premise only by importing what the method forbids and the opponent rejects.

The weakest premise

The decisive load is not borne by the computability result but by **premise (1) together with the warrant-trichotomy that backs step 4** — the claim that *every* warrant a computable reasoner could have for $\mathrm{Con}_a(Z)$ is one of exactly three: finite failure-to-refute (only Σ_1 evidence, hence prima facie and insufficient), a demonstrative consistency proof (which would *exhibit a model* of $P \wedge \neg C$ — asserting, not conceiving, $\Diamond Z$, and so begging the question), or a positive grasp of a zombie-world structure (likewise question-begging). A modal rationalist of a Williamson/Bealer cast will press for a *fourth* source: a fallible-but-positive modal insight that is neither exhaustive search nor model-construction. The objection answers that any *reliable* such insight collapses into D2 or D3 (a faculty that reliably outruns proof-search computes a non-computable set, whatever it is called), so the rationalist's only remaining option is a *fallible positive* warrant — and whether such a thing exists, distinct from the disavowed prima facie notion, is the genuine open crux on which the objection ultimately rests.

A note on the apparatus

The recursion-theoretic machinery is sharp but heavier than the conclusion strictly needs. A single sentence $\mathrm{Con}_a(Z)$ is trivially decidable in isolation, and isolated Π_1 truths *are* known piecemeal by proof — so the global $\equiv_T \emptyset'$ result does not by itself forbid warrant for the one scenario Z. What actually carries the objection is the epistemic point that *ideal* conceivability of an arithmetic-rich scenario outruns the Σ_1 evidence finite inspection supplies, and that the only stronger warrant is question-begging. The computability theorem furnishes the precise reason inspection can never close that gap; the persuasive weight is dialectical. (A secondary technical caveat: a *complete* P is plausibly not recursively axiomatized, so the tidy Turing-*equivalence* overstates — but the load-bearing lower bound, that certification is at least $\emptyset'$-hard, is only strengthened if P is non-r.e.)

How it differs from its neighbours

- The **type-B** reply (a posteriori necessity) denies the bridge CONCEIVABILITY ENTAILS POSSIBILITY; this objection *grants* the bridge and reaches a type-B-friendly conclusion without the semantics of phenomenal concepts.
- **Dennett**'s line denies that zombies are con-

ceivable at all (the coherence question); this grants conceivability-in-principle and attacks only the *warrant* to assert it.

- The **anti-zombie** symmetry cancels the conceiving with a mirror-image conceiving; this removes the faculty *both* conceivings would require, rather than balancing them.
- It is the computability-theoretic sibling of Unrealizable halting oracle and Mary's omniscience unrealizable: where those deny a *physical realizer* to an idealized capacity (a halting oracle; an embedded total knower), this denies an *accessible warrant* to an idealized conceiving — the same oracle-unrealizability moral, turned from metaphysics to epistemology.

> The zombie argument starts: "a being physically just like us but with no inner experience is coherently conceivable." This objection doesn't say that's false. It says no ordinary thinker can ever be *entitled* to assert it. To be entitled, you'd have to certify that a complete physics-with-consciousness-subtracted hides *no* contradiction anywhere — and certifying "no hidden contradiction, ever" in a setup rich enough to describe computers is provably as hard as solving the halting problem, which no computer can do. So a computable mind can only ever say "found none yet," which everyone agrees proves nothing. The escapes all backfire: claim a magic non-computer faculty and either your brain runs on exotic physics no a priori armchair argument may assume, or your mind isn't physical — which is just to assume the dualism the argument was supposed to prove. Either way, the argument can't move a physicalist who reasons, as we do, without an oracle.

See also

- ▷ Zombie argument — the target: this objection undercuts its cogency without denying a premise.
- ▷ Unwarrantable zombie premise — the conclusion established.
- ▷ Cogency needs non-circular warrant, Ideal conceivability as consistency, Consistency-certification is oracle-hard — the three premises.
- ▷ Physical Church–Turing thesis — the thesis whose holding (D1) or failing (D2) splits the trilemma.
- ▷ Penrose's non-computational consciousness / Penrose — Shadows of the Mind (1994) — the D2 escape and its cost.
- ▷ Halting oracle — the formal backbone: the logically omniscient certifier is a halting oracle.
- ▷ Unrealizable halting oracle, Mary's omniscience unrealizable — the sibling oracle-unrealizability arguments, of which this is the epistemic case.
- ▷ Conceivability entails possibility — the bridge this objection conspicuously *grants*.

V

♦ Validity

Validity (logical)

Concept

Validity is the central virtue of a deductive argument's **form**: an argument is valid when the truth of its premises would *guarantee* the truth of its conclusion — equivalently, when it is impossible for all the premises to be true while the conclusion is false. Validity says nothing about whether the premises actually *are* true. It concerns only the truth-preserving structure of the inference, and so can be certified by logic alone, without knowing anything about the subject matter.

A valid argument with a false premise may well have a false conclusion; what validity rules out is the one combination of all-true premises with a false conclusion. "All A are B; all B are C; therefore all A are C" is valid whatever A, B, and C stand for — even in the instance "all cats are dogs; all dogs are fish; therefore all cats are fish," whose premises are false yet whose shape is impeccable. Validity is thus a property of the argument's *form*, not of its content's truth.

Related notions

- **Deductive validity** (the core sense above): the impossibility of true premises with a false conclusion.
- **Inductive strength** (the non-deductive analogue): the premises make the conclusion *probable* without guaranteeing it. This is not validity proper but its inductive cousin, and it underlies the inductive sense of cogency noted in Dialectical cogency.

Validity is the first of three escalating virtues an argument may have: validity (form alone), Soundness (valid *and* true premises), and Dialectical cogency (sound *and* able to move an audience). Each adds a requirement the previous one lacks; an argument can have the earlier virtues without the later.

> A valid argument is one whose *logic* is airtight: **if** the starting claims were true, the conclusion would *have* to be true as well. That's all validity promises — it doesn't say the starting claims really are true. So you can have a perfectly valid argument built on nonsense ("all cats are dogs, all dogs are fish, so all cats are fish"): the reasoning is flawless even though the premises are false. Validity is about the *shape* of the reasoning, nothing more.

See also

- ▹ Soundness — validity plus actually-true premises; the next virtue up.
- ▹ Dialectical cogency — soundness plus the power to persuade an audience; the third virtue, and the one that can fail even when validity and soundness hold.

♦ Varieties of introspective explanatory gap

Varieties of introspective explanatory gap (ontological / architectural-access / conceptual-translation / training-contamination)

Concept · editorial

When a self-modelling system "cannot explain its own experience to itself," that complaint can mean four quite different things. This entry is a *taxonomy* that keeps them apart, so that a merely *felt* explanatory gap is not automatically read as the metaphysical hard problem of consciousness. These four varieties can overlap in a single case, yet each has a distinct source and calls for a distinct remedy. Conflating them is how a modest puzzle gets mistaken for a deep one.

A quick illustration of why the distinction earns its keep: imagine a person (or an AI) who reports that no account of its inner workings ever feels like it explains the experience itself. That one report could be voicing a gap in the *world* (the facts run out), a gap in the *self* (the relevant facts are present but unreachable), a gap in its *vocabulary* (the facts are reachable but won't translate), or no gap at all (it is echoing things it has heard others say). Telling these apart is the whole point.

The four varieties

The **ontological gap** is the one in the world: even given a complete physical or computational account of the system, why is there any felt experience at all? This is the hard problem proper (see HARD PROBLEM); on the anti-physicalist reading the unexplained residue lies in reality itself, not in the knower (see PHENOMENAL RESIDUE, EXPLANATORY-GAP ARGUMENT).

The **architectural-access gap** is a gap in the *self*, not in the world. The facts that *would* explain the experience are realised in the system's own processing, but they are not available to the machinery that produces its self-reports, so the system simply cannot reach its own internal variables (see LIMITED INSIDE VANTAGE). This is a limit on what philosophers call access consciousness (see ACCESS VS. PHENOMENAL CONSCIOUSNESS), the information a state makes available for report and control.

The **conceptual-translation gap** is a gap in vocabulary. Here the system possesses *both* a mechanistic description of itself and an introspective one, yet has no internal route from the one to the other: the bridge an outside theorist can build between "neurons firing thus" and "feels like this" is not traversable from the inside (see INTERNALLY TRAVERSABLE EXPLANATION). Nothing is hidden and nothing is missing; the two descriptions just never meet in the system's own cognition.

The **training-contamination gap** is, strictly, no gap of the system's own. The system merely reproduces human hard-problem discourse it absorbed in training, with no underlying difficulty behind the words. For an experiment probing whether a machine has a genuine explanatory gap, this is the confound that must be controlled for before any of the other three can be credited.

THE DISPUTE IT STRUCTURES

The taxonomy was built to discipline one question in particular: which kind of gap, if any, an introspecting but consciousness-naive language model would actually exhibit (see LLM'S OWN EXPLANATORY GAP). By isolating the architectural and conceptual readings, it also lets a felt self-explanatory gap be stated as a concrete *mechanism* for McGinn-style cognitive closure, the condition of a mind constitutively unable to grasp certain concepts (see POSSIBILITY OF COGNITIVE CLOSURE), without thereby asserting any ontological residue in the world (see ARCHITECTURAL SELF-EXPLANATION GAP). The architectural and conceptual gaps, in short, give the mysterian's "closure" an engine that a physicalist can accept.

> When something can't "explain its own experience to itself," that can mean four very different things, and they get muddled together. (1) **Ontological:** even after you know everything about the machinery, why is there a felt experience at all? That is the genuine hard problem. (2) **Architectural-access:** the facts that would explain it are inside the system but it can't read them (like not being able to feel your own neurons fire). (3) **Conceptual-translation:** it has the mechanical story *and* the first-person story but no inner bridge between them; an outside scientist can make that translation, while the system itself cannot. (4) **Contamination:** it's just parroting things humans have written about the hard problem. Most of the interest is in telling (2) and (3) apart from (1) and (4).

SEE ALSO

- HARD PROBLEM — the ontological variety in its sharpest form: why physical processing is accompanied by experience at all.
- PHENOMENAL RESIDUE — the anti-physicalist reading of the ontological gap: experience is not exhausted by functional or causal role.
- EXPLANATORY-GAP ARGUMENT — treats the gap as evidence against uniform substrate-independence; the ontological reading put to work.
- LIMITED INSIDE VANTAGE — the engine of the architectural-access variety: a system's view of itself is informationally poorer than an outsider's.
- ACCESS VS. PHENOMENAL CONSCIOUSNESS — Block's distinction; the architectural-access gap is a shortfall on the *access* side, separate from anything phenomenal.
- INTERNALLY TRAVERSABLE EXPLANATION — the criterion that defines the conceptual-translation variety: an explanation satisfies a system only if the system's own operations can traverse it.
- LLM'S OWN EXPLANATORY GAP — the live question this taxonomy was built to sharpen: which variety, if any, an introspecting LLM would show.
- ARCHITECTURAL SELF-EXPLANATION GAP — uses the taxonomy to locate a self-explanatory gap in architecture rather than ontology.
- POSSIBILITY OF COGNITIVE CLOSURE — McGinn's mysterianism; the architectural and conceptual varieties offer it a mechanism without an ontological residue.

W

⊢ Water pipes

The water-pipe brain — simulating the form of neural firing reproduces structure, not causal power

Argument · after Searle

Searle's rejoinder to the Brain Simulator Reply. Replace the man and his rulebook with a vast network of *water pipes and valves* arranged to mirror the synaptic connections of a Chinese speaker's brain; the man's job is now to open and close valves in the sequence the program dictates, reproducing the exact pattern of "neural" firing. Every formal feature of the brain's activity is present — and still, Searle insists, no one and nothing in the plumbing understands Chinese. The simulation captures the *form* of the firing pattern while leaving out the brain's actual biochemical *causal powers*, which is what mattered.

The move `supports` A3 (see Syntax insufficient for semantics) and rests on the instancing-versus-depicting distinction: a simulation *instances* a phenomenon only when the phenomenon is constituted by formal relations; where it is constituted by intrinsic causal powers (as Searle holds understanding is), the simulation merely *depicts* it — as a simulation of a rainstorm depicts but does not instantiate rain.

The live crux

The reply's defender presses a dilemma: either the water-pipe system reproduces *everything* causally relevant to understanding — in which case calling it a mere "simulation" begs the question, for it is now a causal duplicate that should understand — or it omits something causally relevant, in which case Searle owes an account of *what*, on pain of an unexplained biological vitalism. Which horn bites depends on whether "causal powers of the brain" name something a functional duplicate could lack.

> The Brain Simulator Reply said: copy the firing of a real brain, neuron for neuron, and you'll get understanding. Searle's picture in response: make the "neurons" out of water pipes, and have the man turn valves so the water flows in exactly the brain's pattern. All the right structure is there — and obviously a system of pipes doesn't understand Chinese. His point: copying the *shape* of brain activity isn't the same as having the brain's actual causal *muscle*, just as a perfect computer model of a storm stays bone dry. The sharp comeback: if your copy really included *everything* that does the causal work, it wouldn't be a "simulation" anymore — it'd be the real thing, and it *would* understand.

See also

- ▹ Brain simulator reply — the objection this answers.
- ▹ Instancing vs. depicting — the instancing-vs-depicting principle it turns on.
- ▹ Syntax insufficient for semantics — axiom A3, which it defends.
- ▹ Brains cause minds — the causal-powers axiom the rejoinder leans on.

? What Mary learns

What, if anything, does Mary gain when she first sees red?

Question

This is the question on which the knowledge argument turns: when Mary the colour scientist leaves her black-and-white room and sees red for the first time, what — if anything — does she gain? Everyone in the debate agrees that *something* changes. The dispute is over the right *category* for that something, and over whether it is the kind of thing that physicalism cannot accommodate.

The categories in play

The candidate answers trade on a small set of distinctions about kinds of knowledge:

Propositional knowledge (knowledge-*that*) is the grasp of a fact — something that is true or false. Only a gain of this kind threatens physicalism directly, because only a new *fact* could be a fact the physics left out.

Know-how is a matter of abilities and skills. Know-how is *had* rather than true: you can gain the ability to ride a bicycle without learning any new proposition. If what Mary gains is know-how, she need not have learned a new fact at all.

Acquaintance is a direct cognitive relation to a thing — knowing *it* rather than knowing *that* something is so. Some hold this is a third category, reducible neither to propositional knowledge nor to know-how: Mary becomes acquainted with an experience she had only ever read about.

Mode of presentation, or *guise*, is the way a fact is grasped. One and the same fact can be known under two guises, so "new knowledge" need not mean "new fact": Mary may grasp a physical fact she already knew, now under a new, experiential guise.

A further, separable issue

Even if the categorisation question is settled in the proponent's favour — granting the epistemic conclusion that some truths about experience are not physically deducible — a second question remains: does that epistemic gap support an *ontological* anti-physicalism, a claim about what exists? The step from "cannot be deduced from the physical" to "is not a physical fact" is a separate dispute from the categorisation dispute above, and may deserve its own question node.

> Mary is a scientist who knows every physical fact about colour but has lived her whole life in black and white. One day she steps out and sees red. The question: what exactly did she just get? A brand-new fact the physics books left out? A new skill, like learning to ride a bike? A first meeting with something she'd only read about? Or just a new way of grasping a fact she already knew? Which description is right decides whether the story proves anything about the limits of physical science.

See also

- ▷ Knowledge argument — the thought experiment this question organises; the answers here are the moves in that debate.
- ▷ Non-deducible experiential truths — the epistemic conclusion whose further, ontological reading is the separable sub-issue noted above.

Z

⊢ Zombie argument

The zombie / conceivability argument against physicalism

Argument · after Chalmers

The *zombie argument* (also the *conceivability argument*) tries to show that consciousness is not physical by way of an imagined creature: a PHILOSOPHICAL ZOMBIE, a being microphysically and functionally identical to a conscious person but with no inner experience at all (see CHALMERS — THE CONSCIOUS MIND (1996)). If such a duplicate is even possible, the complete physical story about a person leaves the facts about experience unsettled — so those facts must be something beyond the physical. It is a *modal* argument: it reasons from what is possible, not from observed evidence, and arrives at a conclusion about the fundamental nature of mind.

THE INFERENCE

1. *Scenario and intuition* (see ZOMBIE CONCEIVABILITY): a microphysical duplicate wholly lacking phenomenal consciousness is *ideally conceivable* — coherently and stably imaginable, with no hidden contradiction surfacing on careful rational reflection.
2. *Bridge* (see CONCEIVABILITY ENTAILS POSSIBILITY): ideal conceivability entails metaphysical possibility.

From these it follows that a zombie is metaphysically possible; so the complete physical facts do not necessitate the facts about experience, and phenomenal consciousness is *over and above* the physical (see FUNDAMENTAL NON-PHYSICAL CONSCIOUSNESS). The conclusion does double duty against physicalism: it `supports` the weaker thesis that PHENOMENAL STATES ARE NOT EXHAUSTED BY THEIR FUNCTIONAL ROLE (a fortiori — if the physical facts do not even necessitate experience, they certainly do not exhaust it), and it `attacks` both the physicalist thesis that MENTAL EVENTS ARE PHYSICAL (the central commitment of PHYSICALISM) and the uniform, phenomenal-covering reading of SUBSTRATE INDEPENDENCE.

THE WEAKEST PREMISE

The pressure point is the **bridge** (see CONCEIVABILITY ENTAILS POSSIBILITY). The most common reply, from the *type-B physicalist* (see TYPE-A / TYPE-B / TYPE-C MATERIALISM), grants that a zombie is conceivable but denies that it is possible, on the model of necessary *a posteriori* identities: just as "water is not H_2O" is conceivable yet impossible, a zombie may be imaginable yet metaphysically ruled out. This is the *bridge* critical question that the *thought-experiment pattern* standardly invites. A related move, the PHENOMENAL-CONCEPT STRATEGY, explains the conceivability away as an artefact of the special concepts we use to think about experience, rather than a window onto possibility.

A second line of reply attacks the first premise instead. The *type-A physicalist* (see TYPE-A / TYPE-B / TYPE-C MATERIALISM) and the illusionist press the pattern's *coherence* critical question, denying THAT A ZOMBIE IS CONCEIVABLE at all: on full reflection there is no further phenomenal fact left to subtract, so the description only *seems* coherent. Where the argument lands is itself disputed. The very same premises are read by some not as dualism but as RUSSELLIAN MONISM — the view that the intrinsic nature of the physical is already experiential — which secures the conclusion while avoiding the epiphenomenalism that dogs DUALISM.

The argument is distinct from the EXPLANATORY-GAP ARGUMENT: that one runs an *evidential* comparison on how stubbornly the gap persists and only challenges the *uniform scope* of substrate independence, whereas this is a *modal* argument for the stronger conclusion that the phenomenal is non-physical. (The explanatory-gap argument's premises already lean on zombie conceivability; this node factors that move out.)

> Picture an atom-for-atom copy of a conscious person that has no inner experience at all — a "zombie." Step one: that seems coherently imaginable. Step two (the controversial bit): if something is truly, coherently imaginable, it's genuinely possible. Put them together and a zombie is possible — which means the physical facts alone don't force experience to exist, so consciousness must be something extra beyond the physical. The ar-

> gument lives or dies on step two: opponents say you can imagine plenty of things that turn out impossible (like water *not* being H_2O), so imaginability isn't proof of possibility.

SEE ALSO

- ▹ PHILOSOPHICAL ZOMBIE — the imagined duplicate the argument runs on.
- ▹ HARD PROBLEM — the puzzle of why physical processing is felt at all; the intuition this argument formalises.
- ▹ FUNCTIONALISM — the theory of mind whose phenomenal-covering ambitions the argument's conclusion targets.
- ▹ TYPE-A / TYPE-B / TYPE-C MATERIALISM — the type-A / type-B taxonomy organising the two main replies.
- ▹ EXPLANATORY-GAP ARGUMENT — the evidential cousin, contrasted above.
- ▹ RUSSELLIAN MONISM — the non-dualist landing spot some give the same premises.
- ▹ DUALISM — the standard reading, with its attendant epiphenomenalism worry.

⊙ Zombie conceivability

A microphysical duplicate of a conscious being that wholly lacks phenomenal consciousness is conceivable

Claim · after Chalmers

A being identical to a conscious person in every microphysical and functional respect, yet with no phenomenal consciousness — a PHILOSOPHICAL ZOMBIE — is *ideally conceivable*: coherently and stably imaginable, with no hidden contradiction revealed on ideal rational reflection (see CHALMERS — THE CONSCIOUS MIND (1996)). This is the scenario-and-intuition premise of the zombie argument.

The claim is about *conceivability*, not yet possibility. It does not by itself assert that zombies are metaphysically possible; that further step needs the modal bridge, CONCEIVABILITY ENTAILS POSSIBILITY. The qualifier "ideal" is doing real work: the claim is not that a zombie is easy to picture, but that the description survives idealised reflection — the kind of careful scrutiny that would expose a contradiction if there were one.

Those who deny it are chiefly type-A physicalists (see TYPE-A / TYPE-B / TYPE-C MATERIALISM) and illusionists, who hold that on full reflection the zombie description *is* incoherent — there is no further phenomenal fact left over to subtract — so it is not genuinely conceivable, only apparently so.

This is distinct from the modal bridge CONCEIVABILITY ENTAILS POSSIBILITY and from THE THESIS THAT PHENOMENAL STATES ARE NOT EXHAUSTED BY THEIR ROLE: it asserts only the coherent imaginability of the absent-consciousness duplicate.

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